Company Histories and Scenario Simulation

Foundations, failure cases, and proposed methods

AccDSL research monograph

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This reading edition and the governing proposal use the same new chapter sources. Examples distinguish exact calculations, conditional assumptions and unimplemented methods.

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Chapter 1

The same statements can describe different companies

A financial statement describes selected consequences of a company's activities. A simulation must explain how those consequences can change. These are different tasks. A balance of borrowing tells us how much debt is reported under a particular measurement basis and date. It does not, by itself, tell us when debt resets, who may demand repayment, which assets secure it, or whether management can replace it. Two companies can report the same debt and interest expense and respond very differently to the same increase in market rates. Recovering one plausible journal does not solve this problem. Even a complete journal may omit the contractual fact that determines the response.

This part develops a representation of companies for which evidence is incomplete. Its purpose is useful conditional simulation without concealing what the evidence leaves unresolved. The numerical examples are synthetic teaching cases, not estimates of actual company exposures. They make a distinction visible: a calculation can be internally exact while its interpretation remains conditional on an unverified assumption.

1.1 A rate shock exposes the missing information

Consider a company with cash of 95 and debt of 100 at the forecast origin. Its recent annual interest expense was 5. Suppose debt was outstanding throughout that year, interest was paid when accrued, there were no fees or derivatives, and carrying amount equalled principal. A five per cent fixed-rate loan fits these observations. So does a floating-rate loan whose reference rate plus spread happened to equal five per cent. A mixture fits as well.

Let α be the floating fraction of principal. Assume a two-percentage-point increase applies to that portion for the entire next year, with no cap, floor, maturity or hedge. Next year's interest payment is

$$I(\alpha) = 100(0.05)(1-\alpha) + 100(0.07)\alpha \tag{1.1}$$

$$= 5 + 2\alpha, \qquad 0 \le \alpha \le 1. \tag{1.2}$$

The two terms are payments on fixed and floating portions. Historical observations admit every value of α in the interval. They imply interest between 5 and 7 under the shock, but supply no frequency distribution within that range. Uniform weights would be an additional judgment.

If the interest payment is the only cash movement, closing cash is

$$C(\alpha) = 95 - I(\alpha) = 90 - 2\alpha. \tag{1.3}$$

Every result satisfies $C + I = 95$. The marginal intervals $I \in [5, 7]$ and $C \in [88, 90]$ do not retain this relationship by themselves. The pair $(I, C) = (7, 90)$ lies in the rectangle formed by these intervals but is impossible in the company described. An interface may display intervals; the calculation must retain their common dependence.

Suppose a later note establishes $\alpha \in [0.2, 0.4]$. Then

$$I \in [5.4, 5.8], \qquad C \in [89.2, 89.6], \qquad C + I = 95. \tag{1.4}$$

The family has narrowed because evidence changed. This is not an economic event changing the contract. A reproducible historical forecast needs the earlier evidence family, while a current analysis may use the narrower one.

1.2 Journals and economic histories

For a simple interest payment of 5, a journal debits interest expense and credits cash by 5. Both fixed and floating histories can contain that posting. Storing alternative journals while omitting contracts can therefore collapse economically relevant uncertainty. The journal is essential for accounting consequences, but is not a complete description of rights, obligations and exposure.

Depreciation of 10 may arise from one asset or several vintages with different remaining lives. Equal inventory totals may conceal different expiry dates and replacement costs. Receivables of 50 may consist of five customers owing 10 each or one owing 50. A customer-specific disruption distinguishes these histories even when their published totals agree.

Write an economic history as h, its journal under accounting interpretation b as $J_b(h)$, and its report as $R_b(J_b(h), h)$. The second argument permits notes to reveal facts absent from the journal. Compatibility with report y means

$$R_b(J_b(h), h) \in \mathcal{O}(y), \tag{1.5}$$

where $\mathcal{O}(y)$ includes precision and context. A representation should preserve a joint family of histories, contracts, journals and interpretations. Independently choosing from lists of possible cash, debt and policies can create a company no coherent history supports. Shared evidence need not be copied for each alternative.

1.3 Agreement depends on the question

First restrict attention to histories compatible with the same retained reports. Within that family, two histories may agree under one specified simulation or under every permitted future decision. Agreement for the full set of decisions is stronger than agreement for one selected query. Equal next-year cash under one rate path does not establish equal cash after refinancing or early redemption.

For query q, scenario s and policy π, write the requested result as $F_{q,s,\pi}(h)$. Equality

$$F_{q,s,\pi}(h_1) = F_{q,s,\pi}(h_2) \tag{1.6}$$

permits shared calculation for that question if provenance remains distinct. It does not authorize reuse for a different horizon, shock or policy. When learning changes decisions, agreement for fixed action sequences may itself be insufficient. Later chapters include observation laws in the equivalence condition.

The inverse-problem treatment in Stuart (2010) explains why inference requires an observation relation and, for Bayesian conclusions, prior assumptions. We do not transfer its function-space theory wholesale to accounting. The relevant lesson is that a forward relation need not have a unique inverse. Possible-worlds approaches to incomplete data in Libkin (2014, 2016) provide language for answers holding across compatible interpretations. Neither identifies a company's actual transactions from totals alone.

1.4 Model completeness is a separate question

A model containing only fixed and floating unsecured loans may enumerate every feasible assignment while omitting convertible debt, leases and guarantees. Completeness within its variables does not establish completeness of the company description. An omitted instrument may dominate the stress response.

A family therefore needs a declared domain: entities, dates, instruments, bases, documents and event types. Its computational status must distinguish an exact characterization, a sampled subset and an unfinished search. Ten witnesses establish ten compatible cases, not that only ten cases exist.

Finite report compatibility also differs from statistical identification. Giacomini and Kitagawa (2021, Sections 2.1–2.3) studies observational equivalence of probability laws and set-identified parameters. Two loan histories fitting supplied reports need not induce the same law of every possible disclosure. Calling them globally observationally equivalent would overstate the evidence. Here they are report-compatible.

1.5 Selecting a convenient history

A fit criterion may choose the smoothest transaction path or the fewest contracts among equally accurate reconstructions. Such a selection can be useful for illustration. It does not turn smoothness or sparsity into an observed fact. It can erase the maturity concentration that makes a stress consequential.

If $\alpha = 0$ and $\alpha = 1$ both fit interest expense 5, a preference for fewer reset clauses may choose zero. It then predicts interest 5 under the shock; the omitted compatible case predicts 7. No historical residual selects the correct response. Repeating the fit does not resolve the missing information.

Bayesian model averaging gives a probabilistic alternative after models, priors and likelihoods are declared (Hoeting et al., 1999). Its weights are not counts of reconstruction rows. Duplicating a fixed-rate record ten times must not increase its probability. This invariance under representational duplication is a useful implementation test.

1.6 Mechanics and management responses

The rate illustration holds financing decisions fixed. A company may repay, refinance, hedge, sell assets or issue equity. Every response needs resources, counterparties and information. Automatically increasing debt to pay all obligations assumes financing capacity. Refusing all adjustment is also a policy choice, even when it is useful for stress analysis.

Shahnazarian (2004) separates financial identities from behavioral relationships. Caiani et al. (2016) specifies action sequences and financial counterparts in an agent-based economy. These ap-

proaches motivate separate accounting and behavior, but do not supply a universally transferable issuer policy. Responses must be estimated, supplied, bounded or left unresolved.

An output can distinguish historical explanations from future policies. One result may assume a floating share between 20 and 40 per cent and existing facilities only; another admits a stated refinancing offer. Averaging these without justified weights erases the question being answered. Conditional results remain informative even when an unconditional forecast is unavailable.

1.7 Trees are explanations, not compulsory storage

A tree displays mutually exclusive events, information arrival and subsequent decisions. It is useful for a small example. The continuum $\alpha \in [0, 1]$ already contains infinitely many compatible companies, so a finite selection of branches is generally an approximation. An interval and symbolic formula can represent the same uncertainty more faithfully and compactly.

Branches must not imply that a signed contract is redrawn each year. Fixed versus floating may be a persistent unknown fact. Future demand may instead generate fresh realizations under a declared stochastic process. A model must preserve the distinction.

Sometimes a formula suffices; sometimes a split at a covenant threshold is needed; sometimes filtering a probability distribution is justified. In difficult cases the result must retain an unresolved region or bounded approximation. None of these choices requires a separate stored company for every imaginable world.

1.8 A useful simulation explains its dependence

For each result a reader should be able to identify its evidence, remaining histories, contractual and behavioral assumptions, and computational status. Balancing a projected balance sheet is weaker than this requirement. The model must explain why cash is available, why a payment occurs and what changes the answer.

The remainder of this part develops reconstruction, learning, scenarios, financing, accounting transitions and tractable computation. Failure cases accompany the mechanisms they test. Their purpose is to expose a specific mistaken inference and its repair, not to imply that a finite catalogue anticipates every future error.

Chapter 2

Reconstructing joint admissible histories

The useful starting point for simulation is a description of what the evidence permits. That description must connect opening balances, transactions, closing balances, notes and contractual facts across time. Without these links, individually plausible entries can be jointly impossible. With unjustifiably strong links, the model can exclude real alternatives because an analyst supplied a convenient default. Reconstruction is consequently both an accounting problem and an inference problem.

2.1 An observation has a context

A reported amount comes with a concept, entity or consolidated group, period, units, measurement basis and precision. Cash at a date differs from cash generated during a quarter. Continuing operations differ from the whole group. Millions differ from currency units. These distinctions determine which equations may use the number; they are not annotations that can be restored after arithmetic.

Let observation o_i contain value y_i, context c_i, publication time p_i and source location ℓ_i. A reconstruction predicts $g_i(h, b; c_i)$ from history h and accounting interpretation b. If the source rounds to increment d_i using a declared rule, compatibility may take the form

$$y_i - \frac{d_i}{2} \le g_i(h, b; c_i) < y_i + \frac{d_i}{2}. \tag{2.1}$$

The half-open endpoint is justified only by the assumed rounding convention. Other tie-breaking rules, signs and reporting practices require a different observation set. The printed absence of decimal places does not establish an exact underlying value.

A debt balance reported as 100 million might therefore permit nearby amounts. A difference of 0.4 million may be immaterial for a broad exposure chart and decisive for a covenant lying inside the reporting interval. Precision should survive extraction so that the query can determine whether it matters. An arbitrary numerical tolerance chosen for solver convenience is different from reporting precision. The first controls computation; the second describes evidence.

2.2 Stocks constrain net movement

For opening balance B_0, signed movements m_k and closing balance B_1, the stock-flow identity is

$$B_1 = B_0 + \sum_{k=1}^{n} m_k. \tag{2.2}$$

It constrains the sum, not the individual events. Receivables increasing from 40 to 50 can reflect credit sales 60 and collections 50, or sales 110 and collections 100. Revenue disclosures narrow the alternatives, but returns, acquisition, currency, taxes and write-offs may add terms.

For an illustrative single-entity bridge, let A denote gross receivables, S_c credit sales, T_c tax amounts included in receivables, K collections, W write-offs, X currency movements and G acquired or transferred balances. Let L be the allowance and N net receivables. Then

$$A_1 = A_0 + S_c + T_c - K - W + X + G, \tag{2.3}$$

$$N_1 = A_1 - L_1. \tag{2.4}$$

Each movement has a defined period and scope. This is a decomposition for the example, not a universal reporting-standard prescription. If only net balances and revenue are observed, interpreting the change in net receivables as uncollected sales silently sets several terms to zero. A later credit stress then begins from a distorted exposure.

The repair is not to invent missing movements. Retain their domains, constrain them with relevant notes, and identify the quantities the evidence does not determine. A zero can be a verified amount, an explicit scenario assumption or an unsupported shortcut. The record must distinguish these meanings.

2.3 A family of compatible histories

Let z collect unknown historical amounts and categorical choices: transaction dates, principals, customers, asset vintages and accounting interpretations. Given evidence D and additional assumptions A, define

$$\mathcal{H}(D, A) = \{z : C_{\text{obs}}(z; D) \wedge C_{\text{acct}}(z) \wedge C_{\text{contract}}(z) \wedge C_{\text{time}}(z) \wedge A(z)\}. \tag{2.5}$$

Observation constraints reproduce the sources within precision. Accounting constraints preserve valid postings under a specified basis. Contract constraints preserve promises and exercise rights. Time constraints preserve sequence and persistence. Assumptions close parts of the model that the evidence leaves open.

This separation makes contradictions interpretable. An empty family may arise from extraction error, inconsistent reports, an overstrong assumption, an unsupported mapping or an impossible contract interpretation. It does not establish misconduct by the company. The reconstruction can be wrong.

Each constraint should retain its origin. A small conflicting subset may identify an annual movement table, an overlapping quarterly flow and a mistaken sign mapping. That is a focused repair task. Removing arbitrary constraints until a solver succeeds would make the result untrustworthy. Relaxation needs a reason, a changed observation model and an explanation of how answers depend on the relaxation.

2.4 Overlapping periods

If half-year revenue is 120 and first-quarter revenue is 50 under the same scope and basis, second-quarter revenue is 70. Adding both reports as independent events counts the first quarter twice. Reports constrain an underlying flow; they are not themselves transactions.

Let f_t denote flow on elementary intervals. A report for period P_i gives

$$y_i = \sum_{t \in P_i} f_t + \varepsilon_i. \tag{2.6}$$

Here ε_i denotes admitted rounding or discrepancy and is not automatically stochastic noise. Multiple reports can constrain the same f_t. Their source identities remain distinct even when their numerical information is redundant. The duplicate can independently confirm a value or reveal a later revision.

A reconstruction must also distinguish year-to-date arithmetic from seasonal interpolation. Deriving a quarter by subtraction is valid under aligned scopes. Dividing an annual flow by four assumes an intra-year pattern. A cash stress during a seasonal trough is sensitive to precisely that assumption. Even when annual totals are exact, the within-year timing may remain unidentified.

2.5 Joint margins conceal concentrated exposure

Suppose total debt is 100. Foreign-currency and domestic-currency debt are each 50. Debt maturing within a year and later debt are each 50. These two disclosures are different partitions of one debt population. They do not give four independent amounts.

Let x denote foreign-currency debt maturing within a year. Compatibility requires

	within a year	later	total
foreign	x	$50 - x$	50
domestic	$50 - x$	x	50
total	50	50	100

$$0 \le x \le 50. \tag{2.7}$$

A 20 per cent rise in the domestic-currency price of the foreign currency raises the domestic-currency value of foreign principal by 10. The additional repayment due within a year is $0.2x$, ranging from zero to 10. Matching both reported margins does not identify near-term cash exposure.

Choosing $x = 25$ selects a midpoint. Assuming independence also yields 25 in this symmetric example, but independence is an extra assumption about the joint exposure. A concentrated currency maturity wall is a case that this assumption can conceal. One scalar with bounds represents all compatible tables more faithfully than a list of arbitrary examples.

Additional dimensions increase the difficulty. Currency, maturity, collateral, lender and rate type may each have published margins. Their joint table can be large, and feasible completions may be numerous. Maintaining constraints on that table preserves the issue even when enumerating it is impractical. Later query-driven elimination can remove dimensions irrelevant to a specified result, provided it retains the relationships they induce.

2.6 Accounting and economic alternatives

Some uncertainty concerns events: whether a sale occurred, when collection happened or which asset was pledged. Other uncertainty concerns recognition and measurement: applicable basis,

estimates and policy interpretation. Their alternatives cannot be combined freely when premises conflict.

A debt cannot be both repaid and continuing to accrue its old contractual interest without a further event explaining the continuation. A historical carrying amount computed under one interpretation cannot be forecast under another for the unchanged instrument without an explicit transition. A presentation difference need not imply a different cash obligation.

Attach interpretation b to its premises and consequences. Evolve economic state under the scenario, then apply the selected accounting rules. Where the applicable interpretation remains unresolved, report conditional statements for each admitted basis. An average of incompatible book measurements is not automatically a valid financial statement under either basis.

2.7 Historical record and execution state

Future calculation rarely needs every historical event in active memory. It may need principal, accrued interest, reset dates, tax basis, asset age and outstanding rights. The evidence history remains important for provenance and later corrections. These are separate storage needs.

The evidentiary record should remain available and versioned. Execution state should contain quantities sufficient for future transitions. Compressing the latter need not delete the former. A derived state can point to supporting constraints and records, and many alternatives can share common events.

An event log alone is not an inference representation. If an event's occurrence is uncertain, a definitive log entry invents certainty. Conversely, constraints without explainable event witnesses make results difficult to inspect. The proposed representation retains symbolic conditions and constructs concrete witnesses for explanations, regression fixtures and counterexamples.

2.8 Feasible, infeasible and undecided

A solver may find a witness, prove no witness exists, or stop without deciding. An adapter may not support the instrument. These outcomes are economically different. A timeout cannot become infeasibility, and unsupported debt cannot become a zero liability.

For the question whether cash can fall below 20, a compatible witness with cash 18 establishes possibility. Failure to find one establishes safety only if the search is complete and successfully proves the relevant bound. A lower bound above 20 applies to the declared family, horizon and policies. Changing any of those premises can invalidate it.

Floating-point feasibility adds another distinction. A small residual is evidence about a numerical calculation, not necessarily a logical certificate. Exact rational examples help detect semantic errors. Larger models may require validated bounds, solver-specific certificates or an explicit undecided status. The method used to establish a conclusion belongs with the conclusion.

2.9 Reconstruction as a source-research guide

The family can identify useful missing evidence. In the currency-maturity example, another total-debt figure contributes little. A cross-classification can determine x and resolve near-term depreciation exposure. The question is which fact changes the decision, not which source is easiest to extract.

Without probabilities, candidate disclosures partition the family and give possible reductions in the answer set. With an admitted observation model, one may compute expected value of information. Assigning equal weights merely to obtain an expectation would confuse these two exercises.

A practical sequence is to construct the important joint constraints, locate uncertainty crossing a decision threshold, and inspect sources capable of resolving it. The source may not exist or may remain inaccessible. The process still identifies the economic consequence of the gap and prevents extraction volume from being mistaken for identification.

2.10 Provenance and multiplicity

Two records can support the same assertion, and one record can support several constraints. Provenance needs to preserve these distinctions without treating duplicate records as additional worlds. The semiring treatment in Green et al. (2007) concerns algebraic provenance for database queries; its value here is a disciplined account of how source contributions combine. It is not a probability assignment to economic histories.

If a source is withdrawn, a derived claim supported by another independent source may survive. If the withdrawn source supplied the only premise for a maturity date, the maturity should become unresolved. Recomputing only displayed totals can miss the effect on future liquidity. Dependencies should extend through contract state and policy eligibility to scenario results.

A stable identity for an economic fact is also different from the identity of a particular source assertion. Two filings can refer to the same loan with different labels. Merging them solely because the labels resemble each other risks combining different facilities; failing to merge verified duplicates can double debt. Entity and instrument reconciliation therefore needs evidence and reversible decisions.

2.11 A method proposal for AccDSL

The proposed reconstruction service accepts an evidence view, a declared domain and an assumption set. It returns a persistent constraint family, its known contradictions or unresolved mappings, and optional witnesses. Queries request possible outcomes, certain conclusions, bounds or distributions under separately admitted measures. A request for one witness should not trigger exhaustive enumeration.

Constraints should be grouped by their source and economic role. Observation intervals and unit conversions can often be checked before expensive solving. Linear bridges and discrete interpretation choices can then be combined, with nonlinear valuation isolated when possible. If a nonlinear method is needed, its completion status remains distinct from the accounting identities it evaluates.

The service should preserve both a stable source snapshot and a reusable symbolic family. A later disclosure can produce a refined version without overwriting the earlier one. A scenario uses a particular version and records which assumptions it adds. This is a concrete way to carry multiple possible journals and economic histories without creating a full independent database for each possibility.

Chapter 3

Learning, persistent uncertainty, and who knows what

A future disclosure may resolve an uncertainty about today's company. It may also reveal nothing relevant. Simulation must allow both outcomes. It must distinguish a change in the company from a change in the analyst's knowledge, and distinguish the analyst's knowledge from information available to managers, lenders and investors. These distinctions determine which actions are possible and which apparent forecasting successes are merely uses of future information.

3.1 Five dates attached to one fact

A loan can be signed on 15 January, take economic effect on 1 February, appear in a report covering the quarter ending 31 March, and become public on 10 May. A system may ingest the report on 12 May. The signature, economic, reporting, publication and ingestion dates answer different questions. A forecast made on 30 April cannot use the 10 May disclosure simply because it describes a March balance.

For a report-based forecast origin t, let D_t contain assertions available by t under the declared availability rule. If the analysis uses public information, publication time is relevant. If it models a lender with private contractual access, the information rule differs. Ingestion time controls reproducibility of the system's own operation; it need not equal the earliest public availability.

A historical replay should therefore specify both the economic date and evidence cutoff. A corrected extraction performed today may accurately recover a number that was publicly available before the old cutoff. A new restatement published today was not available then. The first can repair a data-processing mistake in an old evidence view; the second changes the information set and must not be silently backfilled into the old forecast.

3.2 Persistent unknown facts

Let θ denote a contract characteristic fixed at inception, such as whether a loan is fixed or floating. Let X_t denote changing economic state, including cash, accrued interest and remaining principal. A simple transition is

$$X_{t+1} = T(X_t, \theta, a_t, \varepsilon_{t+1}), \qquad \theta_{t+1} = \theta_t. \tag{3.1}$$

Here a_t is a decision and ε_{t+1} a future innovation under the specified scenario or stochastic process. The second equation is essential. Uncertainty about a fixed contract does not license redrawing it each period. Refinancing or amendment can change the terms, but those are dated events with their own permissions and cash consequences.

Suppose a loan is either fixed at five per cent or floating at the market rate, and the market rate is seven per cent for each of two future years. The possible interest sequences are $(5, 5)$ and $(7, 7)$. Independent yearly resampling adds $(5, 7)$ and $(7, 5)$. Those additional paths describe contract changes that the model did not authorize. Their inclusion can alter liquidity minima, covenant duration and refinancing behavior.

The same persistence issue affects customer identity, asset vintage, tax basis, priority and collateral. A compact state must retain persistent variables even if current totals do not depend on them. Eliminating a variable because it is irrelevant today is unsafe if it changes later observations or events.

3.3 Set-valued evidence updates

When no probability measure is justified, retain a feasible family $\mathcal{H}_t$. If a new compatible hard observation y supplies condition C_y, updating the same historical scope gives

$$\mathcal{H}_t^+ = \{h \in \mathcal{H}_t : C_y(h)\}. \tag{3.2}$$

This operation narrows the old family. It does not advance the economic clock. Forward propagation then maps each remaining history into future possibilities under the transition relation. A smaller historical family can still produce a wider future range when the horizon lengthens or new shocks are admitted.

An empty intersection calls for investigation. A source may conflict with another source, the observation mapping may be wrong, or a model assumption may be too narrow. Selecting one convenient history to restore nonemptiness would hide the contradiction. The result should identify the conflicting premises and preserve the previous view.

A restatement is not always an additional constraint on the old asserted amount. It may replace that assertion in a current evidence view. The system should preserve the original source and link the correction to it. Historical replay can then reproduce the original view, while current reconstruction uses the revised one. Conjoining both as exact truths would manufacture infeasibility.

3.4 Probabilistic filtering requires a model

If a prior measure μ_t and likelihood $L(y \mid h)$ are admitted, Bayesian updating has the form

$$\mu_t^+(dh) = \frac{L(y \mid h)\mu_t(dh)}{\int L(y \mid u)\mu_t(du)}. \tag{3.3}$$

The denominator must be finite and positive. Its role is normalization, not an opportunity to substitute an arbitrary fallback when the observation has zero modeled probability. A zero denominator can reveal a misspecified likelihood, an impossible observation under the model or inappropriate handling of a continuous conditioning event.

For continuous exact observations, ordinary density multiplication must be understood relative to the declared dominating measure and conditional distribution. Adding a tiny Gaussian error

can make a computation run, but it changes the observation model. Its scale should reflect a justified measurement or reporting process, or be explicitly treated as a sensitivity assumption. Numerical convenience does not establish the company's measurement error distribution.

The POMDP formulation in Kaelbling et al. (1998, Sections 3.1–3.4) provides a formal treatment when transition and observation probabilities are known. Its belief state summarizes the history needed for control under those assumptions. An unweighted feasible family is not a probabilistic belief state merely because it is stored in the same data structure. Supporting both requires distinct semantics.

3.5 A finite filter that preserves contract identity

Let θ be an unknown but fixed contract type, x_t a changing hidden operating state, and y_t the analyst's observation. A declared probabilistic model supplies transition probabilities $P_\theta(x' \mid x)$ and observation likelihoods $\ell_t(y \mid \theta, x')$. With joint posterior weights $w_{t-1}(\theta, x)$, the prediction and update are

$$w_t^-(\theta, x') = \sum_x P_\theta(x' \mid x) w_{t-1}(\theta, x), \tag{3.4}$$

$$Z_t = \sum_{\theta, x'} \ell_t(y_t \mid \theta, x') w_t^-(\theta, x'), \tag{3.5}$$

$$w_t(\theta, x') = \ell_t(y_t \mid \theta, x') w_t^-(\theta, x') / Z_t. \tag{3.6}$$

There is no summation that changes θ into another contract type. A contract modification needs an explicit event and a different transition model. If $Z_t = 0$, the result is an evidence conflict under the declared model; normalizing arbitrary fallback weights would answer another question.

An uninformative observation has constant likelihood across states, so its update does not change the predicted relative weights. A missing disclosure is uninformative only when the observation model makes it so. If no prior is justified, propagate a feasible set of joint (θ, x) pairs through the transition relation and evidence restrictions. That procedure produces support, not posterior probabilities. A declared set of priors can instead be updated member by member; zero-likelihood members need an explicit conditioning rule, and one must not replace the result by independently chosen period marginals.

A source correction can remove an earlier restriction. Retaining only the last posterior may then be insufficient: a zeroed type cannot be recovered by multiplying it by another likelihood. Keep the original prior or feasible family, the evidence versions and the transition model, and replay all currently active evidence. Checkpoints can accelerate replay only when their evidence dependencies remain valid. The reference filter compares this recursion with direct path enumeration for four priors and all 27 length-three observation sequences. It also checks rare positive types, uninformative observations, conflict, and a correction that reopens a discarded type. This is an exact finite method; estimating a defensible issuer likelihood and controlling particle approximation remain separate problems.

3.6 An informative and an uninformative disclosure

Return to the fixed-versus-floating loan. Assume, only for this example, prior probabilities of one half each. A report can say "floating" or "fixed." Suppose the report identifies the correct type with probability 0.9 and the wrong type with probability 0.1, independently of other

modeled quantities conditional on the type. After a "floating" report, Bayes' rule gives

$$\Pr(F \mid y = F) = \frac{0.9(0.5)}{0.9(0.5) + 0.1(0.5)} = 0.9. \tag{3.7}$$

These probabilities are supplied assumptions, not estimates from the loan balance. The example isolates the mechanics of updating. If the report were equally likely under both types, its likelihood would cancel and the prior would remain unchanged. More pages of disclosure need not imply more information about the exposure of interest.

In the set-valued version, an exact contract note may eliminate one type. An ambiguous phrase compatible with both eliminates neither. A probabilistic false-positive model is a different treatment from a set of possible interpretations. The analyst should choose the treatment supported by the evidence process and preserve its assumptions.

3.7 Why some ambiguity does not disappear

A long sequence of reports can remain uninformative about a parameter. In the rate example, if fixed and floating rates coincide throughout the observed period, additional interest totals may never distinguish the contract types. Sample size alone does not resolve an unidentified direction.

The set-identification analysis of Giacomini and Kitagawa (2021, Sections 2.1–2.3) separates parameters identified by the observable law from aspects of the prior that data cannot revise. The accounting application is conditional: one must first establish which observations depend on which hidden quantities. It is not enough to invoke a general consistency slogan and expect all uncertainty to vanish.

Persistent ambiguity can also become irrelevant to one query without being resolved. After final repayment, the old rate clause may no longer affect future cash. It may still matter for explaining historical interest, estimating management behavior or evaluating a counterfactual refinancing decision. Removing it from an active cash-flow state is different from asserting that the evidence identified it.

3.8 Managers need not share the analyst's ignorance

Public disclosures may leave θ unknown to an analyst even though management signed the contract. A management policy can legitimately depend on that private fact. If managers refinance only floating-rate debt after a rate shock, their actions differ across histories that the analyst cannot distinguish. That is not foresight; it is a specified information advantage.

Let I_t^a denote actor a's information history and let π_t^a map it to a decision. Then

$$a_t^a = \pi_t^a(I_t^a). \tag{3.8}$$

Two simulated worlds with the same I_t^a must receive the same action under a deterministic policy. For a randomized policy, they must receive the same conditional action distribution. The simulator's access to hidden state does not entitle the actor to use it.

Different actors can have different information. A lender may observe covenant certificates before public investors. A subsidiary may know its operating cash before a parent receives a report. A regulator may receive nonpublic prudential data. The model should specify these information channels where they affect behavior; assuming universal knowledge can make financing appear available too early.

3.9 A simple foresight counterexample

A manager must choose at date zero between two production plans. Plan A yields 10 if next year's demand is high and zero if low. Plan B yields 6 in either state. Assume high and low demand are equally likely and the manager learns demand only after choosing. The expected values are 5 and 6, so the expected-value policy selects B.

A simulator that first draws demand and then chooses the best plan in that realized world obtains 10 in the high state and 6 in the low state, averaging 8. It has valued perfect advance information. The arithmetic is correct for a different decision problem. The target problem permits an average of 6.

The discrepancy is not repaired by drawing more scenarios. It is caused by an information violation. A nonanticipativity check must require the same date-zero action across worlds sharing date-zero information. In a tree formulation this often appears as equality constraints on decisions at indistinguishable nodes. In a symbolic representation it can be enforced by restricting the policy's inputs.

3.10 Signals can have economic value

Suppose the manager can pay 1 at date zero for a perfectly informative demand signal before selecting A or B. With the signal the gross expected payoff is 8 and the net payoff is 7. Without it the payoff is 6. The signal is worth 2 gross and 1 net in this example. These calculations use a declared distribution and feasible action set.

The same logic applies to obtaining a debt schedule before committing liquidity, but real disclosures may be noisy, delayed or nonpublic. A signal's value depends on whether the actor can act after receiving it and whether the relevant action remains feasible. A maturity note arriving after repayment cannot improve that repayment decision, even if it improves a retrospective explanation.

Without probabilities, one can compare worst-case payoffs or regret before and after a partition of the family. The result is not an expected value of information. An interface should name the criterion because different criteria can recommend different research priorities.

3.11 Simulated learning does not revise actual evidence

A scenario may include a hypothetical disclosure at a future date. Inside that run, an actor updates its information and changes its action. The hypothetical report must not enter the analyst's actual evidence store as a fact. Otherwise running a scenario can contaminate later reconstructions with imagined information.

Separate identifiers should distinguish actual source assertions, simulated observations and analyst assumptions. A scenario observation can refer to the source model that generated it and to the hidden state of that run. It cannot acquire the authority of a filed report merely because its text looks like one.

The distinction also matters for conversational systems. An explanation generated from a selected witness should say that it is one compatible history. If the explanation is later ingested as evidence without its conditional status, a possible event becomes a purported fact. Provenance and type checks should prevent that feedback.

3.12 Learning can change the appropriate compression

Two histories may have equal current cash and equal next-period payments, yet differ in the chance of a revealing report. If the report affects a later refinancing choice, their future decision values can differ. A merge based only on current accounting totals discards this difference.

Suppose two contract states both pay 5 next year. In one state, a scheduled certificate will reveal a restrictive covenant before the next refinancing date; in the other, no such information arrives. If management or lenders condition their offers on that certificate, equal immediate cash does not imply equal future financing. The observation process belongs in the transition model used for compression.

This is why exact aggregation later requires equality of the joint law of successor state, observations and requested outputs for each allowed action. Preserving a marginal cash distribution is too weak. A graph may share current arithmetic while retaining separate information states for future decisions.

3.13 Recursive ambiguity and persistent contracts

Bayesian conditioning and a fresh worst-case choice at every date answer different questions. Suppose a persistent binary type θ is either zero or one, and tomorrow a fair binary signal X unrelated to that type is observed. The admitted laws are Q_0 and Q_1: under Q_i, $\theta = i$ with certainty and X remains fair. For the loss $Z = \mathbf{1}\{X = \theta\}$, either law gives expected loss $1/2$, so

$$\max_{Q\in\{Q_0,Q_1\}} \mathbb{E}_Q[Z] = \tfrac{1}{2}, \qquad \mathbb{E}_X\left[\max_{i\in\{0,1\}} \mathbb{E}_{Q_i}[Z \mid X]\right] = 1. \tag{3.9}$$

The second expression selects type zero after signal zero and type one after signal one. It therefore evaluates a new law with $\theta = X$. That law is absent even from the convex hull of the original two laws: every mixture retains independence between the signal and type. No type has physically changed along a realized path, yet dependence has changed across possible paths. Preserving path support alone would miss the error.

Shapiro defines rectangularity through equality between static worst-case evaluation and its nested conditional counterpart (Shapiro, 2016, Definition 2.1, Theorem 2.1 and proof, Section 5). His finite construction pastes conditional distributions across histories; it can enlarge the family. A dynamic program for that enlarged family solves a changed robustness question. Its convenience cannot justify claiming that it solved the original persistent-type problem.

Terminology also requires care. Pichler and Shapiro call a product family a rectangular setting, but explicitly show that its original worst-case functional and its composite conditional functional can differ (Pichler and Shapiro, 2021, Section 4.2, Example 4.7). Computing a conditional law-based risk measure can also differ from conditioning an unconditional ambiguity set. Take six equiprobable states and an observed three-state group with losses $0, 1, 2$. Conditional expected shortfall at level $1/2$ within that group is $5/3$. The unconditional six-state risk envelope permits probabilities at most $1/3$ per state. It can assign $1/6$ to the loss-2 state, zero to the other two group members, and $5/18$ to each state outside the group. This is an admissible probability law; after conditioning on the group it assigns probability one to loss 2. The conditional worst case over that envelope is therefore 2. The difference is between two precise conditional risk definitions, not an arithmetic error (Pichler and Shapiro, 2021, Example 3.2 and equations (3.1)–(3.5)).

Policy consistency and equality of risk values are distinct as well. Xin and Goldberg give moment-constrained inventory examples in which an initially optimal policy ceases to be

optimal after reoptimization, and another in which the optimal policy sets agree although the two optimal values differ (Xin and Goldberg, 2021, Definitions 3.1–3.3, Observations 5–7, proofs in Sections 6.3–6.5). The company model must specify whether management can commit to a policy, what information each later decision uses, and which family of joint laws is retained. For a finite reference problem, enumerate complete admissible laws and observation-measurable policies before choosing an approximation. For larger problems, a set of beliefs over persistent types or an explicit precommitment formulation can retain those restrictions, but its recursive sufficiency and computational cost need a separate derivation. Repeated worst-case optimization remains useful as a stress test when the changed family is stated explicitly.

3.14 Persistent parameters and particle approximation

A particle approximation can reduce computation after a probability model is chosen. It does not create that model. A type with probability 0.01 has probability $0.99^{100} > 1/3$ of disappearing from 100 independent multinomial resampling draws. No new evidence is involved in this loss. Rare but material types therefore need explicit support checks, stratification or separate conditional computations, with the resulting approximation stated.

For state-space models with persistent unknown parameters, SMC^2 combines parameter particles and state filters, with particle-MCMC rejuvenation (Chopin et al., 2013, Section 3.2, equation (7), Proposition 1, Appendix A). Its extended target has the desired parameter posterior as a marginal under the source assumptions. Finite particles still approximate that posterior; exactness of a target marginal does not mean that one finite run represents every possible type. The unbiased full particle likelihood estimate used in this construction must be distinguished from an arbitrary estimated likelihood ratio.

The inspected author-linked modern implementation creates a state filter for each parameter particle, runs that filter through the current observation date when evaluating a proposed parameter, and incorporates a likelihood correction when increasing the state-particle count. These operations have real time and memory costs. Retaining only filtering arrays differs from retaining complete genealogies. This is a candidate for explicitly probabilistic filtering, not a replacement for symbolic unweighted historical ambiguity.

3.15 Changing particles without changing the target

Increasing the number of particles can change an inference result even before new evidence arrives. The reason is that a likelihood estimate and its random construction belong to the state used by particle MCMC. Replacing the estimate with another unbiased estimate does not automatically preserve their joint distribution. A test that examines only current company-type probabilities can miss the error.

Fix an evidence revision v and a persistent type θ. Write its prior as $p(\theta)$ and likelihood as $L_v(\theta)$. A particle representation of size N uses random variables u with law $m_N(u \mid \theta)$ and produces a nonnegative unbiased estimate $Z_{N,v}(\theta, u)$ of that likelihood. Assume the normalizing constant C_v is finite and positive. The joint target weights the random construction by the estimate it produces:

$$\widetilde{\pi}_{N,v}(\theta, u) = \frac{p(\theta) m_N(u \mid \theta) Z_{N,v}(\theta, u)}{C_v}. \tag{3.10}$$

Integrating over u replaces $Z_{N,v}$ by its expectation L_v, giving the desired type posterior. This is the extended-target argument in the retained SMC^2 source, arXiv 1101.1528v3, 27 January 2012 (Chopin et al., 2013, equation (7), Proposition 1 and Appendix A). Unbiasedness establishes

this marginal relation; it does not make an arbitrary finite particle population equal to the posterior.

Now change representation from N to M and draw fresh randomness u' from $m_M(\cdot \mid \theta)$. The augmented proposal law is $\widetilde{\pi}_{N,v}(\theta, u)m_M(u' \mid \theta)$. With backward law $m_N(u \mid \theta)$, the desired augmented target is $\widetilde{\pi}_{M,v}(\theta, u')m_N(u \mid \theta)$. Their ratio is

$$w(\theta, u, u') = \frac{\widetilde{\pi}_{M,v}(\theta, u')m_N(u \mid \theta)}{\widetilde{\pi}_{N,v}(\theta, u)m_M(u' \mid \theta)} \tag{3.11}$$

$$= \frac{Z_{M,v}(\theta, u')}{Z_{N,v}(\theta, u)}. \tag{3.12}$$

The prior, normalizer and auxiliary proposal laws cancel because the type and evidence revision are unchanged. This recovers the exchange correction in Section 3.6.1 of the inspected source. The ratio requires the relevant support conditions. The following example uses strictly positive estimates, so it does not hide a zero-denominator problem inside that cancellation.

Take two types with prior probability $1/2$ each and true likelihood one. For type zero the old estimate is $1/2$ or $3/2$ with equal probabilities; the new estimate is $1/4$ or $7/4$, again equally likely. Type one's estimate is always one. Each estimator is unbiased. Order the new joint states as $(0, 1/4), (0, 7/4), (1, 1)$. The likelihood-weighted target in equation (3.10) and the distribution produced by unweighted replacement are respectively

$$\pi = \left(\frac{1}{16}, \frac{7}{16}, \frac{1}{2}\right), \qquad q = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right). \tag{3.13}$$

Both assign probability $1/2$ to type one. Immediate marginal agreement therefore passes a test that the joint target fails.

To see the consequence, make one valid particle marginal Metropolis–Hastings move. Propose the other type, draw its new estimate, and accept with $\min(1, Z_{\text{proposed}}/Z_{\text{current}})$. From $(0, 1/4)$ the proposal is accepted certainly. From $(0, 7/4)$ it is accepted with probability $4/7$. From type one, the low-estimate proposal has total accepted probability $(1/2)(1/4) = 1/8$, and the high proposal has probability $1/2$. The transition matrix, with the same state order, is thus

$$K = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 3/7 & 4/7 \\ 1/8 & 1/2 & 3/8 \end{pmatrix}. \tag{3.14}$$

The correct joint law satisfies $\pi K = \pi$ and detailed balance. Starting instead from q, the probability of type one after the move is

$$(qK)_3 = \frac{1}{4} + \frac{1}{4}\frac{4}{7} + \frac{1}{2}\frac{3}{8} = \frac{65}{112} = \frac{1}{2} + \frac{9}{112}. \tag{3.15}$$

The transition is valid for its stated target; the wrongly initialized joint distribution is the problem. This is an exact one-step counterexample, not a claim that a valid PMMH kernel has the wrong stationary distribution. Applying the exchange weight restores the desired joint law in the finite enumeration.

The inspected author-linked implementation includes the exchange correction (Chopin and contributors, 2026, `SMC2.exchange_step`, pinned source). Its new filter is run through the preceding observation before the next incremental weight is added; processing that next observation in both places would count it twice. Rejected Metropolis proposals retain their existing filter state, and repeated resampled selections receive distinct mutable filter copies. These are scoped source-code correspondences, not a run of the full modern package or a performance comparison. The modern waste-free extension requires its own dependence and finite-run analysis.

3.16 A correction can restore an excluded history

Changing the particle representation preserves the likelihood being computed. Correcting a historical disclosure can change that likelihood itself. Suppose two histories have prior probabilities $(1/2, 1/2)$. An old hard observation has likelihoods $(1, 0)$, producing posterior $(1, 0)$. Its replacement has likelihoods $(1/4, 3/4)$. Recomputing from the prior gives

$$\pi_{\text{replacement}} = (1/4, 3/4). \tag{3.16}$$

Multiplying the restricted old posterior by the replacement likelihood and normalizing instead gives $(1, 0)$. The operation conjoins the old observation with the new one. It cannot recover the history the old assertion excluded. The new-to-old likelihood ratio is also undefined on that restored history, where the denominator was zero.

A replacement therefore requires replay from a sufficiently informative preserved checkpoint, or a separately justified bridge with the required support. The inference request must identify the evidence revision and which assertion is replaced. Merely attaching a corrected document to an existing particle population does not establish the corrected posterior. This is the probabilistic counterpart of retaining alternative histories that become admissible after a source correction.

The source's quantitative guarantees require additional premises. Its Appendix B uses uniform transition and observation-density ratio bounds and posterior regularity conditions, and its refreshed-particle idealization is stronger than a finite PMCMC move (Chopin et al., 2013, Proposition 2 and Appendix B). A permanent deterministic type, a hard zero-likelihood accounting observation or an unidentified parameter direction can violate the intended transfer assumptions. Exact finite target calculations establish the distinctions above; they do not establish company-model mixing, scaling, likelihood identification or forecast accuracy.

3.17 Replaying a declared evidence vintage

The preceding correction becomes an executable calculation once the inference request says exactly which information remains active. A document identifier alone is insufficient. Two documents may repeat one observation, and a new document may replace an earlier assertion rather than supply a second independent measurement. The model must retain these relations before it multiplies likelihoods.

Consider a finite set of persistent company types $h \in H$, with declared prior masses $p(h)$. Each active observation occupies a named slot and supplies a likelihood $L_j(h)$. We assume that these observation blocks are conditionally independent given h in this example. Under that assumption, for a selected evidence vintage v with active slots A_v,

$$Z_v = \sum_{h \in H} p(h) \prod_{j \in A_v} L_j(h), \qquad \pi_v(h) = \frac{p(h) \prod_{j \in A_v} L_j(h)}{Z_v}. \tag{3.17}$$

The normalizer must be positive. A zero value means that this finite model assigns zero likelihood to the selected evidence; it does not justify returning an arbitrary distribution. Missing prior probabilities or likelihoods instead leave the calculation unresolved. Neither circumstance supplies probabilities for a family that was specified only as a feasible set.

The selected reference, `FiniteVintageInference/1`, retains the named types, exact rational prior, observation-model identity and factorization assumption. A request also names the actor and information cutoff, the active observation versions and their replacement ancestry. Each observation retains its source content, information identity, slot, likelihood vector, supplied

actor-availability time and any superseded version. Its content identifier changes when any of those fields changes. The semantic information identity answers a different question: whether two records purport to describe the same informational contribution. A replacement keeps that identity even when its likelihood or source changes.

The supplied availability time must be no later than the request's cutoff. An unknown time remains unresolved; a later time is ineligible for that request. The reference uses the existing exact fixed-offset timestamp conversion, including fractional seconds, rather than rounding time to a calendar day. Its convention excludes leap seconds and unspecified offsets. These checks establish consistency with declared availability. Establishing when an actual investor, manager or system could obtain the information still requires a source-selection procedure.

Replacement selects a new head for an existing slot and preserves the chain to the earlier record. Retraction removes that slot's likelihood contribution, equivalently replacing it by one for every type. The request cannot select two heads for one slot, use a missing predecessor, or treat a different information identity as a revision of the same observation. Several unresolved candidate revisions therefore require several conditional requests. Selecting a head does not establish that it is the authoritative version.

The numerical consequences are immediate in the two-type example. After the correction, the posterior is $(1/4, 3/4)$. Suppose a separately declared, conditionally independent signal has likelihoods $(1/2, 1/4)$. Multiplying gives unnormalized masses $(1/8, 3/16)$, whose sum is $5/16$, and hence

$$\pi_{\text{corrected+signal}} = (2/5, 3/5). \tag{3.18}$$

Retracting the corrected observation leaves only the new signal; with the original equal prior the resulting posterior is $(2/3, 1/3)$. Retracting the only observation restores the prior. A type assigned zero prior mass remains at zero unless the prior itself is changed explicitly. Restoring support lost through an erroneous observation is distinct from inventing support absent from the declared model.

Repackaging the corrected observation under another slot would produce a different answer. With the equal prior, counting its likelihood twice gives masses proportional to $(1/16, 9/16)$ and posterior $(1/10, 9/10)$. The reference rejects two active observations that declare the same semantic information identity. An earlier implementation checked distinct slots but omitted this condition: its arithmetic tests passed while the declared information contract remained incomplete. The repaired example makes the resulting overconfidence visible. False distinct identities can still conceal repeated information, just as a false independence assumption can make a well-formed calculation substantively wrong.

A checkpoint retains its complete request and computed result. Reuse requires the same model and prior, actor and exact cutoff, with the old active records forming an unchanged prefix of the new sequence. New independent records can then update that prefix. A replacement that changes the prefix instead causes replay from the preserved prior. This avoids the failed operation that starts from $(1, 0)$ and attempts to recover the excluded type by reweighting it. Corrupt checkpoint contents or incompatible model and information definitions are rejected rather than silently relabelled.

In this finite reference the old checkpoint calculation is itself recomputed before reuse, so the test establishes the posterior relation without claiming a computational saving. Safe reuse from a larger store would require a proved sufficient dependency set. Particle checkpoints would additionally retain the likelihood estimate, proposal and ancestor state required by the extended target. A vector of posterior weights alone is insufficient.

The reference compares complete masses and normalizers over a finite rational test grid and explicit contrary cases, including a correction, retraction, duplicate information, unknown times

and a later fractional-second timestamp. An uninformative observation leaves both types alive. This illustrates a representation in which uncertainty can persist without enumerating a new copy of every historical node. It establishes the exact finite calculation under its declared premises; continuous inference, source completeness and the suitability of the likelihood for a real company remain further questions.

3.18 A correction reaches different decision makers at different times

Replaying the corrected history answers an inference question. It does not yet say which decision maker could use the correction. Suppose the analyst learns on 1 February 2026 that the earlier observation was wrong, while the manager receives that correction on 1 March. On 15 February they possess different information even though they are studying the same company. A simulation of the manager's choice must use the manager's information.

Continue the hypothetical fixed-versus-floating example. Under the original observation the posterior is $(1, 0)$; under the correction it is $(1/4, 3/4)$. Suppose an unhedged exposure costs zero for the fixed type and 20 for the floating type. A perfect hedge costs 6 in either case. These are supplied losses in the same units, and both actions are assumed feasible. They are enough to define this comparison without inventing an observed hedging price or an accounting transaction.

Let p denote the probability of the floating type under the particular actor's available information. The expected losses are therefore

$$L(\text{unhedged} \mid p) = 20p, \qquad L(\text{hedged} \mid p) = 6. \tag{3.19}$$

The original posterior gives an unhedged loss of zero. The corrected posterior gives 15, so the hedge has the smaller declared expected loss. At $p = 3/10$ the two losses both equal 6 and both actions remain minimizers. An implementation must retain that tie unless a separate choice rule is supplied. The manager on 15 February still uses the old posterior, while the analyst can already select the hedge under the corrected one.

The selected information procedure takes an actor and an exact cutoff, selects the available unreplaced observation in each information slot, and passes that request to the finite evaluator. Adding the future correction does not change the manager's stored 15 January request or its decision. Later knowledge can motivate a new decision, but it must not silently replace the information used to reproduce an earlier one.

Unknown availability cannot be treated as known absence. Two incomparable available revisions also leave the request unresolved; choosing the newer file arbitrarily would add an authority rule. The checked examples retain these distinctions, refuse corrupt observation content and fabricated posterior results, and leave unknown action losses unresolved. They select actions from the declared feasible set; they do not execute a hedge or infer its causal effect. Actual actor access, registry completeness, instrument terms and feasible financing still need evidence.

3.19 A proposed information service

AccDSL should represent evidence views and actor information views separately. An evidence view selects actual assertions by availability, revision and scope. An actor view specifies observable state variables, private contractual facts and signal channels at each decision date. Policies receive only that view and permitted memory.

Forward execution first applies events available at the current boundary, constructs observations, updates the relevant information state, and then permits decisions according to the stated timing. Alternative orderings are possible, but they must be explicit and tested. Moving a decision across an observation boundary changes the economic problem.

The service should refuse ambiguous timing where it affects the answer. It can still return a conditional comparison of both orderings. Such a comparison is preferable to silently selecting whichever ordering makes the simulation easier. Exact timestamps, time zones and publication lags matter when events occur on the same calendar date.

3.20 Tests that discriminate information errors

A useful test duplicates a world, changes an unobserved future shock and confirms that the current action remains unchanged. Another changes a privately known contract term and confirms that an informed manager's policy may change while a public analyst's evidence view does not. A third inserts a later restatement and checks that an earlier frozen forecast remains reproducible.

These are semantic tests. They do not require a fitted behavioral model. They establish that information restrictions are implemented as specified. Whether a particular manager actually follows the chosen policy is a separate empirical question. Keeping that distinction prevents a passing information test from being presented as behavioral validation.

A final test removes all informative observations and confirms that persistent uncertainty remains. A system that gradually collapses to a single history merely because a particle filter resampled or an optimizer selected one witness has confused computational concentration with evidence. The retained family and the approximation used for calculation must remain distinguishable.

Chapter 4

Scenarios, interventions, and feasible decisions

A scenario is a specified change to a model, not merely a new column of forecast numbers. It must say what changes, what remains structurally linked, what actors observe and how they may respond. A rate increase imposed on a contract schedule differs from conditioning an empirical model on periods when rates were high. A revenue decline with unchanged prices differs from one caused by price reductions. These distinctions determine the cash flows and the meaning of the result.

4.1 Describe the experiment before calculating it

A scenario request should identify the forecast origin, horizon, evidence view, shock variables, shock timing, transition rules, management policy and requested outputs. It should also specify whether the shock is a deterministic path, a random process or an ambiguity set. A phrase such as "rates up 200 basis points" leaves reset dates, currency curves, spread behavior and duration unresolved.

For a deterministic scenario s, let $u_t(s)$ be the imposed external path. The company evolves as

$$X_{t+1} = T_t(X_t, \theta, a_t, u_{t+1}(s)), \qquad a_t = \pi_t(I_t). \tag{4.1}$$

The transition includes contractual events and operational behavior. The policy uses information available at its decision time. A statement projection is then a reporting map of the resulting events and states. Directly shocking a reported total can be a useful reduced-form approximation, but it requires an explanation of how the corresponding events and balances are reconstructed.

A scenario can also specify a set $\mathcal{S}$ of paths and a collection Π of candidate policies. Let $\mathcal{A} \subseteq \mathcal{H} \times \mathcal{S} \times \Pi$ contain the jointly admitted triples: a compatible history, an allowed shock path and a policy whose information and execution rules are specified for that history and path. The outcome family is

$$\mathcal{Y} = \{F(h, s, \pi) : (h, s, \pi) \in \mathcal{A}\}. \tag{4.2}$$

The joint restriction matters. A lender's offer can depend on the company's contract state; a commodity shock can affect both sales and collateral; a policy may be admissible only for certain information states. Choosing each component independently from its marginal set can violate those dependencies. An admitted triple can still produce a refused funding request, distress or an unresolved result when its declared execution rules allow those outcomes. Admissibility does not mean that every attempted payment succeeds.

4.2 Conditioning and intervention

Suppose sales S and market rates R both respond to an economic factor U. Observing high rates can reveal information about U and therefore about sales. Imposing a rate change while holding the structural equation for sales fixed asks a different question. The distinction is formalized in causal models by separating conditioning from intervention.

In a simple illustration, let $R = U$ and $S = 100 - 10U$, where U is either zero or one. Observing $R = 1$ implies $U = 1$ and sales 90. An intervention setting $R = 1$ replaces the rate equation. It does not, by itself, change U or the sales equation. If $U = 0$ in the starting world, sales remain 100 under that intervention.

This is a local derivation illustrating the distinction in Pearl (2009, Sections 3.2–3.4). It is not an empirical claim that rates have no effect on sales. A richer structural model may include a direct rate effect, feedback and expectations. The point is that the intervention must name which mechanisms change. An observed correlation cannot decide that intervention on its own.

For financial stress work, a reduced-form scenario may intentionally impose several joint paths without claiming a causal intervention. That is a valid conditional exercise if described accurately. It should not be presented as the causal effect of one variable when the exercise changes several variables simultaneously.

4.3 One shock can enter several contracts

A market-rate shock may affect floating debt, deposit income, derivative settlements, pension valuation and refinancing prices. Applying it independently to each reported line can miss offsets or count them twice. The shock should enter the relevant primitive variables, and the affected contracts should generate their consequences.

Suppose floating debt of 100 and a receive-floating/pay-fixed swap of notional 60 share the same reference rate and reset date. Ignoring basis, timing and credit effects, a two-point increase raises debt interest by 2 and swap receipts by 1.2, leaving a net increase in cash interest cost of 0.8. Applying the shock to net finance cost and again to the swap would double count the hedge.

The simplification is material. A different reset date, reference index, currency, collateral agreement or counterparty default changes the result. The correct representation preserves separate contracts and links them through the common rate process. An accounting hedge designation is another relationship; it does not itself create the economic offset.

4.4 Timing changes exposure

For principal N and rate increment Δr applying over year fraction δ, the simple incremental payment is $N\Delta r\delta$. A full-year shock on 100 at two points gives 2. A half-year exposure gives 1. If the debt resets only after maturity, the current contract's payment may not change at all.

The year fraction must use the contract's convention and relevant dates. Replacing actual reset schedules with annual averages can preserve annual interest in a calm period while changing a short-horizon liquidity stress. A payment on the last day of a quarter and one on the first day of the next quarter have different reporting and liquidity consequences even if their annual totals agree.

A simulation should state whether events at a boundary occur before or after a decision.

Refinancing proceeds received after an old debt payment cannot fund that payment unless a bridge or intraday facility is included. Netting same-day totals can conceal this ordering problem.

4.5 Behavioral closure

Accounting identities do not determine production, hiring, investment, dividends or borrowing. A simulation needs rules for these choices. They may be supplied policies, estimated relationships, optimization problems or bounded alternatives. Each has a different evidentiary status.

A fixed payout ratio is easy to calculate but can produce dividends when cash is unavailable or legal restrictions bind. A cash-target policy can call for borrowing beyond committed facilities. An optimal investment policy can require knowledge of demand the manager does not possess. The policy must be evaluated together with feasibility and information, not appended after the financial statements have been balanced.

Let $A_t(X_t, I_t)$ be the feasible action set. A policy is admissible only if

$$\pi_t(I_t) \in A_t(X_t, I_t) \tag{4.3}$$

for the worlds it claims to govern. If the actor cannot observe X_t fully, feasibility may itself be uncertain from that actor's perspective. The model then needs a rule for attempted actions, acceptance by counterparties and consequences of rejection. Assuming successful execution whenever management requests funding is a substantive optimism assumption.

4.6 Attempts, commitments and settlements

An equity issue may be proposed, approved, subscribed, priced and settled at different dates. A loan may be committed but subject to conditions before drawing. A sale of assets may require a buyer and transfer completion. Treating intention as cash receipt collapses these stages.

The event model should distinguish an action request from acceptance, a binding commitment from a contingent one, and settlement from recognition. A financing policy can request 20 while the lender supplies 12. The shortfall remains an economic state requiring a further response. It must not disappear because the request was entered into a forecast cell.

These distinctions also prevent duplicate events. If a commitment is recognized in a schedule and the eventual settlement is recognized again as new borrowing without linking them, principal or cash can be counted twice. Stable contract and event identities support reconciliation from request through settlement.

4.7 Feedback and simultaneous determination

A company's borrowing cost can depend on leverage, while leverage depends on borrowing proceeds and operating decisions. This creates a fixed-point or equilibrium problem. Sequentially evaluating one formula and then the other once is generally not a solution to the simultaneous system.

For illustration, suppose required proceeds are $P = 20$ and face value satisfies $F = P(1 + y)$,

while the yield rule is $y = 0.05 + 0.001F$. Substitution gives

$$F = 20(1.05 + 0.001F), \tag{4.4}$$

$$0.98F = 21, \qquad F = 21/0.98. \tag{4.5}$$

Using the initial guess $F = 20$ in the yield rule and stopping gives a different face value. The discrepancy is small here because the feedback coefficient is small. Near a financing threshold it may change feasibility.

This rule is a teaching assumption, not a market-pricing model. Its purpose is to distinguish simultaneous equations from an event sequence. A proper model must also consider whether a solution exists, whether it is unique, whether the implied offer is feasible, and which equilibrium is selected when several exist. Numerical convergence to one point does not answer all those questions.

4.8 Joint shocks and dependence

A recession scenario can combine lower sales, slower collections, weaker collateral values and tighter funding. Sampling these shocks independently may understate their joint effect. Conversely, forcing every adverse marginal outcome to occur together can describe an intentionally severe stress that is not a probabilistic percentile.

If a joint probability model is used, its dependence structure needs evidence or an explicit sensitivity range. Correlations alone may not determine tail dependence. If a set of scenarios is used, the feasible combinations should be declared. A collection of marginal intervals does not establish that every combination is economically possible.

Historical and future uncertainty can also be dependent. A company with short-term debt may follow a different liquidity policy from one with long-term debt. Treating debt maturity and cash management as independent can generate histories inconsistent with the assumed behavior. The model should either retain the dependence or state that the policies are deliberately varied independently for a conditional stress comparison.

4.9 Counterfactual histories

A counterfactual asks what would have happened under a different action or shock while preserving specified features of an actual or hypothesized world. It requires a rule linking the factual and counterfactual worlds. Independently drawing a new company for each scenario does not preserve that link.

If two policies are compared using the same initial history and exogenous innovations, differences can be attributed to the policies within the model. If histories or shocks are redrawn independently, the comparison includes sampling variation. Common random numbers can reduce variance, but only when the coupling preserves the correct marginal model for each policy.

For unidentified histories, a useful result is the range of policy differences $F(h, s, \pi_1) - F(h, s, \pi_0)$ over the same h. Subtracting an independently minimized baseline from a maximized alternative can combine different companies and overstate the achievable difference. The paired relation should be preserved.

4.10 A threshold example

Suppose cash after interest is $90 - 2\alpha$ and a covenant requires cash at least 89.4. Solving the inequality gives

$$90 - 2\alpha \geq 89.4 \quad \Longleftrightarrow \quad \alpha \leq 0.3. \tag{4.6}$$

After evidence narrows α to $[0.2, 0.4]$, both compliance and breach remain possible. The equality point belongs to compliance under the inclusive covenant in this example. A strict covenant would require a different boundary.

The model can retain two guarded regions without enumerating individual values. The breach region then triggers whatever cure, waiver or acceleration rules the contract specifies. It does not automatically trigger liquidation. An unresolved guard is a reason to preserve conditional outcomes or refine the family, not to choose the midpoint and declare compliance.

4.11 Scenario comparison as a decision aid

A useful comparison reports which conclusions hold across histories, which depend on a particular contract assumption and which depend on a policy. For instance, a rate increase may raise interest for every floating exposure but reduce the value of a fixed-rate liability under a separate valuation basis. These are different quantities and can move in different directions.

The result should separate cash liquidity, book income, book equity, market value and contractual compliance. A positive accounting profit does not guarantee payment capacity; a market-value loss does not necessarily cause an immediate cash outflow. Combining them into an unnamed risk score makes the mechanism difficult to audit.

A proposed dashboard can show a conditional envelope with witness histories at important boundaries and a list of assumptions that change the conclusion. The envelope should retain joint outcomes and timing. A worst cash value at one date and a worst debt value at another may come from different paths; presenting them as one path would mislead.

4.12 Scenario failures and repairs

A scenario can fail because its shock is underspecified, because the model omits an exposure, because behavior uses future information, because financing is infeasible, because the numerical calculation fails, or because the economic assumptions are outside their domain. These failures require different remedies. More Monte Carlo paths cannot repair an omitted contract or an information violation. A more detailed journal cannot repair an unidentified behavioral response.

AccDSL should therefore return a typed result: completed conditional simulation, unresolved history region, unsupported mechanism, infeasible action, economic distress event, or numerical failure. A run may contain several statuses across different regions. Dropping failed regions and renormalizing successful ones changes the target distribution and can systematically remove the worst outcomes.

The next chapters make the financing mechanisms explicit. Debt and equity are especially consequential because they can appear to solve every cash shortage if issuance is represented only as a balancing entry. Their economic terms, counterparties, timing and rights must determine whether the entry may occur.

Chapter 5

Debt is a dated contract, not a balancing amount

Debt issuance joins a cash transaction today to promises about future payments and creditor rights. A journal can record proceeds and a liability without describing those promises. A structural model can describe the promises without supplying the accounting measurement. Scenario simulation needs both, together with a financing policy and a model of whether counterparties accept the proposed issue. Confusing these layers can make a company appear liquid while its funding equation is economically inconsistent.

5.1 Five amounts that should not share one symbol

For a debt instrument distinguish face or promised payment F, gross proceeds P, net cash received $P - f$ after issuance costs f, accounting carrying amount B, and market value V_D. They may coincide in a simple par loan at inception. They need not coincide for discounted debt, transaction costs, accrued interest, modifications or subsequent market changes.

The gross contractual promise is relevant to redemption. Net cash is relevant to liquidity. Carrying amount is relevant to the balance sheet under the selected basis. Market value is relevant to valuation or repurchase. A debt-to-equity covenant may use yet another defined measure. Reusing one variable called debt for all of these quantities creates errors that can remain hidden in a par, no-fee example.

A debt record also needs currency, lender, borrower, issue and settlement dates, maturity, coupon or reset rule, day count, amortization, seniority, collateral, covenants, options and modification rights. Unknown terms remain unknown. A model can support a bounded profile without implementing every clause, but its unsupported terms must not silently default to benign values.

5.2 The one-period pricing equation

Consider a loan issued at date zero and settled at date one. Let D be the date-one default indicator. If there is no default, the lender receives promised amount F. If default occurs, the lender receives actual recovery R, measured in date-one currency units. Let r be a risk-free one-period rate. Under the illustrative assumption that discounting the expected payoff at r is

appropriate, lender break-even is

$$P(1+r) = \mathbb{E}[(1-D)F + DR]. \tag{5.1}$$

This equation is a pricing assumption for the example. Under physical probabilities, discounting at a risk-free rate requires an appropriate risk-neutrality or pricing justification. More generally, with stochastic discount factor M,

$$P = \mathbb{E}[M((1-D)F + DR)]. \tag{5.2}$$

An explicitly justified pricing measure can provide an equivalent formulation. A real-world default probability is not automatically a risk-neutral probability.

Define the promised yield y by $F = P(1+y)$ for positive proceeds and a single final payment. Then

$$P = \frac{F}{1+y}. \tag{5.3}$$

The yield is a ratio of the promised repayment to the price, not the realized return in default. With coupons or multiple dates, a yield-to-maturity equation replaces this simple relation and may have additional conventions. The cash schedule remains the primitive description.

5.3 The convention problem in the supplied note

The user-supplied lecture note by Chak Wong adapts the risky-debt model of Strebulaev and Whited (2012, Section 3.6) to residual-based deep learning (Wong, 2026). It correctly recognizes that equity value, default and debt pricing interact. Its inherited debt convention requires repair before implementation.

The retained December 2012 source's equation (3.26) raises $b'/(1+\widetilde{r})$ at issuance and carries b' into the next period's inherited debt. That use makes b' a promised repayment. Equation (3.27), however, equates $b'(1+r)$ to an expected payoff whose nondefault leg is $b'(1+\widetilde{r})$. That use makes b' cash advanced. Both printed equations were inspected in the retained primary source; the mismatch is not merely a transcription error in the supplied note. The publisher's current correction status has not been cleared.

The consistent face-value convention uses F for the next promised payment and P for issuance cash, as in equations (5.1)–(5.3). Substituting gives

$$\frac{F(1+r)}{1+y} = \mathbb{E}[(1-D)F + DR]. \tag{5.4}$$

Multiplying by $1+y$ yields

$$F(1+r) = (1+y)\mathbb{E}[(1-D)F + DR]. \tag{5.5}$$

The multiplier applies to recovery as well as to the nondefault payoff. Omitting it on recovery changes the equation when recovery is positive. Alternatively one can formulate the entire model in proceeds units, but then future liability state, repayment and recovery arguments must change consistently. Renaming one symbol in the pricing equation is insufficient.

5.4 An exact positive-recovery diagnostic

Hold face at 100, default probability at 0.1, recovery at 40 and the risk-free rate at 0.05. These are synthetic inputs chosen to expose the convention. The expected payoff is

$$0.9(100) + 0.1(40) = 94. \tag{5.6}$$

Thus

$$P = \frac{94}{1.05} = \frac{1880}{21} \approx 89.52381, \tag{5.7}$$

$$y = \frac{100}{P} - 1 = \frac{11}{94} \approx 0.1170213. \tag{5.8}$$

The promised yield exceeds the risk-free rate because the lender does not always receive the promise. The expected payoff discounted at five per cent exactly equals the proceeds by construction.

Under the conflicting printed convention, solving $105 = 0.9(100)(1+y)+0.1(40)$ gives $y = 11/90$. Its implied proceeds are $9000/101$. Applying the actual payoff 94 to those proceeds leaves

$$94 - 1.05\left(\frac{9000}{101}\right) = \frac{44}{101} \approx 0.43564. \tag{5.9}$$

This is a lender break-even residual measured in date-one cash. It is not a rounding error. Zero recovery or no default can conceal the mismatch, so a no-default smoke test is inadequate. Positive recovery is the discriminating case.

5.5 Recovery needs its own definition

Recovery can mean a fraction of face, a fraction of market value before default, a share of realized collateral, or a cash distribution after costs and delays. These meanings are not interchangeable. If recovery is $R = \rho F$, the one-period expected payoff is $F[1 - p + p\rho]$ when default probability is p. If R is an absolute asset distribution, scaling face does not generally scale recovery.

For delayed recovery at date two, discounting it as a date-one payment is wrong. A stochastic discount-factor formulation must use the payment's actual date. Recovery uncertainty can be correlated with default and aggregate conditions; replacing it by an unconditional mean can misprice the claim.

Multiple creditors require an allocation rule. Suppose distributable assets after costs are 70, senior claims are 60 and junior claims are 50. Under an assumed strict priority waterfall without other rights, senior recovery is 60 and junior recovery 10. Paying both creditors 70 per cent would distribute 77 and violate the available asset constraint. The example illustrates conservation and priority, not a legal conclusion for a particular jurisdiction. Actual enforceability, security and restructuring rules need admitted authority.

5.6 Default is a transition, not deletion

An inability or decision not to pay can lead to arrears, cure periods, waiver, acceleration, restructuring, enforcement or liquidation. The contract and legal process determine which transitions are available. A simulator that sets debt and equity to zero at a default flag loses unpaid claims, assets, recovery timing and operating consequences.

A useful state includes scheduled obligation, amount paid, arrears, default status, acceleration status and recovery process. A missed payment can create an arrears balance even if operations continue. A covenant breach can restrict dividends without immediate acceleration. A waiver can restore eligibility subject to fees and revised terms.

Limited liability concerns shareholders' obligations and the value of their claim under the model. It does not require book equity to remain nonnegative. Negative book equity, negative market equity before applying limited liability, cash shortage and legal default are different concepts. Their relationships must be modeled rather than inferred from a single sign test.

5.7 Refinancing creates a new problem

A maturing debt of 100 cannot be repaid by writing a new debt balance of 100 unless a financing transaction supplies the cash in time. Suppose new face 100 sells for 90 and issuance fees are 2. Net proceeds are 88, leaving a shortfall of 12 before considering cash reserves. Treating face as cash hides that shortfall.

If the company requires net proceeds of 100 and the price is 0.9 per unit face with a fee of 2, required face is $(100 + 2)/0.9 = 113.333\ldots$. Increasing face may itself worsen price or violate capacity. The funding calculation is then coupled to the pricing and eligibility problem.

A committed facility may provide capacity but still have draw conditions, currency limits, borrowing-base constraints or expiry dates. An uncommitted facility is not guaranteed funding. A refinancing policy should distinguish request, offer, acceptance and settlement. Failure at any stage leaves a residual funding need and possible distress transitions.

5.8 Cash, debt and net debt

Some structural models use a signed debt variable, with negative debt interpreted as cash. This is economical in a stylized model, but cannot automatically represent a company with cash and debt simultaneously. Restricted cash may not repay a particular obligation. Cash in one subsidiary or currency may not be available to another. Gross debt can affect covenants and refinancing even when net debt is small.

Consider cash 100 and debt 100. Net debt is zero. If cash is unrestricted in the borrowing entity and repayment is allowed without cost, netting may be adequate for a particular valuation question. If cash is pledged to another creditor or held in a subsidiary subject to transfer restrictions, netting destroys liquidity information. The same net balance describes different payment capacity.

AccDSL should therefore preserve gross instruments and cash accounts in the general representation. A net-debt reduction is a query-specific abstraction requiring explicit conditions. It should not become the universal state merely because a numerical paper uses it successfully in a simpler economic model.

5.9 Coupons, accrual and effective interest

Contractual coupon determines a payment schedule. Accounting interest expense can differ because of discount, premium, fees or measurement rules. For a simple illustrative amortized-cost schedule with effective rate i, carrying amount evolves as

$$B_{t+1} = B_t + iB_t - \text{cash coupon}_{t+1} - \text{principal paid}_{t+1}, \tag{5.10}$$

subject to the selected accounting treatment and any additional events. The effective-interest term is not automatically a cash payment.

Suppose a zero-coupon promise of 100 is issued for 90 for one year. With no fees, an effective rate $100/90 - 1$ accretes carrying amount from 90 to 100. Cash is received at issuance and paid at maturity. Charging 10 as both accretion and a coupon payment would double count the financing cost.

Conversely, a floating coupon can reset while an accounting measurement follows its own applicable rules. A simulation should generate contractual cash first from terms, then calculate

accounting consequences under an admitted basis. A generic rate shock on interest expense cannot substitute for these two mechanisms in every instrument.

5.10 Issuance costs and tax timing

A fee can be paid to an intermediary at issuance, deducted from proceeds, financed separately or incurred conditionally. Its cash timing should be explicit. The accounting allocation between expense and carrying amount depends on the applicable standard and facts. A model cannot infer that allocation merely from the fact that cash was paid.

The retained structural source and supplied lecture include a simplifying treatment that credits the present value of a future interest tax benefit at issuance. That can be part of an economic valuation convention. It must not be posted as an actual contemporaneous tax receipt without a separate tax mechanism. A present value is not a cash settlement.

Tax losses, deduction limits, jurisdiction, timing and the availability of taxable income can all affect realization. A scenario that assumes every interest expense produces immediate cash tax savings may overstate liquidity in distress. The proposed design keeps tax cash payments, tax expense, deferred tax and economic tax-benefit valuation separate, then links them under explicit rules.

5.11 Options and amendments

A callable bond gives the issuer an exercise right; a puttable bond gives a holder a right. Their exercise can depend on rates, liquidity, transaction costs and information. Replacing the schedule with the contractual final maturity ignores these options. Exercising automatically whenever it appears favorable assumes a policy and ignores possible constraints.

An amendment can change coupon, maturity, collateral or covenants. It needs an effective date, consent requirements, fees and accounting treatment. A revised schedule should supersede future obligations consistently while retaining the original agreement and past payments. Applying both old and new coupons after amendment creates duplicate obligations; erasing the old contract destroys provenance.

Where a clause is unknown, the family can contain alternative exercise rights or amendment paths. Those alternatives should be activated only by their premises. A model should not give the borrower an option that belongs to the lender, nor assume the lender consents to a concession because it improves borrower liquidity.

5.12 Pricing and policy interact

A debt price can depend on future investment, payouts and refinancing because these affect repayment. The issuer's policy can depend on the price offered. Solving the two separately while holding inconsistent policies fixed can produce a price that is correct for a different company behavior.

The structural-finance survey Strebulaev and Whited (2012, Sections 3.4–3.6) is useful for this interaction. The newer working paper Duarte and Fonseca (2026, Sections 2.1.3–2.1.5 and 3) also solves bond pricing and equity value jointly under its own timing, recovery and tax assumptions. Those assumptions differ from the lecture's model; it is not a drop-in correction.

A general AccDSL financing service can accept a declared price or offer as an input. That supports conditional simulation without claiming equilibrium. An optional equilibrium service may solve pricing and policy together, with separate numerical evidence. The interface should identify which service supplied the offer and what assumptions support it.

5.13 Credit supply and rationing

A fair price does not imply an unlimited quantity is available. Lenders can face capital, liquidity, concentration or mandate constraints. A borrower may be unable to issue enough securities at any admissible terms. A model that solves only lender break-even can miss rationing.

Bank stress and balance-sheet models such as Halaj (2016); Behn et al. (2019) provide examples where financing and adjustment depend on constraints. Their institutional settings should not be transferred mechanically to a corporate issuer. The general lesson is that a financing transaction needs a counterparty and feasible quantity, not only an interest rate.

An offer curve can therefore specify price, maximum quantity, conditions and expiry. A policy chooses among feasible offers or records no acceptable offer. If the curve is uncertain, results remain conditional on alternative credit-supply assumptions. A scenario can deliberately freeze refinancing, but that restriction should be visible as a stress premise.

5.14 What dynamic financing models establish

For longer-lived debt, lender receipts depend on future issuance, investment, priority changes and default choices. A candidate policy π produces a price q^{π}. If management optimizes while holding an incompatible price fixed, the resulting policy may not be the one lenders priced. A simultaneous solution must make the chosen policy optimal under its price and make the price value that policy's promised and default-state payments. A constrained contract-price input is a useful alternative for mechanical stress testing, but its interpretation is conditional on that price and availability.

Benzoni and coauthors make this distinction explicit in a dynamic model with fixed issuance costs (Benzoni et al., 2023, Section 2, Propositions 1–2, Appendix A). Their debt quantity is face value, its price is per unit of face, and shareholders choose default, issuance or inaction. Value matching and smooth pasting generate a candidate barrier policy; off-policy deviations and a supermartingale condition are required for equilibrium verification. The retained model assumes zero default recovery and risk-neutral agents. It therefore illustrates the policy-price consistency problem, but is not the positive-recovery solution required by the preceding example.

The alternative managerial model of Lambrecht and Myers places investment, borrowing, payout and rents in a dated sources-and-uses constraint and imposes a no-Ponzi restriction (Lambrecht and Myers, 2017, Sections 2–4). Its safe-debt solutions depend strongly on managerial preferences and governance. In its exponential-utility example, the liquidation covenant protects lenders under the stated diffusion dynamics. A jump across the boundary, delayed liquidation or costly asset disposal would require a different recovery analysis. A closed-form policy derived under those restrictions is a competitor to consider, not an identified default policy for every issuer.

Dewatripont, Legros and Matthews provide a primary contracting comparison (Dewatripont et al., 2003, Sections 2–6 and Appendix B). Their environment has observable but noncontractible effort, limited liability, monotone investor receipts and efficient renegotiation. Under entrepreneur-offer renegotiation, a debt contract can dominate a more general contract in the stated pure-strategy class. Their appendix also supplies cases in which changed bargaining

power, third parties or mixed messages change that conclusion. Even their investor conversion option need not be an ordinary equity security, as footnote 16 explains. Contract labels cannot replace the exact contingent payments and exercise rights in the company model.

5.15 Equal balances after unequal payment histories

A borrower can recover from a cash shortage before the reporting date. Its closing statements then need not reveal that it failed to pay on time. To isolate the timing, consider a synthetic company with cash 60, inventory 100, debt carrying amount 80 and book equity 80. The contractual promise is 100, payable on 30 June 2027. Assume a zero coupon, no fees or taxes, and one remaining measurement interval that accretes the discount of 20 at maturity. This deliberately simple interval makes the cash and contract comparisons exact; a general effective-interest schedule needs its own measurement rules.

The company issues 40 new shares for one unit of cash each. In the first history, the subscription settles on 29 June. Cash rises from 60 to 100, the discount accretes on 30 June, and the borrower pays the complete promise that day. In the second history, the shares settle on 2 July. The borrower can pay only 60 at maturity. On 1 July an explicit observation records the unpaid 40; the new cash arrives on 2 July and pays the remaining claim on 3 July. No overdraft, emergency loan or retroactive receipt funds the earlier payment.

The following comparison follows the money and claim in the second history. The liability column is positive for presentation, although journal credits use the implementation's signed convention.

After the dated event	Cash	Claim unpaid	Historical miss
30 June: accretion	60	100	0
30 June: partial payment	0	40	0
1 July: missed-payment observation	0	40	40
2 July: share proceeds	40	40	40
3 July: final payment	0	0	40

The zero in the first two historical-miss cells records that the observation has not yet been made. It is not a finding that a payment was timely. The observation changes contract memory without changing cash or posting another expense. The final payment clears the current claim; it leaves the history of the earlier shortfall intact.

Both histories end with the same financial statements under the supplied treatment. Inventory of 100 is the only remaining asset, the debt is zero, and equity is opening equity plus the subscription less the discount expense:

$$A = 100, \qquad L = 0, \qquad E = 80 + 40 - 20 = 100. \tag{5.11}$$

Profit for the illustrated movements is -20. Net financing cash flow is $40 - 100 = -60$ in either history. The late path divides the repayment into $60 + 40$, which changes its timing but not its total. Even agreement across the closing balance sheet, income and cash-flow statements does not answer whether the maturity was paid on time. A simulation query about missed payments must retain the distinguishing history.

The difference also explains why balancing the journal is insufficient. Share proceeds can debit cash and credit capital correctly while a faulty method fails to increase the recorded shares. A missed-payment observation can omit its history update without unbalancing any account. In the checked reference, both omissions pass generic accounting execution and fail the independent

complete-state comparison. Explicit financing conditions subsequently reject these proposed transitions before accepting their successors.

Those conditions have two roles. Snapshot conditions constrain current quantities: claims and arrears are nonnegative, arrears cannot exceed the remaining claim, and under this matured zero-coupon convention the carrying liability equals that claim. Transition conditions constrain how an event changes them. Payment of p gives remaining claim $q' = q - p$, reduces cash by p, and leaves the historical missed amount unchanged. A share subscription must increase both the monetary capital account and the separate share quantity under the supplied issue terms. An internally plausible rewrite of historical miss from 40 to 50 can satisfy snapshot inequalities but violates the transition condition. This is why both kinds of condition are needed.

Opening history creates another distinction. A supplied fiscal opening must satisfy the declared period-start, temporary-balance and flow restrictions; an intermediate state containing this year's payments is not such an opening. Importing a consistent snapshot does not establish that it includes every earlier missed payment. The selected opening method therefore compares it with a separately supplied history premise and preserves unknown completeness. Actual source evidence is still needed to establish that premise.

5.16 A missed payment and the available remedies

The amount unpaid does not by itself identify the creditor's remedy. Apple's 2024 annual report provides a concrete historical description of why the distinction matters. Exhibit 4.1 expressly calls its description incomplete and qualifies it by the 2013 and 2018 indentures; its issuer references concern Apple Inc., rather than automatically every subsidiary (Apple Inc., 2024, Exhibit 4.1, printed p. 3). We use the summary to identify conditions a simulation must represent. It does not establish an actual issuer default, complete operative terms, or the applicability of those terms to the synthetic company.

The described interest-payment event requires default continuing for 30 days after the installment becomes due. Principal and premium use a different condition: nonpayment when due and payable, including maturity, redemption or declaration. Importing the interest grace period into principal repayment would therefore apply the wrong described condition. The other-covenant route has its own exclusions, a continuing breach and 90 days after qualifying written notice. Holder notice uses at least 25% of outstanding principal under the 2013 indenture and at least 33% under the 2018 indenture; the trustee has a separately described notice route (Apple Inc., 2024, Exhibit 4.1, printed p. 9).

The thresholds concern the relevant outstanding series, not a fraction of consolidated debt or a count of creditors. Series mapping, denominator, delivery to the required recipients and validity of notice are substantive inputs. No notice and no evidence about notice lead to different answers. Likewise, day-level data at exactly the 30- or 90-day boundary may leave event order unresolved. The conditional reference retains that uncertainty; it does not assert that the complete contract has no answer at the boundary.

The summary next distinguishes declaration from automatic acceleration. Certain continuing nonbankruptcy events permit a declaration by the specified trustee or holder route. Certain bankruptcy events make the obligations due without that declaration. A negative equity balance establishes neither the specified bankruptcy event nor the required category. Moreover, an event for one series does not necessarily constitute an event for another. Neither automatic contagion nor universal independence follows (Apple Inc., 2024, Exhibit 4.1, printed pp. 9–10).

Cure and rescission must also remain separate. The described rescission of a declaration requires holder action and specified deposits and cure or waiver conditions, with an exception concerning

amounts due solely because of acceleration. Payment can remove current arrears without by itself proving that an earlier acceleration has been rescinded. In the selected synthetic continuation, payment changes the economic claim, reassessment changes the current clause finding, and a separate supported rescission assessment changes the recorded acceleration state. The historical miss remains 40 throughout.

Consent cannot be supplied by a numerical convenience either. The summary requires each affected holder's consent for specified protected changes, including changes to stated maturity and reductions of principal or interest. Its general majority-waiver provision excludes payment defaults and certain individually protected provisions. A waiver concerns the specified past default; it does not license a future one (Apple Inc., 2024, Exhibit 4.1, printed pp. 11–12). A current contractual default may cease under a valid waiver while the economic record still shows that payment arrived late. Keeping those two statements is consistent.

The surrounding qualifications matter economically. Individual payment enforcement differs from collective proceedings; legal discharge differs from covenant defeasance. The description warns that assets sufficient to fund scheduled maturity may be insufficient after acceleration (Apple Inc., 2024, Exhibit 4.1, printed pp. 10–13). A simulator must therefore retain the relevant rights, legal owner, timing and usable funding, even when a consolidated closing statement appears fully funded.

The typed conditional implementation translates the retained examples and records selected findings through the common monetary and memory engine. An unresolved finding produces no successor state. Recording a known finding changes no cash, carrying amount, shares or historical missed amount. This establishes a limited representation and execution result. Multiple concurrent clauses, the full instruments, actual notices and amendments, and the economic consequences of acceleration still require their own admitted methods.

5.17 Implementation proposal

The debt module should maintain immutable contract versions, a dated obligation schedule, current economic state and separate accounting measurements. Events include issue, draw, reset, accrue, coupon, amortize, redeem, amend, breach, cure, accelerate, restructure and recover. Each event has preconditions, amounts, dates and successor state.

Existing repository methods already cover bounded issue, accrual, coupon, redemption and conversion calculations, and newer declared methods preserve timing and debt memory. They are useful foundations. They do not by themselves establish a general pricing model, market capacity, legal default process or complete instrument admission. The extension should compose with their checked boundaries rather than describe those capabilities as absent.

The smallest discriminating financing fixture should include proceeds different from face, positive recovery, a fee paid at issuance, a maturity shortfall and an unavailable refinancing offer. A second fixture should add two creditors with priority and delayed recovery. A third should test unknown rate terms and a covenant boundary. Passing these checks establishes the specified mechanics; empirical pricing validity and legal applicability require separate evidence.

Chapter 6

Equity funding, ownership, and dilution

An equity issue can provide cash while changing ownership and future claims. A model that records only an increase in cash and book equity captures one accounting consequence but misses the allocation of control and value. A structural model that charges an issuance cost captures a financing friction but does not automatically create a share register. Scenario simulation needs quantities, prices, rights, holders and dates, together with an explanation of who is willing to subscribe.

6.1 Book equity, market value and share capital

Book equity is the residual of recognized assets and liabilities under an accounting basis. Market equity value is the value assigned to owners' economic claims. Legal share capital is a legally defined component of capital that may be related to nominal share values. These quantities have different meanings. Dividing book equity by shares produces book value per share, not automatically the market subscription price.

A company can have positive book equity and insufficient cash to pay a maturity. It can have negative book equity and valuable future opportunities. Limited liability can bound the value of a shareholder's claim below by zero in a model without forcing the accounting residual to zero. The simulator should retain these quantities separately and define any relation between them.

Let N be the number of existing ordinary shares, V_E the total economic value assigned to their claim, and $p = V_E/N$ a per-share value when shares are homogeneous and $N > 0$. This division presumes that V_E refers to exactly that class. If preferred shares, options or noncontrolling interests are included in total equity value, allocating all of it to ordinary shares is wrong.

6.2 A simple cash issue

Suppose the company issues n new homogeneous ordinary shares at subscription price $p_s > 0$. Gross proceeds are $G = np_s$. If cash issuance costs are f, net cash is $G - f$. With no simultaneous repurchase or conversion, the post-issue count is

$$N^+ = N + n = N + \frac{G}{p_s}. \tag{6.1}$$

An existing holder with n_i shares has ownership fraction n_i/N before and $n_i/(N+n)$ after the issue, assuming that holder does not subscribe. This is percentage dilution. Its economic consequence depends on the price received and the use of the proceeds.

For example, let $N = 100$, pre-issue equity value be 1,000, and issue 20 shares at 10. If the cash adds 200 of value and there are no costs or other effects, post-issue value is 1,200 and each share remains worth 10. The old holders' percentage falls from 100 per cent to 100/120, but their combined value remains 1,000. Percentage dilution is not necessarily a wealth loss.

If the same 20 shares are issued at 8 while pre-issue assets remain worth 1,000, post-issue value is 1,160 and each share is worth $1,160/120$. Old holders then own $100(1,160/120) = 966.666\ldots$. The transfer arises from the discounted subscription price under these assumptions. A model that equates all share-count dilution with value destruction misses the first case; one that ignores issue price misses the second.

6.3 Raising a target amount after a price shock

Suppose net funding need is K and a fixed cash fee is f. At a supplied price p_s, required shares are $(K+f)/p_s$. A price decline therefore increases the share count required to raise the same cash. If authorized issuance or investor demand limits n, the funding target may be infeasible.

Take $K = 90$, $f = 10$ and an initial subscription price 10. The company needs ten shares. After a price shock to 5 it needs twenty. If only fifteen shares may be issued under the assumed authorization, maximum net proceeds become $15(5) - 10 = 65$, leaving a shortfall of 25. Increasing book equity by 90 would conceal both the share constraint and the cash shortfall.

The fee may instead be proportional, for example $f = \lambda G$. Net proceeds are $G(1-\lambda)$ and required gross proceeds are $K/(1-\lambda)$ for $0 \leq \lambda < 1$. A fixed plus proportional fee gives $G = (K + f_0)/(1-\lambda)$. These equations specify different cost structures. The choice should come from a declared offer or empirical assumption, not a silent generic parameter.

6.4 Pricing is not supplied by an accounting identity

The equation $G = np_s$ is an identity once shares and price are specified. It does not determine investor willingness to pay. A subscription price can be an external offer, an assumed discount to an observed market price, or the result of a valuation and demand model. Each supports a different claim.

A public market quote can be stale, illiquid or refer to a different class. A large issue can affect price. A distressed company may face a discontinuous demand schedule or no feasible offer. A model that uses yesterday's price for unlimited new shares assumes unlimited demand at that price. That may be useful in a deliberately simplified scenario, but should not be presented as a financing solution established by the statements.

The same caution applies to an endogenous equity value function. It may value existing shareholders' future payouts under a particular policy and financing friction. Converting that value into a new issue price requires assumptions about how new investors share those payouts and how issuance changes the state. A scalar cost function $\eta(e)$ for external equity is not a complete security issuance model.

6.5 The rights attached to a share

Shares can differ in voting rights, dividends, liquidation preference, conversion rights and redemption terms. An aggregate share count is sufficient only when the distinctions are irrelevant to the question or explicitly normalized. A dollar of preferred funding is not generally equivalent to a dollar of ordinary funding for future payouts.

A simple preference waterfall illustrates the point. Suppose new investors contribute 30 for a nonparticipating preference claim entitled to 30 before ordinary holders, with the option to convert into 25 per cent of ordinary equity. If distributable value is 80, the preference payment 30 exceeds conversion value 20. If value is 200, conversion value 50 exceeds the preference. The exercise threshold is 120 under these assumptions.

This example requires a declared contract: no accrued preference dividend, no senior claims, no participation after the preference and no other options. Adding any of these changes the waterfall. A flat share-percentage allocation would misallocate value below the conversion threshold. The model should retain class-specific rights and exercise policy.

6.6 Options, warrants and contingent issuance

An option can create future shares when exercised. Exercise can bring cash to the company, transfer existing shares or settle in cash, depending on terms. Counting all potential shares as already issued can distort current ownership; ignoring them can distort future dilution.

Let an option cover m shares at exercise price k. In a simplified physical exercise, the company receives mk and issues m shares. The holder's decision depends on share value, expiry, constraints and possibly tax or employment conditions. A purely intrinsic-value rule is a modeling policy, not a universal description of exercise behavior.

Contingent performance awards add dependence on future operating outcomes. A scenario changing revenue can affect both the company's value and the number of shares delivered. Treating share count as independent of the operating shock loses that dependence. Historical statement totals may not identify all award terms, so the family must preserve unresolved conditions.

6.7 Conversion joins debt and equity

A convertible instrument links a liability schedule to ownership changes. At conversion, principal or another contractual amount may be exchanged for shares according to a fixed ratio, a variable price or a contingent formula. Cash settlement alternatives can change liquidity materially.

Suppose debt face 100 converts into ten ordinary shares. Physical conversion removes the specified debt obligation and increases shares by ten under the assumed terms. It does not also pay cash principal 100 unless the contract requires a separate payment. A simulator that schedules both redemption and conversion double settles the same obligation.

Accounting derecognition and allocation can involve carrying amounts different from face and separate components. Those postings need the applicable basis and facts. The economic event should identify exactly which obligations and rights are extinguished or created before the accounting module measures the consequences. Existing bounded conversion methods in the repository can support this event discipline without claiming a universal convertible classification.

6.8 Repurchases and treasury shares

A repurchase uses cash and changes outstanding shares or treasury holdings according to the legal and accounting treatment. A net increase in equity during a period can conceal both gross issuance and gross repurchase. Their timing and prices matter for liquidity and ownership.

Suppose 20 shares are issued at 10 and ten are repurchased at 12. Net cash inflow before costs is 80, while outstanding shares increase by ten if repurchased shares cease to be outstanding. Reconstructing the period as a simple issue of ten shares for 80 uses an implied price of 8 and loses the actual transactions. A future policy inferred from that synthetic price can be misleading.

Repurchase authorization is not settlement. A board may authorize up to an amount while management executes only part. A scenario should not consume cash for the entire authorization unless the policy calls for it and execution is feasible. Conversely, an already binding purchase commitment may create obligations before the cash settlement date.

6.9 Dividends require both claims and capacity

A dividend policy can specify a target payout, but payment depends on declaration, entitlement dates, cash, legal restrictions, covenants and class rights. Retained earnings are not a cash account. A positive accumulated balance does not establish that funds are available in the paying entity.

A declared dividend can create a payable before payment. The event sequence must avoid expensing it as an operating cost in an illustrative ordinary distribution model, while still recognizing the liability and equity movement under the selected basis. Payment reduces cash and the payable. The exact legal and accounting treatment requires its admitted rules.

A cash-constrained policy may reduce or suspend a prospective dividend. It cannot necessarily cancel an already enforceable obligation. Distinguishing intended, declared and paid distributions is therefore important in stress scenarios. A single annual dividend line cannot capture all of these states.

6.10 Ownership and control are related but distinct

Economic ownership percentages do not always determine control. Voting rights, contractual arrangements, dispersed holdings and substantive options can matter. A scenario that changes ownership may alter the consolidation perimeter, but the threshold and assessment are not supplied by a generic percentage rule.

The security register should record holders and rights, while a separate control assessment uses the applicable facts and accounting basis. When control changes, group reporting can change even without a large contemporaneous cash flow. Conversely, a minority issue can raise cash without changing control. The two calculations should communicate without being collapsed.

Historical disclosures may report total shares but not every beneficial holder. A model can represent ownership categories and unknown allocations where sufficient for the query. It must not invent holder identities or treat an unknown allocation as evenly spread. Concentration can matter for approval, takeover and financing decisions.

6.11 Earnings per share is not the financing price

Basic and diluted earnings-per-share calculations use accounting earnings, weighted share counts and rules for potential ordinary shares. They are reporting calculations. They do not determine market value per share or subscription price. A price-to-earnings heuristic is a separate valuation assumption.

Timing matters even in a simple weighted count. If 100 shares are outstanding for half a year and 120 for the other half, the simple time-weighted average is 110. For positive earnings, using closing count 120 understates basic earnings per share relative to that assumed timing. For a loss, it makes loss per share less negative; a zero numerator remains zero. Actual calculations require the applicable conventions, participation rights and events.

Diluted measures may exclude antidilutive instruments under the reporting rules. Exclusion from diluted earnings per share does not mean the contract cannot create future cash or ownership consequences under another scenario. The scenario engine must retain the instrument even when one reporting calculation excludes it.

6.12 An issuance changes future incentives

Financing can change not only cash but the distribution of gains and losses. Existing shareholders, new investors and creditors may prefer different policies after an issue. A model holding behavior fixed is a conditional mechanical simulation. A model claiming an optimal response must specify whose objective is optimized and what constraints apply.

For example, a deeply subordinated new equity contribution can protect creditors while diluting existing owners. Existing owners may refuse it even when total enterprise value rises. A policy maximizing total firm value can therefore differ from one maximizing existing shareholders' value. Treating these objectives as interchangeable can change default or investment decisions.

The structural literature surveyed by Strebulaev and Whited (2012) makes financing frictions and shareholder objectives explicit in stylized models. Transfer to AccDSL requires matching the actor, security rights and timing. A convenient objective function should remain a hypothesis until its application is justified.

6.13 A complete small financing comparison

Consider an immediate cash need of 40, current cash 10 and no available debt facility. Net new funding need is 30. Existing shares are 100 and an admissible offer supplies ordinary shares at price 5 with a fixed cash cost of 5. Raising 35 gross requires seven shares, gives 30 net cash and permits the payment. Closing cash is zero and shares are 107.

If the offer instead caps issuance at six shares, gross proceeds are 30 and net proceeds 25. Available cash is 35, leaving a payment shortfall of 5. The difference is an economic feasibility result. The journal cannot repair it by crediting equity for another 5 without a transaction.

If management can defer 10 of the payment at a cost, the capped issue may become feasible. That is a different policy and contract path, with future obligations. If a shareholder loan supplies the missing 5, it creates debt and related rights. The model should show these alternatives explicitly rather than use a generic balancing source.

6.14 Proposed register and execution methods

A dated security register should identify issuer, class, units, holder or holder category, rights, issue price where relevant, subscription and settlement state, and links to conversion or cancellation events. Outstanding, authorized, treasury and potentially issuable units should be distinct. Changes should reconcile through events.

An equity-financing method should accept a funding request, eligible offers, authorization limits and actor policy. It returns an accepted transaction or a shortfall. The transaction then produces cash, ownership and accounting events. Price formation may be supplied or solved by a separate model; the result records which.

The repository's simple shares-times-price postings and bounded capital events are useful components. The proposed extension adds offer feasibility, class rights and ownership evolution. It should reuse existing atomic posting and timing checks while adding negative cases: zero or negative subscription price, unsupported class, duplicate settlement, insufficient authorization, failed subscription and inconsistent conversion.

6.15 What the equity checks establish

Conservation checks can establish that cash equals settled subscriptions less costs and that share movements reconcile. Waterfall checks can establish that distributions do not exceed available value under the assumed priority rules. Timing checks can establish that ownership changes occur at the declared event date.

These checks do not establish that an issue price is fair, that investors will subscribe, that a legal approval is valid or that management will choose the modeled action. Those claims need their own evidence. The simulator should be able to give useful conditional calculations while keeping these unresolved questions visible.

A useful adversarial fixture issues shares below an assumed pre-issue value, then compares percentage dilution with wealth transfer. Another changes the price while holding funding need fixed and tests an authorization cap. A third combines conversion and redemption to ensure that the same obligation is settled once. These examples expose mistakes hidden by a simple par issuance with unlimited demand.

Chapter 7

From economic events to coherent statements

A scenario produces a sequence of events. Financial statements summarize selected consequences of those events under a reporting basis. Keeping this direction explicit makes it possible to ask whether an output follows from a plausible company history. Starting with independently forecast statement lines and inserting a balancing amount reverses the logic: it can make the accounts add up while leaving the economic transaction unexplained.

7.1 An event carries more than an amount

An event identifies actors, instrument or resource, effective time, settlement time, currency, quantity, conditions and provenance. A sale can transfer goods today, be recognized under a specified rule today, and be collected next month. A loan draw can be approved today and settle tomorrow. A derivative can change value without an immediate cash transfer.

The event layer should therefore represent economic occurrence separately from recognition, measurement and payment. The accounting layer maps admitted events and estimates to postings. The reporting layer aggregates postings into statements. The mapping may create noncash movements, and some economically relevant commitments may be disclosed without a current journal amount.

Pierson (2020, Sections 3.3–3.5) explicitly separates accrual, payment and reporting accumulation in a system-dynamics setting. Its simple debt-delay mechanism is acknowledged as limited. The lesson for AccDSL is the separation of clocks, not the adoption of that simplified debt mechanism as a general contract engine.

7.2 A credit sale and collection

Consider a synthetic sale of goods for 30, with inventory cost 18, recognized at delivery and collected later. Ignore tax, returns and credit impairment for this illustration. At delivery, the postings are a debit to receivables and credit to revenue of 30, and a debit to cost of sales and credit to inventory of 18. Cash has not increased.

At collection, cash increases by 30 and receivables decrease by 30. Revenue is not recognized again. The two events explain why profit and operating cash can differ during the interval. A scenario that delays collection changes receivables and liquidity even if recognized revenue is

unchanged.

If the sale is instead prepaid, receipt creates a different sequence under the selected recognition rule. The same annual revenue and cash collection can conceal different within-year funding needs. An annual-only model can be adequate for some questions, but it cannot claim to resolve a monthly liquidity trough that its time aggregation removes.

7.3 Balanced entries are necessary, not sufficient

Double-entry balance verifies a conservation relation in the accounting representation. It does not establish that the event occurred, that the account mapping is correct or that the measurement basis is appropriate. Debiting cash and crediting revenue balances even when the money was borrowed.

Let j_e be the vector of signed account movements for event e, under a declared debit-credit convention. A balancing check can be written $\mathbf{1}^\mathsf{T} j_e = 0$. This is one invariant. Additional invariants depend on the event: a loan settlement should reconcile cash to proceeds and fees; a physical inventory sale should not consume more units than available unless backorders or short sales are explicitly allowed; a share issue should reconcile units and ownership.

The event's economic and accounting checks should remain separate. A balanced but economically impossible issue must fail transaction admission. An economically valid event with an unresolved accounting treatment can remain in the economic state while its statement consequences are conditional or unsupported. Forcing one layer to masquerade as the other hides useful information.

7.4 Quantity and valuation

Inventory value can change because quantities change, unit costs change, measurement rules change or currency changes. A model using only a monetary stock can confuse these mechanisms. Production capacity and physical shortages depend on quantities, while reported carrying values depend on measurement.

Suppose ten units cost 5 each. Inventory carrying value is 50 under the illustrative cost basis. A replacement-price increase to 8 does not automatically increase that carrying value to 80. If five units are sold and replaced, cash purchases, cost allocation and ending stock depend on the chosen inventory-flow assumptions. A future demand shock can expose differences among vintages with the same initial total value.

The economic state should retain the granularity needed for production and sale decisions: units, location, quality, expiry and ownership where relevant. Accounting state can aggregate or measure those units under its basis. An aggregate sufficient for valuation may be insufficient for a supply interruption.

7.5 Asset vintages and depreciation

Two firms can report the same property carrying amount and depreciation while holding assets of different ages. The remaining useful lives, maintenance needs and replacement dates can differ. A capital-expenditure scenario therefore requires more than a depreciation ratio.

For a simple straight-line asset with cost K, residual value v and life L, annual depreciation is $(K - v)/L$ under the assumed full-year convention. Several combinations of K, v and L can

yield the same charge. The historical charge does not identify replacement timing.

An asset-vintage representation retains acquisition date, cost, accumulated depreciation, remaining life, capacity and relevant impairment or disposal events. If only aggregates are observed, the reconstruction should preserve a family of vintage allocations. A query about next year's depreciation may admit a compact aggregate; a query about replacement cash over five years may require finer structure.

7.6 Working capital is a set of contracts

Receivables and payables are often forecast using days sales outstanding and days payable outstanding. These ratios summarize historical relationships but do not identify individual due dates or counterparties. A ratio-based projection can be a useful approximation when its scope is stated. It cannot automatically describe payment priority or concentration under stress.

A customer delay can raise receivables while the firm still owes suppliers on fixed dates. If the model shifts payables proportionally with sales without checking contracts, it may manufacture supplier financing. Conversely, a model that fixes all payment dates may omit renegotiation that management could pursue. Both are policy choices.

The proposed design permits ratio methods as explicitly reduced models alongside contract schedules. Their outputs should carry the aggregation assumptions, horizon and validation domain. A breach-sensitive liquidity query can require refinement to invoices or cohorts whose due dates cross the critical boundary.

7.7 Currency conversion and cash availability

A foreign-currency account has both native units and a reporting-currency value. Exchange movements can change reported amounts without changing native cash. A debt payment in foreign currency requires the appropriate currency at settlement, not merely sufficient consolidated reporting-currency cash.

Suppose the company holds 100 domestic units and owes 10 foreign units. At exchange rate 8 domestic per foreign, repayment costs 80; at 12 it costs 120. A consolidated cash total of 100 does not establish payment capacity after the depreciation. A hedge, foreign cash balance or exchange facility may alter the result, but each must be represented.

Translation for group reporting differs from converting cash for settlement. The accounting basis determines exchange differences in profit, other comprehensive income or other components. The economic layer tracks actual currencies and conversions. A rate applied for presentation cannot automatically be used as an executable market price for a large transaction.

7.8 Group boundaries and internal funding

A consolidated group can report cash in one entity and debt in another. Consolidation removes some internal balances but does not necessarily make all cash freely transferable. Legal, tax, contractual, regulatory or operational constraints may restrict movement. A group-level cash plug can conceal an entity-level default.

Represent cash accounts and obligations by legal entity, then model permitted transfers as events with dates, costs and approvals. Intercompany loans create claims in both entities. Dividends,

capital contributions and service payments have different legal and accounting consequences. They cannot be substituted solely because each moves cash.

For consolidated reporting, internal claims may be eliminated under the applicable perimeter and basis. The underlying entity records remain necessary for liquidity and counterparty analysis. Eliminating the records themselves would prevent the simulator from identifying which entity can pay which obligation.

7.9 A two-entity liquidity example

Parent P has cash 5 and external debt due 20. Subsidiary S has cash 30 and no current external debt. Consolidated cash is 35, apparently enough for the maturity. Suppose only 10 can legally and operationally transfer from S before the due date under the declared facts. P can then assemble 15 and has a shortfall of 5.

If a permitted intercompany loan of 15 becomes available in time, P can pay. If a dividend of 15 is declared but settles after maturity, it cannot fund that payment without a bridge. If the subsidiary cash is restricted, even the first 10 may be unavailable. The scenario result depends on transfer rights and timing, not just the consolidated total.

A correct consolidated statement after payment can coexist with an incorrect event sequence if the model silently borrowed the cash before it was available. Tests must therefore examine entity cash at event boundaries, not only the final group balance sheet.

7.10 Taxes and losses

Tax expense, cash tax and deferred tax are different quantities. A profit forecast cannot generally multiply pretax income by one rate and use the result simultaneously for expense and immediate payment. Loss carryforwards, payment lags, jurisdiction, interest limitations and temporary differences can separate them.

A simple teaching model may assume immediate tax payment at rate τ on positive taxable income, giving cash tax $\tau \max(T, 0)$. That is a declared simplification. If losses can offset future income, the state must retain the remaining loss balance and expiry or utilization conditions. Forgetting that state can overstate future tax; treating all losses as immediately refundable can overstate current cash.

Accounting recognition of deferred tax assets requires its own admitted rules and evidence. Their carrying amount is not spendable cash. A stress scenario that turns a deferred tax asset into a funding source without an actual refund or financing transaction is economically wrong.

7.11 Estimates, errors and changes of policy

An estimate revision changes the model's measurement of an uncertain quantity. An error correction revises a prior assertion or application. An accounting-policy change can alter recognition or measurement according to the applicable rules. Their timing and comparative effects differ.

The scenario engine should distinguish an economic shock that changes expected credit losses from a correction showing that the old receivables data were wrong. Both can change reported equity, but only one necessarily describes new economic information at the forecast date. The evidence and accounting version relations should make the distinction explicit.

A correction should not silently rewrite every frozen forecast. A current restated view and an as-published historical view can coexist. Their differences are analytically useful: they separate errors in data available at the time from errors in the forecasting model.

7.12 Consolidation and scenario-dependent perimeter

An acquisition, disposal or change of control can alter the reporting perimeter. The group total before and after the event may not be comparable without a bridge. A simulation that treats acquired revenue as organic growth can distort future behavior and validation.

Perimeter state should identify included entities and the relevant control assessment over time. Acquisition events introduce assets, liabilities and consideration under the selected accounting treatment. Disposal events remove or reclassify them and may create gains, losses or continuing interests. These are not ordinary sales receipts applied to an unchanged company.

Unknown deal terms can produce alternative histories. Earn-outs, guarantees and retained interests may affect future cash after apparent disposal. A compact representation that keeps only closing consolidated balances can lose these rights. The source-to-contract-to-event chain should survive even when the initial reporting effect is aggregated.

7.13 Atomic posting and partial failure

A multi-leg event should either apply all admitted accounting movements consistently or remain unapplied with a recorded failure. Posting cash but failing to create debt because a later validation step failed corrupts the state. Similar risks arise when conversion removes debt but does not issue shares.

Atomicity is an engineering property that supports accounting consistency. It does not certify the event's economic validity. Preconditions should be checked before mutation where possible, and the event should retain an immutable input snapshot. Replay should reproduce the same movements under the same rule version.

A failed accounting mapping should not erase a successfully observed economic event from the evidence store. Instead, the system can retain the event and mark its accounting consequences unresolved. This separation permits later repair without inventing or deleting transactions.

7.14 A finding can become stale within one day

Consider the late-funded borrower in Section 5.15. An observation records an unpaid principal clause as satisfied on 2 July. The company then pays the remaining 40 on that same date. A later command asks whether a continuing event permits a new acceleration. Matching the observation's date to the command's date would accept the old unpaid finding, even though the intervening payment changed its premise. Calendar equality does not establish that the financial state is unchanged.

This is a concrete failure in the earlier selected adapter. Its original trace, with payment on the following day, passed. Moving payment to the same day exposed the defect: the stale satisfied finding remained usable. The successor retains the old observation as history but requires reassessment before using it as a current condition. A fresh paid assessment then supports the corresponding absence of a new trigger. It does not retroactively revoke an acceleration already recorded under earlier valid facts.

To express the dependency, let B be the complete typed book and let $P(B)$ retain its economic contents: balances, cash-flow history, nonremedy memory, context, account and type definitions, reporting period and schema. The projection omits the remedy-status record, the list of seen event identifiers and the current observation date. The date is checked separately. The first omission is essential: recording a finding changes remedy status and must not, by that act alone, invalidate the finding it records. Event identifiers alone do not establish an economic change. Other dependencies are retained conservatively, even when a particular clause might use fewer of them.

An observation stores its clause kind k_o, date d_o, evidence revision v_o, and the projected economic state $P(B_o)$, represented by a digest in the implementation. A proposed use at book B and date d, under revision v, requires

$$d = d_o, \qquad P(B) = P(B_o), \qquad v = v_o, \tag{7.1}$$

together with a compatible clause kind and the separately supplied continuing event finding. The same-day payment violates the middle equality. A status recording with no economic effect does not. A later evidence correction can violate the last equality while leaving every balance unchanged.

The evidence revision identifies a supplied immutable view; it is not a certificate that the view contains all relevant facts. A caller that omits a new notice or supplies a false old view cannot be corrected by comparing digests. Source acquisition, availability and revision selection remain separate requirements. Similarly, checking a conservative whole-economic projection can demand reassessment after an unrelated change. Minimal clause-specific dependency analysis would be a further optimization needing its own correctness argument.

Clause kind prevents a different misuse. A satisfied principal-payment clause cannot supply the missing premise for automatic bankruptcy acceleration. The selected implementation rejects that mismatch; a separately supplied qualifying bankruptcy observation can support its own automatic route. Its truth remains conditional on the applicable terms and supplied event finding. The system preserves unresolved applicability rather than turning an unsupported factual inference into a remedy.

This repair illustrates a broader simulation requirement. Assertions about current state need the dependencies under which they were observed, not only an observation timestamp. Reusing a cached covenant finding after a cash transfer, or a borrowing-capacity calculation after a collateral release, has the same risk. Keeping the previous assertion permits historical replay; withholding its current use prevents a past result from deciding a different present problem. Neither operation requires enumerating a separate tree node for every stored copy. Shared data can retain the dependence explicitly.

The checked continuation preserves complete states and the prior missed amount through payment, reassessment and rescission. Unknown or stale findings return without an accepted successor. The reference covers one selected synthetic series and supplied evidence views. Automated source revision, several simultaneous clauses and general rights-changing financial effects remain additional work, rather than consequences of this freshness check.

7.15 Closing a year without erasing a missed payment

A reporting boundary creates another opportunity to confuse present balances with financial history. Return to the late-funded borrower. In the paid case, the final settlement leaves no principal outstanding. In the unpaid case, the company has received the new equity but retains cash 40, outstanding principal 40 and arrears 40 at year-end. Both cases record new shares of

40 and a historical missed payment of 40. The second case must be able to enter the next year with the unpaid obligation intact; requiring every claim to be settled before carrying the book would exclude the scenario being studied.

Closing transfers interest expense of 20 into retained earnings under the selected report mapping. Current-period flows are archived before the new period begins. Carry then initializes the next reporting period without turning that accounting boundary into a payment, waiver or new issue. The selected policy preserves outstanding principal, arrears, share quantities and the old missed amount. It also preserves the prior clause category, observation date, observed economic-state identifier, supplied evidence revision and acceleration record.

This policy is explicit because a generic typed carry operation may permit different treatment of different memory fields. A type-correct reset of the missed amount to zero would destroy financially relevant history while leaving the opening balances coherent. The selected financing wrapper therefore requires preservation for these fields before admitting the next opening. Other memory, including period-specific counters, can require a different policy; preservation of this selected contract record is not a universal fiscal-closing rule.

Preserving the observation does not make it current. The new period and its flow state change the economic dependency in equation (7.1), and the old observation date remains old. A new proposed acceleration consequently requires a new assessment. Even after that assessment, a changed supplied evidence revision requires reassessment again. Carry must not refresh these fields merely to make a later command pass.

The unpaid example continues into 2028 with an explicit fresh observation and a payment of 40. Payment reduces cash and the outstanding claim together; the historical miss remains 40. The separately recorded acceleration also survives unless an applicable rescission or other supported rights-changing event changes it. A principal-payment observation still cannot authorize the automatic bankruptcy route. Those are different contractual premises even when a reporting boundary lies between their observations.

The exact continuation compares the complete closing entries, archived flows, next opening, subsequent payment and next-period statements on two account charts. The unchanged carry wrapper has separate compatibility evidence for the stronger remedy profile that includes supplied evidence revision and clause category. This is explicit compatibility for the selected records, not an implicit migration between earlier and later schemas. The examples demonstrate why financial histories and observation dependencies must travel together across reporting periods. Actual terms, several simultaneous clauses, guarantees and the economic consequences of acceleration still require their own sources and methods.

7.16 Conversion changes a claim and the people who control it

Preserving rights across a reporting boundary is necessary until an event changes them. A conversion provides a contrasting case: cash may stay fixed while debt disappears, new shares are issued and another holder gains control. A simulator that checks cash and balanced journals alone can miss the decision consequence.

Consider a separately supplied fiscal opening on 1 January 2028. Project assets are 100 and cash is zero. Senior debt has face and carrying amount 40. A convertible claim has face 80 and a supplied carrying amount C. There are 40 old shares and no shares held by the convertible creditor. Opening capital is $60 - C$, so the signed balances reconcile. The examples use $C = 80$ and $C = 17000/441$ as two distinct accounting premises. The latter is a diagnostic amount; selecting it does not establish a measurement rule.

The supplied contract says that the conversion signal extinguishes the claim and issues 120 shares to its holder on that date. Under the selected carrying-value transfer, the journal debits the convertible liability by C and credits issued equity by C. The face becomes zero, new shares become 120 and the observed signal and conversion are recorded. There is no cash flow or profit from this assumed transfer. The financial statements then show assets 100, liabilities 40 and equity 60 in either carrying case. Under the other signal, the debt remains outstanding and no shares issue.

Cash-flow and voting fractions follow from the resulting shares under the assumed equal rights per share. After conversion the new holder owns $120/(40 + 120) = 3/4$ and the old holder owns $1/4$. Under this supplied economic governance rule, the new holder therefore has strict majority control. Before conversion the old holder controls all shares. The separate accounting-control assessment, including potential rights before legal conversion, is explained in Section 26.16. These rights follow from the completed transition, rather than from a controller label retained from the opening. Different voting classes would require a different supplied rights rule.

The complete comparisons retain every account, the claim and share record, unrelated memory, event history, journal and report on two account charts. They also compare the resulting rights with a separately authored payoff table. Omitting claim extinction or share issue fails even when a proposed journal balances. Repeating conversion with a fresh event identifier fails because the event has already occurred. Wrong dates and signals fail before effects are accepted. Reporting the equity increase without explaining its noncash movement also fails, while preserving the already accepted conversion.

This example starts from an explicit opening and assumes the accounting transfer. It does not derive that opening from issuance, select a reporting standard, or establish that a real contract has these terms. Its contribution is narrower and useful: the accounting balances and economic rights change together, and the rights can supply the next decision without counting both converted debt and equity as surviving claims.

7.17 Reporting frequency and event frequency

A company may report quarterly while obligations settle daily. A quarterly engine that nets all flows can miss a temporary liquidity failure. A daily engine can still report quarterly by aggregation. The appropriate event resolution depends on the question and the contracts.

Adaptive resolution is possible. A long period with no state-dependent decisions can be aggregated analytically. A period containing maturity, reset, dividend and covenant dates may require explicit boundaries. The aggregation must preserve ordering and any intermediate constraint that matters.

For example, cash 10, a receipt of 20 and payment of 25 produce closing cash 5 regardless of order. If payment comes first and no overdraft is permitted, the path is infeasible despite the positive closing balance. A minimum-cash query requires the event sequence. A closing-balance query alone would not detect the problem.

7.18 Stock-flow consistency across counterparties

When several entities are modeled, a payment by one is a receipt by another, subject to fees, taxes and timing. Keeping counterpart links helps detect money creation or disappearance. In an external-counterparty boundary, the system can record the flow leaving the modeled domain rather than pretending the counterparty does not exist.

The stock-flow consistent approach in Caiani et al. (2016) gives a useful example of explicit sectors and transaction sequence. Its named cash-flow measures must be interpreted in its own definitions: a quantity called operating cash flow in that model subtracts principal repayments and should not automatically be mapped to an accounting statement category. Reusing labels without inspecting definitions is a semantic error.

AccDSL should therefore give each reported or derived quantity a definition and units. A label match is a candidate mapping, not proof of equivalence. This is especially important when connecting economic simulators, accounting standards and regulatory templates.

7.19 A coherent event-to-report method

The proposed method first admits an economic event under its contractual and behavioral premises. It updates quantities, rights and obligations. It then invokes the accounting rule for the declared basis and context, validates the posting bundle, and updates report accumulators. Source and scenario provenance accompany each stage.

At reporting boundaries, the system reconciles opening balances, movements and closing balances, checks units and perimeter, and preserves unresolved amounts or conditional interpretations. It does not force an unsupported movement into a generic other line merely to close the statement. A residual can be displayed as a diagnostic requiring explanation.

This sequence supports both concrete witnesses and symbolic families. If every member of a region has the same event rule, amounts can remain symbolic. If a guard changes the rule, the region must split or retain a piecewise expression. Accounting coherence then applies within every admissible branch, not just to an average statement that no individual history produces.

Chapter 8

Keeping uncertainty without enumerating every world

Twenty binary unknowns have more than a million assignments. A company can have thousands of contracts, continuous unknown amounts and decades of possible events. Explicit enumeration is therefore an unsuitable default. The alternative is to represent the family and compute the answer demanded by a question. This can save enormous work, but cannot guarantee that every problem has a small exact solution.

8.1 Representation size and answer size

A formula containing twenty Boolean variables can describe all 2^{20} assignments without listing them. A solver may answer whether any assignment violates a cash constraint without printing every world. If the user asks for all distinct journals, the output itself can be enormous. No representation can print an exponentially large answer in fixed small space.

Continuous uncertainty sharpens the distinction. The family $0 \leq \alpha \leq 1$ has infinitely many members, but interest $5 + 2\alpha$ is a finite exact answer. Useful computations include symbolic propagation, optimization over a region and classification of regions by behavior. Enumeration is one method, not the definition of correctness.

Darwiche and Marquis (2002) compares knowledge-compilation languages by succinctness and supported queries and transformations. Its definitions of directed acyclic representations, decomposability and determinism explain why sharing and structural restrictions help. Its tradeoffs also warn against assuming that a compact formula makes every query cheap. Compilation can itself be expensive.

8.2 Three different graphs

An evidence graph links assertions to sources and interpretations. A constraint graph links variables through relations. An execution graph links states through dated events. Their edges have different meanings. Source support is not causation; a constraint edge is not a probabilistic independence statement.

An expression DAG can share a repeated discount factor or subtotal. An immutable history graph can share common event prefixes. A transition graph can merge states only when relevant future behavior is equivalent. Confusing these forms of sharing can turn an engineering

optimization into an unjustified economic assumption.

Economic feedback may remain cyclic while expression storage is acyclic. Debt price and investment policy can be simultaneous unknowns. An acyclic residual evaluator does not eliminate the fixed-point problem. The representation and solution method answer different questions.

8.3 Share common structure without merging worlds

If a thousand histories differ only in one loan's reset clause, their source reports, asset register and most events can be shared. Each alternative stores its clause condition and local differences. This reduces storage without identifying the alternatives economically.

A common event prefix can be executed once when every history has the same relevant state and rule up to a boundary. Execution separates at the first differing reset or decision. Persistent data structures retain ancestors and changed fields for explanation and replay.

Sharing must not allow mutation of one branch to alter another. Refinancing in one history cannot rewrite the common original loan. Immutable contract versions and branch-local successor records prevent that failure. Cache identities must include all relevant inputs and rule versions, not merely a displayed company name.

8.4 Elimination must preserve consequences

If a query uses variables x and the full family is $C(x, z)$, exact elimination of z means

$$C_Q(x) \iff \exists z C(x, z). \tag{8.1}$$

Dropping every constraint containing z is generally not this projection. Constraints $x = z + 1$ and $0 \le z \le 2$ imply $1 \le x \le 3$. Removing both constraints leaves x unrestricted. Correct elimination retains the implied interval.

Projection can create dense relationships among remaining variables. A small original representation does not guarantee a small projected one. A dependency slice must include outputs, event guards, feasible actions and observation rules over the horizon. A variable absent from today's report can affect tomorrow's signal or covenant.

An asset identifier irrelevant to interest expense may matter to collateral eligibility. A customer identifier irrelevant to total receivables may matter to concentration. Query-driven elimination therefore follows economic dependencies rather than simply selecting columns present in the requested output.

8.5 Split where behavior changes

If every member of a region follows the same affine transition, propagate it symbolically. When a guard changes the event rule, partition the region or retain a piecewise expression. For cash $90 - 2\alpha$ and requirement 89.4, the split is $\alpha \le 0.3$ versus $\alpha > 0.3$. The equality belongs to compliance under the inclusive rule.

If breach has no effect on the requested output, its consequences may be deferred while the condition is retained. If it changes funding eligibility, it matters immediately. Repeated guards can still create exponentially many distinct behaviors. Guard-based execution avoids unnecessary expansion; it does not prove universal tractability.

A partition should carry its logical coverage: regions are disjoint where intended and their union covers the admitted parent. A missing equality boundary can silently lose a possible company. An overlapping boundary can double probability mass or duplicate an event. Exact rational threshold fixtures are especially useful here.

8.6 Equal totals are insufficient

Two states with cash 50 and debt 100 can differ in tomorrow's maturity. Two states with equal tomorrow payments can differ in a disclosure that changes the next decision. Neither pair can automatically be merged for the corresponding query.

Givan et al. (2003, Section 3.1) shows why equality under fixed action sequences is too weak for general policy equivalence in an MDP: policies react to intermediate states. Section 3.3 develops stochastic bisimulation through equal rewards and transition probabilities to equivalent blocks for each action. The accounting application below adds observation and requested-output preservation; this extension is a local sufficiency argument.

8.7 A sufficient probabilistic condition

Consider a finite horizon. Let x be full state, a an action and s a scenario control. Let $K_t(dx', do, dy \mid x, a, s)$ be the joint law of next state, observation and requested output. Let $\phi_t(x)$ be the compressed state. States sharing a compressed state must have equal feasible actions and current outputs. For every allowed a, s and measurable set B, also require

$$\begin{aligned} &K_t(\{(x', o, y) : (\phi_{t+1}(x'), o, y) \in B\} \mid x_1, a, s) \\ &\quad = K_t(\{(x', o, y) : (\phi_{t+1}(x'), o, y) \in B\} \mid x_2, a, s). \end{aligned} \tag{8.2}$$

Initial distribution and information must be mapped consistently. Policies use only the retained information history and permitted memory. Under these conditions, the compressed model preserves the requested joint output and observation law. Equal means, or equal marginal laws separately, are weaker. Dependence between observation and successor state matters for learning.

8.8 Finite-horizon proof

At the final date, equality of current outputs establishes the result. Suppose the compressed model reproduces continuation laws from date $t + 1$. At date t, an admissible policy selects the same action distribution for identical retained information histories. Equal feasible actions make this choice meaningful in either state.

Equation (8.2) gives the same joint law of next compressed state, observation and immediate output. At each successor, the induction hypothesis supplies the same continuation law. Integrating equal continuation laws against the equal successor law gives equality of the remaining path law. Backward induction reaches the initial date.

This condition is sufficient, not necessary. The finite helper described next checks explicitly supplied finite models; it is not a general checker for arbitrary company-state representations. If managers privately observe a hidden variable removed by ϕ, their policy may not be expressible in the compressed model. The information condition must be strengthened or the state refined. Incorrect kernels cannot be repaired by an otherwise correct aggregation proof.

8.9 Checking a supplied finite aggregation

A lending decision makes the joint-law requirement concrete. Suppose a company starts with cash 10 and observes a green or red signal before deciding whether to lend. A good outcome adds 2 to cash if it lends; a bad outcome subtracts 4. Waiting leaves cash unchanged. In this hypothetical model, good and bad states each have probability one half, and the signal identifies the state correctly with probability 9/10. The policy lends after green and waits after red. The resulting cash movements $+2$, -4 and 0 have probabilities 9/20, 1/20 and 1/2. For example, the profitable lending path needs both a good state and its green signal, so its probability is $(1/2)(9/10) = 9/20$.

Two copies of the same initial state can be merged without changing those paths. But suppose a second state reverses the signal's association with the good outcome. Green and red still each occur half the time, as do good and bad outcomes. The separate distributions therefore agree, while the policy lends in different circumstances. Comparing the joint successor state and signal detects the difference. A separate checked example keeps the successor state fixed and changes dependence between observation and immediate cash movement; separate marginal checks miss that change too.

The finite conformance helper checks this distinction for an explicitly supplied model and proposed partition. At every date it compares permitted actions and current requested outputs within each block. For every action and scenario it groups successor records by the complete triple of successor block, observation and immediate output. In the probability case it sums explicit rational masses and compares the exact totals; each supplied kernel must have total mass one. In the unweighted case it compares sets of triples. Repeated representations of a possibility never supply a probability.

Initial information needs the same care. The input specifies state and initial observation jointly. Aggregation maps the state while retaining that observation and the current requested output. Each later decision uses the retained history of observations, actions and outputs. A checked first-decision example lends after one initial observation and waits after another, even though their two states belong to the same block. Removing the observation would change the permitted policy. A policy asking directly for the hidden state is outside this information interface.

The finite argument follows the preceding induction. At each successor block, the requested current output is fixed by the block. Equality of the joint triples thus gives equal probabilities for each extended observable history. Applying the same admissible policy and continuing the induction preserves the whole path law. For an unweighted relation, compatible path concatenation and union replace probability products and sums. Every quotient transition must be possible from each member of its source block; otherwise an aggregated path might not lift to a path in the original model.

An independently coded path enumerator checks twelve accepted finite cases, including every deterministic lend-or-wait choice after the two signals, duplicate successor states, initial-information decisions and an unweighted model. Eight proposed merges fail with a specific unequal action, output or joint-kernel component. Two deliberately incorrect representative merges also produce different complete path laws. These checks support the supplied finite examples; they do not establish that a company's chosen state contains every economically relevant variable.

The helper refuses missing transitions, invalid probability mass, unrecognized information and incomplete partitions. A comparison budget exhausted before all obligations are checked returns an unknown result with the unfinished comparisons and no approved quotient. Saved results bind the full model, partition, information, outputs and horizon; reuse rechecks the condition. This is a sufficient condition over all declared states, including unreachable ones. Failure may therefore require a finer partition without proving that every aggregation for a particular initial

policy is impossible. The next example shows why persistent contract type is one dependence worth retaining.

8.10 A larger state can preserve the information structure

The distinction between a process and its information becomes especially clear in the work of Beiglböck, Pflügl and Schrott. Their information process recursively augments today's value by the conditional law of tomorrow's information process. Equal laws of suitable Markov lifts characterize their notion of equivalence of filtered processes (Beiglböck et al., 2024, Definitions 2.6–2.8, Proposition 3.6 and Theorem 3.9). A lift may be large or measure-valued. Its existence does not imply that a small sufficient state can be recovered from financial statements. Moreover, transferring strategies requires attention to independent randomization and the available stochastic basis (Beiglböck et al., 2024, Theorem 5.4 and Remark 5.7). An implementer must decide whether an actor may randomize and which observations are available, rather than deducing these permissions from equal outcome laws.

8.11 Unweighted families require relational memory

For an unweighted family, replace probabilities with a relation of feasible successors, observations and outputs. Exact aggregation must preserve joint possibilities and persistent cross-time constraints. One-step marginal sets alone are insufficient.

A fixed-or-floating loan with two-year interest paths $(5, 5)$ and $(7, 7)$ has one-step marginal set $\{5, 7\}$ at each date. Recombining those sets creates $(5, 7)$ and $(7, 5)$, which require unmodeled contract changes. Retaining the latent type or equivalent memory prevents these paths.

An enlarged family can sometimes provide an outer bound for a worst-case query. It is not an exact family of possible companies. A harmful witness from the enlarged set must be validated against original constraints before being described as economically possible. Conservatism does not excuse mislabeling the target.

8.12 Approximation needs a query-specific error

A cash error of 0.1 may be harmless for a chart and decisive at a covenant. For a smooth output, state error can imply an output bound under regularity. For a default indicator, an arbitrarily small change near a threshold can reverse the classification.

If computed cash lies in $[19.95, 20.05]$ and payment required is 20, a midpoint cannot establish feasibility for the whole region. The correct classification is unresolved unless a stronger method determines the sign. Refinement should focus on this boundary rather than indiscriminately increasing precision.

If f is Lipschitz with known constant L in a stated norm and $\|x-\hat{x}\| \le \epsilon$, then $|f(x)-f(\hat{x})| \le L\epsilon$. This follows from the definition. It requires a justified constant and domain. An unknown or very large L can make the bound unhelpful. No such smoothness claim is made for a discontinuous breach indicator.

8.13 Inner and outer approximations

Let $\overline{\mathcal{H}}$ contain the true modeled family and let $\underline{\mathcal{H}}$ be contained in it. Then

$$\inf_{h \in \overline{\mathcal{H}}} C(h) \leq \inf_{h \in \mathcal{H}} C(h) \leq \inf_{h \in \underline{\mathcal{H}}} C(h). \tag{8.3}$$

The outer minimum is a lower bound on the true minimum; the inner minimum is an upper bound. A low cash value from relaxation can be spurious. A high minimum over sampled witnesses does not prove safety.

If the outer lower bound exceeds the payment requirement, the original family is safe for this query under its assumptions. If an inner feasible witness falls below it, failure is possible. Otherwise the status remains unresolved. This three-way answer is more informative than presenting every approximate extremum as exact.

8.14 Counterexample-guided refinement

Begin with a coarse outer model. If it proves the desired bound, the query may finish. If it returns a harmful candidate, validate that candidate against full constraints. A feasible candidate is a genuine witness; an infeasible one identifies a lost dependence and motivates refinement.

The marginal interest-cash rectangle admits $(7, 90)$, but $C + I = 95$ rejects it. Adding that relation removes the spurious combination without enumeration. Other refinements can restore currency-maturity dependence, collateral ownership or persistent contract identity.

This is a proposed reasoning method, not a universal termination result. Refinement may exhaust resources. The returned bounds and unresolved region should remain available, together with the dependencies still preventing a definitive answer. A resource limit is not an economic default or a proof that the region is empty.

8.15 Caching has an economic scope

A cached answer depends on evidence, assumptions, contracts, policy, observation model, horizon, requested outputs and numerical method. A change in any relevant component invalidates it. Company name and scenario label are insufficient cache keys.

An aggregation checked for one-year interest may fail for three-year refinancing. Adding a new observed covenant can invalidate an information summary. The system should record the query class covered by an abstraction and require revalidation before broader reuse.

Provenance remains separate even when computations are shared. Two sources supporting one fact need not duplicate arithmetic, but their independent support matters if one is withdrawn. Execution identity must not replace source identity.

8.16 Trees as selected explanations

A small tree is useful for showing information arrival, choices and counterexamples. It can be generated from the symbolic family for a question. Nodes should identify whether they are exact worlds, samples, regions or hypothetical observations.

A representative tree is not exhaustive unless the finite model and coverage establish that claim. Branch probabilities require a measure. Pruning a display branch must not silently delete it from the calculation or renormalize remaining probabilities. A region can instead remain visible as an unresolved aggregate.

A hybrid interface can show important branches, formulas and bounds together. Users expand regions affecting their decisions. This preserves explanation while avoiding compulsory exhaustive materialization. The underlying family remains authoritative when a new question requires a different partition.

8.17 Budgeted computation

A query should state resources and acceptable result types. Exact values, certified bounds, feasible witnesses and statistical estimates are distinct deliverables. If exact work exceeds budget, return partial bounds and unresolved regions. Do not substitute an unsupported point merely because the interface expects a number.

Priority can go to liquidity, covenant, default and decision boundaries. This is a query-specific strategy and must not remove rare regions from a tail-risk calculation. The next chapter develops sampling and other approximations while preserving the distinction between retained possibilities and sampled evidence.

The layered proposal is symbolic constraints, shared immutable records, dependency-aware elimination, guard-based splitting, justified aggregation and controlled approximation. Each layer has a correctness condition. Choosing a graph instead of a tree is useful engineering, but the mathematical and economic obligations remain.

Chapter 9

Approximate inference and simulation without false certainty

Exact symbolic calculation will not solve every useful company model within available resources. Approximation is therefore necessary. Its legitimacy depends on naming the quantity being approximated and preserving the distinction between numerical evidence and economic possibility. A finite sample can estimate an expectation under a declared distribution. It cannot, by itself, characterize every history compatible with disclosures. A fitted surrogate can interpolate a function well while giving the wrong answer at a default boundary.

9.1 Choose the target first

Different questions require different methods. An expected cash balance, a worst-case cash balance, a breach probability, a feasible breach witness and a policy value are different targets. A method appropriate for one can be misleading for another. Sample averaging targets an expectation; minimizing over samples supplies a witness-based bound on a minimum, not an exact worst case.

Let $Y = F(H, S, \pi)$ under a declared joint measure for history H and scenario S. Monte Carlo draws $Y_1, \ldots, Y_n$ give the sample mean

$$\widehat{m}_n = \frac{1}{n} \sum_{i=1}^{n} Y_i. \tag{9.1}$$

Under independent identical sampling and finite variance, its variance is $\mathrm{Var}(Y)/n$. Those conditions are part of the claim. Correlated paths, importance weights or adaptive sampling require a different variance treatment. The formula does not apply merely because the output file has n rows.

Without a probability model, a finite collection of histories can still be useful for explanation and testing. It should be called a set of witnesses or selected cases. Equal frequencies in that collection do not establish probabilities in the underlying company family.

9.2 Sampling a persistent parameter

If a contract type is fixed but unknown, sample it once for a simulated world and retain it through time. Draw future innovations according to their conditional process. This produces the

intended joint law of parameter and trajectory. Redrawing the type each period approximates a different model.

The distinction remains important after filtering. Resampling particles may duplicate a parameter value as a computational operation; it does not mean the company changed contract. Particle identity and world identity must not be confused. A lineage record can preserve which parameter and past events each particle represents.

If the uncertainty is set-valued rather than probabilistic, selecting parameter values for numerical exploration should retain the original feasible region. Sampled values can reveal failure cases or guide refinement. They should not replace the family with a smaller set and thereby turn unexplored histories into impossible ones.

9.3 Particle filtering under known model assumptions

A particle filter approximates sequential distributions in a specified state-space model. Let x_t be the latent state at date t and y_t the observation then available. The transition density is $f(x_t \mid x_{t-1})$, the observation likelihood is $g(y_t \mid x_t)$, and a proposal q generates the next particle. First consider a sequential importance-sampling step that extends particle i without resampling it. Its updated weight satisfies

$$w_t^{(i)} \propto w_{t-1}^{(i)} \frac{g(y_t \mid x_t^{(i)}) f(x_t^{(i)} \mid x_{t-1}^{(i)})}{q(x_t^{(i)} \mid x_{t-1}^{(i)}, y_t)}. \tag{9.2}$$

The fraction is the incremental weight; multiplying it by the previous weight gives the new weight. It follows by dividing the joint posterior update by the proposal update, as in Doucet and Johansen (2008, Section 3.3, equations (29)–(30)). The proposal must cover the relevant target support. A proposal that never visits a feasible high-risk region cannot recover it by weighting.

Resampling changes this calculation because selecting a parent already uses its weight. Let $W_{t-1}^{(j)}$ be the normalized weights before selection. Suppose child i selects ancestor $A_t^{(i)} = j$ with probability $r_{t-1}^{(j)}$, then draws its next state from $q(\cdot \mid x_{t-1}^{(j)}, y_t)$. Relative to this finite weighted approximation, the joint target mass for ancestor j and new state x is proportional to $W_{t-1}^{(j)} f(x \mid x_{t-1}^{(j)}) g(y_t \mid x)$; the proposal mass is $r_{t-1}^{(j)} q(x \mid x_{t-1}^{(j)}, y_t)$. Their ratio therefore gives

$$\widetilde{w}_t^{(i)} \propto \frac{W_{t-1}^{(A_t^{(i)})}}{r_{t-1}^{(A_t^{(i)})}} \frac{g(y_t \mid x_t^{(i)}) f(x_t^{(i)} \mid x_{t-1}^{(A_t^{(i)})})}{q(x_t^{(i)} \mid x_{t-1}^{(A_t^{(i)})}, y_t)}. \tag{9.3}$$

Both ancestor and state proposals must have positive mass wherever this target has positive mass. For ordinary multinomial resampling, $r_{t-1}^{(j)} = W_{t-1}^{(j)}$, so the first ratio cancels. Equivalently, the resampled children start with equal weights $1/N$ before propagation; the old parent's weight is not multiplied again. The always-resampled and adaptive algorithms in Doucet and Johansen (2008, Sections 3.4–3.5) make that reset explicit. If resampling is skipped, the previous unequal weights are retained. These conventions concern numerical particles, not changes to the company's contract.

For example, suppose two persistent contract types have current weights 3/4 and 1/4, and a new disclosure has likelihoods 1/3 and 1. Bayes' rule gives unnormalized masses 1/4 and 1/4, hence equal posterior probabilities. A four-child resampling allocation with three copies of the first type and one of the second gives aggregate incremental weights $3(1/3) = 1$ and 1, also equal. Multiplying by the old parent weights again would instead give 3/4 and 1/4. This exact

allocation exposes the double counting; a random finite resampling draw need not reproduce the posterior exactly.

Doucet and Johansen (2008, Sections 2–3) explains sequential importance sampling, resampling and degeneracy. Its filtering setup conditions on a specified model. Static unknown parameters are not automatically handled well by adding them to the state and repeatedly resampling. That extension can suffer severe impoverishment.

For AccDSL, particle filtering is appropriate only after prior, transition and observation laws are admitted. It can be an optional numerical view of a history family. The family and source constraints remain available to detect support loss and distinguish probabilistic approximation from exact admissibility.

9.4 Weight degeneracy and genealogical collapse

If nearly all weight concentrates on one particle, a nominal population of ten thousand does not provide ten thousand effective alternatives. A common weight diagnostic is

$$N_{\text{eff}} = \frac{1}{\sum_i (\bar{w}_i)^2}, \tag{9.4}$$

where normalized weights sum to one. It summarizes weight concentration, not every form of approximation error. Uniform weights after resampling can coexist with many duplicate ancestors.

Genealogical collapse matters for historical reconstruction and smoothing. Particles can share the same early history even while their current states differ. If an old unknown contract type is lost, later data cannot revive it without a rejuvenation method or retained symbolic support. A high current effective sample size does not establish historical diversity.

Diagnostics should therefore include unique persistent parameter values or clusters, ancestor diversity, support checks against constraints and sensitivity to proposal design. These are explanatory or veto diagnostics under a stated protocol. No single scalar certifies that the posterior or scenario tail is accurate.

9.5 Rejuvenation and conditional calculation

Move steps can restore diversity if they preserve the intended target distribution. Their acceptance probabilities, conditional variables and constraints must be correct. Arbitrarily perturbing a loan rate or maturity can create a history inconsistent with evidence. Projection back into a feasible set can also alter the target distribution unless its effect is accounted for.

Analytic integration of conditionally tractable components can reduce sampling variance. Doucet and Johansen (2008, Section 4.6) discusses Rao–Blackwellization. In a company model, some accounting accumulations may be deterministic conditional on contract type and shock path, so they need not be sampled separately.

This benefit depends on a valid conditional decomposition. Integrating a variable out while forgetting the dependence it induces on remaining quantities gives the wrong target. The same projection discipline used for symbolic families applies probabilistically: marginalization preserves dependence, whereas independent replacement usually does not.

9.6 Rare events and empty samples

If a breach has probability p and paths are independent, the probability of observing no breach in n draws is $(1-p)^n$. For $p = 0.001$ and $n = 100$, this is about 0.905. Zero observed breaches are therefore unsurprising and do not establish safety.

For an ideal Bernoulli experiment with zero events, a one-sided 95 per cent upper bound solves $(1-p)^n = 0.05$, giving $p = 1 - 0.05^{1/n}$. This is a statistical bound under the sampling assumptions, not a statement about model misspecification. If the simulator omits a default mechanism, more draws cannot detect it.

Tail estimates need enough events and a suitable uncertainty analysis. Extreme quantiles and maxima have high sampling variability. A lower observed maximum in one run is descriptive evidence, not a defensible ranking of methods. The target, sampling design and uncertainty calculation should be specified before interpreting a comparison.

9.7 Importance sampling

Importance sampling deliberately visits rare regions more often. If target density is $p(x)$ and proposal is $q(x)$, an expectation can be estimated using weights $p(x)/q(x)$ where the proposal covers target support. Oversampling distress is useful only if weights and support are handled correctly.

For event indicator $B(x)$, the ordinary importance estimator is $n^{-1} \sum_i B(X_i)p(X_i)/q(X_i)$ with $X_i \sim q$. Self-normalized weights define a different finite-sample estimator. The method should record which is used and how uncertainty is assessed.

A proposal chosen after observing a preliminary sample can be effective, but adaptive procedures require care in interpretation. Selecting only paths that breach and averaging them estimates a conditional severity, not the unconditional breach probability. A report should separate frequency from loss given breach.

9.8 Stress testing without probabilities

A deterministic stress can be useful even when its likelihood is unknown. It asks what happens under the specified path. Reverse stress can search for conditions that produce a shortfall. Neither exercise needs invented probabilities.

For example, solve for the smallest rate increase causing cash below a threshold under each compatible debt history. The answer is a family of thresholds. A minimum across histories identifies a vulnerable compatible case; a maximum identifies how much stress is needed to affect every admitted history, subject to monotonicity and feasibility.

The search should include joint constraints on shocks. Allowing sales, exchange rates and funding spreads to vary independently over arbitrary ranges can produce an economically incoherent reverse stress. A broad stress set may be intentionally conservative, but its interpretation is then a designed adversarial exercise rather than an estimated likelihood.

9.9 Scenario reduction

A large finite scenario set can be reduced by selecting representative paths or aggregating similar ones. Similarity in Euclidean shock distance does not automatically imply similarity in financial

consequences. Two rate paths close in norm can lie on opposite sides of a covenant or option boundary.

The proposed AccDSL method should define reduction error in terms of requested outputs, feasible policies and information timing. If a reduction is intended to preserve an optimization value, it needs evidence for that target. Preserving a few unconditional moments is insufficient for a path-dependent refinancing decision.

Probability-metric reduction provides a precise method when a measure and suitable regularity are available. Its usefulness depends on the financial quantity and decision being approximated. The following derivations distinguish fixed-policy expectations, threshold probabilities and policy optimization.

9.10 What probability-metric reduction preserves

A finite probability model permits a more precise reduction than deleting paths and renormalizing. Let scenarios s_i have admitted probabilities p_i, with $\sum_i p_i = 1$. Choose a smaller retained index set K and map each original scenario to a retained representative $s_{k(i)}$. The reduced probability of representative j is

$$q_j = \sum_{i:k(i)=j} p_i, \qquad j \in K. \tag{9.5}$$

All probability mass remains accounted for, although the distribution has changed. The method approximates a specified measure; it supplies no probabilities for an unweighted family of unknown company histories.

Dupačová et al. (2000, Section 3, Theorem 3.1 and its proof) derives optimal redistribution for a fixed retained subset under the paper's probability-metric assumptions: deleted mass moves to a nearest retained scenario under the admitted cost. The choice of which scenarios to retain remains a separate combinatorial problem. Algorithms 3.6 and 3.7 are backward-reduction and forward-selection heuristics. Their sequential choices do not generally establish a globally optimal subset.

A short local derivation shows what a distance can buy for a financial query. Suppose a fixed-policy cash outcome $f(s)$ satisfies $|f(s) - f(u)| \le Ld(s,u)$ for all admitted pairs, where d is a stated scenario distance and L is a justified constant. Then

$$\begin{aligned} |\mathbb{E}_P f - \mathbb{E}_Q f| &= \left| \sum_i p_i [f(s_i) - f(s_{k(i)})] \right| \\ &\le L \sum_i p_i d(s_i, s_{k(i)}). \end{aligned} \tag{9.6}$$

The first equality substitutes the redistributed masses; the inequality applies the triangle inequality and the assumed regularity. Thus the weighted movement of scenarios bounds this expected-output error. A distance small in interest rates is useful only when the required output responds regularly in that distance.

For example, let the original states be 0, 1 and 2 with probabilities $1/4, 1/2, 1/4$. Keep 0 and 2, and resolve the middle state's distance tie by assigning it to 0. Reduced masses are $3/4, 1/4$. Under absolute distance, weighted movement is $1/2$. For $f(s) = s$, the original mean is 1 and the reduced mean is $1/2$, exactly attaining the bound with $L = 1$. Mass conservation alone does not preserve the mean.

A default indicator is discontinuous. Put equal mass at $1 - \eta$ and $1 + \eta$, where $\eta > 0$, and retain only the first state. Weighted movement is η, tending to zero, but the probability of

$s > 1$ changes from one half to zero. Therefore a small distance does not by itself give a useful default-probability guarantee. One needs additional control of probability near the boundary, a distance tailored to the event, or explicit preservation of the relevant region.

9.11 What sequential transport theorems require

Nested and adapted transport distances address the temporal issue by restricting couplings so that matching paths cannot depend on information that is unavailable at the relevant date. Pflug and Pichler's stability argument uses convex decision sets, convex loss in decisions and uniform regularity in uncertainty (Pflug and Pichler, 2012, Definition 5.1, Theorem 6.1 and proof). A projected action is formed by conditional averaging. If the only permitted issuance sizes are 0 and 1, an average of 1/2 is infeasible. The theorem therefore does not automatically justify merging discrete issuance choices.

Backhoff-Veraguas and coauthors obtain related stability results for financial optimization with bounded predictable strategies and regular payoffs (Backhoff-Veraguas et al., 2020, Theorem 1.5; Lemma 3.7; Theorem 3.10). Their weak transfer permits a different strategy in the other model. Reusing the same strategy needs additional regularity. Their semimartingale distance also has a different cost functional from a simple sum of discrete path deviations; the paper explains why time refinement matters. Its examples of discontinuous strategies and exact superhedging delimit the guarantee. These are useful reasons to formulate the company decision problem precisely, rather than reasons to attach one distance to every accounting query.

9.12 Reduction before choosing a policy

An approximation adequate for one fixed policy may fail after optimization selects a different one. Suppose, more strongly, that every policy in the same feasible, nonanticipative class satisfies

$$|V_P(\pi) - V_Q(\pi)| \le \delta. \tag{9.7}$$

If maximizers π_P and π_Q exist, adding and subtracting their reduced-model values gives

$$\begin{aligned} V_P(\pi_P) - V_P(\pi_Q) &= [V_P(\pi_P) - V_Q(\pi_P)] \\ &\quad + [V_Q(\pi_P) - V_Q(\pi_Q)] + [V_Q(\pi_Q) - V_P(\pi_Q)] \\ &\le 2\delta. \end{aligned} \tag{9.8}$$

The middle term is nonpositive because π_Q optimizes the reduced problem. This elementary bound needs uniform control over the policy class, identical feasibility and consistent information. A bound established only at the final selected policy cannot justify the argument.

Dupačová et al. (2000, Section 2) give stability results for convex stochastic programs under explicit solution-set, integrability and regularity assumptions. Remark 2.2 adds dependence conditions for its multistage setting. Those conditions do not automatically cover discrete default, share conversion, nonconvex financing or private information in a company model. Nor does a close distribution over complete paths establish that intermediate observations and admissible decisions are preserved.

An AccDSL reduction should therefore state the admitted measure, retained scenarios, redistribution rule, distance and query guarantee. It should separately check liquidity and default boundaries and the actor's observation structure. The original joint history family remains available for refinement. Reducing a numerical probability approximation must not erase a

structurally possible contract or turn an unobserved future path into information available today.

9.13 Surrogates and boundary errors

A surrogate can approximate expensive valuations or transitions. Its input domain, training distribution, output definition and constraints must be explicit. A small average prediction error can coexist with large errors in rare distressed states. If those states determine issuance feasibility, the surrogate may be a poor component despite good average accuracy.

Suppose true funding capacity is $c(x)$ and issuance is allowed only when $c(x) \geq K$. An approximation error bound $|\hat{c} - c| \leq \epsilon$ permits safe classification only outside $[K - \epsilon, K + \epsilon]$. Inside that band, refine or return uncertainty. A point prediction alone gives no such protection.

A surrogate should also respect units and economic constraints. Negative principal, duplicated shares or cash creation may be detectable regardless of predictive error. Enforcing constraints can improve mechanical validity, but does not prove that the underlying economic mapping is correct. Both checks remain necessary.

9.14 A quantitative boundary argument

Default, covenant breach and a margin call are threshold events. Let X be exact cash at a decision date, Y its approximation under a specified joint coupling, and c the contractual threshold. All three are in the same currency and date basis. A cash approximation of $-\epsilon$ by $+\epsilon$ has error 2ϵ, yet changes $\mathbf{1}\{X < 0\}$ from one to zero. Small cash error alone does not imply small default-probability error.

Proposition 9.1. *If $\mathbb{E}|X - Y| < \infty$, then for every $\delta > 0$,*

$$|\mathbb{P}(X < c) - \mathbb{P}(Y < c)| \leq \mathbb{P}(|X - c| \leq \delta) + \frac{\mathbb{E}|X - Y|}{\delta}. \tag{9.9}$$

Proof. The absolute difference in event probabilities is at most the probability that their indicators disagree. If $|X - c| > \delta$ and $|X - Y| < \delta$, X and Y lie on the same side of c. Consequently disagreement is contained in $\{|X - c| \leq \delta\} \cup \{|X - Y| \geq \delta\}$. The union bound and Markov's inequality give (9.9). □

The first term is probability mass close to the covenant; the second is the probability allowed for a sufficiently large approximation error. This local derivation is independent of any claim that a particular algorithm supplies the required coupling error. Write $e = \mathbb{E}|X - Y|$. If $e = 0$, then $X = Y$ almost surely and the threshold probabilities agree. For $e > 0$, if X has a density bounded by $M > 0$ near c, the first term is at most $2M\delta$. Choosing $\delta = \sqrt{e/(2M)}$, when that choice lies within the neighborhood where the density bound holds, gives $2\sqrt{2Me}$. Atoms at the threshold and uncontrolled conditional distributions prevent this simplification.

For several dates, a union bound can bound the probability that any breach indicator differs by the sum of the per-date bounds, provided the relevant coupling and cash errors are supplied. If a different breach decision changes subsequent actions and cash flows, those cash errors must already account for that feedback; an open-loop cash comparison does not establish them. When there are no weights over histories, the appropriate output is instead an uncertain boundary region or certified feasible-set containment. A probability bound cannot be manufactured from the proportion of enumerated witnesses.

9.15 A common-policy regret argument

Let $J(\pi)$ and $\widetilde{J}(\pi)$ be expected costs in two models, with the same admissible policy class Π, finite expected costs and finite optimal values. This comparison presupposes that a policy can be implemented with the information and feasible actions in both models. Suppose a uniform bound $\sup_{\pi\in\Pi} |J(\pi) - \widetilde{J}(\pi)| \le \varepsilon$ holds. If $\widehat{\pi}$ is η-optimal for the approximate problem, then

$$\begin{aligned} J(\widehat{\pi}) &\le \widetilde{J}(\widehat{\pi}) + \varepsilon \\ &\le \inf_{\pi\in\Pi} \widetilde{J}(\pi) + \eta + \varepsilon \\ &\le \inf_{\pi\in\Pi} J(\pi) + \eta + 2\varepsilon. \end{aligned} \tag{9.10}$$

The last step follows by applying the uniform bound to an arbitrarily near-optimal policy of the exact problem, so attainment is unnecessary. The result is a regret bound for a common feasible policy. If one only knows that the two infima are close, the first inequality is missing. If a policy lift changes the policy class, a bound for that lift and the relevant comparison class is needed. This is why reporting a small optimized residual or close portfolio value alone does not establish that a compressed financing policy will work in the original company.

9.16 When sampled paths disclose their own futures

With continuous first-period observations, independent sampled paths typically have distinct first values. If each stored first value has only its single sampled continuation, that value becomes an identifier for the sampled future. An optimizer can exploit this unintended information. More paths do not by themselves repair the information structure of the empirical measure. Backhoff and coauthors give an explicit optimal-stopping counterexample (Backhoff et al., 2021, Example 7.1). Their adapted empirical measure quantizes observations so that paths share histories. Strong consistency applies to iid complete paths on a fixed finite horizon and bounded state space; quantitative rates additionally require Lipschitz conditional kernels (Backhoff et al., 2021, Definitions 1.1–1.2, Theorems 1.3, 1.5, 1.7).

A Markov modification improves the horizon exponent only when the process is in fact Markov (Backhoff et al., 2021, Theorem 6.1); constants still depend on the horizon. A company's current balance sheet generally is not a sufficient Markov state when hidden debt terms or asset vintages matter. Nor are reports from one issuer an iid sample of complete company paths. The method suggests how a deliberately generated simulation sample can retain information. Its rates cannot simply be assigned to an observed issuer panel.

9.17 A finite coupling method for discrete decisions

A stronger but directly checkable condition can avoid averaging actions. Consider a finite horizon of T transitions. Keep a persistent contract type θ fixed and let a history record the actor's observations and past actions. In both the full model and a reduced model, require the same feasible actions at paired histories. Specify a relation between their states and couple the initial states so that they belong to this relation, their initial observations agree, and every requested current output agrees almost surely. An equal opening signal alone does not establish these other conditions.

At each paired history and common action, suppose a joint coupling of the next state, observation and reported output leaves the states in the relation and the observation/output records equal

with conditional probability at least $1 - \delta_t$, where $0 \le \delta_t \le 1$. This condition includes every history and action reachable before a mismatch; fitting one historical policy is insufficient. A common deterministic policy chooses the same action while its retained information histories agree. Randomized policies can use the same independent random draws if both models permit that randomization. Neither policy may depend on a discarded private signal.

Let E_t mean that the initial pairing and all comparisons through transition t have succeeded. Under the initial condition, $\mathbb{P}(E_0) = 1$. Integrating the conditional success bound over the surviving histories gives $\mathbb{P}(E_t) \ge (1 - \delta_t)\mathbb{P}(E_{t-1})$. This unconditional recurrence also applies when no histories survive; it does not condition on an event of zero probability. Iterating it gives

$$\mathbb{P}(E_T^c) \le 1 - \prod_{t=1}^{T}(1 - \delta_t) \le \sum_{t=1}^{T} \delta_t. \tag{9.11}$$

The requested path loss must also be preserved by a successful comparison. Assume that the two models use the same measurable loss functional of the retained observations, actions and requested outputs, with values in $[0, B]$. Its two path evaluations therefore agree on E_T and can differ by at most B otherwise. Applying the triangle inequality to the expectation under the chosen coupling yields

$$|\mathbb{E}L - \widetilde{\mathbb{E}}L| \le B\mathbb{P}(E_T^c) \le B\left\{1 - \prod_{t=1}^{T}(1 - \delta_t)\right\}. \tag{9.12}$$

The two expectations refer to the same loss functional under the respective model laws. No convexity or fractional issuance is required. A default indicator is covered when default is one of the compared outputs, or is determined by the matched retained records. A loss depending on a hidden recovery allocation that was discarded is not covered merely because its values are bounded.

Initial and within-pair approximation errors can be included explicitly. If $\mathbb{P}(E_0) \ge 1 - \delta_0$ and the two losses differ by at most ρ on E_T, with $0 \le \rho \le B$, define

$$\Delta_T = 1 - (1 - \delta_0)\prod_{t=1}^{T}(1 - \delta_t), \tag{9.13}$$

$$|\mathbb{E}L - \widetilde{\mathbb{E}}L| \le \rho + (B - \rho)\Delta_T. \tag{9.14}$$

Indeed, with $p = \mathbb{P}(E_T^c) \le \Delta_T$, the coupled expected absolute loss difference is at most $\rho(1 - p) + Bp$. The earlier bound is the special case $\delta_0 = \rho = 0$. These terms have distinct meanings: δ_0 measures failure of the initial pairing; ρ is a loss discrepancy even when the paired path remains valid. With no transitions, identical signals but different opening cash already show why transition errors cannot cover an omitted initial report. Likewise, matching cash paths does not bound an unreported creditor-allocation loss. The initial relation and requested loss must therefore be chosen before the comparison.

For finite kernels with common outcome labels, the smallest one-step mismatch probability is their total-variation distance $\frac{1}{2}\sum_z |K(z) - \widetilde{K}(z)|$. Pairing full states with reduced states requires a coupling whose successful pairs remain in the declared relation; marginal observation agreement is insufficient. An implementable finite procedure first partitions states by persistent type, available information, feasible actions and required outputs. It then compares joint successor kernels into the current blocks for every action, splitting a block whenever these differ. Repeat until no block changes. Each split reduces the number of states per block, so a finite partition stabilizes after finitely many splits. This yields an exact quotient for the declared finite model, not a claim that the model contains every economically possible company.

Approximate sharing uses the same comparison with an explicit error allowance and a checked pairing. Threshold regimes and different feasible actions are kept separate unless their mismatch has been bounded. An unresolved comparison is retained as unresolved, not assigned zero error. For a fixed common policy class, the uniform cost bound in (9.12) also gives the common-policy $2\varepsilon + \eta$ regret bound derived in this chapter. If the persistent types are unweighted, apply the bound within each type. A uniform type-wise bound also bounds the worst-type value, since $|\sup_\theta J_\theta - \sup_\theta \widetilde{J}_\theta| \le \sup_\theta |J_\theta - \widetilde{J}_\theta|$. The adversary does not redraw the contract type at each date.

The exact reference uses three periods, cash observed by management, a hidden fixed type, binary borrowing, and a default penalty. It compares complete paths for all 64 policies in the explicitly restricted class of date-and- zero-cash rules. Removing an irrelevant tag preserves every path. Merging types is rejected. A synthetic change of 1/20 in each period's bad-expense probability gives the conservative cost bound $12\{1 - (19/20)^3\} = 3423/2000$. The largest observed value discrepancies are 4433/8000 and 1/2 in the two types, within the bound. These finite checks support the stated method and expose its conservatism. They do not supply a capacity estimate or a bound for an unenumerated continuous company model.

9.18 Error propagation through time

Small one-step errors can accumulate or amplify through nonlinear feedback. For an illustrative recursion with Lipschitz transition constant L, one-step approximation error at most ϵ and initial error e_0, a bound satisfies

$$e_{t+1} \le L e_t + \epsilon. \tag{9.15}$$

Iterating gives $e_T \le L^T e_0 + \epsilon \sum_{j=0}^{T-1} L^j$. When $L < 1$, the sum is bounded by $1/(1 - L)$; when $L > 1$, it can grow rapidly. This is a local mathematical bound under its assumptions, not a claim that a company model is globally Lipschitz.

Default and option transitions can be discontinuous. Near them, a small state error can select a different regime and invalidate a smooth bound. Validation should therefore include boundary perturbations and multi-period rollouts, not only one-step residuals on typical states.

9.19 Combining exact and sampled components

A hybrid method can keep discrete contract alternatives exact, solve linear accounting relations symbolically and sample future demand conditional on each alternative. It can also bound unresolved histories while estimating conditional means for selected regions. This is often more informative than forcing every uncertainty into one Monte Carlo distribution.

The result must retain which component is exact, bounded or estimated. An exact accounting identity inside a sampled simulation does not make a sample quantile exact. A certified bound on one contract region does not cover omitted regions. Mixed results should be combined using rules appropriate to their evidence type.

A useful report can state that breach is impossible in one region, possible with a verified witness in another, and probabilistically estimated in a third under an admitted measure. These are compatible forms of information. Reducing them to one unlabeled scalar would lose their meaning.

9.20 Stopping rules

A numerical budget should specify what makes an approximation adequate for the decision. A tolerance on expected cash may be appropriate for planning; a confidence interval for breach probability may be needed for a probabilistic statement; a certified lower bound may be needed for a robust safety claim.

Stopping because the sample mean appears stable can miss a rare region. Repeatedly checking a confidence interval without accounting for the stopping design can alter its coverage. A deterministic residual tolerance does not establish statistical accuracy. The stopping criterion should match the target and method.

If the required evidence cannot be obtained within budget, the answer remains provisional or unresolved. This is not a reason to discard the model's useful conditional results. It is a reason to state what is known, what remains uncertain and which computation would discriminate the remaining cases.

9.21 Proposed approximation records

Each approximation should record its target, retained family, measure if any, sampling or discretization rule, random seeds, support assumptions, error estimates, failed regions and method version. A run manifest supports reproduction; the reader-facing result should explain the consequences rather than expose every internal field.

Tests should include a small exact reference, a rare-event fixture, a persistent-parameter fixture, a boundary-sensitive fixture and an intentionally unsupported region. Comparison should use the same model and query. A fast method solving a simplified target is not a faster solution to the original target unless the difference is quantified.

The objective is controlled approximation, not universal exactness. The system can remain useful while refusing an unjustified probability or exact bound. Retaining the original family ensures that future evidence and better methods can revisit uncertainty that today's computation could not resolve.

Chapter 10

Decisions when probabilities and histories remain ambiguous

A simulator can describe possible consequences without selecting an action. A decision method needs an objective, a feasible policy class and a treatment of uncertainty. These are substantive choices. Worst-case liquidity, expected shareholder value and minimax regret answer different questions. No accounting identity selects among them. When uncertainty persists over time, the choice also determines how plans respond to new information.

10.1 Evaluation before optimization

For a fixed policy π, let $V(\pi, h, s)$ be a defined payoff or cost. It may be terminal cash, discounted owner distributions, service continuity or another quantity. Its units and horizon should be stated. Combining several objectives requires weights or constraints that should remain visible.

Evaluating a supplied policy is often the safest first task. It separates economic mechanics from policy selection. Once those mechanics are checked, optimization can search over an admissible class. A numerical optimizer should not be allowed to choose actions outside contractual rights or use information unavailable to the actor.

The distinction also helps interpret historical behavior. A policy that would maximize a modeled objective need not describe what management actually did. Simulation under that policy is a normative or counterfactual exercise unless behavioral evidence supports its use as a forecast.

10.2 Expected value under a declared measure

If a joint probability measure P is admitted, a policy can be evaluated by $\mathbb{E}_P[V(\pi, H, S)]$. The distribution must preserve dependence between history, shocks and signals. A posterior over report-compatible histories is one possible input, but its prior and likelihood assumptions remain part of the result.

If two histories have equal likelihood under all retained evidence, their posterior odds depend on prior odds. More observations of the same uninformative kind do not remove that dependence. The robust Bayesian analysis of Giacomini and Kitagawa (2021) makes this issue explicit for set-identified models. Its application here motivates sensitivity to unresolved prior choices, not an automatic probability for each journal.

A report can show policy rankings under several admitted priors. If every prior gives the same

ranking, that conclusion is more robust to the specified ambiguity. If rankings differ, a single average conceals a material decision sensitivity. The next useful evidence may be the fact that distinguishes those priors' consequences.

10.3 Worst-case evaluation

For an unweighted feasible family $\mathcal{U}$, a robust payoff is

$$W(\pi) = \inf_{u \in \mathcal{U}} V(\pi, u). \tag{10.1}$$

A maximin policy maximizes W. This criterion protects against the worst admitted case. Its conservatism depends on the family and objective. If the family includes implausible combinations introduced by a loose relaxation, the policy may respond to a company that cannot exist.

Robust feasibility is often more interpretable than robust optimization of a single scalar. For example, require every admitted history to maintain cash above zero, then compare expected or conditional costs among feasible policies. If no policy satisfies the requirement, the result should identify the conflicting cases or needed assumptions. It should not silently shrink the family until feasibility appears.

A worst-case result does not need probabilities, but it does need a defensible scope. An unlimited shock set can make every policy fail. That may reveal a badly posed decision problem rather than an economic conclusion. Bounds on shocks and actions require provenance and sensitivity analysis.

10.4 Regret compares actions within the same world

Regret measures the loss relative to the best feasible action for the same realized world. For a one-time action a, define

$$R(a, u) = \sup_{b \in A} V(b, u) - V(a, u). \tag{10.2}$$

Minimax regret chooses the action with the smallest worst regret. The comparator uses the same u, so it should not combine an optimistic world for one action with a pessimistic world for another.

Consider two actions and two states with payoffs $A = (10, 0)$ and $B = (6, 6)$. Maximin selects B because its minimum is 6 rather than zero. Statewise best payoffs are $(10, 6)$, so regrets are $A = (0, 6)$ and $B = (4, 0)$; minimax regret also selects B here. Other payoff tables can separate the criteria. Agreement in one example does not establish equivalence.

In sequential settings, the hindsight comparator may use information the actual actor did not have. That can be intentional in regret analysis, but it must be stated. A claim about implementable policy improvement should compare policies under the same information restrictions.

10.5 Multiple priors

A set $\mathcal{P}$ of probability measures can represent ambiguity about a stochastic model. A lower expected payoff is

$$\underline{V}(\pi) = \inf_{P \in \mathcal{P}} \mathbb{E}_P[V(\pi)]. \tag{10.3}$$

This differs from an unweighted family of deterministic worlds. A prior set can retain restrictions on event probabilities, dependence and learning. Replacing it by all distributions over its support generally enlarges ambiguity and can change the decision.

A singleton prior set reduces to ordinary expected value. A set containing every point mass over a finite family reduces lower expectation to the minimum payoff over that family. Intermediate sets express intermediate restrictions. The representation should preserve which restrictions came from evidence and which express preferences or modeling caution.

Updating each prior by Bayes' rule requires positive conditioning probabilities or a declared extension for null events. Removing priors solely because they yield inconvenient decisions changes the ambiguity model. A likelihood-based restriction can be justified under a specified method, but should not be confused with an arbitrary pruning heuristic.

10.6 Why dynamic consistency matters

A policy selected today may cease to be preferred tomorrow after information arrives. Some changes are rational responses to information. Others arise because the decision criterion evaluates future uncertainty differently from the original plan. To model behavior, one must specify whether the actor commits to the original policy or reoptimizes.

Epstein and Schneider (2002, Definition 3.1 and Theorem 3.2) connects recursive multiple-priors preferences, dynamic consistency and rectangular prior sets under its stated axioms and information filtration. Rectangularity permits specified combinations of one-step marginals and conditional distributions. It is a substantive restriction on the prior family relative to information, not a harmless implementation trick.

The source's theorem is not a statement that every economically natural uncertainty set is rectangular. Nor does it authorize changing a fixed unknown contract at each date. Applying a recursive worst-case operator without examining the allowed cross-time combinations can solve a different decision problem.

10.7 A concrete pasting counterexample

Consider two dates of losses and a hidden type θ that remains fixed. If $\theta = A$, losses are $(0, 10)$. If $\theta = B$, they are $(10, 0)$. The total loss is 10 in either type. Thus the worst total loss over the original two-world family is 10.

If a stagewise procedure independently selects the larger loss at each date, it constructs $(10, 10)$ and reports total loss 20. That path is absent from the original family. The procedure has allowed the adversary to switch type across dates. It computes a worst case over an enlarged family.

Calling the enlarged result conservative does not make it equal to the original target. It may be useful as an outer bound, but the target must be labeled. Retaining θ in the state prevents the unintended switch. This simple example is the deterministic analogue of the dependence issues that rectangularity addresses for prior sets.

10.8 Rectangular enlargement changes the model

A rectangular hull can facilitate recursive evaluation by admitting combinations of conditionals that were not jointly present in the original prior set. The source explicitly discusses enlargement

and its relation to the information structure. In an economic application, added combinations may represent an adversary whose choices adapt over time.

If the original ambiguity concerns one fixed but unknown loan contract, allowing adaptive adversarial changes to that contract is usually a different physical model. One can retain the contract type as part of the state and formulate conditional uncertainty appropriately. If a robust policy intentionally protects against changing terms, those changes should be modeled as amendments, refinancing or legal events.

AccDSL should therefore record whether a robust method uses the original joint ambiguity set, a rectangular enlargement or a commitment formulation. The result should compare the enlargement's effect on important outputs where feasible. A convenient Bellman recursion does not justify hiding that change.

10.9 Commitment and reoptimization

A commitment policy is selected at the initial date as a full rule for future information histories. A reoptimizing actor selects a new continuation policy at each date. These can differ under nonrectangular ambiguity or changing preferences. The simulator should not alternate between them silently.

For a treasury policy approved by a board, some actions may be committed while others remain discretionary. A hedge may be binding after execution; a future dividend target may be revisable. The feasible policy class should represent these commitments. Reoptimizing an already executed hedge as though it never occurred gives the actor an unavailable option.

A useful analysis can compare commitment and reoptimization explicitly. Differences then reveal the value or cost of flexibility under the chosen uncertainty model. They should not be attributed solely to forecasting skill.

10.10 Distributional ambiguity and historical ambiguity

Uncertainty about which debt contract exists differs from uncertainty about next year's demand distribution. Both can enter a decision, but they should retain separate origins. A contract family may be constrained by notes; a demand ambiguity set may be estimated from data or chosen for stress robustness.

Combining them independently can be unjustified if financing structure and demand exposure are related. The joint ambiguity set should preserve known dependence. Conversely, imposing a relation merely because it makes optimization easier can exclude valid histories. The default should be explicit uncertainty, not an invented joint law.

A layered result can report policy performance conditional on contract region and across demand models. This makes it possible to ask whether better contract evidence or better demand estimation would change the decision. A single blended uncertainty score obscures that distinction.

10.11 Risk constraints and probability thresholds

A chance constraint such as $P(C_t < 0) \leq \epsilon$ requires a probability model and a specified time event. A bound on failure at each date differs from a bound on failure at any date. Summing marginal probabilities gives a union bound, not generally the exact path probability.

If failure events are strongly dependent, treating them as independent can distort the probability of ever breaching. Persistent contract uncertainty is one source of this dependence. A company with a floating loan may remain vulnerable in every high-rate period; redrawing the type can alter duration and clustering.

A robust chance constraint requires the probability bound under every measure in an admitted set. It is stronger than a constraint under an averaged measure. Numerical estimation must account for sampling error near the threshold. A point estimate just below the limit is not sufficient evidence of compliance.

10.12 Value of information under ambiguity

A proposed disclosure partitions the history family. Under a probability model, expected information value compares optimized expected payoffs before and after the signal, accounting for its cost and timing. Under maximin or regret, the criterion changes. Information can matter because it enables different feasible actions in different regions.

A signal that arrives after a financing commitment may have little operational value even if it greatly narrows a posterior. A signal about an irrelevant parameter may reduce statistical uncertainty without changing decisions. The useful target is the decision consequence, not entropy reduction alone.

Without defensible probabilities, one can report the range of improvements across possible signal outcomes and criteria. A guaranteed improvement may be zero even when some outcomes are valuable. This is a meaningful conclusion and avoids assigning probabilities merely to produce an expected number.

10.13 Selecting a method for a user question

For disclosure explanation, preserve possible and certain answers. For a supplied stress, evaluate conditional mechanics. For planning under an admitted distribution, estimate expected values and relevant risks. For a safety requirement across compatible histories, use robust feasibility and bounds. For competing actions without probabilities, consider dominance and explicitly chosen robust or regret criteria.

Dominance is particularly useful: if one feasible policy performs at least as well in every admitted world and strictly better in some, it can be preferred under many monotone criteria. If policies cross, the choice depends on preferences or probability assumptions. The simulator should show that dependence rather than invent a universal ranking.

A method-selection interface should ask for a criterion only when the user has requested a decision that the evidence alone cannot determine. It can still complete mechanical comparisons and identify dominant or infeasible options. This preserves progress while keeping value judgments explicit.

10.14 Implementation and validation

A decision record should specify actor, objective, information, feasible actions, uncertainty set or measure, commitment convention, horizon and terminal treatment. Terminal assumptions matter: liquidating assets at book value at the horizon can create an artificial reward and change earlier decisions.

Small exact finite examples should compare direct policy enumeration with recursive methods. Include the fixed-type pasting counterexample, an information-revelation example, a capacity-constrained financing example and a case where expected value and robust choice disagree. These fixtures discriminate semantic errors before expensive numerical optimization.

Passing them establishes the specified criterion and timing mechanics. It does not establish that the criterion describes management or that the uncertainty set is empirically adequate. Those remain separate assumptions or research questions. The system can support several methods without declaring one a universal default.

Chapter 11

Solving the economic equations that the model actually states

A correct financial representation can still be evaluated by the wrong numerical method. Conversely, a numerical method can converge accurately to equations that describe the wrong financing problem. The first obligation is to state the target equations, their timing and their feasible domain. Only then does it make sense to choose a solver or train a neural approximation. The risky-debt convention discussed earlier illustrates why a small residual is insufficient: an inconsistent cash convention can be solved very accurately.

11.1 Separate identities, conditions and approximations

Accounting identities reconcile stocks and flows. Contract equations determine obligations from terms. Behavioral conditions describe choices. Equilibrium conditions connect counterparties and prices. Numerical approximations evaluate these relations with finite resources. They should not be treated as interchangeable equations in an undifferentiated loss.

An accounting identity can often be enforced exactly by construction. A lender pricing equation may involve an expectation and an endogenous default rule. A management optimum may require inequalities and boundary conditions. A reporting standard may require a categorical treatment selected by facts. Each type needs a suitable validation method.

A residual vector can combine them for computation, but the result should still report each component in meaningful units. A small average residual can hide one economically decisive violation, especially when large accounting amounts dominate the scale. A pricing residual of one currency unit and a share-count residual of one share are not directly comparable without a declared normalization.

11.2 The Bellman problem

For a finite-horizon shareholder problem, let x be state, a a feasible action, $d_t(x, a)$ the current net distribution to the modeled owners, and β a discount factor under the chosen valuation formulation. A Bellman equation is

$$V_t(x) = \sup_{a \in A_t(x)} \{d_t(x, a) + \beta \mathbb{E}[V_{t+1}(X_{t+1}) \mid x, a]\} . \tag{11.1}$$

The expectation uses the declared transition law and information. If an action depends on private information or a belief state, the state must be defined accordingly. The terminal value V_T is an assumption with economic consequences, not merely a numerical boundary condition.

A liquidation-at-book-value terminal rule can make asset accumulation artificially attractive. A zero terminal value can induce premature payouts or abandonment. A continuation approximation can be useful, but its accuracy and domain should be checked. Changing terminal treatment while comparing solvers changes the target.

In a stylized limited-liability problem, continuation value may be compared with default or exit value. The choice and its consequences must be specified. A maximum operator alone does not provide a legal restructuring model or creditor recovery schedule.

11.3 Pricing and value must agree on policy

Suppose debt price $P(x, F)$ depends on repayment under future policy π. The policy depends on funding proceeds and therefore on P. A joint solution must satisfy both the policy-value problem and lender pricing for that same policy.

One can write residuals $R_V(V, \pi, P) = 0$ and $R_P(V, \pi, P) = 0$, supplemented by feasibility and optimality conditions. Solving R_V with a fixed old price and R_P with a different policy leaves a mismatch. Alternating iteration can be a candidate algorithm, but convergence and target consistency need checking.

The retained Duarte and Fonseca (2026) working paper uses a lagged value network in part of its pricing training and a subsequent local grid refinement. These are specific numerical choices in its model. They do not supply a general convergence or global-optimality guarantee for AccDSL. Transferring them requires a target-specific protocol and matched timing, recovery, taxes and state definitions.

11.4 A residual can target the wrong expectation

Let $r(x, \varepsilon; \theta)$ be a random equation contribution with parameterized functions θ. Suppose the desired residual is the conditional expectation $m(x; \theta) = \mathbb{E}[r \mid x]$. Squared expected residual is $m(x; \theta)^2$. Squaring a single random contribution and averaging gives

$$\mathbb{E}[r^2 \mid x] = m(x; \theta)^2 + \mathrm{Var}(r \mid x). \tag{11.2}$$

The second objective adds a variance term. If that variance depends on θ, minimizing the two objectives can select different functions. This is a mathematical target mismatch, not merely noisy estimation of the same loss.

With conditionally independent draws $\varepsilon_1, \varepsilon_2$ under the same x,

$$\mathbb{E}[r(x, \varepsilon_1; \theta) r(x, \varepsilon_2; \theta) \mid x] = m(x; \theta)^2. \tag{11.3}$$

Independence is conditional on the common state. Reusing the same draw restores the unwanted variance term. Maliar et al. (2021, Definitions 2.5–2.9 and Algorithm 1) discusses objective formulations for dynamic economic equations, including this distinction. The local derivation explains why the sampling convention must match the intended residual.

11.5 A small objective counterexample

Let $r(\theta, \varepsilon) = \theta - 1 + \theta\varepsilon$, with $\mathbb{E}[\varepsilon] = 0$ and variance one. Then $m(\theta) = \theta - 1$, so the desired squared residual is $(\theta - 1)^2$, minimized at one. The single-draw squared objective is

$$(\theta - 1)^2 + \theta^2 = 2\theta^2 - 2\theta + 1, \tag{11.4}$$

minimized at one half. At that point the expected equation residual is negative one half. The optimizer has correctly minimized the wrong objective.

This exact example is a useful unit test for residual construction. It also shows why training loss alone cannot certify economic equations. Validation should directly evaluate the target conditional residual using independent quadrature or sufficiently controlled sampling.

11.6 Inequalities and complementary slackness

A borrowing or cash constraint can bind. Interior first-order conditions do not apply unchanged at a boundary. Consider minimizing a differentiable cost $f(a)$ subject to $a \geq 0$. With multiplier $\lambda \geq 0$, the Karush–Kuhn–Tucker conditions under appropriate regularity include

$$f'(a) - \lambda = 0, \qquad a \geq 0, \qquad \lambda \geq 0, \qquad a\lambda = 0. \tag{11.5}$$

If $a > 0$, the multiplier is zero and the interior derivative vanishes. If $a = 0$, the derivative need not vanish. A residual enforcing only $f'(a) = 0$ can demand an infeasible negative action.

For $f(a) = (a + 1)^2$, the constrained minimum is $a = 0$, with derivative 2 and multiplier 2. An interior solver seeks $a = -1$, which violates the constraint. Clipping its output to zero may recover the action in this example, but does not establish a general constrained solution or correct derivatives through the clipping operation.

Complementarity formulations, active-set methods and direct constrained optimization are possible approaches. Their conditions are necessary or sufficient only under their stated assumptions. Nonconvex financing problems may have local optima and discrete choices; a small KKT residual does not prove global optimality.

11.7 Discrete choices and nondifferentiability

Default, conversion, refinancing and investment shutdown can be discrete. Their value functions may have kinks or discontinuities. Differentiating a smooth surrogate through these decisions can optimize a smoothed problem rather than the original one.

A smooth approximation may be useful as a warm start. Its smoothing parameter changes the target and should be varied, with final policies evaluated under the original event rules. A soft default indicator that pays a fractional blend of full repayment and recovery does not describe an actual realized default event unless explicitly interpreted as an expectation under a probabilistic model.

Near a threshold, classification error can dominate value error. A network can fit values closely while moving the default boundary enough to affect debt pricing. Validation should include state pairs on both sides of the boundary and pricing consequences under the resulting policy.

11.8 Partial and total derivatives

A financing choice can affect a price directly and through future policy. If $P(a) = P(a, \pi(a))$, the total derivative includes both paths:

$$\frac{dP}{da} = \frac{\partial P}{\partial a} + \frac{\partial P}{\partial \pi}\frac{d\pi}{da}. \tag{11.6}$$

Holding a learned policy fixed computes a partial derivative. It may be the intended derivative in an alternating algorithm, but it is not automatically the total derivative of the equilibrium price.

Automatic differentiation differentiates the implemented computation. It does not decide which economic dependencies should be present. Stop-gradient operations, target networks and detached iterations can deliberately omit paths. Their use should be justified relative to the algorithm and validated against the final target.

An implicit derivative can be derived when a smooth equilibrium equation has a nonsingular Jacobian, but default boundaries and multiple equilibria can violate those conditions. Numerical finite differences can diagnose implementation mistakes away from such boundaries; they do not resolve the economic choice among equilibria.

11.9 Scaling and units

Residuals measured in millions of currency, fractions, shares and utility can differ by orders of magnitude. A loss that sums their squares without scaling effectively assigns arbitrary importance based on units. Changing currency units from dollars to millions can change the trained solution if the formulation is not invariant.

Normalization should be documented and checked. One can scale a cash residual by a meaningful transaction or balance scale, a share residual by outstanding units, and a probability residual by its natural range. Zero or near-zero denominators need explicit treatment. An adaptive scale can itself alter the objective over training.

A unit-rescaling test is useful: transform all monetary inputs consistently and confirm that the implied economic policy and normalized diagnostics transform appropriately. Failure reveals a numerical or modeling dependence on representation rather than economics.

11.10 Training distribution and the queried domain

A learned solver is trained on selected states. If training emphasizes typical solvent states, it may be unreliable near liquidity crises, maturity walls or default. Uniform sampling in raw coordinates is not automatically balanced in economic importance. A high-dimensional box may contain mostly infeasible states.

The training protocol should identify feasible-state generation, boundary coverage, tails, heldout regions and distribution shift. Data from simulated policies can omit states reached by a different policy or scenario. Updating the policy can therefore change the relevant state distribution.

No architecture, learning rate or loss weighting should become a default merely because it worked on another model. Transfer can provide a baseline or warm start. A substantially different target needs its own capacity, optimization, budget and validation protocol. This manuscript proposes no promoted neural default without that evidence.

11.11 Causal generators and feasible policy transfer

Acciaio, Eckstein and Hou construct a time-causal variational autoencoder whose encoder and decoder at date t use only input coordinates available by t. Their retained preprint bounds a *directional* causal transport discrepancy by reconstruction error and a term involving latent relative entropy (Acciaio et al., 2024, Lemma 2.4, Theorem 2.5 and Corollary 2.6). The relative-entropy step requires the reconstructed and generated distributions to have the stated bounded support. Normalizing the observed sample does not establish that a learned decoder satisfies that support restriction for every generated input. A finite training estimate is also different from a population bound.

There is a further issue in transferring the control result. Theorem 3.1 of the retained version states its comparison for a closed convex subset $\mathcal{H}$ of adapted strategies. Its proof takes a policy G on the second process and constructs $\widetilde{G}(x) = \mathbb{E}[G(Y) \mid X = x]$. Causality establishes adaptedness, but adaptedness alone does not establish $\widetilde{G} \in \mathcal{H}$. Here is a direct counterexample to that stated generality. There is one date, $\mu = \delta_1$, $\nu = \delta_0$, and

$$\mathcal{H} = \{h\}, \qquad h(x) = \min(1, \max(0, x)), \qquad Q(x, a) = a. \tag{11.7}$$

The singleton class is closed and convex, and Q is convex in a and uniformly zero-Lipschitz in x. Nevertheless, $V(\mu) - V(\nu) = 1$, whereas the proposed right-hand side with $L = 0$ is zero. The transferred action at $x = 1$ is zero, outside the singleton policy class. This disproves the unrestricted policy-class statement in that retained version. It does not refute the argument when the feasible class is explicitly closed under the conditional transfer. The published edition has not been matched equation by equation, so this is not asserted to be a journal erratum.

For AccDSL, state-dependent borrowing limits, integer issuance sizes and contractual exercise restrictions make closure substantive. A repaired transfer argument must verify feasibility of the mapped policy, or use a method that retains the original feasible actions. The one-sided value comparison is not itself a bound on the performance of the same learned policy after deployment. The inspected paper-linked code has forward LSTM encoder/decoder paths, but also external conditioning variables; those variables require their own availability dates. Its selected conditional decoder uses ReLU and a final linear layer, so the architecture does not enforce the theorem's unit-ball support. These observations identify additional obligations before adoption; no training result is inferred from source-code inspection.

11.12 Controlled transition laws and off-policy support

Saldı and Yüksel distinguish weak and strong kernel-embedding topologies and give a square-wave counterexample to their equivalence (Saldı and Yüksel, 2025, Theorem 4.1 and proof). Let Y be uniform on $[0, 1]$. On alternating half-cells of a fine partition, make a binary successor U deterministically 1 or 0. As the partition becomes finer, smooth integrated tests can approach those of a fair binary kernel. At every particular input, however, the conditional law remains a point mass. For the Kronecker kernel on $\{0, 1\}$, its squared embedding discrepancy from the fair law is $1/2$. The local exact calculation checks this discrepancy and the vanishing error for a linear test, rather than claiming that a few tests prove weak convergence.

Their discounted-control application uses bounded continuous costs, finite actions, continuous transition kernels and a common dominating measure, including for the initial distribution. One route requires a strong essential-supremum condition; the weaker route adds bounded equicontinuous densities (Saldı and Yüksel, 2025, Assumptions 2–4, equation (11), Theorems 5.1–5.2). A default discontinuity, hidden creditor information or an action-dependent financing constraint therefore needs a separate argument. An equally important problem is support: models that

agree under the historically chosen action may disagree under a new action. With action losses $(1, 100)$ in the real model and $(1, 0)$ in an approximate model, historical use of the first action gives identical observations. Optimizing the approximation chooses the second action and incurs regret 99. Agreement under the historical policy alone cannot establish a policy-uniform approximation claim.

The original robustness result used in that application makes another condition explicit. Kara and Yüksel distinguish continuity of optimal values from robustness of the policies attaining them. Their discounted-control argument uses continuous kernel convergence, bounded continuous costs and compact actions. Policy transfer additionally requires each approximate policy to be optimal from every initial state, as when it satisfies the discounted Bellman equation throughout the state space (Kara and Yüksel, 2020, Assumption 1.1, Theorems 4.2–4.4 and Appendix A.5). An optimizer trained only from a historical starting state can leave its actions elsewhere arbitrary; the true transition can reach those neglected states. The paper's counterexample makes precisely this distinction. Its finite-state learning result also assumes continuing coverage of every state–action pair (Kara and Yüksel, 2020, Corollary 5.1). Financial statements generated by one management policy do not by themselves provide that coverage.

11.13 Classical reference methods

Small discretized problems can be solved by direct enumeration, grid search, backward induction or carefully controlled quadrature. These methods can be slow, but they provide transparent references for testing learned or accelerated methods. The reference should be strong enough to answer the same target question.

A weak coarse grid can make a surrogate appear superior merely because the comparator is underresolved. A fair comparison includes convergence or refinement evidence for the reference and matched terminal conditions, constraints and state domains. Comparing different models is not a solver benchmark.

Exact rational examples are especially useful for cash identities and simple pricing. They are not substitutes for numerical validation of high-dimensional expectations, but they can expose sign, timing and unit errors before expensive training. The positive-recovery debt example belongs in this first layer.

11.14 Multiple solutions and selection

A nonlinear system can have no solution, one solution or several. A solver initialized differently may converge to different roots. Reporting the first root as the economic equilibrium requires a selection argument or an explicit conditional interpretation.

A financing market can have feedback in which perceived distress raises spreads and higher spreads increase distress. A stylized fixed point may admit low- and high-spread solutions. Stability under a chosen iteration is a numerical property, not necessarily an economic equilibrium-selection criterion.

The method should search or bound multiplicity where material, report dependence on initialization and preserve unresolved alternatives. If only one local root is found, describe it as such. A failed solver does not establish that no financing equilibrium exists.

11.15 Downstream validation

A learned value function is useful only if its errors are acceptable for the decisions and reports it drives. Validation should evaluate feasible actions, debt price consistency, default classification, cash paths, ownership changes and accounting outputs under heldout scenarios. Low prediction error alone is insufficient.

A pricing error can alter proceeds, which changes investment, which changes default and feeds back into price. A one-step residual screen may miss that loop. Multi-period policy evaluation under an independent method is needed for a claim about the coupled computation.

Stochastic comparisons require uncertainty evidence. One seed or a few short runs can nominate a candidate but cannot establish superiority from descriptive averages alone. Hard failures, descriptive differences and statistically supported rankings should remain separate. This is particularly important when rare defaults dominate economic conclusions.

11.16 A proposed solver interface

The solver receives a typed economic problem: state domain, timing, objectives, constraints, transition and observation laws, price equations and requested accuracy. It returns functions or policies together with target residuals, feasibility diagnostics, unresolved regions and reproducibility information.

A solver should not mutate the economic model to obtain convergence without recording the change. Smoothing, truncation, penalty relaxation and altered terminal values define modified problems. Their relation to the original target must be evaluated. A successful run on a modified problem is not an implicit solution of the original.

AccDSL can support several solvers behind the same problem definition. The interface should preserve method-specific limitations rather than force every solver to claim identical guarantees. A certified finite result, a local numerical solution and a learned approximation can all be useful when labeled accurately.

11.17 The smallest justified numerical program

Begin with a one-period debt model whose face, proceeds and recovery are distinct. Verify cash-leg pricing and limited-liability conventions exactly. Add a finite-state dynamic model with explicit feasible policies and compare enumeration with backward induction. Then add borrowing limits, equity issuance and default transitions one at a time.

Only after these targets are stable should a learned solver be compared on the same equations. The comparison needs a predeclared primary criterion, veto diagnostics, budget ladder and heldout downstream checks. Training settings remain hypotheses until supported on this target.

This staged approach does not postpone the difficult economic questions. It exposes them early in forms that can be checked. The purpose of deep learning is to accelerate a well-defined computation, not to make an ambiguous financing model appear resolved through a low loss value.

Chapter 12

Validation, adversarial cases, and the limits of a simulation

A credible simulator should fail intelligibly when its premises are insufficient. It should also be tested against cases where a plausible implementation gives a confident but wrong answer. The purpose of validation is not to accumulate a large count of passing assertions. It is to distinguish errors in evidence, representation, economics, accounting, information, numerical calculation and interpretation. Each class requires a different test and a different repair.

12.1 What is being validated

There are at least three separate questions. Does the implementation compute the specified model? Is the specified model mathematically and economically coherent? Does the model usefully describe the company or decision under study? A passing unit test can answer part of the first question. It cannot answer the third by itself.

For example, a simulator can correctly implement unlimited borrowing at a fixed rate. That implementation may pass every accounting reconciliation. It remains unsuitable for a claim about refinancing risk unless unlimited borrowing is an appropriate premise. Conversely, a realistic financing model can be implemented with the wrong proceeds convention. The two failures should not be blurred into a generic low accuracy score.

A validation record should state the claimed target, quantity actually computed, comparison method and interpretation. If the target is a worst-case cash bound but the method reports a sampled minimum, the mismatch is explicit. The sampled result may still be useful as a witness; it should not inherit the stronger claim.

12.2 An extensible failure taxonomy

No finite catalogue can prove that every possible simulation failure has been anticipated. New contracts, standards, behaviors and software interactions can create new cases. A systematic taxonomy is nevertheless valuable because it exposes the dimensions along which failures arise and guides new tests.

The taxonomy used here follows the calculation: evidence admission, historical reconstruction, economic state, actor information, scenario definition, feasible behavior, financing, accounting, aggregation, approximation and reporting. Cross-cutting dimensions include time, units, entity,

dependence, precision and provenance. A failure can occupy several categories. A maturity error, for example, affects evidence, timing, financing and liquidity.

Coverage should be expressed by mechanisms and boundaries, not only by scenario count. One thousand random rate paths do not test a missing conversion clause. A single carefully designed positive-recovery case can expose a proceeds-versus-face error that many no-default paths miss.

12.3 Evidence admission failures

A value can be extracted from the wrong row, assigned the wrong sign, scaled by the wrong unit or associated with the wrong period. A note can be omitted because it is a scanned image. A cross-reference can point to an incorporated document outside the downloaded filing. An amended filing can be mistaken for the original.

Tests should include duplicate labels with different contexts, negative values in parentheses, mixed currencies, overlapping periods and rounded totals. The expected result should preserve uncertainty where the source is ambiguous. A parser that always returns a number may look complete while inventing precision.

A source coverage report should distinguish retrieved, parsed, interpreted and modeled content. These are different achievements. Retrieving every page does not establish semantic completeness. A passage that cannot be formalized should remain linked to the model as unresolved material rather than disappearing from the scope claim.

12.4 Reconstruction failures

Historical balances can reconcile while hidden gross flows are wrong. A net borrowing change can conceal large issuance and repayment. A net share change can conceal issuance and repurchase at different prices. An aggregate receivable can conceal concentration. Tests should use pairs of histories with the same observed totals and different future responses.

The family should include both histories when evidence permits them. A selected witness should be labeled as selected. If a regularizer removes one, the result should identify the selection assumption and its consequences. A duplicate encoding of one history must not alter probabilities or set membership.

Infeasibility tests should distinguish source contradiction from solver failure. Deliberately conflicting exact observations should produce an interpretable conflict. Deliberately difficult but feasible constraints should not be reported impossible merely because the budget expires. Unsupported instruments should remain outside the solved profile with their economic significance visible.

12.5 State sufficiency failures

A state can omit a quantity that matters later: accrued interest, remaining loss carryforward, asset age, covenant cure clock, collateral ownership, unpaid dividends or option exercise status. Current statements may still match. A multi-period replay reveals the omission.

A useful test starts from two histories with equal proposed compressed state but different omitted quantities. Apply the same future policy and shocks. If outcomes differ, the compression is insufficient for that query. If observations differ and later decisions change, the information component is insufficient even when immediate cash agrees.

The repair is to retain the missing state or restrict the compression's claimed domain. Adding a balancing adjustment after the divergence hides the cause. A state variable should be justified by future dependence, not by its convenience in a data schema.

12.6 Information failures

Look-ahead can enter through restated data, future prices, a sampled shock passed to a policy, or a compressed state containing private information the actor lacks. It can also enter through training labels if a purported real-time decision uses features calculated from the full future path.

A nonanticipativity test changes future unobserved data while holding current information fixed and checks current actions. A publication test shifts a disclosure after the decision cutoff and checks that it no longer influences that decision. A private-information test permits an informed manager to respond to a known contract while preventing a public analyst from treating it as observed.

These tests must inspect policy inputs and execution, not merely documentation. An interface can claim to hide future information while a cached feature leaks it. Provenance should connect each decision input to an information rule and availability time.

12.7 Scenario-definition failures

A shock can be applied twice, applied to the wrong rate, or applied for the wrong fraction of a year. A currency depreciation can be inverted. A rate shock can affect a fixed coupon that should not reset. A scenario can change both demand and price while being described as a pure volume shock.

Metamorphic tests are useful when exact full outcomes are hard to specify. A zero shock should reproduce the baseline under unchanged policies and information. Splitting one identical contract into two proportional pieces should preserve aggregate cash when no nonlinear clause is introduced. Changing currency units consistently should scale monetary outputs without changing dimensionless decisions.

These invariances have premises. Contract splitting can change fees, concentration limits or legal rights. A test should enforce invariance only where the model says it holds. Otherwise the test itself becomes an incorrect economic requirement.

12.8 Financing failures

Face can be mistaken for proceeds, net cash for gross cash, coupon for effective interest, or book debt for market value. Refinancing can be assumed available without an offer. A fee can be charged twice. Recovery can be distributed beyond available assets. A conversion can coincide with an unlinked redemption.

The positive-recovery diagnostic in Chapter 5 is a primary fixture because it distinguishes two conventions that agree in simpler cases. Additional tests should use discounted issuance, delayed settlement, binding capacity, failed subscription, priority and recovery delay.

A no-funding result is economically meaningful only after the method has correctly evaluated the admitted offers and actions. A numerical pricing failure should be reported separately. If

the model omits a financing channel, the conclusion is conditional on that omission; it is not proof that the real company has no financing option.

12.9 Accounting failures

A journal can balance while mapping borrowing to revenue, treating a valuation change as cash, or recognizing a transaction twice. A reporting aggregation can combine different perimeters or bases. A cash-flow label from an economic model can be mapped incorrectly to a standard's category.

Tests need event-specific expected postings under explicitly supplied rules. They should reconcile physical quantities, cash, obligations and accounting balances. Where authority or facts are unresolved, the expected result may be a refusal or conditional treatment, not a forced entry.

A standard-specific test establishes behavior under the retained authority version and admitted facts. It does not establish universal compliance across jurisdictions and dates. The rule package should preserve applicability conditions, source passages and unresolved interpretations.

12.10 Aggregation and dependence failures

Independent marginal ranges can create impossible combinations. Stagewise worst cases can switch persistent types. A mean state can cross a nonlinear threshold differently from every original state. A scenario reduction can remove the path causing a maturity shortfall.

Tests should retain a full small reference family and compare exact joint paths with the compressed representation. Check outputs, observations, feasible actions and policy values, not just means. Include a case where fixed-action sequences agree but adaptive decisions differ.

A failed exact aggregation test can still motivate an approximate method with bounds. It should not be repaired by weakening the test until the original claim passes. State whether the target changed and quantify the consequence for the decision.

12.11 Approximation failures

Particles can lose a persistent contract type. Importance sampling can omit support. A surrogate can extrapolate beyond training. A numerical tolerance can classify a covenant incorrectly. A Monte Carlo sample can contain no rare defaults while the true modeled probability is positive.

A small exact reference provides a baseline. Coverage checks verify that rare or boundary regions are visited or bounded. Replications and uncertainty intervals assess stochastic estimates. Distinct seeds are not independent evidence if every run shares the same misspecified transition or omitted mechanism.

A computational diagnostic should be assigned a role before interpretation. It may veto validity, explain behavior, trigger repair or support promotion under a declared criterion. A low loss can nominate a candidate; it does not automatically promote it to a default solver.

12.12 Terminal and horizon failures

A model can postpone default beyond the horizon, refinance indefinitely without repayment, or liquidate assets at an unrealistic terminal value. Extending the horizon can reveal that an

apparently safe policy merely shifts the problem.

Tests should compare adjacent horizons under consistent continuation assumptions and inspect obligations remaining at the end. A terminal balance sheet should retain outstanding debt, contingent rights and future commitments. A clean closing cash number is not a complete account of solvency.

The terminal rule should be part of every policy comparison. Using a favorable terminal value for one method and zero for another is an unfair comparison. A finite-horizon model can still be useful, but its conclusion should state what remains outside the horizon.

12.13 Interaction failures

Individually correct modules can compose incorrectly. A debt amendment can change future interest while a cached tax calculation uses the old schedule. A share conversion can update ownership but leave diluted earnings per share on the old count. A group transfer can be recorded in both entities but omitted from elimination.

Composition tests should follow complete event sequences across modules. They should check shared identities, dates, units and versions. A module's standalone test suite cannot certify every interaction. The most valuable sequences are those crossing boundaries: refinancing plus fees plus tax, conversion plus ownership plus reporting, or default plus collateral plus recovery.

A failed interaction should be localized to its violated relation. Re-running a broad test suite without understanding the failure can obscure whether the model, adapter or shared state is wrong. The repair should preserve unaffected evidence and explicitly update dependent results.

12.14 A hierarchy of reference calculations

Start with hand-reconstructible exact cases using rational arithmetic. Then use finite-state enumeration for learning and policy tests. Add independent numerical quadrature or refined grids for continuous small models. Only then compare scalable approximate methods on the same target.

A reference calculation must itself be audited. Coarse grids, truncation and arbitrary terminal values can bias it. Agreement between two implementations sharing the same mistake is not strong validation. Independent derivations and different computational routes are valuable where the economic conclusion is important.

The hierarchy is not a requirement to solve every real company exactly. It establishes local semantics and numerical behavior before applying methods to larger unresolved settings. The large application retains its own evidence and domain limitations.

12.15 Empirical validation

Forecast validation requires a defined prediction target, information cutoff, evaluation period and benchmark. A statement reconstruction evaluated against the same statements it was constrained to match is not an out-of-sample forecast test. A restated target can differ from the information that was observable at prediction time.

Probabilistic forecasts need scoring methods appropriate to their target. The proper-scoring literature represented in the inherited bibliography by Gneiting and Raftery (2007) motivates

evaluating distributions rather than only point means. Applicability still requires a correctly defined observed outcome and a forecast distribution produced without leakage.

Set-valued predictions need measures of coverage and informativeness under a declared evaluation design. A very broad set can cover every outcome while offering little decision value. Narrowness without coverage is also inadequate. The tradeoff should be stated before comparing methods.

12.16 No empirical result is claimed here

The examples in this part are exact or analytic diagnostics of proposed semantics. They are not fitted company forecasts, evidence of profitability or rankings of simulation methods. No neural training campaign or stochastic benchmark is performed as part of this writing task.

This distinction keeps the next research step clear. A future empirical study must choose companies, evidence cutoffs, scenarios, baselines, outcomes and uncertainty analysis. It should preserve the question separately from the implementation checklist and state what result would invalidate the harness rather than merely reject a candidate method.

A negative result can reveal a bad implementation, insufficient tuning, an inadequate observation model or a weak economic mechanism. These explanations require different follow-up. Treating every failure as rejection of the entire direction would waste the diagnostic information.

12.17 Failure behavior is part of the specification

The system should return different statuses for unsupported instrument, contradictory evidence, undecided feasibility, infeasible action, economic default, numerical nonconvergence and completed approximation. A result can contain several statuses across regions.

Unsuccessful regions must not disappear from aggregate results. Renormalizing probabilities after dropping numerical failures changes the distribution, often favorably if difficult cases are distressed. A report should state unresolved probability mass when a measure exists, or unresolved feasible regions otherwise.

A completed run also needs a trace to its assumptions and evidence. Reproducibility requires exact versions of source views, rules, methods and numerical settings. A content hash identifies a file; it does not prove the file's correctness. Both provenance and substantive validation are needed.

12.18 From catalogue to implementation work

The proposed failure catalogue should be machine-readable enough to link each mechanism to a fixture, expected outcome and relevant chapter, while the manuscript explains why the failure matters. New cases can be added without claiming that the catalogue is exhaustive.

Implementation can proceed by vertical slices. A slice reconstructs a bounded history family, applies a scenario and policy, generates events and journals, produces statements and reports unresolved regions. This is more discriminating than implementing isolated formula modules without a source-to-result case.

The first slice should preserve ambiguous debt type, issue proceeds distinct from face, a binding financing limit and actor information. Later slices add ownership rights, group transfer constraints, persistent tax state and approximate inference. Each slice should retain the wider

open scope rather than redefine the mission to match what is easiest to execute.

12.19 What successful completion would mean

For this document, success means an explicit and traceable argument, checked illustrative calculations, preserved prior work, inspected source support and a clear account of proposed methods and their failure cases. It does not mean that all methods are implemented or that the product is ready for unrestricted use.

For the simulator, success is query-specific. It means the implementation computes the declared target within its stated evidence and numerical guarantees, preserves unsupported regions, and explains the economic dependence of the result. Whole-company applicability requires coverage of the actual company's material contracts, events and reporting basis.

The following worked chapters apply these principles to complete small problems. Their purpose is to make the design reconstructible: the reader can follow observations through competing histories, decisions, financing, journals and outcomes, and see exactly where a different assumption changes the answer.

Chapter 13

A complete small company: from reports to divergent stress outcomes

The earlier examples isolate individual mechanisms. This chapter joins them in one synthetic company so that every projected balance can be traced to an event. The company is deliberately simple: one currency, one legal entity, no tax, no dividends, no acquisitions and no other comprehensive income. These exclusions make the arithmetic inspectable. They do not describe a complete real-company model. The point that carries over is the need to preserve compatible economic histories and their joint statement consequences.

13.1 The opening statement

At the beginning of the historical year, the company has cash 40, receivables 30, inventory 50 and property carrying amount 80. Its liabilities are debt 100 and trade payables 40. Equity is 60. All amounts use one currency unit.

Assets	Amount	Liabilities and equity	Amount
Cash	40	Debt	100
Receivables	30	Trade payables	40
Inventory	50	Equity	60
Property	80		
Total	200	Total	200

The equality is an accounting starting point. It does not identify customers, inventory units, asset vintages or debt terms. We initially admit the balances as exact to focus on hidden composition. A later section reintroduces rounding and source uncertainty.

The company operates during the year and reports sales 120, cost of sales 70, cash operating expenses 25, interest expense 5 and depreciation 10. With no tax in this teaching model, profit is 10. The cash-flow information reports customer receipts 130, supplier payments 50, operating-expense payments 25, interest payments 5 and capital expenditure 20. No debt is issued or repaid.

13.2 One compatible event history

One history divides sales into cash sales 80 and credit sales 40. Collections on receivables total 50: 20 from opening receivables and 30 from current-year credit sales. Customer receipts therefore total $80 + 50 = 130$. Purchases on credit total 60. Supplier payments are 50.

The receivables and inventory bridges are

$$A_1 = 30 + 40 - 50 = 20, \tag{13.1}$$

$$I_1 = 50 + 60 - 70 = 40. \tag{13.2}$$

Trade payables become $40 + 60 - 50 = 50$. Property becomes $80 + 20 - 10 = 90$. Cash becomes

$$C_1 = 40 + 130 - 50 - 25 - 5 - 20 = 70. \tag{13.3}$$

Profit of 10 increases equity from 60 to 70. Thus closing assets are $70 + 20 + 40 + 90 = 220$, equal to debt 100, payables 50 and equity 70. Every amount follows from a stated movement.

This is one witness. The reports may not disclose the split of collections between opening and new receivables, and they do not disclose which customers remain unpaid. Other event histories can generate the same statements. The model should not turn the chosen witness into an assertion about what actually happened.

13.3 The journal bundle

Under the illustrative recognition rules, cash sales debit cash and credit revenue by 80. Credit sales debit receivables and credit revenue by 40. Collections debit cash and credit receivables by 50. Purchases debit inventory and credit payables by 60. Consumption of inventory debits cost of sales and credits inventory by 70.

Supplier payments debit payables and credit cash by 50. Operating expenses debit expense and credit cash by 25. Interest payment debits interest expense and credits cash by 5. Capital expenditure debits property and credits cash by 20. Depreciation debits expense and credits accumulated depreciation or the summarized carrying-value account by 10.

Each posting bundle balances. More importantly, the bundles reconcile the economic movements. The inventory purchase is not expensed a second time when cash is paid. Credit sales are not recognized again on collection. Capital expenditure is not simultaneously charged as the full current expense and depreciated. These familiar errors are easy to introduce when separate forecasting formulas are joined without event identities.

The summarized property account hides gross cost and accumulated depreciation. That is acceptable for this bounded calculation, but a later disposal or impairment query may require those components. The representation should retain the option to refine rather than declare the summary universally sufficient.

13.4 What the reports leave unknown

At the forecast origin, receivables of 20 can be concentrated in one customer or spread among ten customers owing 2 each. The debt of 100 can be fixed-rate at five per cent or floating-rate with a historical rate of five per cent. Both alternatives reproduce the historical interest payment and closing debt.

Let c indicate concentration: $c = 1$ for one customer and $c = 0$ for ten equal customers. Let f indicate floating debt: $f = 1$ for floating and $f = 0$ for fixed. For this exercise we admit all four combinations because the specified evidence provides no linking restriction. This is an explicitly declared feasible family, not a claim that real financing structure and customer concentration are statistically independent. No probabilities are assigned.

Other hidden quantities remain fixed in this first stress calculation. Inventory and property do not change during the short stress horizon, and there are no new sales or purchases. This isolates collection and financing exposure. An operating forecast would add production, demand and replenishment policies rather than assume these balances remain static indefinitely.

13.5 Define the shock precisely

At the start of the stress horizon, payables of 50 must be paid. Next, one largest customer (any one of the tied customers in the diversified case) defaults and pays nothing on its outstanding receivable; every other customer pays in full. At the end of the horizon, one full year's interest is paid after market rates rise by two percentage points. There is no hedge, cap, floor or principal maturity during the horizon.

The customer loss is 20 in the concentrated history and 2 in the diversified history. Write

$$L(c) = 2 + 18c. \tag{13.4}$$

Collections are $20 - L(c)$. Interest is $5 + 2f$. After paying suppliers, collecting surviving receivables and paying interest, cash is

$$C_2(c, f) = 70 - 50 + [20 - L(c)] - (5 + 2f) \tag{13.5}$$

$$= 33 - 18c - 2f. \tag{13.6}$$

The formula preserves the joint dependence on concentration and rate type. It is an exact result for the declared event sequence.

13.6 The four outcomes

The resulting cash and equity are

Receivables	Debt	Closing cash	Closing equity
Diversified	Fixed	33	63
Diversified	Floating	31	61
Concentrated	Fixed	15	45
Concentrated	Floating	13	43

Equity falls by customer loss and interest, so $E_2 = 70 - L(c) - (5 + 2f) = 63 - 18c - 2f$. Receivables and payables both become zero. Inventory remains 40 and property 90. Each history satisfies

$$C_2 + 40 + 90 = 100 + E_2. \tag{13.7}$$

The accounting relation is shared, but the economic outcomes differ by 20 across the family. A single historical witness could conceal that range.

The worst cash outcome is concentrated floating debt. The best is diversified fixed debt. These are feasible joint witnesses, not independent marginal extrema assembled from different worlds. If later evidence rules out one combination, the family and its extrema should update accordingly.

13.7 A liquidity threshold changes the question

Suppose a policy requires closing cash at least 14. Three histories meet it; the concentrated floating history ends at 13. Without probabilities, the correct statement is that the policy is not robustly feasible across the admitted family. It is wrong to say the probability of failure is one quarter merely because there are four rows.

The row count depends on representation. The diversified case could be split into ten different customer-name assignments without changing its economic exposure. Equal weighting would then change the apparent failure probability. A measure must be attached to economic states or events with justified weights, not to a convenient enumeration.

If the threshold is a contractual covenant, its consequences require the actual covenant definition and remedy process. If it is an internal treasury target, a shortfall is a policy deviation rather than legal default. The same numerical comparison can support different meanings depending on the source of the threshold.

13.8 A funding response

Suppose a committed facility can supply up to 2 at the final payment date, with no fee or interest within this short example, and management draws only enough to reach cash 14. The concentrated floating case draws 1. Its closing cash becomes 14 and debt 101; equity remains 43. The balance sheet still reconciles: $14 + 40 + 90 = 101 + 43$.

The draw improves current liquidity but creates a future obligation. It does not reverse the credit loss or interest expense. A model that raises equity by the draw amount would misclassify financing. A model that treats the draw as free permanent cash would overstate future capacity.

If the facility is unavailable after the customer default, the response is infeasible. If it settles after interest is due, it cannot fund that payment without a bridge. If its covenant depends on equity falling below 44, the worst history may lose eligibility precisely when it needs funding. These are distinct contract variants that should be modeled as conditional cases.

13.9 A dependence hidden by aggregate data

Suppose a private lender report establishes that the floating loan was granted only when the receivables were diversified. Then the concentrated floating combination is excluded. The remaining minimum cash is 15, so the threshold 14 is satisfied by every retained history.

This new conclusion comes from a joint restriction, not from narrower marginal ranges. Concentration still can be either zero or one, and debt still can be fixed or floating. Marginal sets are unchanged. A representation keeping only those marginals cannot express why the worst combination disappeared.

The example demonstrates the value of constraints over named variables. The restriction can be written $f = 1 \Rightarrow c = 0$. A new source should be attached to it, with availability and scope. If the source is later rejected, the excluded combination returns in the revised family.

13.10 Learning before the payment

Suppose management knows the loan type but the public analyst does not. Management can plan a precautionary draw in the floating history without using future information. The

analyst's forecast must integrate or bound those conditional management actions. Forcing management to use the analyst's ignorance would understate its information.

Now suppose a customer deterioration signal arrives before supplier payment. A policy may delay discretionary capital spending or seek financing in response. If the signal arrives only after payment, those actions may no longer be available. The date and actor access matter as much as the signal's accuracy.

The current stress example has no discretionary spending. Adding one should identify its benefit, cancellation cost and commitment status. An apparently simple cash-saving policy can be infeasible if the expenditure is already contractually committed. The event sequence prevents that distinction from being lost in an annual forecast.

13.11 Historical journal ambiguity beyond contracts

The historical reports also permit different sales and collection timing within the year. One company may collect most cash early and pay suppliers later; another may pay first and collect near year-end. Annual cash flow and closing cash can agree while minimum cash differs.

For example, take opening cash 40 and aggregate receipts 130 with payments 100 in total. If payments of 100 occur before receipts, a company without credit cannot execute the sequence. If receipts arrive first, it can. The annual closing cash of 70 does not distinguish them. A feasible historical reconstruction must respect intermediate cash and financing events where those are within scope.

This does not mean every annual disclosure allows precise daily reconstruction. It means the model should retain timing uncertainty or state that it cannot answer an intra-year liquidity question. Inventing a uniform monthly schedule gives a point path whose apparent smoothness comes from the interpolation assumption.

13.12 Inventory alternatives

Closing inventory value 40 can represent eight units costing 5 or four units costing 10. If a future order requests six physically interchangeable units, the first history has sufficient stock and the second does not. If products are not interchangeable, even total unit count is inadequate.

A monetary demand shock should therefore not be converted automatically into units using an average historical cost. Selling price, cost, product mix and capacity are different quantities. A future price decline can reduce revenue per unit without changing available physical stock. A supply shock can reduce units while increasing unit cost.

The present company case holds inventory fixed to isolate credit and rates. Extending it requires product or cohort state, procurement lead times and sales rules. The aggregate carrying amount remains a reporting consequence of those states. A richer query justifies refinement of the family rather than invalidating the earlier bounded calculation.

13.13 Property and replacement cash

Property of 90 after depreciation can comprise young assets with long remaining lives or old assets near replacement. Equal historical depreciation does not identify next year's capital needs. A policy that sets capital expenditure equal to depreciation assumes a replacement relation.

Suppose one history has a machine requiring replacement next year for 30 and another can continue for five years with maintenance 2. Their current carrying amounts may agree under different acquisition costs and accumulated depreciation. A shock to financing availability has different operating consequences.

A complete extension should retain capacity and maintenance as well as book values. Replacement can change productivity and costs, so it is not only a cash outflow. The model needs a production or service relation and a management policy. The accounting bridge alone cannot supply them.

13.14 Rounding widens the family

The exact amounts used above make arithmetic transparent. If reported receivables 20 are rounded to the nearest unit, the underlying amount may lie in a reporting interval. Customer loss and collections then become functions over that interval. A cash threshold at 14 can be crossed by values that a point reconstruction misses.

Suppose the concentrated floating case's opening cash is itself rounded from an amount between 69.5 and 70.5 under the declared convention. Its stressed cash lies correspondingly between 12.5 and 13.5 if other terms are exact. A facility of exactly 1 may no longer guarantee reaching 14. The required draw can range from 0.5 to 1.5.

Rounding intervals can also be jointly constrained by a total. Treating each line's rounding error independently may admit totals outside the reported balance-sheet interval. The feasible family should retain the total relation rather than merely widen every number separately.

13.15 An explanation the system should produce

For the worst witness, an explanation should identify the concentrated receivable history, floating debt, customer default, rate increase, supplier payment and lack of an eligible funding response. It should show cash 13 and equity 43 with their bridges. It should identify which facts are observed and which are alternative assumptions.

The explanation should not say that the real company will fail. It should say that a compatible history produces the stated shortfall under the specified shock and policy. If the threshold is internal, it should call it a target shortfall. If a covenant consequence is modeled, it should name that consequence and its premises.

A second explanation can identify the source that would exclude the witness, such as a verified customer concentration note or fixed-rate contract. This turns the simulation into a guide for focused investigation. It also makes disagreement productive: a reviewer can challenge a premise rather than merely distrust the final number.

13.16 What this complete case establishes

The case connects reports, compatible histories, events, journals, cash and equity. It shows why identical statements do not identify shock response, why joint restrictions matter, why financing needs a transaction and why timing matters. All numerical relations are small enough for exact checking.

It does not establish a calibrated corporate model, a universal customer-default process or a reporting-standard treatment for every event. Those would require additional facts and authority.

The bounded example is valuable because it exposes the mechanisms without pretending those wider questions are resolved.

An implementation fixture should store the input observations, the four initial history cases, the optional dependence restriction, the event sequence, expected journals and final statements. It should also test the rejected variants: equal branch probabilities without a measure, independent marginal output selection, a late facility draw and a cash-balancing equity plug. These are concrete failures of the target semantics.

Chapter 14

A refinancing crisis: dates, offers, covenants, and recovery

A year-end debt ratio can look manageable while a particular morning's payment is impossible. This chapter follows a synthetic treasury problem through dated cash events. The company has several potential responses, but each has conditions. The purpose is to show how financing feasibility differs from choosing a debt amount in an optimization formula. All offers and legal consequences below are assumptions of the example; they are not estimates or statements of general law.

14.1 The payment calendar

At the opening of day 1, unrestricted cash is 12. A customer receipt of 18 is due on day 5, wages of 10 on day 3, an old loan principal of 30 on day 4 and its coupon of 2 on day 4. Inventory worth 20 in the accounting records can be sold, but only through a specified offer. There is no automatic overdraft.

If all receipts and payments are netted over five days, closing cash is

$$12 + 18 - 10 - 30 - 2 = -12. \tag{14.1}$$

The total shortfall is 12. But the timing problem is larger: after wages, cash is 2 before the day-4 debt payment of 32. The company needs 30 by day 4, even though 18 is expected the next day. A facility that supplies only the final net shortfall cannot bridge the maturity.

The receipt is also uncertain. In history H1 it is contractually due and collected on day 5. In H2 it is subject to customer acceptance and may be delayed to day 9. Both histories fit an opening receivable of 18. The financing decision must distinguish the customer's contractual obligation, actual collection policy and the analyst's uncertainty.

14.2 Three offers

Offer A is a bridge facility settling on day 4 before the old payment, with maximum principal 30 and a fee of 1 withheld at draw. Its repayment of principal plus 1 is due on day 6. Offer B is a term debt issue settling on day 5, priced at 0.9 per unit face, with a fixed fee of 2 and maximum face 40. Offer C is an inventory sale settling on day 4, paying 15 for inventory carried at 20.

These offers have different cash, accounting and future consequences. A supplies short-term cash but creates another near maturity. B can supply longer funding but arrives too late to

pay the old loan without a bridge or deferral. C supplies timely cash and incurs a disposal loss under the illustrative carrying-value treatment, while removing inventory needed for operations.

The offers are not interchangeable columns in a financing plug. Their settlement dates and quantity constraints determine feasible sequences. The company can choose combinations only if the contracts permit them and management accepts their economic costs.

14.3 Why the bridge alone fails

Drawing maximum bridge principal 30 supplies net cash 29 after the withheld fee. Cash before old debt payment becomes $2 + 29 = 31$, one short of 32. A model that treats principal as proceeds would declare the payment feasible and miss the fee-induced shortfall.

Required bridge principal to supply net cash 30 would be 31 under this fixed fee, but the cap is 30. The transaction is infeasible at the needed amount. This conclusion does not require a default-probability model; it follows from offer terms and dates.

If the company can defer wages by 1 under a separately permitted agreement, the bridge becomes sufficient for day 4. But that deferral creates another obligation and may have costs. It must be represented as a negotiated event, not as a spreadsheet reduction of operating expenses.

14.4 The inventory sale plus bridge

Suppose management accepts the inventory sale. After wages, cash 2 plus sale proceeds 15 gives 17. The remaining need for old debt payment is 15. A bridge principal of 16 supplies 15 net after its fee, so day-4 cash reaches 32 and the old obligation is paid exactly.

At day 5, customer receipt 18 arrives in H1. The bridge repayment due day 6 is 17. Paying it leaves cash 1. The old debt is extinguished, bridge debt is extinguished, inventory is lower by 20, and the company has recognized the illustrative sale loss 5 and bridge cost 2 in total under the assumed expense convention.

This path solves the payment problem at an economic cost. The inventory sale can reduce future production or require replacement. A short-horizon model ending on day 6 would miss that operational consequence unless the terminal state retains the missing inventory and relevant commitments.

14.5 Delayed collection changes the result

In H2, the customer does not pay until day 9. After the old debt payment, cash is zero. Bridge repayment 17 is due day 6. The inventory-plus-bridge sequence therefore creates a later shortfall unless another offer or deferral is available.

The term debt issue B can settle on day 5. To raise net cash 17 after a fixed fee 2 at price 0.9, required face is

$$F_B = \frac{17 + 2}{0.9} = \frac{190}{9} \approx 21.1111. \tag{14.2}$$

This is within the face cap 40. If B remains eligible, its proceeds repay the bridge and leave a new longer-term promise. Customer collection on day 9 then raises cash 18. The company has transformed a timing mismatch into longer funding at a cost.

The financing path depends on the continued availability of B. A policy that assumes B simply

because it is numerically sufficient confuses an offer with an executed transaction. Eligibility, acceptance and settlement must be checked at the relevant dates.

14.6 A covenant can remove the rescue

Assume offer B requires inventory carrying amount at least 10 at settlement. The company initially has exactly 20 and the sale C disposes of all of it. After C, the condition fails. B is unavailable in the very path that needed it to refinance the bridge.

A modular simulator can miss this interaction if the asset-sale module reports cash but does not update the collateral or eligibility state used by financing. The cash schedule appears feasible until day 6, then fails. The root error is a missing cross-module dependency, not arithmetic.

If C can sell only half the inventory for 7.5, inventory remains 10 and B stays eligible. After wages and partial sale, cash is 9.5. To reach 32 requires net bridge 22.5, so principal 23.5 is within cap. Bridge repayment is 24.5. In H1, receipt 18 leaves a day-6 need of 6.5; in H2 it leaves a need of 24.5. B can supply either need subject to its own cap.

14.7 Face caps and net funding

Under B, maximum face 40 supplies gross proceeds 36 and net cash 34 after the fee. In H2 the need 24.5 is feasible. Required face is $(24.5 + 2)/0.9 = 265/9$, about 29.4444. In H1 the need 6.5 requires face $85/9$, about 9.4444.

If B's price falls to 0.6 after a market shock, maximum net proceeds become $40(0.6) - 2 = 22$. The H2 need 24.5 is no longer feasible. H1 remains feasible. Thus identical opening statements and the same price shock produce different financing outcomes because collection timing differs.

A sensitivity table should vary offer price and collection history jointly. Reporting only the average required face can hide the infeasible H2 path. A midpoint collection date may produce a third path that neither contract history permits.

14.8 A symbolic feasibility boundary

Let q be B's price per unit face and K the net funding need on day 6. With fixed fee 2 and cap 40, feasibility requires

$$40q - 2 \geq K \quad \Longleftrightarrow \quad q \geq \frac{K + 2}{40}. \tag{14.3}$$

For H1 after partial inventory sale, $K = 6.5$, so the threshold is $8.5/40 = 0.2125$. For H2, $K = 24.5$, so it is $26.5/40 = 0.6625$. These thresholds are exact for the offer structure.

The price may itself depend on face or the asset sale. If so, the linear threshold no longer applies without modification. The symbolic result should carry its premises. A new price model is a new financing query, not merely a different number substituted into an unchanged guarantee.

14.9 Who knows the collection history

Suppose management knows whether customer acceptance has occurred, while the public analyst sees only the receivable total. Management can condition its financing plan on H1 versus H2. The analyst should evaluate that informed policy across the two histories.

If management does not know the eventual delay, it must choose a plan before observing collection. A robust plan must remain feasible in both histories, or specify a contingent action after the day-5 observation. A term issue requested only after nonreceipt may settle too late if the offer has a notice period. The observation date and action lead time determine the policy's feasibility.

A simulator that waits to see H1 or H2 and then retroactively chooses the best day-1 plan gives management foresight. A correct tree branches at the observation time and shares decisions before that branch. A symbolic policy representation enforces the same restriction through its allowed inputs.

14.10 Default after a failed refinancing

Suppose H2 occurs, B is unavailable and bridge repayment 17 cannot be made on day 6. Under a stipulated contract, a two-day cure period begins. The day-9 customer receipt arrives after the cure deadline. Whether the lender can accelerate, waive or restructure is specified by additional terms.

The economic state on day 6 contains unpaid principal and fee or interest according to the contract. It does not set debt to zero. The company may continue operations during cure. If a waiver is granted on day 7 for fee 0.5 and repayment on day 9, the receipt can resolve the arrears. If not, enforcement may proceed.

These branches are not numerical solver failures. They are economic transitions. Conversely, inability to solve a pricing equation for B is not itself the bridge default event. The simulator should preserve the distinction even if both ultimately leave the funding need unresolved.

14.11 A recovery waterfall

Assume enforcement yields distributable assets of 25 after costs. The bridge has senior claim 17 and another unsecured creditor has claim 20. Under the example's strict priority rule, the bridge receives 17 and the unsecured creditor 8. Total recovery equals available assets 25.

If the bridge is secured only on an asset that realizes 12 and has no further recourse under the assumed contract, its recovery may instead be 12. The remaining distribution depends on ownership and priority of the other assets. A single aggregate recovery percentage cannot represent both cases without additional assumptions.

If recovery occurs one year later, creditor valuation discounts it from that date. The accounting treatment of the debtor's unresolved obligation and eventual extinguishment requires its own rules. Cash recovery to creditors and book debt removal at the debtor need not occur at the same instant.

14.12 Why aggregate net debt fails here

At opening, net debt may be calculated as debt less cash. That statistic does not contain the day-3 wage date, day-4 maturity, day-5 receipt or day-6 bridge repayment. It cannot decide feasibility of the sequence.

After the partial sale and term refinancing, net debt may improve when the customer finally pays. Yet the company may have lost productive inventory and paid substantial financing costs.

A net-debt chart alone omits these consequences. A useful result reports cash timing, remaining obligations, inventory capacity and owner value separately.

A compressed state can still be possible. For this bounded problem it must retain the ordered obligation calendar, collection state, inventory eligibility amount and offer terms. The full historical journal need not be replayed at every step, but its economically relevant consequences must survive.

14.13 Discrete sale choices and continuous financing

The inventory offers give three sale choices: none, half or all. They do not make the financing policy set finite. A principal cap permits an interval of bridge draws unless the contract also specifies denominations. To compare policies, retain those amounts as variables and first separate the cases in which each offer is used. For each feasible policy, record minimum cash, final cash, fees, remaining debt and inventory. A policy minimizing fees may differ from one preserving inventory or maximizing the lowest cash balance.

The action domain can be derived directly for this example. Let z be the fraction sold, with $z \in \{0, 1/2, 1\}$, and let a be bridge principal. Every sale choice still needs the bridge before day 4. Write b for term face and u for whether B is issued: $u = 0$ requires $b = 0$, while $u = 1$ requires $0 < b \leq 40$. Hold the supplied price $q > 0$ fixed. For this calculation, both sale and bridge proceeds arrive before the old debt payment; any day-5 customer receipt arrives before B settles, and B's fee is withheld from its proceeds. These are explicit settlement-order assumptions. They do not give the manager permission to decide an issue after its contractual notice deadline.

Let R be the cash collected by day 6: $R = 18$ in H1 and $R = 0$ in H2. Cash just after the old debt payment is

$$C_4 = 12 - 10 + 15z + (a - 1) - 32 = a + 15z - 31. \tag{14.4}$$

Between that payment and bridge repayment, the company receives R and the term issue supplies net cash $qb - 2u$. Repaying principal plus one therefore gives

$$C_6 = C_4 + R + (qb - 2u) - (a + 1) = 15z + R + qb - 32 - 2u. \tag{14.5}$$

The bridge principal cancels from C_6 because the company repays the extra amount it borrowed. It still changes C_4. Under the stated event order, nonnegative cash throughout days 1–6 is equivalent to

$$31 - 15z \leq a \leq 30, \qquad 15z + R + qb - 32 - 2u \geq 0, \tag{14.6}$$

together with the issue/no-issue domain and offer eligibility. To see why no other cash constraint is missing, cash after wages is 2 and the sale and bridge then add nonnegative proceeds. After day 4, the receipt adds cash; cash after B settles must be $C_6 + a + 1$, which is positive whenever $C_6 \geq 0$. The bridge repayment is the final outflow in this comparison. H2's day-9 receipt increases cash later; it cannot repair a negative prefix on day 6. Any term debt remains an obligation beyond this cash horizon.

With the inventory covenant, $u = 1$ also requires $20(1 - z) \geq 10$. For half sale, the bridge interval is $[23.5, 30]$. If B is used, its face must satisfy $qb \geq 8.5$ in H1 or $qb \geq 26.5$ in H2, recovering the earlier price thresholds at the face cap. No sale would require $a \geq 31$, so that entire branch is infeasible. Full sale permits $a \in [16, 30]$; with the covenant it must use $u = 0$, giving $C_6 = 1$ in H1 and $C_6 = -17$ in H2. Without the covenant, the term issue remains a separate possible response.

These intervals describe the continuous amounts; a few selected amounts are only witnesses. For a fixed sale choice and term issue, taking the least bridge principal $a = 31 - 15z$ preserves

feasibility under these assumptions, but it can change the best liquidity policy. In H1 with full sale and no term issue, $a = 16$ leaves cash 0 after the old debt payment and cash 1 after bridge repayment. Taking $a = 17$ instead leaves cash 1 after both payments. Both paths pay the same two units of bridge cost and end with the same cash and no bridge debt, yet their minimum cash balances are 0 and 1. The latter is maximal within this full-sale/no-term comparison because every path ends with cash 1. Restricting the comparison to minimum funding would lose that conclusion. An interest charge depending on the amount borrowed would require a different calculation.

Settlement order also matters for the domain itself. Without the inventory covenant, take full sale, H1, $a = 16$, $q = 0.9$ and $b = 10/9$. Issuing B yields net cash -1 after its fixed fee. Receipt before issue gives a feasible path ending at cash 0. Issue before receipt instead asks the company to pay that one unit while cash is still 0. That order requires the additional prefix condition $C_4 + qb - 2 \geq 0$. The small issue is not proposed as an attractive policy; it exposes an action that a daily net total would mistakenly admit under the opposite order.

A finite enumeration is therefore complete only for an explicitly finite action menu or after a justified reduction for the stated objective. Here one can enumerate sale choices and offer activation, then retain or solve the continuous inequalities inside each branch. With fixed q these are linear constraints. If price depends on face, inventory or the later policy, that continuous problem changes. Across H1 and H2, decisions must still share the same values until the manager's observations permit them to differ. Comparison with another optimizer requires the same amounts, information, settlement order and objective; a better result under different choices answers a different financing question.

14.14 Accounting consequences of the sequence

The old loan repayment reduces cash and the old obligation. Bridge issuance increases cash by net proceeds and creates the specified liability and fee treatment. Inventory disposal removes the asset and records proceeds and loss under the illustrative basis. Term issuance creates a new obligation with proceeds distinct from face.

The exact treatment of debt discounts and transaction costs requires the accounting basis. A cash schedule can be checked without prematurely choosing that treatment. The accounting layer should then reconcile carrying amounts separately from contractual face.

A journal test should ensure that the old debt is repaid once, the bridge is extinguished only when settled or legally discharged, and the term debt remains after bridge repayment. Netting all financing flows into one closing debt amount would make these checks impossible.

14.15 Sensitivity that preserves the same company

Vary B's price while holding the collection history, asset sale and contractual caps fixed. This produces a meaningful price sensitivity for each policy. Vary the collection date while retaining the same receivable identity and amount. This produces a timing sensitivity.

Changing both at once can be a joint stress, but it should be named as such. Comparing a high-price H1 case with a low-price H2 case and attributing the entire difference to price would confound two mechanisms. Paired scenario comparisons should preserve the dimensions not intended to change.

The same discipline applies to recovery. A lower recovery value can affect price at issuance if anticipated, but a surprise enforcement loss after issuance is a different information path. The

model should specify whether creditors knew the recovery change when making the offer.

14.16 A practical implementation slice

This case suggests a bounded event scheduler with exact rational amounts and ordered timestamps. It needs offer admission, net-proceeds calculation, inventory eligibility, collection observation, policy action and arrears transitions. It does not need a neural network to expose the important errors.

The fixture should include both successful and refused paths, with reasons. The refused bridge-only path fails by one due to the fee. The full-sale H2 path loses term eligibility. The half-sale H2 path fails when price falls below 0.6625. The cure-period variant distinguishes late collection from timely cure.

These cases provide a concrete basis for extending the repository's financing methods. They also show what remains outside a small implementation: actual market offers, legal enforceability, operational cost of inventory sale and empirically supported management preferences. The model can explain those dependencies without pretending to have resolved them.

14.17 The price must anticipate who chooses after conversion

The refinancing calendar tells us when an offer can settle. A second issue arises when the security changes the person who chooses later company policy. A price based on the old owner's future choice can be inconsistent with a conversion that transfers control. The following finite example isolates that dependency.

There are three dates: financing on 1 January 2027, a signal and policy choice on 1 January 2028, and terminal distributions on 1 January 2029. Investment costs 60 at the first date. Existing senior face is 40 and new convertible face is 80. The gross discount factor is 21/20 in each of the two intervals. All cash recoveries occur at the terminal date. The equal signal probabilities and the conditional probabilities below are a supplied hypothetical pricing law, also used to evaluate expected cash utility.

Under the debt signal, the new claim remains debt and the old owner chooses policy. Safe policy produces terminal estate 100. Risky policy produces 20 or 220 with equal probabilities. Senior debt is paid first, then the new claim up to its face, and old equity receives the residual. The old owner consequently prefers risky expected cash 50 to safe cash zero. The new creditor's expected cash under that choice is 40.

Under the conversion signal, the new claim is extinguished and its holder receives 120 shares alongside 40 old shares, as in Section 7.16. The new holder now controls policy. Safe estate is again 100; risky estate is 20 or 140 with equal probabilities. After senior priority, safe policy gives the new holder $3(100 - 40)/4 = 45$. Risky policy gives 75/2, so the controlling holder chooses safe.

To make the control distinction consequential, assume the old owner receives a noncash private benefit worth 4 terminal utility units when risky policy is chosen in this conversion branch. Old-owner risky cash has expectation 25/2, giving utility $25/2 + 4 = 33/2$, compared with safe utility 15. Thus a stale old-controller assumption would select risky policy. The benefit is deliberately hypothetical and never enters estate, creditor recovery or accounting equity. At zero private benefit the old owner also prefers safe policy, which checks the mechanism's limiting case.

The consistent continuation is therefore risky under the debt signal and safe after conversion. Let P denote date-zero proceeds paid by the new holder. In this selected competitive-quote comparison, proceeds equal the discounted expected cash the holder will receive:

$$P = \frac{\frac{1}{2}(40) + \frac{1}{2}(45)}{(21/20)^2} = \frac{17000}{441}, \qquad 60 - P = \frac{9460}{441}. \tag{14.7}$$

The two discount intervals matter: the signal date is not a payment date. The second term is the cash the old owner must contribute at financing. Available cash 20 is insufficient, while cash 22 is sufficient.

Affordability and willingness to invest are separate conditions. Under this continuation, the old owner's expected terminal cash is $(50 + 15)/2$. Its date-zero value is $13000/441$. Subtracting the contribution gives $3540/441 > 0$, so participation exceeds the assumed outside utility zero. That positive value does not supply the missing liquid funds in the cash-20 case.

If old control were incorrectly retained after conversion, the new holder's conversion-branch cash would instead be $75/2$. The corresponding quote would be $15500/441$, requiring owner cash $10960/441$, more than 22. Charging the correct-control quote while executing that stale-control policy leaves the lender short by $1500/441$. The discrepancy is a failure to keep rights, continuation and price consistent. It does not establish that conversion is economically superior.

Backward induction agrees with a separate enumeration of all four pure strategies mapping the observed signal to a policy, including every local deviation. A tie variant retains both optimal policies. Equal votes without a supplied governance rule remain unresolved. Policies cannot depend on the terminal realization, which is not known when the choice is made.

The owner's financing contribution is sunk by the policy date, so the continuation ranking in this example is independent of the posted price. This is a finite consistency calculation, not a general fixed-point or market-clearing result. The typed conversion example starts at a separately supplied fiscal opening. The following sections connect the dated funding exchange and identify the source, measurement and control-history conditions needed before issuance, conversion and final liquidation can form one accounting history.

14.18 Funding the project does not yet determine its accounts

The price calculation leaves two different questions. Can the parties deliver the promised cash in the required order? If they can, which amounts should their financial statements recognise between issue and conversion? The first question has a complete answer under the supplied synthetic cash balances. The second still depends on the contract, reporting basis and measurement evidence. Keeping the questions separate prevents a feasible transaction from silently becoming an unsupported accounting history.

At the financing date the new holder pays $17000/441$ to the company and the old owner contributes $9460/441$. Their receipts sum exactly to the project price 60. The company then pays 60 to the seller. The transfer records identify both parties, the date and the same amount in the two legal books. In this example the company starts with no available cash for the purchase, so both receipts must precede payment to the seller. There are six possible orders of these three transfers. The two orders with the seller paid last succeed; the other four fail at the first payment that exceeds available company cash. Positive final net cash is not a substitute for an affordable cash prefix.

The four-party reference retains the company, old owner, new holder and seller books separately. It checks two account charts and both owner cash 22 and cash exactly equal to the required contribution. A refused later transfer preserves the accepted prefix. Counterparty cost-basis

entries are explicit synthetic premises. They demonstrate reciprocal transfers and funding feasibility, not the appropriate measurement of a real convertible instrument or the seller's tax and profit consequences.

The limitation becomes visible if the 5% pricing discount is reused as an accounting yield. Suppose one incorrectly treats the face 80 claim as a plain two-year zero-coupon liability and discounts its entire face at that rate. The resulting liability and residual allocation would be

$$L_0^{\text{forced}} = \frac{80}{(21/20)^2} = \frac{32000}{441}, \qquad P - L_0^{\text{forced}} = -\frac{5000}{147}. \tag{14.8}$$

This negative residual is a diagnostic of that attempted mapping. The plain unconditional cash claim has different payoffs from the contingent instrument used to obtain P. Conversely, simply accreting P at 5% over both intervals gives 85/2, not face 80. Neither calculation supplies a replacement accounting policy. The rate and cash flows appropriate to a liability component require their own evidence.

For the conventional issuer compound-instrument case, the inspected AASB 132 text measures the comparable liability first and assigns the residual to equity. Its application guidance refers to instruments of comparable credit status, substantially the same cash flows and terms, without the conversion option (Australian Accounting Standards Board, 2022b, paragraphs 31–32 and AG31). Those conditions are not established by the economic quote. The typed conversion demonstration instead starts with a supplied carrying amount C. Its balanced transfer from liability to equity remains conditional until classification, intervening measurement and the conversion treatment are supported.

14.19 Issuer and holder classify the same contract differently

The same contract is a claim issued by the company and an investment held by the creditor. Those perspectives need not have matching classification components. The source research therefore begins with the complete settlement terms: who can affect the conversion signal, whether either party can elect cash settlement, the currency and share terms, any redemption provisions, and agreements involving other members of a group. The example supplies none of these facts merely by stating that 120 shares will be issued.

Under the inspected AASB 132 provisions, fixed-for-fixed analysis is subject to the definitions and exceptions for qualifying own-equity instruments. A genuine contingent liability settlement beyond both parties' control raises the separate conditions in paragraph 25. A derivative with settlement choices has its own rule in paragraph 26. The non-derivative compound-instrument assessment in paragraphs 28–32 cannot be inferred from a fixed share count alone (Australian Accounting Standards Board, 2022b, paragraphs 15–16, 22–26 and 28–32). These passages identify the questions to answer; they do not classify the underspecified fixture.

The holder's analysis is different. For an in-scope financial-asset host, AASB 9 assesses the hybrid as a whole. Its fixed-number convertible-bond example links the return to issuer equity and fails the solely-principal-and- interest condition (Australian Accounting Standards Board, 2026i, paragraphs 4.3.2 and B4.1.14, Instrument F). Issuer liability/equity components therefore cannot simply be copied into the holder's asset book. Initial fair value, transaction costs, subsequent measurement and relevant exceptions remain separate dependencies. Likewise, AASB 132 AG32 expressly addresses conversion at maturity; it alone does not justify the fixture's 2028 conversion before its 2029 terminal date.

The resulting holder interest also raises a reporting-perimeter question. Section 26.16 follows the separate-statement, consolidation and business-acquisition alternatives, including the possibility

of control before legal conversion. Its repaired dependency analysis preserves uncertain control histories rather than choosing whichever journal balances. The retained source compilations do not yet establish the complete accounting authority for the fixture's 2027 issue and 2028 conversion. Classification, measurement, control history and applicable edition therefore remain linked questions with distinct evidence requirements.

14.20 Evidence over an interval is stronger than two snapshots

To support uninterrupted control, an analyst needs evidence covering the intervening period. An empty event list supplies no such evidence. The selected interval reference requires an explicit assertion that a control status holds throughout a specified half-open interval, together with its subject, reporting basis, source identity and availability to each actor. This is a representation of an asserted coverage premise. It does not certify that the assertion is true or that the source collection is complete.

For example, let one source assert no control from 30 December to 31 December, and another assert control from 31 December to 1 January. A half-open interval includes its starting instant and excludes its ending instant, so the shared boundary is assigned once. If both assertions are eligible for the selected actor view, they conditionally support a gain at that boundary. If the second source is missing, the second interval remains unresolved. If it conflicts with another eligible assertion, the conflict remains visible rather than being resolved by whichever record was loaded last.

Effective time and information time do different jobs. A correction can describe control during December yet become available to an analyst in January and to a manager only in February. The analyst's January view can use it; the manager's January view cannot. A later available revision replaces the corresponding earlier assertion only within its declared subject and interval. Unknown actor availability remains unresolved. The selected profile refuses partial-interval replacement, whose meaning would require an additional rule.

The implementation reuses the exact fixed-offset timestamp parser and partitions the query at every eligible interval boundary. It retains every covered, uncovered, unknown or conflicting piece. Equivalent timestamp offsets denote the same instant, and adding a future correction leaves an earlier view's control conclusions unchanged. The checked cases include all two-cell combinations of control, no control and unknown control, as well as revision conflicts, coverage gaps, missing ancestry and corrupted source identities.

This gives the simulator a precise way to carry unresolved control history alongside otherwise balanced company books. It does not yet connect an admitted real history to acquisition measurement, changing group membership or complete financial statements. The next financial composition must supply those methods and their dated authority while preserving the source and actor-view distinctions. Neither a successful funding transfer nor a complete conditional interval partition can supply the missing accounting evidence.

Chapter 15

A recapitalization: cash raised, rights transferred, and incentives changed

A financing plan can save a company while reducing the value held by its existing owners. It can also dilute their percentage without reducing their value. Distinguishing these outcomes requires a security-level calculation. This chapter follows a synthetic recapitalization with ordinary shares, a preference offer and a convertible claim. The terms are stipulated so the arithmetic can be reconstructed. They illustrate mechanisms rather than legal or valuation conclusions for a real issuer.

15.1 The company before financing

There are 100 ordinary shares. Existing operating assets have a stipulated economic value of 600, cash is 20 and a senior debt payment of 80 is immediately due. For the first comparison, assume asset value is unchanged by financing and there are no other claims. Before paying debt, residual economic equity value is $600 + 20 - 80 = 540$, or 5.4 per ordinary share.

The company nevertheless cannot pay 80 from cash 20. It needs 60 of additional cash. Positive economic equity value does not eliminate this liquidity problem. Selling assets, borrowing, issuing equity or negotiating deferral are different possible responses. We first study an ordinary-share offer.

Book assets may have a different carrying amount. Nothing in the calculation requires book equity to equal 540. If the model uses book equity to set a subscription price of 5.4, it would be making a separate valuation assumption. Here 540 is a supplied economic valuation used only for the illustrative allocation.

15.2 A fairly priced ordinary issue

Suppose investors subscribe at 5.4 with no fees, and the company raises exactly 60. New shares are $60/5.4 = 100/9$, about 11.1111. Cash becomes 80 and is paid to the senior creditor. Remaining assets are 600, there is no debt, and shares total 1000/9.

Post-transaction value per share is

$$\frac{600}{1000/9} = 5.4. \tag{15.1}$$

Existing holders retain value 540, and new holders own value 60. Their percentages change, but

no value is transferred under these assumptions. The creditor receives its contractual 80. The cash injection solves liquidity without creating extra value by itself.

This result relies on the supplied fair price, homogeneous rights, no costs and unchanged operating value. It is a useful benchmark because deviations can be attributed to specific mechanisms. It is not a prediction that a distressed company can always issue at fair value.

15.3 A discounted issue

Now suppose the offer price is 3, with the same 60 cash target and no fees. Twenty shares are issued. After paying debt, value remains 600 but shares total 120. Each share is worth 5 under the stipulated valuation. Existing holders own 500, a loss of 40 relative to the benchmark; new investors paid 60 for claims worth 100.

The transfer is not an additional operating loss. It is an allocation effect of selling claims below the assumed post-transaction value. A journal recording cash and equity does not reveal it. The ownership register and valuation assumptions are needed.

If investors' required price of 3 reflects information that operating assets are worth less than 600, the benchmark valuation is wrong. The calculation should not accuse the offer of unfairness without validating the valuation premise. Scenario analysis can show transfers conditional on alternative asset values and prices.

15.4 Fees change both quantity and value

Add a fixed cash issuance fee of 6. To net 60, the company must raise 66. At price 3, it issues 22 shares. After fees and debt payment, operating assets remain 600, cash is zero and shares total 122. Existing owners' value is $600(100/122)$, about 491.8033.

The fee destroys 6 of total value relative to an otherwise identical financing if it buys no additional benefit in this model. The discounted price transfers value between old and new owners. These effects should be separated. Calling both dilution obscures which mechanism could be changed by renegotiation or a different financing channel.

If the fee is paid later, current funding need differs and a payable remains. If it is financed by issuing additional shares to an intermediary, share count changes without the same cash payment. The offer must state the form and timing of compensation.

15.5 An authorization constraint

Suppose only fifteen new ordinary shares may be issued under the admitted authorization. At price 3 and fee 6, maximum net cash is 39. Existing cash 20 then supplies only 59 toward debt 80, leaving a shortfall of 21. The ordinary issue cannot solve the maturity under these terms.

At price p, feasibility requires $15p - 6 \geq 60$, so $p \geq 4.4$. This is an exact offer threshold. It does not establish that investors will pay 4.4. A market model or a new offer is needed to answer that question.

An optimizer that increases shares beyond fifteen has changed the feasible set. An accounting routine that credits capital for the missing 21 without issuing a security has invented financing. A correct result records the shortfall and considers other admitted actions.

15.6 A rights offer

Suppose existing holders have a right to subscribe proportionally at the discounted price. If every holder subscribes in proportion, their relative ownership remains unchanged, but they contribute new cash. The issue can still have fees and execution risk.

A holder who does not subscribe may sell the right if trading is permitted and a buyer exists. The value of the right depends on issue terms and market value. A model that treats all existing holders as equally able and willing to contribute assumes away liquidity and participation differences.

For AccDSL, a rights offer needs entitlement, subscription, transfer and expiry events. Unexercised rights may lapse or be allocated under a backstop arrangement. The company receives cash only from settled subscriptions or the backstop, not from the mere existence of rights. Historical statements may not reveal the eventual participation pattern.

15.7 A preference alternative

Consider instead an investor providing 60 for a nonparticipating preference claim. At a future exit, the investor may take 90 before ordinary shareholders or convert into 20 per cent of ordinary equity. Assume no senior debt remains, no interim dividends, no other rights and distributable value V. The investor receives

$$P(V) = \max\{\min(90, V), 0.2V\}. \tag{15.2}$$

Ordinary holders receive $V - P(V)$. The first term respects limited available value: a preference of 90 cannot pay 90 if only 50 exists. Conversion dominates the full preference when $0.2V > 90$, or $V > 450$.

For $V = 50$, the investor takes 50 and ordinary holders receive zero. For $V = 300$, the investor takes 90 and ordinary holders receive 210. For $V = 600$, conversion pays 120 and ordinary holders receive 480. The same 60 of initial funding creates very different allocation across outcomes from a simple ordinary issue.

15.8 Participation changes the waterfall

A participating preference can take a preference amount and then share in remaining value. Suppose the investor receives up to 90 first and then 20 per cent of the residual, with no cap. Its payoff is

$$P_{\text{part}}(V) = \min(90, V) + 0.2\max(V - 90, 0). \tag{15.3}$$

At $V = 300$, this gives $90 + 42 = 132$, compared with 90 for the nonparticipating offer. At $V = 600$, it gives 192, compared with 120 under conversion in the earlier contract.

These are different securities, not two calculation methods for one security. A parser that extracts only the word preference and the amount 90 will miss the distinction. The contract admission model must preserve participation, conversion and caps where they affect the query.

If a participation cap applies, another region boundary appears. The engine can represent the payoff as a piecewise function and split only where the active right changes. A sample of a few exit values is useful for illustration but not a complete characterization of the payoff.

15.9 Value conservation in each state

For every exit value, distributions to all claimants must not exceed distributable assets under the assumed waterfall. In the nonparticipating example, $P(V) + [V - P(V)] = V$ exactly. This is a conservation check independent of whether the offer is fairly priced.

A common implementation error pays the preference and also allocates 20 per cent of total value as though conversion were simultaneous. That double counts mutually exclusive rights. Another pays 90 when $V < 90$, creating cash. Both can be detected with low-value boundary cases.

Fair pricing requires discounting and a distribution or valuation operator over future payoffs. The waterfall itself supplies state-contingent claims, not their current price. An investor's willingness to pay 60 can be taken as a declared offer without claiming that a model has proved its fairness.

15.10 A convertible debt alternative

Suppose an existing note has face 80 due at exit and can convert into sixteen ordinary shares, with no other debt. If conversion is optional for the holder and the company has 100 ordinary shares before conversion, conversion gives a fraction 16/116 of total equity after conversion. The holder compares that claim with the debt payoff under the specified default and priority rules.

If distributable asset value is high enough that debt is paid in full, conversion value is $16V/116$. It exceeds 80 when $V > 580$. Below that threshold, the holder may prefer debt. At low asset values, debt recovery is limited by available assets and seniority. The conversion threshold can change if other securities exist.

A mistaken calculation using $16V/100$ ignores the new shares in the denominator and gives a threshold of 500. That is a dilution error. A second mistake pays debt 80 and also issues sixteen shares. The conversion event must extinguish the specified debt claim if that is what the contract says.

15.11 Cash settlement and issuer options

Some instruments permit cash settlement of conversion value or allow the issuer to choose a settlement form. That changes liquidity and ownership. A model that always assumes physical shares may understate cash need; one that always assumes cash may overstate dilution avoidance.

The choice belongs to the contract and the authorized actor. If the issuer can choose, its policy must consider cash capacity and economic cost. If the holder chooses, management cannot select the favorable form merely to balance the forecast.

An unknown settlement clause should produce conditional outcomes. The system can show a physical-settlement branch and a cash-settlement branch with their premises. It should seek the clause when the distinction changes a liquidity conclusion. This is a high-value source-research target.

15.12 Control and approvals

A recapitalization can shift voting control even when economic ownership changes modestly. A preference investor may receive veto rights or board appointment rights. Those rights can affect future financing, dividends and asset sales. Treating them as irrelevant because they have no immediate journal amount can distort later policies.

The model need not simulate every governance detail for every query. It should retain material rights and declare when behavior assumes their exercise or nonexercise. If a future equity issue requires investor consent, the feasible action set must include that condition.

A public-data analyst may not know whether consent will be given. This is future behavioral uncertainty, distinct from uncertainty about whether the consent right exists. Combining both into one unspecified branch makes it harder to identify useful evidence or policy assumptions.

15.13 Incentives after financing

Suppose ordinary holders choose between a safe project with exit value 300 and a risky project yielding 100 or 600 with equal probability. Under the nonparticipating preference, ordinary payoff from the safe project is 210. Under the risky project, ordinary payoffs are 10 and 480, with mean 245.

Total expected exit value of the risky project is 350, above 300, so both total value and ordinary value favor it in this particular example. Change the low outcome to zero and high outcome to 500. Total expected value is 250, below 300, while ordinary expected payoff is $[0 + 400]/2 = 200$, also below 210. The rankings still agree. These examples should not be used to claim a general conflict.

To exhibit a conflict, consider a senior fixed claim of 250 and ordinary residual value. A safe project with value 300 gives ordinary holders 50. A risky project with values zero and 500 equally likely has total expected value 250 but ordinary expected payoff 125. Ordinary holders prefer the risky project while total expected value is lower. The contract allocation creates the divergence under the stated objective and probabilities.

15.14 Why the actor objective matters

A policy maximizing total enterprise value selects the safe project in the final example. A policy maximizing ordinary equity selects the risky one. A lender pricing debt must anticipate the policy actually modeled. Pricing under the safe policy and then letting shareholders choose the risky one is inconsistent.

Covenants or control rights may constrain the choice. Their presence changes the feasible set and possibly the price offered initially. This joins the security register to behavioral modeling. The register is not merely a reporting table; it determines who receives gains and losses and who can authorize actions.

The example is a local derivation of incentive conflict, not a calibrated prediction. It gives a discriminating fixture for a structural solver: the actor objective, priority and feasible actions should reproduce the stated ranking. A solver returning the same policy for both objectives without a binding constraint is suspect.

15.15 Book entries and economic allocation

The cash issue creates cash and accounting equity under the applicable treatment. The preference or convertible may require classification and component allocation that cannot be inferred from the simplified payoff alone. Those accounting decisions need the standard, jurisdiction and complete terms.

Economic valuation can proceed conditionally on a specified payoff while accounting classification remains unresolved. Conversely, a classified instrument can still have uncertain market price. The system should allow these statuses to coexist. Forcing both layers into one known or unknown flag loses useful work.

A book-equity bridge should reconcile issue proceeds, fees under the chosen treatment, retained earnings, distributions and other movements. A separate ownership bridge reconciles units and rights. A valuation bridge explains changes in economic claims. The three bridges communicate but should not be substituted for one another.

15.16 Historical ambiguity in a capital movement

Suppose a filing reports net cash from equity transactions of 60 and an increase of twenty ordinary shares. This could be a simple issue at 3 with no costs. It could also be an issue of thirty shares at 4 and repurchase of ten at 6, again netting to 60 and twenty shares. The closing share count and net cash agree.

The histories differ in gross funding, repurchase behavior and prices. If future policy extrapolates issuance discounts or repurchase commitments, the distinction matters. A note disclosing gross proceeds and repurchases can narrow the family. Without it, a single implied price of 3 is not established as an actual transaction price.

This example parallels the receivables and debt bridges. Net movements constrain sums but do not identify gross events. The security register should therefore permit unresolved transaction families while preserving observed opening and closing counts.

15.17 An end-to-end fixture

A useful fixture starts with the 100-share company, the debt maturity and the cash need. It evaluates ordinary issuance at fair and discounted prices, fixed fees, authorization limits, preference rights and conversion. Each transaction updates cash, liabilities, shares and rights before producing statements.

Expected failures include using book value as an unstated subscription price, ignoring the authorization cap, counting conversion shares against the old denominator, paying mutually exclusive preference and conversion rights together, and treating an unsigned offer as settled cash. Each failure has a specific input that exposes it.

The fixture should also retain unresolved accounting classification as a legitimate result where full terms are absent. A financing calculation can be mechanically complete while a statement presentation remains conditional. That distinction is preferable to inventing a universal accounting rule to make the example look finished.

15.18 What the case adds to simulation design

Equity funding is a transaction among actors with claims, rights and constraints. Its effect cannot be reduced to an external-equity cost scalar when ownership and future decisions matter. Nor can share count alone establish value transfer. Prices, fees, rights and valuation premises jointly determine the result.

The proposed AccDSL design therefore keeps a dated security register, offer and settlement states, class-specific payoff rules and actor objectives. It can use a simple homogeneous-share profile when appropriate, but retains the boundary that tells a user when richer terms are required. This makes dilution and financing consequences explicit without pretending that every security must always be simulated at maximum detail.

Chapter 16

A decision before and after disclosure: an exact learning example

Learning matters only through a specified relation between hidden facts, observations and decisions. This chapter constructs that relation explicitly. The example is finite, so every policy and probability can be checked by enumeration. It demonstrates why two companies with equal current statements can require different information states, and why preserving only unconditional future cash distributions is insufficient for adaptive decisions.

16.1 Two hidden exposures

A company must choose a hedge before a future settlement. Its exposure type is either A or B. Under type A, hedge A avoids a loss of 10 while hedge B avoids none. Under type B, hedge B avoids a loss of 10 while hedge A avoids none. Each hedge costs 2. Doing nothing costs nothing and leaves the loss 10.

For the first calculation, assume type A and B each have probability one half. This probability is a supplied teaching assumption. The company has enough cash to execute either hedge and can choose at most one. Current financial statements do not distinguish the types.

Net loss after the correct hedge is 2, after the wrong hedge is 12, and after no hedge is 10. Without further information, either hedge has expected loss 7, so a risk-neutral expected-loss policy chooses either hedge rather than doing nothing. Symmetry gives two equally optimal actions. The model should not invent a preference between them.

16.2 A noisy signal

Before choosing, the actor can receive a signal labeled A or B. Let the signal be correct with probability q, symmetrically across types, with $q \geq 1/2$. Bayes' rule with equal priors gives posterior probability q for the signaled type. Following the signal yields expected loss

$$L_{\text{signal}}(q) = 2 + 10(1 - q) = 12 - 10q. \tag{16.1}$$

The first term is hedge cost. The second is the loss when the signal is wrong. At $q = 0.9$, expected loss is 3; at $q = 0.5$, it is 7, matching no informative signal.

The gross value of the signal relative to the best uninformed hedge is $7 - (12 - 10q) = 10q - 5$. A signal costing k is worthwhile under this criterion when $k < 10q - 5$. At $q = 0.9$, the maximum

willingness to pay is 4. This calculation requires the signal to arrive before the action and the actor to be able to respond.

16.3 A complete joint probability table

For $q = 0.9$, the four type-signal combinations have probabilities

Type	Signal	Probability	Loss following signal
A	A	0.45	2
A	B	0.05	12
B	A	0.05	12
B	B	0.45	2

The expected loss is $0.9(2) + 0.1(12) = 3$. The signal is marginally A or B with equal probability. Type is also marginally A or B with equal probability. The information lies in their dependence.

If a compression retains these two marginals but makes them independent, it changes each joint cell to 0.25. Following the signal then has expected loss 7. The compressed model has destroyed the signal's information while preserving both marginal distributions. This is why the joint successor-observation law appears in the exact aggregation condition.

16.4 Two observation technologies with equal current reports

Consider company state U, which will receive the informative signal with $q = 0.9$, and state V, which will receive an uninformative signal with $q = 0.5$. Their current cash, debt and income are identical. Their hidden type priors are identical. Before signal arrival, a fixed hedge has the same expected loss 7 in both.

An adaptive policy following the signal has expected loss 3 in U and 7 in V. Merging U and V because current reports and fixed-action values agree loses a decision-relevant difference. The future observation technology is part of state for this question.

This finite example gives an accounting-oriented counterpart to the policy-versus-action-sequence distinction in Givan et al. (2003, Section 3.1). The numerical construction is local. The source supplies the general warning that policies react to intermediate information and therefore require a stronger equivalence than fixed sequences.

16.5 What happens when the signal arrives late

Suppose the hedge must be committed at date 1 and the signal arrives at date 2. The informative technology no longer improves that hedge decision. A policy that follows the date-2 signal but executes the hedge at date 1 is infeasible.

The expected-loss value returns to 7 for the best uninformed hedge, even though the analyst eventually learns much more about the type. Information has retrospective explanatory value but no value for this already committed action. A simulation must represent action deadlines and settlement lead times, not only observation dates.

If the contract permits cancellation or adjustment at date 2 for cost c, some signal value can return. The calculation then compares adjustment benefits with c and available liquidity. The ability to revise is a contract term or policy permission, not a default property of every action.

16.6 A cancellation calculation

Suppose hedge A is chosen initially. After receiving the signal, the actor may switch to B at extra cost c. With signal B, posterior probability of type B is q. Keeping A has conditional expected loss $2 + 10q$. Switching has conditional expected loss $2 + c + 10(1 - q)$.

Switching is preferred when

$$2 + c + 10(1 - q) < 2 + 10q \quad \iff \quad c < 20q - 10. \tag{16.2}$$

At $q = 0.9$, the threshold is 8. The switching opportunity is exercised only after signal B under the symmetric setup. Its unconditional value must also account for the probability of that signal.

If the initial hedge cannot be unwound and the second hedge creates an offsetting or additional exposure rather than replacing it, the formula changes. A model should not assume that a new action erases an old contract. This is the same event-persistence issue seen in debt amendments.

16.7 Unknown signal quality

Now suppose evidence supports only $q \in [0.6, 0.9]$. The expected loss from following the signal lies in $[3, 6]$ under the same prior and symmetry assumptions. The gross information value lies in $[1, 4]$. A signal priced at 0.5 improves expected loss for every admitted q; one priced at 3 is worthwhile only for some qualities.

These are bounds over a parameterized probabilistic family. They are not probabilities over signal quality. Assigning a uniform distribution to q would add another assumption. A robust decision can use the worst admitted quality, while a Bayesian decision can use a prior over it if justified.

If the quality depends on exposure type, one scalar q is insufficient. Let sensitivity and specificity differ. Posterior probabilities then depend on the prior and both error rates. A report that is usually correct overall can still be misleading for a rare type. The observation model should preserve the relevant asymmetry.

16.8 Unequal prior probabilities

Let prior probability of A be p and signal accuracy remain symmetric at q. After signal A, the posterior is

$$p_A^+ = \frac{pq}{pq + (1 - p)(1 - q)}. \tag{16.3}$$

After signal B, it is

$$p_B^+ = \frac{p(1 - q)}{p(1 - q) + (1 - p)q}. \tag{16.4}$$

The actor chooses hedge A when the posterior probability of A exceeds one half, hedge B when below, and is indifferent at equality under the symmetric loss structure.

If p is very high, even signal B may not overturn the prior. Following the signal mechanically is then suboptimal. For example, $p = 0.95$ and $q = 0.9$ gives posterior A after signal B equal to $0.095/(0.095 + 0.045)$, about 0.6786. Hedge A remains preferred.

This illustrates why a signal label is not itself a decision. The policy combines the observation with prior information and payoffs. In an unweighted reconstruction, there may be no justified

prior at all, so a posterior calculation would be inappropriate until that modeling choice is admitted.

16.9 An unweighted version

Suppose both types remain possible and no probabilities are assigned. An exact signal identifying the type allows the correct hedge in every world, producing loss 2. Without a signal, either hedge has worst-case loss 12, while doing nothing has worst-case loss 10. A minimax-loss policy therefore chooses no hedge before information and the correct hedge after exact information.

The criterion differs from the earlier expected-loss decision, which chose a hedge without a signal. This is not a contradiction. The uncertainty representation and decision rule changed. A system should report the dependence rather than label one answer universally optimal.

If a noisy signal can be wrong in either type and only support is known, both types may remain possible after either signal. The worst-case benefit can be zero. A probability of correctness above one half is a probabilistic restriction; it cannot be inferred from support alone.

16.10 Management and public investors

Suppose management already knows the type from internal contracts, while public investors see only aggregate statements. Management can choose the correct hedge with loss 2 even before public disclosure. The analyst should simulate that informed behavior if it is the declared policy.

A public announcement can then affect investor beliefs and financing offers without changing management's hedge choice. The same information event has different economic consequences for different actors. A single global belief state cannot represent this distinction.

If management's action itself is observable, investors may learn from it. A hedge choice can signal the hidden type under an assumed policy. That inference depends on the policy model. If management sometimes hedges for other reasons, the action is not a perfect signal. Endogenous disclosure and signaling require a richer model than an exogenous likelihood.

16.11 Endogenous observations

A company can choose when and what to disclose within its obligations. A lender can request information before making an offer. An auditor can discover discrepancies with probability depending on effort. These observations depend on actions and hidden state.

The observation kernel should then be written conditional on action as well as state. Treating disclosure as independent can distort the value of monitoring or the selection of companies that reveal information. A missing report may itself be informative only under an explicit model of why reports are missing.

AccDSL need not assume a strategic disclosure game in every application. It should allow exogenous observation models as a bounded profile and state when they are inadequate. If the query concerns financing responses to disclosure, the endogenous channel may be material and should remain an open modeling requirement.

16.12 Persistent type and repeated signals

Suppose two conditionally independent signals of accuracy 0.9 arrive about the same fixed type with equal prior. Two A signals give posterior A

$$\frac{0.9^2}{0.9^2 + 0.1^2} = \frac{81}{82}. \tag{16.5}$$

An A followed by B returns posterior one half under the symmetry and independence assumptions. These updates concern one persistent type, not a newly drawn type for each signal.

If the signals are copies of the same underlying source, conditional independence is false. Counting a duplicated report as independent evidence would overstate confidence. Two identical copies should not square the likelihood unless the observation process actually supplies independent information.

Source provenance therefore matters statistically. A news summary quoting a filing is not necessarily another independent measurement of the loan type. The system should preserve derivation links so that a likelihood model can avoid double counting.

16.13 Correlated errors

Suppose both signals share a common extraction error. With probability 0.1 the type label is reversed in the source, and both downstream reports repeat it. Two matching reports then have accuracy 0.9, not 81/82. The agreement supplies no additional independent confirmation.

A model with separate independent error terms would produce unjustified confidence. The remedy is a shared latent error or a source-dependence relation. If the dependence is unknown, sensitivity over plausible observation models is more defensible than assuming independence because it is convenient.

The same issue arises when several financial databases reproduce one filing or when multiple ratios use the same extracted amount. More records do not necessarily mean more evidence. The model should count informational content through its observation relation, not file count.

16.14 A belief-state calculation

In the finite probabilistic example, the posterior probability p_t of type A summarizes the evidence relevant to the hedge decision, provided the signal model and loss structure remain fixed. Expected losses are $2 + 10(1 - p_t)$ for hedge A, $2 + 10p_t$ for hedge B and 10 for no hedge.

The policy chooses the smallest of these expressions. Hedge A is preferred to B when $p_t \geq 1/2$. It is preferred to no hedge when $p_t \geq 0.2$. Hedge B is preferred to no hedge when $p_t \leq 0.8$. Thus at least one hedge is preferred throughout $[0, 1]$ in this particular cost structure, with the choice switching at one half.

This compact belief is sufficient because the future depends on the past only through the fixed type posterior under the declared model. If signal quality is itself unknown or past actions create commitments, p_t alone may cease to be sufficient. The state must then include those additional quantities or their joint belief.

16.15 Compression tests from the example

A proposed compression should preserve the joint type-signal distribution, action availability, costs and timing. It should reproduce expected loss 3 for informative signals and 7 for uninformative signals. It should refuse to merge those technologies for an adaptive policy query.

A separate test can merge states differing only in an irrelevant customer label when neither payoffs nor observations depend on that label. This establishes a valid reduction within the small model. Positive and negative examples are both needed: a system that never merges is safe from one error but does not solve the computational problem.

Changing the query to one fixed hedge can make a coarser compression sufficient. The approval should be scoped to that query. Reusing it for information acquisition or adaptive hedging without rechecking would repeat the original mistake.

16.16 What the case teaches about trees

The full tree has hidden types, signals and actions. Hidden type branches need not be visible to the actor. Signal branches define information partitions, and actions before a signal must be shared across indistinguishable worlds. A drawing that places separate decisions under each hidden type can accidentally imply forbidden knowledge.

A belief-state or symbolic implementation can encode the same problem without storing every path. It must still preserve the information partition. The tree is a useful explanatory projection of that structure, not its only possible storage form.

The central requirement is therefore not one point per possibility. It is preservation of the joint relation among persistent facts, observations and feasible decisions. This example gives exact numbers against which a proposed representation can be tested before it is applied to a much larger company.

Chapter 17

Continuous history families: projection, optimization, and threshold refinement

A finite list of histories is often an inadequate description even of a small company. Unknown rate exposure, customer allocation, asset age and settlement timing can vary continuously. The right question is frequently whether a conclusion holds over a constrained region. This chapter develops several exact small calculations and shows where they cease to be adequate. The methods are proposals for bounded profiles, not a claim that every company reconstruction is a linear program.

17.1 An affine exposure family

Let x be foreign-currency debt maturing soon and a the floating share of total debt. Suppose disclosed margins imply $0 \le x \le 50$ and $0 \le a \le 1$. A simplified next-period cash requirement under currency depreciation d and rate increase r is

$$Q(x, a) = Q_0 + dx + 100ra. \tag{17.1}$$

If d and r are nonnegative and the two unknowns are jointly unconstrained beyond their bounds, the maximum is $Q_0 + 50d + 100r$ and minimum is Q_0. These extrema occur at feasible corners. No grid is needed.

Now add a disclosure that the floating portion is entirely within the foreign-currency debt maturing soon, so $100a \le x$. The feasible family is no longer a rectangle. The minimum remains Q_0, but the maximum floating share is one half, giving maximum $Q_0 + 50d + 50r$. The relation changes the worst case even though debt total remains 100.

The condition is a stipulated example, not an inference from the margins alone. It illustrates why exact bounds must optimize over the joint constraints. Independently maximizing every marginal amount can overstate or understate the relevant result depending on signs and relationships. A constraint-aware bound carries the assumptions that make its corner feasible.

If the query includes a hedge whose receipt increases with x, the coefficient on x can change sign. Monotonicity should be derived from the full expression, not assumed from a variable's label. A larger foreign-currency exposure can have a smaller net cash effect when a linked hedge changes at the same time.

17.2 Linear programs as a bounded exact method

When observations, bridges and query are linear in continuous unknowns, a bound can be formulated as a linear program. Let z collect amounts and let $Az \le b$, $Ez = e$ define the family. The minimum cash query is

$$m = \inf\{c^{\mathsf{T}} z + c_0 : Az \le b, Ez = e\}. \tag{17.2}$$

The formulation separates a mathematical target from a solver. A solver can return an optimum, infeasibility, unboundedness or an unresolved numerical status. Each requires interpretation. Unboundedness may reveal a missing economic constraint rather than a truly unlimited company exposure.

A dual certificate can support a bound when its assumptions and numerical accuracy are checked. For a simple example, constraints $x \ge 0$, $y \ge 0$ and $x + y \le 10$ imply $2x + y \le 20$ because $2x + y \le 2(x + y) \le 20$. Equality is attained at $(10, 0)$, so the maximum is exactly 20. The inequality proof and feasible witness establish both sides.

A reported optimum without a feasible witness or trustworthy bound is weaker. Floating-point tolerances can matter near a covenant. Rational reconstruction or validated interval checks may be appropriate for a small certificate. The economic model should not change its threshold merely to accommodate solver tolerance.

Linear programming does not resolve categorical interpretations, nonlinear prices or integer security units automatically. Those additions can require mixed-integer, nonlinear or hybrid methods. The profile should identify which relations remain outside the exact solver and how their unresolved regions are retained.

17.3 Projection can preserve a useful small relation

Suppose receivables satisfy $A_1 = A_0 + S - K - W$, with nonnegative write-offs W, collections K and known credit sales S. If only A_0, A_1, S matter to a future query, eliminating K and W leaves $K + W = A_0 + S - A_1$. That sum is determined while its allocation is not.

A credit-loss stress depends on W or surviving customer exposure; a pure cash-history query depends on K. Eliminating both into an unnamed residual can make neither query answerable. A better compressed record retains the identified sum and the feasible allocation interval.

If $A_0 = 30$, $S = 40$ and $A_1 = 20$, then $K + W = 50$. With a note limiting write-offs to at most 5, collections lie in $[45, 50]$. A cash-flow disclosure of collections 48 then identifies write-offs 2 within this simplified bridge. Each additional source narrows the joint relation.

If currency movements or acquisitions are later admitted, the relation changes. A cached inference $W = 2$ must be invalidated unless those terms are established zero. This is an example of model expansion changing an earlier identified quantity. The system should preserve the old inference with its premises rather than silently rewriting its meaning.

17.4 Intervals and dependence

Interval arithmetic can bound expressions quickly, but repeated variables can cause overestimation. If $x \in [0, 1]$, the expression $x - x$ is exactly zero. Treating its two occurrences independently gives interval $[-1, 1]$. The dependency loss is entirely computational.

For the financial expression $C = 95 - I$ with $I \in [5, 7]$, interval evaluation correctly gives $C \in [88, 90]$. But evaluating $C + I$ from separate intervals gives $[93, 97]$ instead of exact 95. A relational representation or symbolic substitution preserves the identity.

Overestimation can be acceptable for an outer bound if it is labeled and used with the correct direction. It can become useless near a threshold or create spurious defaults. Refinement can retain shared symbols, affine forms or polyhedral constraints rather than independently widening every intermediate amount.

Nonlinear functions add further issues. A naive interval for $x(1 - x)$ on $[0, 1]$ is $[0, 1]$, while the exact range is $[0, 1/4]$. Completing the square, $x(1 - x) = 1/4 - (x - 1/2)^2$, identifies the maximum. A small analytic reformulation can be more valuable than a large random sample.

17.5 Piecewise contract payoffs

A floating coupon with floor l and cap u can be written $N\delta \min(u, \max(l, r + s))$ under a stated schedule. Unknown reference rate r partitions into floor, interior and cap regions. The function is continuous but has slope changes.

Suppose principal is 100, year fraction one, spread zero, floor 3 per cent and cap 6 per cent. For $r \in [0.02, 0.08]$, interest ranges from 3 to 6. A linear approximation using the midpoint rate five per cent and an unconstrained slope predicts 2 to 8, missing both contractual limits.

If the query is only the interest range, monotonicity gives exact endpoints. If the query includes a refinancing choice triggered by interest above 5.5, a split at $r = 0.055$ is needed in the interior region. If the cap changes after a covenant event, contract state must retain that dependence.

A piecewise formula can therefore represent many paths compactly. The implementation needs complete and disjoint guards, correct boundary inclusion and a record of the clause version. A missing cap is an economic-model error; an incorrect inequality at the cap boundary is an implementation error. Their repairs differ.

17.6 Nonmonotone outcomes

Many financial outputs are not monotone in a primitive shock. A rate increase raises floating interest but can increase income on cash, reduce a fixed-rate debt’s market value or trigger a favorable option exercise. A customer demand increase can improve revenue while creating a working-capital funding need.

Consider cash after a profitable sale with collection delay. Producing q units costs $4q$ immediately, while revenue $6q$ is collected later. Current cash falls as q rises even though eventual profit rises. A solver maximizing profit may select a quantity that violates interim liquidity.

If financing capacity is B and opening cash C_0, feasible production satisfies $4q \le C_0 + B$ before collection. The eventual profit objective $2q$ does not remove this constraint. Aggregating the period so that revenue and cost net immediately changes the feasible set.

A query-driven method must therefore include intermediate constraints and not rely on a generic sign assumption. Monotonicity can be proved locally within a regime and fail when a guard changes. The method should partition or return a qualified bound rather than extrapolate the local result globally.

17.7 Integer quantities and minimum lots

Shares, bond lots and physical units may be discrete. If a company needs net cash 10 and shares sell for 3 with no fees, a continuous calculation requires 10/3 shares. If only whole shares can be issued, four shares raise 12. The excess cash and dilution are real consequences of the lot constraint.

A minimum issuance size can make a small funding need expensive or infeasible. A fixed fee can make splitting one issue into two change net proceeds. These effects break the proportional invariance that holds for a simple linear loan.

The model should specify whether quantities are continuous approximations or exact units. For large public share counts, a continuous approximation may be immaterial for some queries. For a closely held company or indivisible asset, it may dominate the decision. The approximation needs a bound tied to the question.

17.8 Branch-and-bound interpretation

A mixed discrete-continuous family can be explored through bounds on regions. If a region's best possible result cannot improve the incumbent for an optimization query, it can be pruned under the method's correctness conditions. If a region is infeasible, it can be removed with evidence.

Pruning for computational convenience is different. A low-probability region cannot be removed from a worst-case query merely because it is unlikely. A region far from current data may still be admissible if the observation model permits it. The pruning reason belongs in the result record.

For a safety query, an outer lower cash bound above the requirement can close a region as safe. A feasible witness below it establishes possible failure. Regions whose bounds straddle the threshold remain unresolved and receive refinement priority. This approach can concentrate work where it changes the answer.

The procedure may still face exponential growth. Resource exhaustion should preserve the incumbent witnesses, certified bounds and unresolved regions. Reporting only the best found point hides the remaining uncertainty and can reverse the meaning of a worst-case result.

17.9 Sensitivity of identified sets

A small source revision can change a feasible family discontinuously when constraints meet at a boundary. Suppose exact observations require $x = 1$ and an interpretation requires $x \leq 1$. The family is nonempty. Revising the observation to $x = 1.001$ makes it empty if no tolerance or other interpretation is admitted.

This does not mean the company changed abruptly. It may reveal that exact equality was too strong for a rounded report. Observation precision and numerical tolerance should be audited before interpreting a discontinuity economically.

Conversely, a small genuine change can cross a contractual threshold and have a large effect. Smoothing it away would hide a real mechanism. The analyst must distinguish measurement uncertainty from contractual discontinuity. Both can occur near the same number but require different treatment.

A useful sensitivity report varies source precision, interpretation choices and economic shocks

separately. It identifies whether a conclusion is driven by evidence, modeling assumptions or a threshold in the contract. This is more informative than a single generic tolerance parameter.

17.10 A bounded implementation profile

An initial exact profile can support rational linear constraints, finite interpretation choices and piecewise affine event rules. It can return feasible witnesses, exact or certified bounds, and guarded outcome formulas. Nonlinear valuation and unsupported clauses remain explicit extensions.

The profile should reject incompatible units, unbound variables and missing event premises before solving. It should preserve unknown amounts as variables with justified domains. Arbitrary large bounds inserted to make a solver finite should be labeled as computational assumptions and tested for sensitivity.

A later nonlinear profile can use interval methods, constrained optimization or sampling, but must retain the original query target and distinguish local from global results. A local minimum of cash is not a global worst case. A successful solver exit code is not a proof of completeness.

The practical advantage of a bounded profile is that its claims can be checked. It does not redefine the whole mission to exclude difficult contracts. Instead, it provides a reliable core and a precise account of which additional methods are needed for broader company simulation.

17.11 Sharing a rate family without sharing away a borrowing right

Return to the capped floating coupon, but now ask whether the company can pay it. The company starts with cash of 5.5, other assets of 94.5 and debt of 100. Its equity is initially zero. One admissible history has a fixed coupon of 5; another has the floating coupon

$$I(r) = 100 \min\{0.06, \max\{0.03, r\}\}, \qquad r \in [0.02, 0.08]. \tag{17.3}$$

At the earlier observed rate of five per cent, both histories report interest of 5. A later rate of six per cent separates them: fixed interest stays at 5 while floating interest reaches 6. A journal history that retains only the earlier interest amount cannot answer the new payment question.

Each history also has two possible contractual interpretations. A committed facility may be available, or no facility may be available. In this stipulated example the facility lends 1, charges an immediate fee of 0.25 and can be drawn only when interest exceeds 5.5. These terms define a small test case; they are not inferred from the earlier statements. The four combinations of coupon type and facility availability remain separate possibilities. No probability is attached to them.

There is one decision date. Interest accrues before the decision, and the manager submits either a payment order or a combined draw-and-pay order. These are indivisible instructions. An instruction that fails a prerequisite is refused. Contingent instructions, such as *draw only if the bank permits it*, would enlarge the action set and require a fresh analysis.

17.11.1 The amount and the right determine different conditions

A payment order requires enough existing cash. A draw-and-pay order requires the contractual right, the strict trigger and enough cash after the fee. Write $b = 1$ when the facility is committed

and $b = 0$ otherwise. Feasibility is therefore

$$\text{pay:} \quad 5.5 - I \geq 0, \tag{17.4}$$

$$\text{draw and pay:} \quad b = 1, \quad I > 5.5, \quad 5.5 + 1 - 0.25 - I \geq 0. \tag{17.5}$$

Because interest never exceeds 6, an eligible draw leaves at least 0.25 cash. At interest exactly 5.5, ordinary payment remains possible and the draw is forbidden. Replacing the strict trigger with a weak inequality would change the contract at that point. At interest 6, the two facility interpretations have identical interest amounts but different feasible actions: the committed company can draw, while the company without the facility can execute neither order.

Accrual is independent of that execution result. The company first debits interest expense and credits interest payable by I. If payment refuses, cash remains 5.5 and interest payable remains I; equity has fallen to $-I$. A failed payment does not erase the expense or the creditor's claim. If payment succeeds, it removes the payable and cash becomes $5.5 - I$. If the draw succeeds, borrowing adds 1 of cash and debt, the fee reduces cash and equity by 0.25, and payment removes the payable. The resulting balance sheet satisfies

$$\underbrace{(6.25 - I) + 94.5}_{\text{assets}} = \underbrace{101}_{\text{debt}} + \underbrace{(-I - 0.25)}_{\text{equity}}. \tag{17.6}$$

These journal rules are supplied premises for the example. They demonstrate how refusal and contractual memory enter the comparison; they do not determine a reporting treatment for an actual instrument.

17.11.2 A continuous comparison can be exhaustive

For floating debt, the coupon changes formula at rates 0.03 and 0.06, and the payment trigger changes at 0.055. Suppose also that a coarse signal tells the manager whether the rate is at most 0.04. The six points

$$0.02, \quad 0.03, \quad 0.04, \quad 0.055, \quad 0.06, \quad 0.08 \tag{17.7}$$

then delimit every relevant change. Keep each point as a singleton and each interval between successive points as an open interval. The eleven resulting regions are disjoint and cover the entire closed rate domain. The fixed-coupon alternative can use fewer regions because it has no rate-dependent coupon or payment trigger.

Within each open region, every financial amount is affine and every signal and eligibility condition is constant. Two implementations therefore agree throughout that region when their affine coefficients and discrete results agree. At singleton regions, compare their values directly. If one implementation uses a coarser partition, compare both on a common refinement and establish that each refined region lies entirely inside one region of each implementation. Sampling a midpoint can identify a known affine piece, but it cannot establish that the list of pieces is complete.

The checked reference makes this comparison for the full journal, cash flows, balance sheet, payable, facility right, observations and action outcomes. Its independent comparator explicitly enumerates all coupon types, facility interpretations and the eleven rate regions. The smaller representation retains 32 mode-specific regions and shares 14 identical calculations. Every original mode and region still points to its calculation. These counts describe this example: sharing reduces duplicated results here, but supplies no general bound on memory use or runtime.

Even this small construction exposed a consequential implementation mistake. Solving the draw-liquidity equation using the interior formula $I = 100r$ gives a root at $r = 0.0625$. That formula applies only below the cap at 0.06. The root is irrelevant to the capped region and must not be inserted as though the interior formula still governed there. A root finder must check the domain of the formula that generated its candidate boundary.

Open regions also require careful treatment of extrema. On $0.055 < r < 0.06$, cash after drawing and paying is $6.25 - 100r$, so

$$\inf C = 0.25, \qquad \sup C = 0.75. \tag{17.8}$$

Neither endpoint is attained within that open region. The point $r = 0.06$ attains cash 0.25, while $r = 0.055$ belongs to the ordinary-payment region. Reporting an interval without its endpoint status loses a distinction needed for strict contractual conditions.

17.11.3 Information can prevent an otherwise affordable action

For the moment retain only histories with a committed facility. Every detailed history admits one of the two orders. This does not imply that the manager can select a feasible order using the information available at the decision date.

Under the coarse signal, *high* means only that $r > 0.04$. It includes floating rates with interest below 5.5, where payment is possible but drawing is forbidden, and rates above 0.055, where drawing is possible but ordinary payment fails. The manager must issue the same order at indistinguishable histories. Neither order works throughout this signal class.

Now suppose a second, explicit pre-decision signal reveals whether interest exceeds 5.5. The manager can pay when the trigger is false and draw and pay when it is true. This rule is feasible throughout the committed-facility family. Its feasibility comes from the additional information, not from a better numerical approximation. If unavailable facilities are restored to the family, some high-rate floating histories admit neither order. Even complete information cannot create borrowing capacity.

The exact condition is elementary in this finite action setting. Let $H(o)$ contain all retained histories with observation o, and let $A(h)$ be the actions feasible at history h. A deterministic policy feasible throughout the family exists exactly when

$$\bigcap_{h \in H(o)} A(h) \neq \varnothing \quad \text{for every observed class } H(o). \tag{17.9}$$

Necessity follows because one observation selects one action, which must work at every compatible history. For sufficiency, choose an action from each nonempty intersection. The finite collection of signals then defines a total policy. This argument depends on the supplied observation map, two atomic orders and robust feasibility across the family. It is not a conclusion about contingent instructions or a probability-weighted objective.

17.11.4 What remains safe when calculation is unfinished

A computation budget can stop evaluation of individual regions without removing them from the family. In the reference, the partition is constructed first and transitions are evaluated when requested. If only one region has been evaluated, the others remain explicitly unresolved. The coupon still has the valid outer bound $3 \leq I \leq 6$, but that bound alone does not identify a feasible payment policy. Eager construction of the partition itself can become expensive when more continuous variables or contractual alternatives are added.

Approximation needs a separate justification. Suppose an independently established bound gives $|I - \widehat{I}| \leq \varepsilon$ throughout a region. A strict trigger $I > K$ is certainly true when $\widehat{I} - \varepsilon > K$, and certainly false when $\widehat{I} + \varepsilon \leq K$. In the remaining band the bound permits both answers, so the region must be refined or left unresolved. The conclusion follows by comparing the entire allowed interval with K; sampled errors do not supply its uniform premise.

Observation thresholds need their own error bounds. Preserving the interest trigger while changing which rate signal the manager sees can still change the feasible-policy set. A reused calculation must therefore retain its rate domain, contract interpretation, clause version, horizon, query, observation map, action class and precision. Changing any of these requires reconsidering the grouping. The example provides an exact one-rate, one-date reference for that discipline. Persistent hidden terms across several dates, higher-dimensional regions, nonlinear guards and defensible approximation bounds remain further method problems.[1]

17.12 Repayment changes tomorrow's feasible payments

The previous payment calculation ends as soon as interest has been settled. A facility used to make that payment may have to be repaid before the next coupon. To see why the horizon matters, carry the same company forward for one more date. The fixed or floating coupon interpretation persists, as does the existence or absence of the facility. The two reference rates may vary independently within the earlier interval. This rectangular rate family is an assumption for the example, not a probability model.

The facility permits one draw of 1 during the two-date horizon and charges the same immediate fee of 0.25. A first-date draw must be repaid on the second date, after an operating cash receipt and before the second coupon. Repayment extinguishes the facility debt but does not restore the right to draw. The original principal of 100 matures beyond the horizon. We omit tax and interest on the facility to isolate the timing of cash and the persistence of the drawing restriction.

Let I_1 and I_2 denote the two coupons and R the operating cash receipt on the second date. In the fixed interpretation both coupons are 5. In the floating interpretation each is calculated from its own reference rate using the earlier floor and cap. Receiving one coupon of 5 does not by itself reveal the debt type: floating debt can produce the same amount. Nor does it reveal whether the facility exists.

17.12.1 The first order leaves a different second opening

Ordinary payment on the first date leaves cash $5.5 - I_1$. Drawing and paying leaves $6.25 - I_1$ and facility debt of 1. On the second date the operating receipt increases cash and income by R, after which any outstanding facility debt is repaid. Cash available before paying the second coupon is therefore

$$C_2 = \begin{cases} 5.5 + R - I_1, & \text{after ordinary first-date payment,} \\ 5.25 + R - I_1, & \text{after a first-date draw.} \end{cases} \tag{17.10}$$

The second line follows by adding R to $6.25 - I_1$ and subtracting the repayment of 1. Although the facility improved first-date liquidity, its fee leaves less cash available over the complete two-date sequence.

[1]The derivation and independently authored comparison are retained in `data/accounting-master/continuous-guards-20261026/`. The accepted result is in `attempt-02/`; the earlier failed partition and its repair remain available separately.

Compare two supplied receipt amounts, $R = 6$ and $R = 7$. When both coupons are 6, the first payment requires a draw and leaves cash 0.25. The consequences at the second date are:

Operating receipt	Cash after repayment	Coupon due	Payment result
6	5.25	6	Refused; payable 6 remains
7	6.25	6	Paid; cash 0.25 remains

The receipt of 6 cannot fund the second coupon. Omitting repayment would incorrectly show 6.25 cash. Allowing another draw would also reverse the answer: the extra 1 less its 0.25 fee exactly fills the 0.75 shortfall. Both mistakes erase a consequence of the first decision.

Before either payment order, the second coupon accrues as interest expense and interest payable. A refused order leaves that payable and the earlier receipt, repayment and fee entries intact. If a first-date order already refuses, this particular reference returns that incomplete prefix without inventing second-date states. Modeling later default, recovery or renegotiation requires additional events.

17.12.2 The relevant boundary now joins both rates

Following a first-date draw, ordinary second payment requires $C_2 - I_2 \geq 0$. Substituting the cash expression gives

$$I_1 + I_2 \leq 5.25 + R. \tag{17.11}$$

For a receipt of 6, the boundary is $I_1 + I_2 = 11.25$. Within the interior floating-coupon pieces it becomes a diagonal boundary in the two rates, $100r_1 + 100r_2 = 11.25$. Separate bounds on the two marginal rates do not describe which joint histories cross it. Equality permits payment; larger coupon totals cause refusal.

For a receipt of 7, the corresponding bound is 12.25. Each coupon is at most 6, so their sum is at most 12 and ordinary second payment is feasible throughout the first-draw region. After ordinary first payment, $I_1 \leq 5.5$; even the receipt of 6 then supplies at least 6 before the next coupon. The failure with the smaller receipt is thus concentrated in histories that needed an earlier draw and then face a sufficiently large combined coupon.

The exact comparison first takes the products of the earlier point and open rate intervals. It then splits those regions at the later linear cash conditions. Every resulting balance and journal amount is affine in (r_1, r_2), while each action condition has a fixed truth value. The independent financial calculation uses the closed-form cash expressions above. It checks that accepted regions cover exactly the feasible domains, that refusals cover the remainder, and that both calculations produce the same complete journal and financial state.

The geometric calculation also retains strict boundaries. On a bounded region described by rational linear inequalities, its closed relaxation is generated by finitely many vertices. The average of all those vertices satisfies a required strict inequality whenever at least one vertex satisfies it strictly: every other vertex has nonpositive slack, so their average remains strictly negative. If all vertices lie on its boundary, no point of the closed region can satisfy that inequality strictly. Applying this reasoning to every strict constraint decides whether the original region contains a point. Extremum attainment is checked by adding the optimizing equality and testing the resulting face in the same way. These arguments apply to the supplied bounded linear regions; a nonlinear or unbounded problem needs another method. The financial comparator shares these geometric primitives, so its agreement does not constitute an independent solver validation.

17.12.3 More cash does not reveal a hidden contract

With a committed facility and a receipt of 7, the refined first-date signal supports the same initial rule as before: pay when the trigger is false and draw when it is true. On the second date the manager observes whether the coupon exceeds 5.5 and whether available cash covers it, together with the earlier signal and chosen order. Ordinary payment is feasible throughout this family.

Some histories also permit a second-date draw: the first coupon was paid without drawing, the facility remains unused and the second coupon exceeds 5.5. There are two such reachable observation classes in this example, and either ordinary payment or drawing is feasible in each. Consequently four policies satisfy the stipulated feasibility conditions. This count does not rank their value. Optional borrowing incurs the fee and leaves debt beyond the current horizon.

Under the coarse first-date signal, the manager still cannot distinguish histories requiring opposite first orders. Raising the later receipt does not repair that information problem. Retaining histories without a facility also leaves some first payments unfunded. A correct continuation therefore preserves contractual memory, cash timing and the information available to the actor together. Renewing the facility, redrawing the debt type or revealing an unobserved trigger would change the problem being solved.[2]

17.13 Connecting the history to accounting execution

A financial path can be correct in a small mathematical model and still be translated incorrectly into a company's accounting system. The preceding example treats equity as a single amount after income and expenses. An accounting book usually distinguishes capital, operating income, interest expense and fees. Those distinctions matter when the simulator must produce an income statement, cash-flow statement and retained claim history as well as a closing balance sheet.

The checked connection uses the existing common accounting executor with explicit definitions of the selected events. It does not put the earlier Python posting calculation inside that executor. The comparison instead asks whether independently specified methods produce the same full accounts, journals, cash-flow origins and contractual memory. Two different account charts test whether a method depends on accounting roles rather than convenient account names.

17.13.1 Equity agrees only after the accounts are mapped

Let B_k be an account's signed balance, positive for a debit and negative for a credit. Assets and expenses then normally have positive balances, while liabilities, capital and income normally have negative balances. In this example the reduced equity E used in the earlier calculation corresponds to

$$E = -B_{\text{capital}} - B_{\text{income}} - B_{\text{interest expense}} - B_{\text{fee expense}}. \tag{17.12}$$

For example, an operating receipt of 7 gives the income account a credit balance of -7. Two coupons of 6 give interest expense a debit balance of 12, and a facility fee adds expense of 0.25. With no opening capital, the reported equity after profit is $7 - 12 - 0.25 = -5.25$. This equals the reduced model's equity without hiding the income statement inside the capital account.

[2]Complete symbolic states, accepted-domain checks, geometric diagnostics and the retained journal-identity repairs are in `data/accounting-master/two-date-continuous-memory-20261026/`. The accepted run is `attempt-04/`. This is an exact small reference; demand-driven two-date compression, checkpoint/resume guarantees, source revision invalidation and integration with the common typed executor remain separate work.

An interest accrual I adds I to expense and $-I$ to interest payable. Under the mapping, equity falls by I and the creditor's claim rises by I. Paying the coupon reduces cash and removes the payable, leaving profit unchanged. A combined draw-and-pay instruction has the following signed account changes:

$$\Delta B_{\text{cash}} = 0.75 - I, \qquad \Delta B_{\text{facility debt}} = -1, \tag{17.13}$$

$$\Delta B_{\text{fee expense}} = 0.25, \qquad \Delta B_{\text{interest payable}} = I. \tag{17.14}$$

Their sum is zero. The cash-flow record separately retains borrowing of 1, the fee payment of 0.25 and the interest payment of I. Netting those amounts in the journal's cash entry must not erase their origins in the cash-flow statement.

Repayment removes the facility liability and reduces cash by the same amount. Its accounting balance becomes zero, but the separate memory that the facility has already been used remains true. Each method also checks that interest payable agrees with accrued coupons less paid coupons, that the event occurs at the declared phase and date, and that its prerequisites are satisfied. A refused payment leaves the accepted prefix unchanged, including the coupon already accrued.

For the concrete execution comparison, the two decisions are assigned to 30 June and 31 December 2027 within one supplied reporting period. The coupon amounts remain the one-period charges of the earlier example. These dates test ordering; they do not supply an actual instrument's day-count convention. Each comparison checks the complete account balances, income statement, balance sheet, gross cash flows, journals and memory. Applying the equity mapping also recovers the earlier independent financial calculation.

17.13.2 A checked path is not a checked continuous region

The common executor accepts concrete rational amounts. The selected comparison therefore uses exact points at the coupon cap, at the first trigger, and at and immediately to either side of the later joint cash boundary. Those executions exercise important failure mechanisms. They do not by themselves establish that every rate in a continuous region produces the correct accounting result.

Within a region where the guards are fixed, the amount relations above are affine. If the opening accounts map correctly, each event has the stated affine effect, and the same guard, timing and memory conditions govern both descriptions, induction over the accepted events gives the same mapped financial quantities. The conditions carry the argument: a wrong guard or lost drawing right can change which event occurs even when its monetary formula is correct. A general checker must verify those conditions and every supported method, rather than replace that obligation with a larger sample of rates.

The concrete evidence at this stage has two parts. The exact continuous reference establishes its financial formulas and feasible domains. The concrete common-engine comparison checks complete accounting execution at declared points. The explicit mapping and local event derivation motivate the selected symbolic certificates in Section 17.14; extending them to general methods remains a separate problem. Source-backed terms and actual accounting applicability also remain distinct from agreement between two synthetic representations.

17.13.3 Stopping computation must preserve what remains unknown

Suppose the simulator has executed the first coupon and then reaches its computation budget. A useful saved result contains the complete accepted book and journal, the next command, and

the commands still pending. It must also retain the alternative histories and regions that have not been evaluated. A successful payment on one path supplies no reason to delete a different path that may fail tomorrow.

The selected checkpoint reference validates a saved concrete prefix by replaying it from its declared opening and method registry. It compares the complete transition records and current state before continuing. The tested interruptions recover the same result as uninterrupted execution, including the first-date and second-date refusal prefixes. Altering cash while keeping the book balanced, changing paid-coupon memory or removing the used-facility flag fails that comparison. Merely recomputing a checksum on the altered record would not establish that the altered history followed from its opening.

A zero budget attempts no new financial event. It can still incur the cost of checking the saved prefix. More generally, the budget in this reference counts new event attempts, while replay validation costs additional work. A practical implementation would need a separately justified way to reduce that cost. The present result establishes a small exact recovery comparison, not an efficient or crash-durable storage system.

The queue also distinguishes a completed concrete program from a resolved continuous region. For the two-date example, the declared roster contains four persistent contract combinations, 11×11 rate-atom pairs, four order pairs and two receipt hypotheses:

$$4 \times 121 \times 4 \times 2 = 3{,}872 \quad \text{region and instruction obligations.} \tag{17.15}$$

A finite witness cannot mark any of those whole regions complete. Their identities remain unresolved under this concrete checkpoint even when individual concrete jobs succeed or refuse; Section 17.14 supplies separate certificates for the selected whole regions. Within the finite job queue, pending, completed and refused jobs form an exact partition. A refused job retains both its payable and its unexecuted commands; it is a completed assessment of that particular instruction sequence, not proof that every alternative has failed.

Finally, reuse depends on what was asked and what information was available. The saved record names the source revision, clause, actor's observation map, available actions, horizon, query, precision and method. A changed scope refuses reuse under the old record. An earlier query can still use its earlier explicitly supplied scope, preserving the distinction between later knowledge and information available at the time. Those bindings do not authenticate the source or establish when a real manager received it. Source admission, extension beyond the selected symbolic domains, efficient refinement and durable recovery require their own evidence.[3]

17.14 Proving a result for an entire rate region

The second coupon gives a useful test of what a simulation has actually established. Suppose the company drew its once-only facility at the first date. After the operating receipt and repayment, cash available for the second coupon is $5.25 + R - I_1$. A payment succeeds exactly when

$$I_1 + I_2 \le 5.25 + R. \tag{17.16}$$

At $I_1 = I_2 = 6$, a receipt of 6 leaves only 5.25 before the payment, so the coupon remains unpaid. A receipt of 7 leaves 6.25 and permits payment, with 0.25 left over. A simulator that happens to test only the latter case has learned nothing about the former. Even a large set of

[3]The complete common-engine and checkpoint comparisons are retained in `data/accounting-master/guarded-memory-runtime-20261026/`, respectively under `attempt-01/` and `checkpoint-attempt-01/`. The former checks concrete accounting execution; the latter checks replay and queue preservation. Neither promotes a concrete witness to a whole-region certificate.

successful sample executions does not exclude an omitted boundary or an incorrect condition between the samples.

The selected example now has a stronger, exact connection to the declared accounting methods. It proceeds in three stages: establish each method's equations, substitute those equations through the dated sequence, and divide the rate domain wherever a condition changes. This division concerns computational work. It does not assert that a company, a manager or an observer learns which historical explanation is true.

17.14.1 Compare the method with an equation, not another sample

An affine monetary expression is a constant plus a weighted sum of quantities. For this example, those quantities include account balances, accrued and paid coupons, event amounts and the supplied parameters I_1, I_2, R. Write an expression as

$$f(x) = c + \sum_j a_j x_j, \tag{17.17}$$

where each coefficient and the constant are exact rationals. The accounting language expresses these calculations through constants, references, addition and subtraction. A constant contributes to c; a reference contributes one coefficient; addition and subtraction combine the coefficient lists. Induction on these four constructions shows that the resulting list represents precisely the original expression. Two identical lists therefore give the same amount for every known typed valuation, rather than merely for the values tested in a run.

For instance, the combined borrowing and payment changes cash by $3/4 - I_t$. The corresponding coefficient of the due coupon is -1 and the constant is $3/4$. Doubling both interest expense and the credit to interest payable would still balance the journal. It would change these independently specified account relations and fail the comparison. Balance is a necessary accounting constraint, but it cannot establish that the right transaction was recorded.

The comparison also preserves the conditions on the transaction. A numerical comparison is represented by its exact left-minus-right affine expression and its comparison sign. The facility trigger $I_t > 11/2$ retains a strict sign; replacing it by $I_t \geq 11/2$ fails even if a chosen set of sample rates misses equality. Boolean conditions retain their operands, and memory assignments retain their source and destination. The same check covers cash-flow origins: a borrowing receipt, fee and interest payment cannot be replaced by one net cash movement.

Parameters enter through a separate selected template. Its coupon and receipt amounts, contract mode and facility right become ordinary typed literals before execution by the common runtime. Exact placement checks bind the parameters to the opening state, event inputs and selected-amount conditions. The surrounding method records retain their account maps, units, dates, schemas, declared dependencies and refusal policies. This is a certificate for the inspected selected profile. It does not extend the runtime to arbitrary symbolic programs or prove an unrestricted compiler correct.

17.14.2 A correct event still needs a correct predecessor

Local agreement leaves a temporal question. When a method reads the payable or the facility balance, does it receive the state produced by the preceding accepted event? Substitution answers that question for the selected sequence. Every posting and memory assignment reads the same pre-event state; the resulting changes become visible together. The next event receives the complete new state, including accrued and paid coupon history.

After a first-date draw and payment, cash is $25/4 - I_1$. The receipt adds R, and repayment subtracts the borrowed unit. Substituting successively gives

$$C_{1+} = \frac{25}{4} - I_1, \tag{17.18}$$

$$C_{\text{receipt}} = \frac{25}{4} - I_1 + R, \tag{17.19}$$

$$C_{2-} = \frac{21}{4} - I_1 + R. \tag{17.20}$$

The second accrual changes expense and the payable, leaving this cash unchanged. Substituting C_{2-} into the payment condition $I_2 \le C_{2-}$ yields equation (17.16). Repayment removes the facility liability while leaving the used-facility flag true. Omitting that memory would admit a second draw which the contract in this example forbids.

The checked substitution compares all signed accounts, memory fields, gross flows and conditions with independent complete formulas. It includes both refusal positions. A first refusal leaves the first coupon accrued; a second refusal retains the accepted first-date events, receipt, repayment and second accrual. It does not replace the final payable by zero merely because the proposed payment appeared in the command sequence.

At this stage a branch can be written down even when no rate makes it feasible. For example, the formal second-draw branch can be constructed after a first draw, but its unused-facility condition is false. Distinguishing an algebraically described branch from a possible history is the task of the next stage.

17.14.3 Why an empty branch needs a proof

Within each rate atom the capped coupon is constant or linear in the corresponding rate. A point atom uses its exact constant coupon. Substituting those pieces into the conditions gives inequalities in two variables, r_1 and r_2. Each comparison divides the current region into the points satisfying it and the points satisfying its exact negation. For $f \le 0$, the other child is $-f < 0$; for $f < 0$, it is $-f \le 0$. Equality belongs to one child, never to both and never to neither.

A discarded child carries a short arithmetic reason why its inequalities cannot hold together. Suppose the child requires

$$a_i^{\mathsf{T}} r + c_i \le 0, \qquad i = 1, \dots, m, \tag{17.21}$$

with some signs possibly strict. Choose rational weights $\lambda_i \ge 0$ that cancel both variable coefficients:

$$\sum_i \lambda_i a_i = 0. \tag{17.22}$$

Adding the weighted inequalities then requires $\sum_i \lambda_i c_i \le 0$. If that constant is positive, the requirement is impossible. A zero constant also proves impossibility when at least one strict inequality receives positive weight: the sum would require $0 < 0$. These implications follow directly by multiplying and adding the original inequalities. The verifier checks the weights against those original rows rather than trusting a solver's declaration that a branch is empty.

The distinction matters even in one dimension. The pair $x < 0$ and $-x \le 0$ has no solution: equal unit weights produce $0 < 0$. Weakening the first inequality to $x \le 0$ admits $x = 0$. A certificate that discards the strictness would incorrectly remove that boundary point.

The reference producer eliminates the two variables with exact rational arithmetic and retains the weighted combinations it uses. A separate checker verifies each proposed contradiction. A retained leaf instead carries a point satisfying every inequality, including every strict sign. That

point proves the leaf is nonempty. The affine method and sequence argument, together with the predicates along the branch, establishes the financial relation throughout the leaf; the witness alone does not.

17.14.4 What can now leave the unfinished-work list

For the declared two-date model, all 3,872 rate-region and instruction obligations have a checked partition. The partition contains 918 nonempty accepted leaves and 2,960 nonempty refusal leaves; 4,772 additional children are discarded with checked contradiction certificates. These are computational counts, not probabilities. They depend on the selected atom refinement and cannot be interpreted as the likelihood of repayment or default.

The saved proof archive is re-read before its jobs enter a new certified queue. Each entry binds the source revision, clause, method and sequence certificates, domain, query and horizon. A pending job leaves the queue only with the corresponding verified proof. Zero-budget and interrupted JSON continuations retain the same final partition as uninterrupted processing. Checking the saved proof registry costs work in addition to the budget for newly completed jobs.

The earlier concrete checkpoint correctly kept these regions unresolved under its own evidence. The new result supplies separate whole-region certificates; it does not reinterpret the earlier witness as a proof. The distinction is useful beyond this example: completing a computation should mean satisfying the evidence obligation attached to that job, not merely reaching its last instruction.

This result concerns known exact values, the supplied coupon and once-only facility terms, two bounded rates and the selected event sequence. It leaves a further decision problem. An actor can choose differently only when the actor's information distinguishes the relevant histories. Consequently, a list of financially feasible region-specific actions is not yet an admissible policy. Section 17.15 binds these leaves to the supplied observation map and checks decisions that must agree across indistinguishable histories. Section 17.16 supplies a conditional source-revision and actor-access binding for the selected public-document profile; actual company evidence and broader information structures remain separate requirements. Higher-dimensional and nonlinear relations, unknown-value semantics, actual accounting authority, numerical quantisation and crash-durable recovery require additional evidence.[4]

17.15 A feasible action for each world is not yet a policy

Return to the first coupon, when cash is 5.5 and the once-only facility is available. Two companies can produce the same coarse observation that the market rate is high. In one, a fixed coupon is 5; in the other, the floating coupon at the cap is 6. Paying is feasible for the first company, while drawing is barred by the strict trigger. The second company cannot pay from existing cash and needs the facility. If the manager sees only the coarse high signal, the two individually feasible instructions do not form a decision rule the manager can follow. The rule asks for information that the manager does not have.

This is the difference between a financially possible action and an information-feasible policy.

[4]The local derivations and immutable comparisons are retained under `data/accounting-master/symbolic-method-certificate-20261026/`, `symbolic-path-composition-20261026/` and `exact-region-certificates-20261026/`. The last directory includes failed diagnostics, the exact partition archive, an independent weighted-sum checker and separate saved-archive/queue validation. No new literature proposition is asserted by these local calculations.

Let w denote a possible history and $O_t(w)$ the manager's observation history when the date-t instruction is chosen. An admissible policy must satisfy

$$O_t(w) = O_t(w') \quad \Longrightarrow \quad \pi_t\big(O_t(w)\big) = \pi_t\big(O_t(w')\big). \tag{17.23}$$

Here O_t includes remembered earlier observations and the manager's own earlier instructions. The condition gives one decision to each information cell. It does not require that the underlying histories become identical or that the manager eventually learns the contract type.

17.15.1 Refining an observation without inventing knowledge

The selected first coarse signal reports whether $r_1 \le 0.04$. The refined signal adds whether the first coupon exceeds 5.5. In the pair above, refinement distinguishes the two required first instructions. It still does not reveal the facility right or generally identify a persistent coupon type. At the second date the manager remembers the first signal and instruction, then observes whether the second coupon crosses the trigger and whether the cash then available covers it. The receipt hypothesis is held fixed for each comparison.

A financial region need not have a single observation label. For example, the condition that drawing plus cash covers a coupon is weaker than the condition that existing cash alone covers it. A region on which a draw is financially possible can therefore straddle the manager's ordinary cash-coverage observation. The checked construction splits that region on the cash-coverage inequality and its exact complement. Empty children carry the contradiction proofs of the preceding section; nonempty children retain their symbolic financial outcomes. Whole-atom proofs likewise establish that the first low/high signal and each coupon-trigger signal are constant where they are used. The observation is thus valid throughout the cell, rather than assigned from one representative rate.

This timing is consequential. The first decision does not receive the second rate or a later cash balance. The second decision uses cash before its proposed payment. Using terminal cash would let the result of an action determine what the manager supposedly observed before choosing it. Including the hidden facility right in the signal would change the information problem, even if every resulting journal still balanced.

17.15.2 Which instructions survive every indistinguishable history?

For each possible first decision rule, retain every history consistent with that rule. A first-date refusal rejects the rule. At each reachable second-date information cell, intersect the allowed instructions across the retained histories. An empty intersection means there is no action that works in every history the manager still considers possible. Otherwise choose one instruction from each cell's intersection. Cells that cannot be reached under the first rule need no arbitrary decision in this comparison.

An independent formulation expresses the same condition as Boolean restrictions. Let X_o say whether the first instruction at signal o is draw, and Y_h the second instruction at remembered history h. A certified first-date refusal under instruction a forbids $X_o = a$. A certified second-date refusal under first instruction a and second instruction b forbids their joint selection:

$$(X_o \neq a) \ \vee \ (Y_h \neq b). \tag{17.24}$$

Enumerating the satisfying assignments and retaining only reachable second-date choices gives the complete policy set. This independently authored constraint calculation agrees with the action-intersection calculation and with the earlier finite-policy reference. Agreement concerns whole policies, not only the number found.

Across the eight supplied problems—receipts 6 or 7, coarse or refined observations, and committed or mixed facility rights—only the receipt-7, refined-observation, committed-right problem has robustly feasible policies. It has four. The first instruction is fixed by the refined trigger. After a first draw, the exhausted facility cannot be used again and the second instruction must be pay. After a first pay, two reachable second-date cells allow either pay or draw, giving $2 \times 2 = 4$ reachable policies. The model supplies no preference or objective that ranks them.

The failures have different causes. Coarse information combines incompatible first instructions. A family containing an unavailable facility includes a high-coupon world with no feasible first instruction. Receipt 6 leaves the double-high path short after repayment, despite knowing the trigger and having the original right. Removing such a world or ignoring its refused branch changes the robust question; it does not solve it.

These results concern the explicit family of possible histories and the supplied observation maps. They assign no probabilities to that family, and a feasible robust policy is not a forecast or an optimal policy. Applying the calculation to an actual manager next requires evidence about source revisions, access times and observations. That evidence must preserve what was available when a decision was made rather than replacing it with a later corrected account.[5]

17.16 Which source could this actor use at this date?

The four feasible policies in the preceding example depend on two supplied interpretations: which receipt and facility-right family is possible, and which signals the decision maker observes. A revised document can change either interpretation. It can also leave both interpretations unchanged while correcting the evidence supporting them. The simulator needs to distinguish those cases before it decides which calculation to repeat.

Consider a correction released on 3 January to a document used for a decision on 2 January. The correction may improve a retrospective account of the company, but it cannot be inserted into what the actor knew on 2 January merely because the simulator downloads it later. Conversely, a document may have been available to the actor on 1 January but entered the simulator on 3 January and passed internal review on 4 January. On 4 January the system can reconstruct the earlier information set if the retained evidence supports that access. It has not established that the reconstruction itself was run on 2 January.

These are distinct dates with distinct subjects. Publication concerns the document's release. Access concerns a particular actor. Ingestion and review concern the information system. Effective time concerns the economic period or event to which the interpretation applies. A publication timestamp alone does not establish personal access; a download timestamp does not establish either the original publication or the document's economic effect.

17.16.1 Select revisions before selecting the policy problem

The selected construction uses an immutable record for each supplied interpretation. The record identifies the source bytes, the interpretation's subject, the previous revision it replaces, its effective time, and separate observations of publication, access, ingestion and review. The caller supplies the evidence accepted for these timing observations. Hashes preserve the identity of those bytes and observations; they do not establish their truth.

[5]The complete observation refinements, whole-atom proofs, two policy formulations, source locks and eight complete comparisons are retained in `data/accounting-master/certified-observation-policy-20261026/attempt-01/`. The reference checks all 3,872 saved financial jobs, with 1,766 complementary observation refinements. These are exact calculations for the selected synthetic information structure.

For a request, retain revisions that are effective for the chosen economic event, available to the actor by the decision time, and ingested and reviewed by the system's reconstruction cutoff. Known later revisions are excluded from that earlier view. A missing relevant time leaves the selection unresolved. Within a subject, select a revision only if it is the unique remaining head of its replacement chain. Two incomparable available corrections preserve a conflict. Retraction of a required interpretation leaves a missing input rather than silently restoring its earlier assertion.

This procedure is deliberately restricted to a declared source scope. An unrelated management note cannot change the coupon interpretation merely because it appears in the same registry. A relevant correction must be inspected again on each request, however. Comparing a cached view identifier without checking for newly available revisions could miss exactly the change that should invalidate the result.

In the reference example, one required subject supplies receipt 6 or 7 and the mixed or committed facility-right family; the other supplies the coarse or refined observation scheme. These choices map to the eight previously certified problems. They preserve the same possible persistent contract types, complete financial rows and rejected actions within the selected family. No convenient prior or likelihood is introduced to make a source selector callable. The question remains whether one policy works in every history the actor cannot distinguish.

The timing checks reuse the existing source and policy-information mechanisms. The selected adapter accepts explicit whole-second UTC instants and public-document evidence with a separate supplied actor-access observation. It refuses finer or incomplete clock representations instead of rounding them into eligibility. Section 17.17 extends this selected reference to exact clock ranges and an explicit private-delivery route; broader information practices remain separate. An unresolved access record is therefore a substantive result about this request, not an instruction to substitute the publication date.

17.16.2 Reuse a calculation without reusing an obsolete source binding

Suppose a replacement document supports exactly the same receipt, facility-right family and observation scheme. The set of robust policies can remain unchanged. Nevertheless, a report that still cites the replaced source would give the wrong explanation of which evidence supported the current request. Numerical equality and evidential identity answer different questions.

Let F denote the retained financial relations and certified rows, s the selected receipt and facility family, and o the observation scheme. Write the complete reachable policy calculation as

$$\mathcal{P}(F, s, o). \tag{17.25}$$

Its numerical inputs do not include a document's filename or the system's ingestion date. Those records determine whether the interpretation (s, o) is available for the actor and request. The reported result therefore needs both $\mathcal{P}(F, s, o)$ and the source, revision, actor and cutoff records that justify using these inputs.

The checked dependency relation makes this separation explicit. Certified financial evidence supports the chosen financial scenario; that scenario and the observation scheme support the policy set. Separate source bindings attach the selected financial and observation interpretations to their revision histories. The final actor binding joins the numerical policy set to both source bindings and to the decision, reconstruction and effective cutoffs.

A same-meaning correction changes a source binding and the final actor binding while preserving the numerical policy set. A change from coarse to refined information preserves the financial relations but requires a new observation and policy calculation. A changed receipt or facility

family changes the financial scenario and its dependent policy problem. Changing only the actor or cutoff requires a new actor binding even when the selected interpretations and numerical answer happen to be identical. A correction known to arrive after the old cutoff changes none of that earlier view's selected dependencies.

Reuse is justified by those dependencies, not by observing that the new and old policy counts coincide. The reference compares the entire reused policy set and its complete world-row population with a fresh calculation. It also requires the saved inputs and method bytes to match expected identities supplied outside the transported bundle. Removed or duplicated rows, changed hidden rights or signals, altered policy references and rewritten prior results are rejected. Recomputing an inner checksum after changing a saved result does not make it the previously accepted result.

The scope remains conditional. The caller must establish which source and timing evidence is accepted, and a complete history family for an actual company may require many more subjects than these two. The result shows how revision-sensitive selection can coexist with numerical reuse while preserving the robust question. It establishes neither authenticated storage nor crash recovery, and the small exact comparison supplies no performance ranking.[6]

17.17 A date range can leave several information views possible

A filing marked as published on 2 January does not establish whether a manager had read it at the beginning of that day. Nor does a correction received within a narrow interval around a decision establish whether the original or corrected interpretation was available for that decision. These are questions about the order of events. Assigning midnight or the interval's midpoint would settle them by assumption.

The selected successor retains that uncertainty explicitly. An exact timestamp is a point; a calendar day in UTC is a half-open interval from its midnight to the next midnight. A supplied time range has the same half-open convention. An unknown access time remains unknown. Exact rational seconds preserve differences smaller than a microsecond and equivalent fixed-offset spellings without passing through a floating-point or microsecond clock.

If the access time lies in $[a, b)$, a decision at b is certainly after access. A decision at a can coincide with access or precede it: the lower endpoint belongs to the interval. The distinction disappears if an implementation replaces the interval with a single representative time. It also disappears if a strict upper bound is inadvertently changed to a closed one.

17.17.1 Availability depends on the route and on joint chronology

A public-document retrieval and a private delivery impose different conditions. In the first route the source must have been released publicly before the claimed access. In the second, an actor-addressed delivery can establish access before any public release. The private route neither invents a public timestamp nor makes the document available to other actors. Both routes still require evidence about that actor's access; publication alone cannot supply it.

The earlier whole-second reference did not distinguish these routes. One of its retained cases places actor access before a later public release. Under the new, explicit public-retrieval premise, that chronology is inconsistent. It can be meaningful under a private-delivery premise.

[6]The immutable source-view cases are retained in `data/accounting-master/certified-source-views-20261026/`; the separate dependency and complete cold-versus-reused comparisons are in `data/accounting-master/source-policy-invalidation-20261027/`. Both preserve their supplied source scope and withhold actual-source, accounting, real-actor and forecast admission.

Preserving the difference is necessary: changing the access mechanism changes the question rather than improving timestamp precision alone.

Clocks are also dependent. Authorship precedes ingestion, ingestion precedes review, and authorship precedes access. Public retrieval additionally follows publication. The selected revision profile requires nondecreasing authorship, ingestion, review and same-actor access along a replacement chain, while permitting retrospective effective dates. These are declared premises of this bounded source model. A document practice that violates one of them requires another profile; a corrected economic effective date is not forced forward merely to preserve a convenient time ordering.

An unknown clock need not make every conclusion unknown. If a private access time is supplied before the decision and the document was necessarily authored before that access, the chronology can establish the relevant authorship inequality without identifying its timestamp. The result must retain the unknown-time marker. Logical implication from other premises does not turn an unobserved time into an observed fact.

17.17.2 Partition clock possibilities without enumerating timestamps

Let x_e be the time of event e, measured in exact UTC seconds on the retained calendar axis. Point and interval observations, chronology and cutoff conditions become difference inequalities. A non-strict edge represents

$$x_v - x_u \le c, \tag{17.26}$$

with a strict variant when the right endpoint is excluded. A fixed origin converts an absolute bound into the same form. The source identities, actors, access routes and original precision remain attached to these mathematical restrictions; the inequalities alone do not establish the source facts.

For example, publication before access contributes $x_{\text{publication}} \le x_{\text{access}}$, while access by the decision contributes $x_{\text{access}} \le t_{\text{decision}}$. Keeping both inequalities in one system prevents the calculation from accepting a combination that separate per-clock classifications would mistakenly permit.

For each revision, eligibility is a conjunction of its required actor, system and effective-time guards, $g_1, \ldots, g_m$. The complementary cases can be written as a disjoint first-failure partition:

$$\neg(g_1 \wedge \cdots \wedge g_m) = \neg g_1 \ \vee \ (g_1 \wedge \neg g_2) \ \vee \ \cdots \ \vee \ (g_1 \wedge \cdots \wedge g_{m-1} \wedge \neg g_m). \tag{17.27}$$

Each branch preserves the earlier satisfied guards. Repeating the construction for the relevant revisions covers every clock assignment admitted by the original ranges and chronology. Eligibility and the selected revision heads are constant within each surviving branch. The procedure therefore represents families of timestamps symbolically; it does not need an explicit tree point for every possible instant.

An empty branch requires a checked contradiction. Summing original difference inequalities around a cycle cancels all clock variables. A negative right-hand sum is impossible; a zero sum is also impossible if at least one inequality in the sum is strict. A nonempty branch instead carries exact rational clock values satisfying every original inequality. The checker verifies those values directly and reconstructs every complementary branch before accepting the partition.

The witness establishes nonemptiness, while the guards and complete partition establish the scope of the conclusion. A few feasible timestamp examples would establish neither exhaustive coverage nor the absence of an omitted information view. Conversely, reaching the computation budget leaves an unfinished temporal family. The selected reference then reports incompleteness and does not present the already examined views as the full set.

17.17.3 Different timing views can support different policies

Return to receipt 7 and the committed facility. Suppose a correction changes the available observation scheme from coarse to refined, but the actor's access interval straddles the first decision. There are two remaining timing views. In the view where the correction has not yet arrived, the coarse signal leaves incompatible first instructions in the same information cell and no robust policy is feasible. In the view where it has arrived, the refined signal supports the four previously derived policies. The complete financial row population is retained in both calculations.

The timing evidence supplies no probability for either view. Selecting the four-policy answer would discard the other possibility. Reporting the two answers separately is therefore a substantive conclusion. Constructing one decision rule that is admissible when even the observation mechanism is uncertain requires a common specification of what the actor sees and how actions are represented across those mechanisms; a naive intersection of differently labelled policy tables is insufficient.

The exact comparison verifies interval boundaries, equivalent offsets, unknown and untrusted clocks, public and private access, correlated correction times, conflicting revisions and retractions. The saved partitions retain unresolved source outcomes as well as selected ones. Their scope is the supplied chronology and access model, not an empirical statement about how a real company releases information. General revision practices, authenticated evidence and durable recovery remain separate questions; Section 17.18 develops complete-family reuse and common decisions across possible observation mechanisms.[7]

17.18 Preserving possible views does not yet choose a common policy

The correction interval leaves two questions. A calculation must first preserve every information view consistent with the supplied clocks. The decision maker must then choose instructions that can actually be followed when the relevant view may itself be unknown. Keeping both policy tables answers the first question. It does not, by itself, answer the second.

17.18.1 Share calculations while retaining their different supports

A complete temporal family carries the original clock restrictions, its checked partition, every nonempty branch and the source outcome on that branch. Several branches can select the same revision; several different revisions can also imply the same financial and observation problem. The numerical calculation can then be stored once, provided that all the distinct source and timing supports remain attached to it. Removing a repeated numerical answer must not remove a possible history.

The selected binding reference therefore compares complete policy sets and their full financial-row populations. A same-meaning correction can reuse the numerical result while changing the source binding. A changed financial assumption or observation scheme requires its corresponding numerical problem. Even two empty policy sets retain their different scenario meanings.

[7]The derivation, original-edge checker, complete saved temporal partitions, independent finite order comparisons and full policy sets are retained in `data/accounting-master/source-time-intervals-20261027/`. The reference preserves its failed namespace diagnostic, the explicit public-route difference from the whole-second profile, and the six-revision and computation-budget limits. No clock probabilities or actual actor-access admission are introduced.

Their common cardinality, zero, is not sufficient evidence that the underlying questions are interchangeable.

The request and saved family are checked against expected identities supplied outside the transported record. The request includes the actor, original sources, clock observations, caller's admitted evidence and computation budget. The family retains the full temporal proof and each unresolved outcome. These checks detect alteration and missing support relative to that supplied record. They do not authenticate a document or establish that the actor actually received it.

A budget stop needs special care. The earlier temporal procedure does not save a partial partition that can be resumed. The binding reference therefore preserves the entire original clock domain as unresolved and supplies no completed-view policy answer. This conservative remainder keeps all possibilities, at the cost of repeating work later. Describing it as an incremental frontier would claim a computational capability that has not been implemented.

17.18.2 Decisions must agree on the histories the actor can distinguish

Let z denote a possible observation mechanism and w a possible financial history. At decision date t, the actor receives a message $m_t(z, w)$ from that mechanism. The message may differ across histories, and the meaning of a given message may differ across mechanisms. Let $\eta_t(z)$ record a mechanism identifier only when the model explicitly makes it observable by that date. Without such an observation, η_t has the same null value for every mechanism.

For the two-date example, the information used by a deterministic policy is

$$h_1(z, w) = \big(m_1(z, w), \eta_1(z)\big), \tag{17.28}$$

$$h_2(z, w, a_1) = \big(h_1(z, w), a_1, m_2(z, w, a_1), \eta_2(z)\big). \tag{17.29}$$

The second history retains the first message and the actor's own instruction. It contains no future message, hidden financial state or internal proof identifier. The action at each date is a function of the corresponding history, $a_t = \pi_t(h_t)$. Consequently,

$$h_t(z, w) = h_t(z', w') \quad \Longrightarrow \quad \pi_t\big(h_t(z, w)\big) = \pi_t\big(h_t(z', w')\big). \tag{17.30}$$

This is the common-policy restriction: indistinguishable available histories require the same instruction. It applies across possible mechanisms as well as across financial histories within one mechanism. Feasibility must then hold for every retained world under the selected stress question. No probability distribution is needed for that requirement.

Internal source hashes can identify z for the analyst without identifying it for the actor. Putting such a hash into h_1 would add information and could change the answer. Conversely, if a real message identifies its source or interpretation, omitting that information would solve a different problem. The observation declaration must say which case is intended and why.

17.18.3 Four policies in each view can leave no common instruction

Consider again receipt 7, the committed facility and the refined financial problem. Its complete row population supports four policies. For a concrete test, call the first messages L, H and X. Mechanism A sends these messages for, respectively, low conditions without a trigger, high conditions without a trigger, and high conditions with a trigger. Mechanism B exchanges the meanings of H and X. These are deliberately supplied synthetic encodings; they are not an empirical claim about a company's communications.

Each known encoding preserves the original four policies by relabeling their observations. The difficulty appears when the actor sees H but does not know which encoding applies. Under A, the instruction must be pay; under B, it must be draw. A common policy cannot satisfy both requirements. The two individually feasible problems therefore have no jointly feasible policy when the mechanism is hidden.

If the encoding is identified before the first decision, the two message histories are distinguishable. One of four policies can be selected for each encoding, giving $4 \times 4 = 16$ feasible joint policies. If the encoding is identified only at the second date, the earlier contradiction remains. Later learning cannot repair a first instruction that had to satisfy incompatible requirements at the time it was chosen.

The conclusion depends on when the incompatibility arises. In a separate elementary liquidity example, the only first instruction is wait. At the second date, a company with cash 1, an amount due of 1 and no facility must pay. Another possible company has cash 0, the same amount due and a facility of 1, so it must draw. Two mechanisms exchange the meanings of the messages describing those circumstances. Hidden message meaning leaves no common second instruction; revealing the mechanism at the second date restores one feasible policy because waiting had created no earlier conflict. If the second message itself has an unambiguous meaning, learning can resolve the uncertainty without revealing a mechanism's name. This example concerns stipulated cash feasibility, not the accounting recognition of a facility or a complete company report.

A different case shows why late revelation can expand a policy set even when the mechanisms have the same financial meaning. Two possible correction heads can both select the same refined problem. All four of its policies share the same first instructions; only their later choices differ. Learning the source identity at the second date then permits four continuations for each source, giving 16 joint policies. When that identity is never observed, the continuations must agree and only four policies remain. A first attempt incorrectly expected four in both cases; the complete-set comparison exposed the omitted late distinction, and the corrected independent relabeling preserves the failed attempt as evidence.

17.18.4 What a complete conditional answer must retain

The original coarse/refined correction example remains different from the swapped-encoding example. The coarse member already has no feasible policy. Identifying that member does not supply the refined signal it lacks. The joint policy problem therefore stays infeasible even if the actor is told which mechanism applies. An unresolved source outcome for which no observation map is specified instead leaves the policy question unresolved; it is not another zero-policy result.

The checked finite method retains the full financial population for every mechanism and solves the shared-history constraints in two ways: enumeration of feasible instructions and an independent formulation of forbidden action combinations. It also compares the complete policy sets with explicit relabelings of the original policies. The checks distinguish a missing action row from an available safe action, and preserve the original temporal supports when equal numerical answers share storage.

These results explain how unresolved uncertainty can persist without requiring a separate explicit node for every time or transaction possibility. Symbolic temporal restrictions represent many possible times; shared numerical results avoid repeated calculation; common histories impose the information limits on decisions. Each operation preserves a different relation and needs its own justification. General observation languages, longer horizons, empirical actor access and durable recovery require further work. The saved partial-frontier extension for the temporal profile is developed in Section 17.20.[8]

17.19 A visible label need not identify the observation mechanism

In the preceding example, a first message H requires pay under one interpretation and draw under another. An additional label can resolve the conflict if it tells the actor which interpretation applies. Suppose, however, that both interpretations attach the same label, *reviewed*, to that message. The actor sees a label but still cannot distinguish the two cases. Declaring a label visible and declaring a mechanism identified are different assumptions.

The distinction changes the feasible policy set. With distinct labels supplied before the first decision, the two four-policy problems admit 16 joint policies. With the same label supplied in both, the conflicting instructions remain and no common policy is feasible. A simulator that automatically inserts its internal mechanism identifier into the actor's message would report the first answer while pretending to solve the second problem.

The selected observation declaration therefore records the actual token presented to the actor. It permits distinct mechanisms to emit the same token. Source identities and temporal proof labels remain attached to the analyst's explanation, but do not enter the decision history unless the declaration supplies a separate observation with that meaning. The numerical comparison keeps these cases apart even though both could informally be described as a labeled source.

[8] Complete-family binding, 26 family-change cases, 26 adversarial controls and saved independent replay are retained in `data/accounting-master/temporal-family-binding-20261027/`. The 21 cross-mechanism policy cases, independent full-set comparisons and corrected late-revelation expectation are retained in `data/accounting-master/cross-mechanism-policy-20261027/`. All message mappings and visibility regimes are explicit synthetic assumptions; neither reference introduces probabilities, actual actor knowledge or accounting permission.

17.19.1 A message map must use information available at that date

Hiding internal identifiers is insufficient if the message itself encodes an unavailable fact. A map that takes the entire financial history as its input could send one message when a hidden borrowing right exists and another when it does not. The resulting policy would exploit that right even if the source observations never revealed it. Giving the message an innocuous name would not remove the additional information.

The finite reference avoids that ambiguity by restricting what each map can read. At the first date, the coarse view supplies the market signal; the refined view supplies that signal and the current coupon-trigger observation. At the second date, the map receives the already specified trigger and cash-coverage observations. The actor retains the first message and its own earlier instruction. Hidden contractual rights, complete world identifiers and later observations cannot be introduced as new first-date map inputs.

For each selected source outcome, the declaration must map every supported observation key to a message. A missing key is an unfinished declaration, not permission to remove the corresponding financial history. Two entries for the same key are also invalid. These restrictions establish a closed finite interpretation of the existing observations. They do not establish that those observations describe a particular company's actual information practices, nor do they define a general language for arbitrary messages.

This ordering is useful when designing richer models. First specify which quantities can be observed and when; then specify how the observation is represented to the actor; finally constrain decisions to agree on indistinguishable histories. Reversing the order can make a desirable policy feasible by designing messages around the hidden state that policy would need to know.

17.19.2 Source, actor and time remain separate premises

A claimed visible tag carries an actor, the selected source revision, its content identity, a token and an observation time. The first decision must agree exactly with the temporal family's cutoff; the second occurs strictly later. A tag claimed at the first decision must have been available by that instant. Equivalent fixed-offset spellings describe the same instant, whereas a tag received one ten-millionth of a second afterward is late under the selected exact-time convention.

Unknown tag availability leaves the declaration unresolved. So does a supplied observation whose chronology cannot be established from the retained authorship bounds. The current reference treats this conservatively: it does not discard inconvenient clock possibilities or silently refine the source family with a new tag observation. A later method may condition or refine that family under explicitly admitted additional evidence, but it must preserve the distinction between the original question and the revised one.

These unresolved cases differ from a complete observation model with no feasible policy. In the latter, the available histories are specified and every possible instruction fails somewhere in the retained financial family. In the former, the model has not established which histories the actor may distinguish. Returning the same empty policy list for both would hide a missing information premise behind an apparently decisive financial conclusion.

A content identity detects alteration relative to expected bytes; it does not show that the actor received them. Likewise, the synthetic tag records used in the comparison test the consequences of supplied assumptions. They are not observations of a real company's managers or investors. That distinction remains necessary even when every mapping and timestamp check passes.

17.19.3 Reuse the calculation only when the decision problem is unchanged

A corrected source can preserve the same financial meaning and messages. The numerical policy calculation can then be reused while its source explanation changes. A different actor can also have the same supplied observation map; the actor binding must change even if the resulting mathematical problem is identical. Neither case licenses removal of the old and new source supports from the retained temporal family.

Changing a message label is different. A policy is a rule for responding to the labels the actor receives. Relabeling H as another token changes that rule's interface even if four policies remain feasible and a mathematician can construct a one-to-one relabeling between the sets. The selected reference recomputes this case. It does not claim to have proved and implemented a general translation between equivalent policy representations.

The reuse comparison consequently includes the complete financial and action population, the actual first and second messages, the meanings of the actions, and the numerical methods. Identical repeated constraints can share computation while keeping every source outcome and temporal proof support attached. Distinct problems with zero feasible policies retain distinct identities. A future correction that is unavailable at the earlier cutoff remains in the source record without replacing the earlier view.

The independent checks reconstruct the messages from the supplied declarations and original observations, compare full policy sets from two formulations, and reject missing or duplicated keys, hidden-state inputs, wrong actors or times, altered sources and forged prior results. They also check changed second-date labels without changing first-date messages. These examples make the dependency rule concrete: identical counts are not enough, but a changed source explanation need not force an unchanged numerical problem to be solved again.

For this observation-binding profile, the remaining computational difficulty is how to continue an unfinished temporal search while retaining its unexamined possibilities. Section 17.20 develops that extension for the selected temporal domain. The conservative full-domain remainder already prevents a partial result from masquerading as complete; a checked, resumable partition would additionally preserve useful completed work. Actual message provenance, richer observation languages and refinement by new tag evidence remain distinct extensions.[9]

17.20 Stopping a search without losing the possibilities it has not examined

A correction may have reached the manager before a payment decision, or afterward. An interval for the delivery time can preserve both possibilities without assigning either a probability. If the source record contains several corrections, public releases and private deliveries, the exact temporal calculation may exhaust its work allowance before checking every combination. The unfinished calculation still has a useful result, provided it keeps the completed conclusions together with an exact description of what remains unexamined.

The conservative earlier method returned the whole original clock domain whenever its search stopped. That answer preserved uncertainty but discarded useful completed proof work. A resumable calculation can retain more. It must save the partition it has already established, including every unfinished part. Saving only the branches that produced interesting financial

[9]The finite declarations, complete mapping comparisons, 22 main cases and 30 adversarial controls are retained in `data/accounting-master/observation-map-binding-20261027/`. A separate saved audit rechecks all main cases and four additional source, exact-time and token cases. The reference retains full source supports under numerical sharing and makes no claim of actual actor knowledge or authenticated evidence.

views would change the question on the next run.

17.20.1 Each split must cover the entire timing domain

Consider one cell D of admissible event times under the supplied chronology and access assumptions. For the next source revision, let $g_1, \dots, g_m$ be its eligibility conditions. A condition may require that publication or actor access occurred no later than the decision, or that ingestion occurred no later than the system's reconstruction date. The revision is eligible only if all these conditions hold.

A complete split keeps one eligible cell and one cell for each possible first failed condition:

$$D_{\text{yes}} = D \cap \bigcap_{j=1}^{m} g_j, \tag{17.31}$$

$$D_{\text{no},k} = D \cap \bigcap_{j<k} g_j \cap g_k^{\text{c}}, \qquad k = 1, \dots, m. \tag{17.32}$$

Here a condition also denotes the set of time assignments satisfying it, and g_k^{c} denotes its exact complement. Every assignment in D either satisfies every condition or has a unique first failure. These alternatives therefore cover D and are pairwise disjoint. Their union is the original cell, not a sample of it. Some children may be empty, and a contradiction proof can establish that fact later.

The word *exact* matters at a decision boundary. If g_k says that access occurred at or before time d, its complement says strictly after d. Replacing that complement by *at or after* would put the boundary in two children. Replacing the original condition by a strict inequality would remove the boundary from the eligible child. The selected reference retains rational event times and strict inequalities, so the preceding submicrosecond and fixed-offset examples keep their original meanings.

The search resolves one pending cell at a time. It first checks whether the cell is empty. A nonempty cell with no remaining revision to classify yields a selected or unresolved source view. Otherwise it installs the complete family of children above. Only after all those children exist may the calculation pause. This logical rule prevents a work limit from turning a complete split into a partial list of favorable branches.

17.20.2 Pending does not mean feasible, and known does not mean complete

A saved node has one of four meanings: its domain remains unchecked; a contradiction proves it empty; a witness establishes a terminal source view; or a witness supports an expanded cell whose children form the complete partition. An unchecked child need not contain a possible time assignment. It may already be inconsistent with an earlier chronology restriction, but the calculation has not yet established that. Retaining it is conservative; deleting it without checking is unsupported.

In the selected example, resolving the root installs eight children. Seven are later proved empty. With a one-node work allowance, all eight remain explicitly pending. The result therefore neither asserts eight feasible worlds nor pretends that the single eventual view is already established. The eight cells are a representation of unresolved work under known restrictions.

Once some terminal views are known, they can be displayed together with their source supports. Agreement among them still does not establish that the whole family has that view. An unexamined cell could contain a conflicting correction, a retraction or a different access history. Completion requires that no pending cell remains, including cells that eventually prove empty.

The same distinction controls policy interpretation. A policy that fails in an already established view fails for any larger family containing that view. A policy that succeeds in every established view may still fail in an unexamined one. Thus an incomplete temporal search can sometimes establish a counterexample before it can establish universal feasibility. Connecting that asymmetry to the numerical policy calculation requires an explicit treatment of the pending remainder; a saved search alone does not supply it.

17.20.3 Resume the original question and preserve its proofs

The checkpoint binds the complete source request, actor, trust declarations, timing assumptions, method identities and execution allowance. On reading it, a separate checker reconstructs each branch's constraints from the original domain and its path. It does not accept a saved mutable constraint list as a substitute. It verifies all completed witnesses or contradictions, the exact complement structure, source outcomes, support paths and work counts.

The reference uses a fixed left-first depth-first order. This makes the resumed tree directly comparable with the existing uninterrupted calculation. It also allows the checker to reject a purported completed branch placed after an earlier unfinished one. The order is a comparison convention, not a claim that this schedule is the fastest way to find useful financial conclusions.

A step allowance specifies how much new work one call may perform. A separately bound total execution cap determines when further work must stop. Reaching that cap retains the pending tree and reports a capped calculation; it does not turn unresolved cells into completed results. The test with a cap of two preserves the same original question and performs no repeated solver work when resumed at that cap. Changing the cap or the source assumptions requires an explicitly different, checked request to the continuation interface.

Across the 47 selected temporal requests, step allowances of one, two, seven and 4,096 produce exactly the same final proof trees, outcomes and source supports as uninterrupted search. Earlier certificates remain unchanged at every pause. A separate reread checks 1,060 saved checkpoints and rejects altered partition cells, proofs, source premises and completion declarations. These comparisons establish the selected finite continuation property.[10]

This representation avoids a separate explicit node for every real-valued event-time assignment. It does not remove exponential growth in the number of distinguishable revision combinations. Nor does installing all children before a logical pause establish crash-safe file storage, authenticated evidence or actual actor knowledge. Larger revision families, better schedules, justified approximate reduction and durable storage require their own arguments and comparisons. The useful advance here is narrower: unfinished computation retains both what has been proved and every domain that has yet to be checked.

17.21 Fixing the policy question before the search finishes

A manager who may pay an obligation or draw on a facility needs a rule for acting on the information available at each decision. The preceding search can establish some possible information histories while leaving others unexplored. To compare policies during that search, their meaning must remain fixed. If each newly discovered history changes what a policy is allowed to observe, an apparent reduction in the feasible set may instead reflect a change in the

[10]The implementation, independent structural checker, 188 complete comparisons, 43 main adversarial controls and five additional saved-record controls are retained in `data/accounting-master/temporal-frontier-20261027/`. The main comparison makes 10,146 assertions; the saved-record audit makes 3,191. The source locks and manifests preserve the exact inputs, methods and comparison scope.

question.

The first requirement is therefore a declaration of possible messages before temporal search. In the finite example, start with the original registry of financial assertions and observation schemes. For each assertion-valued observation scheme, declare the first- and second-decision messages for every raw observation obtainable from the original financial assertions. Cross those financial and observation assertions conservatively. Some combinations may later prove temporally impossible; including them at this stage avoids selecting a favourable source history in advance.

The declaration binds the manager, decision dates, source records and meanings of pay and draw. A second-decision key also retains the manager's own earlier action, as in the preceding policy example. Internal source-head identifiers remain evidence of support and do not become messages to the manager. A retraction supplies no replacement assertion payload, but it does not justify removing every message associated with the original assertion before its timing has been examined. This is a declared finite vocabulary, not a finding that a real manager received every listed message.

17.21.1 A fixed domain makes partial restrictions comparable

Let U denote the declared finite set of decision keys, including the decision stage and the required own-action memory. A total rule assigns either pay or draw to every key in U. Let Π_U be the set of these rules. Assignments to keys that are never reached still belong to a total rule; they have no effect on that rule's realised path. Consequently, several total rules can represent the same policy on the reachable histories.

At search stage t, let H_t contain only the selected financial and information histories supported by completed temporal branches. For a history h, write $V_h(\pi)$ when following rule π selects no refused action in that history. This condition uses the complete finite action rows for that selected history, including first-decision refusal and second-decision refusal after the first action. The rules surviving the known restrictions are

$$C_t = \{\pi \in \Pi_U : V_h(\pi) \text{ for every } h \in H_t\}. \tag{17.33}$$

These are sets of decision rules, not monetary balances or probability distributions. The relation is a local definition under a fixed message declaration and the supplied action semantics. Unresolved source outcomes are retained separately; they cannot supply a selected history or establish its financial constraints.

Resuming the same checked search retains the previously completed histories. Thus $H_t \subseteq H_{t+1}$. A rule satisfying every condition for the larger set also satisfies every condition for its subset, so

$$C_{t+1} \subseteq C_t. \tag{17.34}$$

This inclusion explains the asymmetric conclusions. Once known constraints eliminate every rule, adding more histories cannot restore a rule under the same declaration. If some rules survive, a pending branch may still eliminate them. In the coarse/refined correction example, the known constraints already eliminate all rules after four of the 25 temporal nodes have been resolved. The baseline instead retains four total rules and establishes conditional feasibility only when the search completes with selected source outcomes throughout.

17.21.2 Three reasons that a surviving rule may still be unresolved

Unfinished search, unresolved source meaning and an inadequate domain allowance are different limitations. A two-node search cap retains 15 pending branches in the baseline. Those branches

have not all been shown to contain possible histories, but they must remain represented until checked. The surviving rules therefore remain candidates.

A search can also finish while a required assertion remains unavailable, retracted or in conflict with another head. Unknown access leaves four rules surviving the known constraints, yet one source outcome remains unresolved. Retraction and conflicting-head cases likewise do not become feasible merely because all temporal branches have been processed. An inconsistent clock system is different again: it admits no temporal assignment. The 512 unconstrained rule assignments in that example do not establish a feasible policy for a possible company history.

Finally, the conservative domain may exceed the stated computational allowance. The finite profile permits at most 12 binary decision keys, hence at most 4,096 total rules. Renaming only the low message in one correction produces 12 keys and eight surviving total rules. Renaming all correction messages requires 18 keys, so the reference retains the original request and returns an unresolved domain budget without truncating the keys or enumerating a preferred subset. The extra rule count in the smaller example is not evidence of better information or better economic performance.

The checked partial calculation binds each result to the original request, declaration, domain, methods and temporal frontier. It retains the supporting branches, unresolved source outcomes and pending temporal domains. Changing a message, source, actor or domain cannot reuse the old result as the answer to the new question. A future extension rule may justify some reuse, but that relation must itself be stated and checked.[11]

This construction joins a fixed decision problem to an unfinished search without treating uncertainty as missing arithmetic. Longer horizons, additional actions, changing declarations and broader observation languages still require their own comparisons.

17.22 Changing a message without confusing the decision problem

A manager may receive a differently worded notice carrying the same information, or a notice that distinguishes two situations previously described alike. These changes have different consequences for a policy. The preceding requirement to fix message meaning does not prohibit either change. It requires the simulator to state which relation connects the old decision problem to the new one, and to identify the new result accordingly.

A consistent one-to-one renaming preserves the original distinction between every pair of messages. The finite comparison checks that renaming across every raw observation in every original source head, including heads not selected at the current frontier. It also preserves actor, decision dates, action meanings, source identities and second-stage memory of the manager's first action. Each total rule then has exactly one renamed counterpart. The first- and second-action refusal conditions have the same truth values after the key positions are permuted, so the full surviving rule sets correspond exactly. This is the same finite decision problem under different spellings; it is not an expansion of what the manager knows.

[11]The original-registry domain comparison is retained in `data/accounting-master/prospective-policy-domain-20261027/`; its 10 cases include explicit overflow and conflicting-head cases. The partial comparison in `data/accounting-master/prospective-partial-policy-20261027/` checks 12 cases at 110 frontiers, with independent full-rule enumeration and 31 refusal controls. The candidate takes no complete-family comparison as an input. Exact sets and endpoint projections, rather than the counts alone, support these finite conclusions.

17.22.1 A refined notice allows the old distinction to be recovered

Now suppose a correction distinguishes a low message that was previously shared with another source history. The new message identifies which of those situations occurred. A manager can ignore that distinction and follow the old rule, or potentially act differently in the two situations. To make this precise, each refined message must map to one old message, consistently across all original raw observations. Combining this message map with the unchanged decision stage and own-action memory gives a map from new policy keys to old keys.

Write U_o and U_n for the old and new finite key domains, and $e : U_n \to U_o$ for this erasure map. Require e to reach every old key. An old rule π_o assigns pay or draw to each old key. Its lifted rule $L_e\pi_o$ chooses $\pi_o(e(u))$ at a new key u: the manager acts as if the additional distinction had not been received. New keys that erase to the same old key form a *fiber*. A rule constant on each fiber makes the same choice at all of those keys.

For the same checked temporal frontier, let C_o and C_n be the rules satisfying the old and new known constraints. The comparison preserves each underlying action row and its refusal meaning, and checks that erasing the new forbidden assignments gives exactly the old forbidden assignments. Under those conditions,

$$\{L_e\pi_o : \pi_o \in C_o\} = \{\pi_n \in C_n : \pi_n(u) = \pi_n(v) \text{ whenever } e(u) = e(v)\}. \tag{17.35}$$

To see the first direction, a lifted rule makes the old choice at each corresponding raw observation. A refused first action or refused pair of successive actions would therefore also have been refused under the old rule. Conversely, a new rule constant on fibers defines a unique old rule because every old key has a preimage. The same correspondence of refused assignments then shows that this old rule satisfies the old constraints. The relation concerns exact finite rule sets under supplied message semantics; it assigns neither probabilities nor monetary values to information.

17.22.2 Additional rule assignments need a reached-action comparison

In the correction-specific low-message example, the old declaration permits four total rules and the refined declaration permits eight. Lifting preserves all four old rules. The other four also produce distinct sets of reached action rows across the complete selected family. Here the new distinction therefore permits additional feasible behavior under the stipulated information design. In another example, collapsing the first messages leaves no feasible rule, while restoring the refined messages permits four. Neither comparison establishes that those messages are actually available to a real manager or that any policy maximises value.

The source-unknown examples show why counting total rules is insufficient. With conflicting source heads, a refinement increases the unconstrained assignments from 32 to 2,048, but there is no selected history on which to demonstrate a new action. Every rule has the same empty reached-row signature. Under inconsistent clocks, the corresponding counts are 32 and 512, yet no admissible time assignment exists. An empty signature is a set used for comparison, not a realised or feasible company behavior. With unknown access, additional choices can be exhibited on the selected completed histories while the overall source family remains unresolved.

The comparison therefore retains both the full rule sets and their reached-row signatures. Pending branches, unknown source outcomes and inconsistent clocks keep their original meanings. The refined result has its own declaration and domain identity, even when its feasible rules correspond exactly to an older result.[12]

[12]Exact renaming is checked in `data/accounting-master/policy-domain-transport-20261027/`: 22 frontier comparisons and 12 refusal controls. Erasure and lifting are checked in `data/accounting-master/`

This gives two precise ways to compare changed declarations. Broader changes to sources, decision times, actions or the observation language need additional arguments before earlier policy work can be reused.

17.23 A correction arriving between two decisions

A manager can make the first payment before receiving a correction, then consult the corrected information before the next financing decision. The first action belongs to the earlier information history. Replacing both observations with the latest source view would rewrite that history; treating the two dates independently could invent a sequence that no delivery time permits. The simulator needs a joint sequence of source views before it can interpret the two-stage policy.

Consider the supplied correction from a refined observation scheme to a coarse one. Let the first decision be on 2 January and the second on 3 January. If the correction is received before both dates, both select the coarse scheme. If it arrives between them, the first selects refined and the second coarse. Arrival after both leaves refined at both dates. When its access-time interval spans both decisions, the joint family therefore contains

$$\{(\text{coarse}, \text{coarse}),\ (\text{refined}, \text{coarse}),\ (\text{refined}, \text{refined})\}. \tag{17.36}$$

Each pair lists the first and second selected source schemes. These are possible sequences under the supplied chronology, not probabilities. Each marginal date admits both schemes, so their Cartesian product would also include coarse followed by refined. That fourth pair is impossible here: the same correction cannot have arrived before the first decision yet remain unavailable at the later one. A later retraction or a different replacement would be another source event requiring its own representation.

17.23.1 Keep the source clocks shared while changing the decision cutoff

The joint calculation uses the same source records, actor, access clocks and chronology at both dates. Its system reconstruction cutoff is fixed and may lie after both decisions; that cutoff describes the evidence inventory used by the analyst, not information available to the manager at the time. Each stage separately tests authorship, publication and actor access against its own decision date and applies its effective cutoff. The first increment keeps one financial assertion fixed while observation schemes can change.

The exact temporal partition first classifies eligibility at the first date and then at the second, carrying the accumulated clock constraints throughout. A completed branch records both selected or unresolved source views. An independent checker reconstructs the constraints and both head selections. The resulting joint pairs agree with the authored before, between and after cases, including exact decision boundaries and a submicrosecond arrival immediately after the first decision. Their projections agree with independently checked marginals, while the joint counterexample above tests the dependence that marginal agreement alone cannot establish.

Late retraction and conflicting heads leave the first view selected and the second unresolved. Unknown access retains every timing sequence admitted by the supplied chronology. It does not assert when the manager actually received the correction. Inconsistent clocks admit no joint

`policy-observation-refinement-20261027/`: 17 comparisons and 10 controls, with whole-set equality and explicit reached-action comparisons. Split or merged labels are outside the renaming relation; a false refinement direction, altered source or dropped pending branch is refused. Both profiles remain finite conditional design evidence.

sequence. If the joint search exhausts its allowance, this first profile retains the entire original clock question as unresolved; it does not yet save a resumable joint frontier.[13]

17.23.2 Use the selected map at each stage

A prospective policy domain must now cover more than same-head combinations. Before inspecting completed joint outcomes, the declaration crosses the original financial head with every assertion-valued first-stage observation head and every assertion-valued second-stage head. It constructs the first message from the first selected map and the second message from the second selected map. The second decision key retains the actual first message and the manager's own first action. Internal head identities remain support records, not messages visible to the manager.

In the inherited raw observation schema, coarse versus refined changes only the first observation. The second raw observation remains the trigger and cash-coverage pair. Selecting a different source head at the second date therefore does not, by itself, add second-stage information. An explicit change to a second-message token tests whether the correct stage map is used, while remaining a supplied interface assumption.

That changed-token case needs 15 keys in the conservative original domain, exceeding the 12-key allowance. It remains unresolved without dropping mixed-stage combinations. A separate diagnostic collapses the first messages and uses 7 keys. Its encoded second messages differ when the wrong first-stage head is substituted for the second-stage head, so the test detects that wiring error. The correctly wired diagnostic nevertheless has no feasible policy. Correct source-to-message binding and policy feasibility are separate conclusions.

With the original mappings, arrival between the decisions or after both leaves 16 total rules on the conservative 11-key domain. Some assignments concern keys not reached in the selected family; these are not 16 demonstrated reached behaviors. Arrival before both dates supplies the coarse first view and leaves no feasible rule. The interval containing all three sequences also has no common rule because it includes that coarse-first history. Unknown access preserves the same conditional family and its infeasibility, without turning unknown access into an observed fact.

A required retracted or conflicting source view leaves the overall policy question unresolved, even when candidate assignments satisfy every known numeric constraint. Inconsistent clocks and either temporal or policy-domain budget exhaustion supply no feasible-policy answer.[14]

These results bind a complete joint source family to two-stage decisions. Continuing an unfinished joint search, changing financial rules between dates and admitting actual actor information remain separate tasks.

17.24 What a paused joint search can establish

Suppose the search stops after checking some of the correction's possible arrival times. A financing rule may survive those cases and fail in an unexamined case. To interpret the partial

[13] The exact shared-clock comparison in `data/accounting-master/two-decision-source-views-20261027/` checks 11 cases and 12 refusal controls, preserving full joint partitions, source support and separately checked marginal results. Its 208 assertions include the impossible Cartesian-product sequence and the conservative budget remainder.

[14] `data/accounting-master/joint-stage-policy-20261027/` retains 16 cases,204 assertions and 7 refusal controls. A separate raw-observation mapper checks complete encoded rows; an independent policy oracle checks full rule sets. Same-head cases reproduce the earlier mapping, and the stage-specific token diagnostic tests preserved first-action memory. The 15-key case and the separately collapsed 7-key case are both retained.

answer, the modeller needs both the source sequences already checked and the clock assignments still awaiting examination. The shared-clock construction in the preceding section supplies the latter without choosing a separate history at each date.

Each unfinished branch retains the constraints accumulated on the original clock variables. At a branch concerning a particular source and decision date, the alternatives are that every eligibility condition holds or that a named condition is the first to fail. These alternatives cover the parent branch without overlap. The search creates all of them before pausing. It can therefore resume from the first pending branch while preserving every other pending branch, the first decision's eligibility choices and the second decision's remaining tests. A short allowance changes how much has been examined, not the question being asked.

The saved state distinguishes a per-call allowance from a total node cap. Reaching the total cap leaves a capped, incomplete result with its pending domains. A checker reconstructs each branch's constraints from its path, verifies the rational feasibility or contradiction evidence and recomputes both selected source views at completed leaves. Checking only the number of leaves would miss a lost or duplicated support region. In the bounded examples, resuming one node at a time, seven at a time or without intermediate pauses produces the same complete tree, joint outcomes and support paths.

17.24.1 Surviving rules can shrink before the search is complete

Keep the prospective message domain fixed before examining these outcomes. Let Π be its finite set of total two-stage decision rules: each rule assigns an action at every declared first- or second-stage message key, including keys that a particular source sequence might never reach. After k units of search work, let K_k contain the distinct fully selected source pairs found so far. For a pair h, let $F(h) \subseteq \Pi$ contain the rules satisfying every financial row supplied by that pair. The candidates consistent with the known pairs are

$$C_k = \Pi \cap \bigcap_{h \in K_k} F(h), \qquad K_k \subseteq K_{k+1} \implies C_{k+1} \subseteq C_k. \tag{17.37}$$

This is a local set-theoretic consequence of adding restrictions in the same rule universe. Initially no source pair may have been selected, in which case the empty intersection imposes no restrictions and $C_0 = \Pi$. That initial survival says nothing about financing feasibility. Each newly selected pair can remove a rule; none can restore a rule already rejected by a known pair. The index k measures search work, not elapsed financial time, so this relation does not describe how a manager's information evolves.

If the known restrictions leave no rule, no completion that merely adds further source pairs can supply a rule satisfying all of them. Conversely, a nonempty C_k remains a partial answer while any source region is pending. In the correction example, a completed coarse-first pair can already remove all candidates, whereas observing only refined-first pairs cannot establish that no coarse-first pair remains. A changed source declaration or message domain would alter the comparison and requires a separate justification for reusing old results.

The returned conclusion also preserves the reason a family is unresolved. A retracted or conflicting required source remains an unresolved source view; the overall result keeps that status even if the known restrictions separately rule out every candidate. A complete timing family with unknown actual access is instead evaluated conditionally on all its supplied possible arrival sequences. Inconsistent clocks supply no admissible joint sequence and are not treated as a feasible empty family. Exceeding the prospective key allowance leaves the policy population unassessed. Only a completed search with selected source content and surviving rules supports conditional feasibility under these declarations.

17.24.2 Check the reason for each rejected rule

A saved list of surviving rules is insufficient evidence by itself. For every excluded total assignment, the result names a known financial row that refuses it. A first-stage refusal must match the rule's first action at that row's message. A second-stage refusal must also match the second action, using the second key that retains the actual first message and action. The checker validates these witnesses and requires the surviving and rejected assignments together to cover the entire fixed rule universe. It also reconstructs the conclusion, unknown source pairs, pending domains and support paths from the checked source search.

The finite comparison checks every saved endpoint and selected partial results against an independently evaluated policy population. Altered refusal witnesses, premature feasible conclusions, missing pending evidence and changed identities are rejected. The result checker still uses the established source-to-row mapping and policy oracle; it is not an independent redevelopment of all accounting and observation semantics.[15]

This establishes logical continuation for the supplied two-date question. It does not test crash-safe file storage or establish actual information delivery. Transporting a result to changed observation declarations, extending the horizon and admitting company evidence require further work.

17.25 Changing the observation declaration

The fixed-domain result raises a practical question when a modeller revises the observation declaration. A message may acquire a new label, or a formerly pooled observation may be separated into more informative messages. Both changes alter the encoded decision keys, but they have different consequences. A new spelling can preserve the same choices exactly. A new distinction can let a manager choose different actions where the old declaration required the same action. The comparison must say which change has occurred before reusing a computed policy set.

17.25.1 A different label must preserve the entire decision key

Consider replacing every first-message label with a distinct new label and doing the same for the second messages. The substitution must be global across source heads: two equal old messages must remain equal, and distinct old messages must remain distinct. A substitution chosen separately for each source could reveal the source's identity and therefore change the information available to the manager.

The second-stage key contains the first message and the manager's first action as well as the current message. Consequently, renaming the first message also changes its occurrence inside second-stage memory. The joint comparison generates this full key correspondence from all original financial and observation-head triples, before inspecting completed source outcomes. If the correspondence is a bijection, the old total action assignment can be permuted to the new key order. Inverting the permutation recovers every assignment.

The refusal evidence travels with the assignment. A rejected rule keeps the same financial-row identity, but the checker must confirm that the renamed rule still takes the actions that row

[15]The complete evidence is retained in `data/accounting-master/joint-temporal-frontier-20261027/`. Attempt 02 checks 11 requests, 33 completed search sequences, 662 saved checkpoints and 848 policy prefixes in 16 cases. Attempt 03 checks the 16 saved policy endpoints and 47 selected prefixes with 33 alteration controls, including wrong refusal witnesses and false feasible conclusions. The earlier failed import is preserved separately. These counts describe coverage of the supplied finite cases, not a general error rate.

refuses. The transported surviving population agrees with fresh evaluation under the new declaration, and the source supports, pending clock regions and unresolved source outcomes remain unchanged. An overflowing policy domain stays unassessed; label equivalence does not remove its computational limit.[16]

17.25.2 A new distinction permits an old rule to ignore it

Now suppose a refined declaration distinguishes messages that a coarse declaration pools. The financial rows, action meanings, source chronology and decision dates are unchanged. Let D_c and D_f be the coarse and refined sets of full decision keys. A surjective map $e : D_f \to D_c$ erases the new distinctions. It must also erase the first-message distinction wherever that message occurs in second-stage memory. Surjectivity ensures that every coarse decision key still has a refined counterpart.

A coarse rule π assigns an action to each key in D_c. Its lift $\pi^{\uparrow}$ on D_f ignores the added distinction: at a refined key z, it takes the coarse action at $e(z)$. Let C_c and C_f be the candidate populations for the same selected source pairs under the two declarations. With unchanged financial refusal conditions, the checked relation is

$$\pi^{\uparrow}(z) = \pi(e(z)), \qquad \{\pi^{\uparrow} : \pi \in C_c\} = \{\rho \in C_f : e(z) = e(z') \Rightarrow \rho(z) = \rho(z')\}. \tag{17.38}$$

The right-hand side selects refined rules that take the same action whenever the new messages collapse to the same coarse key. To see why the equality holds, substitute the coarse action at $e(z)$ for each refined action in a refused financial row. Erasing that row's decision keys produces exactly the coarse refusal condition. A lifted coarse candidate therefore passes every refined row. Conversely, a refined candidate constant on each erasure class defines a unique coarse rule, and the same substitution shows that it passes the coarse rows. The equality concerns the same finite decision problem under two declared information restrictions.

In the between-date correction case, restoring the original first-message distinctions from an artificially collapsed first message changes the candidate population from zero to 16 total rules, with four additional reached-row signatures. Restoring the second-message distinctions changes two total candidates to 16, with three additional signatures. A signature records the financial rows reached by a rule's actions; it is not a probability distribution or an observed company path. The comparison verifies the reached rows of each individual lifted rule, so equality of aggregate signature sets cannot conceal a wrong pairing.

The distinction matters particularly when required sources are unresolved. In the late-conflict example, refinement expands eight total assignments to 2,048, yet no financial row is selected and no additional reached-row signature appears. Those assignments do not establish feasible company behavior. Inconsistent clocks produce no assessed policy population at all. If a refined domain exceeds the key allowance, its policy comparison remains unassessed even when the coarse domain has been evaluated.[17]

These are conditional comparisons of supplied observation declarations. The extra distinctions must still be justified as information available to the actor. Their finite effects on permitted actions do not measure utility or predictive accuracy. Both relations also retain one financial source view across the two decisions; allowing a financial correction to change that view requires an account of what was believed earlier and which economic history continues afterward.

[16] The bounded joint comparison in `data/accounting-master/joint-domain-transport-20261027/` checks 34 comparisons across nine declarations and 16 alteration controls. Eight comparisons retain unassessed domain-budget results. The assessed comparisons check full populations, preserved refusal witnesses and inverse roundtrips, including the first message inside second-stage memory.

[17] `data/accounting-master/joint-observation-refinement-20261027/` retains 46 comparisons across 12 declaration pairs, with 14 alteration controls. Nine comparisons remain unassessed. Assessed cases verify full lifted populations, constant-class equality, erased refusal conditions and per-rule reached rows. First-stage, second-stage and simultaneous refinements are separate cases; exact renaming is an equality control.

17.26 A revised report and the economic history it describes

A financial report can change between the two decisions without changing the economic event it describes. Suppose an earlier assertion reports an operating receipt of 7 and a later assertion corrects it to 6. Selecting the available assertion at each date answers an information question. It does not yet determine whether the receipt was always 6, whether an agreement actually changed, or what information the manager received. Those questions must be separated before the selected assertions can support a simulation.

The joint source calculation can now retain a different financial head at each decision. A correction available between the dates selects the old assertion first and the corrected assertion second. With an interval-valued availability time, the supported sequences can be old/old, old/new and new/new. The reverse new/old sequence is impossible under this one-way correction history, even though it would appear in an unconstrained Cartesian product of the two marginal selections. Conflict and retraction retain unresolved content. None of these source selections, by itself, specifies a financial policy.[18]

17.26.1 Continuity concerns the complete carried state

The financing example gives a precise way to distinguish an assertion revision from a change in economic history. Opening cash is $11/2$. The first coupon I_1 is accrued and settled; the operating receipt R arrives next; any first-period facility borrowing is repaid; and the second coupon I_2 is then accrued and settled. A draw supplies 1 and immediately costs a fee of $1/4$. Let d_1 and d_2 indicate a draw at the respective decisions. The once-only right and the other payment guards still determine whether an action sequence is permitted.

After the first payment, cash is $11/2 - I_1 + 3d_1/4$. Receipt and repayment change it to $11/2 - I_1 + R - d_1/4$: repaying the principal removes the first draw's unit of cash, leaving its fee. The second settlement therefore gives

$$C_{\text{final}} = \frac{11}{2} + R - I_1 - I_2 - \frac{d_1}{4} + \frac{3d_2}{4}. \tag{17.39}$$

The last term includes the second draw's principal because the sequence ends at the second settlement. It does not erase the associated liability. Under receipt 6, fixed coupons 5 and 5 with pay/pay give final cash $3/2$ and no facility fee. Floating coupons 6 and 5 with draw/pay give final cash $1/4$ and a fee of $1/4$. Both examples satisfy the inherited pre- and postconditions. The equation also describes other action sequences algebraically; their guards must be checked separately. All these monetary amounts are supplied synthetic units.

Before the receipt arrives, the complete states under $R = 6$ and $R = 7$ coincide in this example. That includes the book, contractual memory and recorded flows. Attaching the receipt-6 continuation to the identical pre-receipt prefix therefore reconstructs exactly the complete receipt-6 path. Rejecting every change of a receipt parameter would miss this valid correspondence. The correspondence can support an explicitly assumed correction of the same event; it cannot establish that the revised report is true.

After receipt settlement, the two states differ. Replacing 7 by 6 then removes one unit of cash, changes the income credit and rewrites the recorded operating receipt. Both books balance, but the complete states no longer join. A different contractual right is visible even earlier: the

[18]The checked financial-source selection companion is retained in `data/accounting-master/joint-financial-source-20261027/`. It reproduces 11 protected prior cases and checks 13 new cases, with 341 assertions and 16 alteration controls. Its declared financial-policy binding remains unassessed.

right is carried in opening memory, whether or not it changes the numeric accounts. An actual amendment consequently needs an explicit event and its accounting treatment.[19]

17.26.2 An explicit interpretation selects a complete world

For a supplied same-event correction, the analyst can declare the corrected head, a fixed receipt and a contractual-right family. This declaration supplies a hypothesis about the whole economic history. The selected heads at both dates must lie in the corrected head's receipt-only ancestry, and the right family must remain unchanged. Retaining the superseded assertion as evidence does not impose both 7 and 6 as simultaneous exact facts about one receipt; imposing those incompatible facts would leave no world.

The binding then selects the complete financial rows for the declared world while retaining the dated source pairs, their supports and any unresolved temporal remainder. Receipt 6 with the committed right binds 980 action-region rows; receipt 7 binds 968. These are complete action-region row families, not counts of feasible policies. Their populations agree with the protected reference problems. Unknown interpretation, actual amendment, changed rights, uncovered ancestry, conflict or retraction, inconsistent clocks and incomplete search receive no financial row set.[20]

The corrected head may be available to the analyst who specifies the hypothesis without having been available to the earlier manager. Neither that head's identity nor the assumed world's identity is automatically an observable message. An adaptive policy still needs an explicit account of the information available at each decision and must choose the same action whenever that information is the same. The next comparison must therefore preserve the distinction between the analyst's economic hypothesis and the manager's dated observations.

17.27 Dated observations and a common decision rule

The analyst's receipt assumption does not tell us what the manager can observe. Two explicit alternatives make the distinction testable. The first retains the original numeric observations: the first coupon signal and the later trigger and cash-coverage signal. The second pairs each numeric message with the receipt reported by the financial assertion selected at that date. These are supplied information assumptions. A reported future receipt is neither cash already received nor proof of the eventual amount.

For a correction arriving between decisions, the supplemented first message reports 7 and the second reports 6, even when the analyst evaluates a complete receipt-6 world. Putting the later correction into the first message would give the manager information too early. At the second decision, the key still includes the first message and the manager's first action. That memory is needed even if the current numeric signal coincides with one reached after a different earlier choice.

Several supplied source/world alternatives can then be evaluated together. The alternatives share an actor, dates and action meanings. Their identities remain internal: equal visible decision keys require the same action in every alternative. Finding a policy separately in each

[19] `data/accounting-master/financial-assertion-continuity-20261027/` checks 16 symbolic cases against an independent full-state recurrence: 112 boundary comparisons, 594 assertions and 48 rejected post-receipt joins. It also checks the two viable receipt-6 examples.

[20] The `SuppliedReceiptCorrectionWorld/1` declaration and its checks are retained in `data/accounting-master/financial-source-world-binding-20261027/`: 20 cases, 289 assertions and 25 alteration controls. Twelve cases bind rows and eight remain unresolved or inconsistent. The declaration records its provenance as a supplied synthetic assumption.

world is insufficient. For example, if one world requires pay and another requires draw at the same first message, each has a policy in isolation but there is no common rule. The retained economic-world declaration still requires a correction ancestry compatible with its supplied world; this comparison pools independently specified alternatives rather than inferring several truths from one corrected report.

The prospective domain is formed before selecting favorable source outcomes, using the original source heads and the declared complete worlds. Adding the reported receipt can expand this domain beyond the twelve-key enumeration allowance. The larger domain is retained intact. A size-limit result says that the complete policy population has not been enumerated; it says nothing about whether a policy exists. The original same-world observation-correction example retains exactly its 16 policies, and pairing its messages with a constant reported receipt preserves that number.[21]

17.27.1 An action assignment or a contradiction can settle existence

The two available actions give a smaller logical question than enumerating every rule. Associate a Boolean variable x_i with each complete decision key: 0 denotes pay and 1 denotes draw. A financial row refused at the first decision forbids one assignment $x_i = a$. A row refused at the second decision forbids the joint choice of its first and second actions, $x_i = a$ and $x_j = b$. A rule avoids those refusals exactly when it satisfies, respectively,

$$x_i \neq a, \qquad (x_i \neq a) \ \vee \ (x_j \neq b). \tag{17.40}$$

The second clause says that at least one of the actions leading to the refused row must change. It is equivalent to the implications

$$(x_i = a) \Rightarrow (x_j = 1 - b), \qquad (x_j = b) \Rightarrow (x_i = 1 - a). \tag{17.41}$$

A unary refusal similarly implies $(x_i = a) \Rightarrow (x_i = 1 - a)$. These clauses preserve the original financial rows and observation keys. They change how existence is decided, without changing which decisions are allowed.

A proposed total assignment is checked directly against every refused row. If none is reached under that assignment, it supplies one common policy for the resolved financial family. It does not enumerate other policies or choose an optimal one. Conversely, suppose implications justified by refused rows lead from $x_i = 0$ to $x_i = 1$ and also from $x_i = 1$ to $x_i = 0$. Either value contradicts a consequence of choosing it. The two paths therefore certify that no assignment can avoid all those refusals. A graph calculation can discover the paths, but verification checks each edge against its cited financial row.

In the between-date receipt-6 example, a three-edge contradiction has a direct decision interpretation. At the first numeric message X paired with reported receipt 7, paying is refused, so the rule must draw. After that action, a later message reports no trigger, inadequate coverage and receipt 6. Paying is refused there, and drawing is also refused. The two second-stage refusals make the required first draw impossible. This identifies incompatible requirements in the supplied decision problem. It does not yet tell us whether a particular economic path is impossible even with full information, or whether indistinguishable paths demand different actions.

[21] `data/accounting-master/revised-source-actor-messages-20261027/attempt-02/` preserves 47 comparisons, 394 assertions and 20 alteration controls. Independent reconstruction checks the dated reports, complete first-stage memory, protected policy population and shared-action restrictions. Unresolved sources and worlds retain their separate statuses.

The certificate comparison decides thirteen domains above twelve keys, including a satisfying rule for a 30-key reverse-receipt declaration and contradictions for receipt-6 declarations with 38 keys and a concurrent-correction declaration with 54 keys. The previous enumeration results remain unchanged. Missing world rows receive no certificate. A contradiction found within a resolved subset refutes that subset, while a satisfying assignment there cannot establish feasibility for unresolved alternatives.[22]

These results make the next question more specific. A rejected common rule calls for examining the financial histories responsible for its contradiction. That diagnosis must distinguish economic infeasibility from an observation that hides a consequential difference before proposing a repair to the model or its information assumptions.

17.28 When the financial history itself cannot be settled

The receipt-6 contradiction contains an economic impossibility that additional information cannot remove. Its two second-stage refusal rows share the exact rate point 19/320 and 43/800. Under the existing floating-coupon formula, these rates give $I_1 = 95/16$ and $I_2 = 43/8$. Checking all four action sequences at this one point avoids confusing two different financial histories that happen to produce the same message.

Opening cash $11/2 = 88/16$ cannot pay the first coupon 95/16, so either sequence beginning with pay is refused immediately. Drawing supplies one unit and costs 1/4, leaving first-settlement cash 5/16. Receipt 6 followed by repayment of the borrowed principal leaves 85/16, while the second coupon is 86/16. Draw/pay therefore fails by 1/16. Draw/draw also fails: the once-only facility has already been used, and the second coupon is below its strict draw trigger. All four sequences fail even for a manager who knows both coupons in advance.

At the same rate point with receipt 7, draw/pay succeeds and ends with cash 15/16. This is an explicit counterfactual, not a replacement for the assumed receipt-6 world. The earlier viable receipt-6 examples also remain valid. The diagnostic establishes one impossible point in the original family, rather than impossibility at every point. Later algebraic cash from a sequence whose first payment was refused is not an executed outcome.[23]

17.28.1 An exact boundary on the committed-right coupon domain

Keep the committed right, opening cash, once-only borrowing rule, fee and repayment timing unchanged. Let both coupons range over $[3, 6]$, and take R to be either 6 or 7. If $I_1 \le 11/2$, the first payment can be made without borrowing. Cash after the receipt is then $11/2 - I_1 + R \ge R \ge 6$, enough to pay every permitted second coupon. Thus pay/pay is viable throughout this first region.

If $I_1 > 11/2$, paying first is impossible, while the trigger permits a first draw. Since $I_1 \le 6$, the net draw of 3/4 leaves nonnegative cash after that payment. Receipt and repayment then leave

$$C_{\text{before second}} = \frac{11}{2} - I_1 + R - \frac{1}{4} = R + \frac{21}{4} - I_1. \tag{17.42}$$

[22] `data/accounting-master/policy-existence-certificates-20261027/` checks all 47 retained source/world comparisons and 260 small Boolean formulas against exhaustive assignments, with 1,083 assertions and 22 alteration controls. The 31 certificates comprise six assignments and 25 contradictions; two contradictions apply only to resolved subsets. Each small-domain result agrees with the protected full policy population.

[23] `data/accounting-master/policy-failure-diagnosis-20261027/` retains ten exact path comparisons, 337 assertions and seven alteration controls. Complete states and all guards agree with the independent recurrence; the two second-stage refusal rows use the same rational witness.

The first borrowing has used the facility, so the second coupon must be paid from this cash. Draw/pay is therefore viable exactly when $I_1 + I_2 \le R + 21/4$. Combining the two regions, the economically infeasible set is

$$\mathcal{N}_R = \left\{ (I_1, I_2) \in [3,6]^2 : I_1 > \frac{11}{2}, \quad I_1 + I_2 > R + \frac{21}{4} \right\}. \tag{17.43}$$

For receipt 6, the sum boundary is 45/4. Equality is viable: the second payment leaves exactly zero cash. Equality at the first trigger is also viable, because $I_1 = 11/2$ can be paid without drawing. For receipt 7, the sum boundary is $49/4 > 12$, above the largest possible coupon sum, so $\mathcal{N}_7$ is empty on this domain.

The result concerns the continuous coupon square. Exact affine certificates check every inherited guard of pay/pay in the first region and draw/pay in the second, and exclude every action sequence in $\mathcal{N}_R$. The three regions are exhaustive and disjoint because they split first on $I_1 \le 11/2$ and then on the sum inequality. Rational boundary examples illustrate the result, while the region certificates establish it.[24]

On the viable subset, the original refined first trigger suffices for a common rule: draw first exactly when that trigger is true, then pay second. The manager does not need the future coupon to choose the first action. This rule cannot settle the excluded points, which remain part of the original model. The result does not cover absent or mixed borrowing rights, arbitrary receipts, or a changed facility contract.

A financially impossible settlement is still a possible company history to represent. The unpaid coupon, the earlier draw and repayment, and the dates of failed payments must remain available to subsequent accounting and recovery events. Refusing a payment instruction must therefore preserve the outstanding obligation and the failed history rather than remove it from later statements.

17.29 Keeping an unpaid coupon through close and later payment

The refused second payment leaves a complete financial state to report. In the receipt-6 draw/pay history, the company has cash 85/16 and owes the full second coupon $43/8 = 86/16$. The difference 1/16 measures the additional cash needed to pay in full; it is not the outstanding liability. No part of the refused payment was executed. The accepted history ends after the second coupon accrual, with the facility principal repaid and the once-only borrowing right still used.

The retained report contains income 6, expenses 185/16 and loss 89/16. Assets are 1597/16, liabilities 843/8, and equity including the loss is $-89/16$. These reconcile because $843/8 - 89/16 = 1597/16$. Operating cash movement is $-3/16$. Financing cash movement is zero, but its gross record contains both a draw of 1 and repayment of 1. Netting those entries would conceal the use and repayment of the facility. The refused event does not enter the accepted-event history; its diagnostic and all unexecuted commands remain separately available.[25]

[24] `data/accounting-master/financing-viability-boundary-20261027/` checks two receipt domains, 366 guard implications, eight action-region exclusions and eight boundary examples, with 315 assertions and ten alteration controls. Emptiness certificates are verified independently of their producer.

[25] `data/accounting-master/refused-payment-reporting-20261027/` preserves the failed first verifier attempt and the corrected successful second attempt: ten cases, 396 assertions and twelve adverse controls. The method wrapper retains a refusal diagnostic, but the refused transition contains no after-state, journal or cash flow. Complete states and reports agree with independent arithmetic and the symbolic recurrence.

17.29.1 Closing the period preserves the debt and the missed decision

The stopped state is dated 31 December 2027, the last day of the declared period. A selected closing operation transfers the temporary income and expense balances into equity and opens the next period on 1 January 2028. Under the signed-book convention, the loss produces a debit capital balance 89/16, representing negative equity −89/16. Cash remains 85/16, the signed interest-payable balance remains −43/8, and the second coupon's amount paid remains zero. The facility balance stays zero and its used flag stays true.

The opening period has no cash flows yet. Earlier flows remain in the closed book, and an associated archive retains the original refusal, the unexecuted commands and the identity of the new opening. Carrying balances alone would lose the distinction between a missed payment and an ordinary unpaid accrual. Conversely, automatically resetting the used flag would invent a fresh borrowing right. The generic typed carry engine can express such a reset, but this selected coupon-preservation rule rejects it. States stopped on 30 June are left unassessed by this year-end operation; they are not silently advanced to December.[26]

17.29.2 A new instruction can reduce the claim without rewriting history

Suppose an explicitly supplied instruction pays 85/16 on 2 January 2028. This amount uses the carried cash; it is not supplied by a new loan or inferred managerial policy. The selected late-payment method reduces cash and the unpaid coupon by the same amount, and adds that amount to cumulative payment toward the second coupon. Let C be current cash, U the positive unpaid coupon and P its cumulative amount paid. For an instructed partial payment q, the accounting and memory changes are

$$C' = C - q, \qquad U' = U - q, \qquad P' = P + q, \qquad 0 < q < U, \quad q \le C. \tag{17.44}$$

Here $U = I_2 - P$, with the originally accrued coupon I_2 unchanged. Thus $U' + P' = I_2$: payment discharges part of the existing claim and creates no second expense. The strict inequality $q < U$ gives this selected method a genuinely partial settlement domain. Full settlement would require a distinct supported instruction and sufficient cash.

For the supplied instruction, $C' = 0$, $U' = 1/16$ and $P' = 85/16$. The earlier action remains `none`, because the original second payment was refused. A new `partial2` phase records the later settlement, while the used right and original refusal remain intact. Reusing the old policy solver with this new phase would require a separate compatibility argument; cumulative payment does not turn the original failed policy into a successful one.

Two supplied payments of 85/32 on 2 and 3 January produce the same final balances and cumulative payment. They produce two dated cash flows and a different last-event date, so equality of financial balances is not equality of histories. The report classifies these payments as operating under the inherited synthetic convention. Complete comparisons on both account charts retain the distinction between the two instruction schedules.[27]

The three operations answer different questions: a refusal states what was not executed, a close carries the accepted balances into the next period, and a later instruction supplies a new

[26] `data/accounting-master/coupon-carry-20261027/` compares six final-day carries and four nonfinal refusals, with 406 assertions and sixteen adverse controls. The checks cover the entire closing journal, closed and opening books, report and refusal archive. The close is provisional recorded-balance carry, not evidence of complete fiscal reporting.

[27] `data/accounting-master/coupon-partial-settlement-20261027/` contains eight complete-state, journal, flow and report comparisons, 446 assertions and twenty adverse controls, with native financial fallback disabled. Source identities, selected-method identities and the full original refusal archive are checked. These finite cases support the stated selected implementation; they are not a universal conformance proof for every partial-payment schedule.

event. None supplies missing contractual terms. Penalties, waivers, legal default, additional funding and actual accounting applicability require their own evidence and selected rules. Event completeness and external source authenticity also remain unresolved. The simulator can therefore preserve the distressed company history without inventing its recovery.

17.30 Which payment histories have the same future consequences?

The residual claim is a property of the cash-only payment rule, not an accident of dividing the payment into one or two instalments. Let $S_i = \sum_{j=1}^{i} q_j$ be cumulative payment after i accepted instructions. Each positive amount q_j reduces cash and the unpaid coupon equally and increases the paid amount by the same quantity. Starting from $C_0 = 85/16$, $U_0 = 86/16$ and $P_0 = 0$, repeated substitution gives

$$C_i = \frac{85}{16} - S_i, \qquad U_i = \frac{86}{16} - S_i, \qquad P_i = S_i, \qquad U_i - C_i = \frac{1}{16}. \tag{17.45}$$

The zero-payment prefix satisfies these identities. If they hold after i instructions, subtracting q_{i+1} from cash and claim, and adding it to payment, replaces S_i with $S_i + q_{i+1}$ in all three expressions. This proves the recurrence by induction. It also preserves $U_i + P_i = 86/16$, the originally accrued coupon.

Because the unpaid claim always exceeds cash by $1/16$, any positive affordable payment $q \le C_i$ is strictly below U_i. For a finite schedule of positive payments, affordability is therefore equivalent to its total being at most $85/16$: every earlier prefix sum is smaller. This statement concerns the economic amounts. Dates, identities and runtime support impose separate conditions. At cash exhaustion the company still owes $1/16$; further cash-only payment cannot remove that claim. A separate cash inflow or a supported change to the claim would be needed.

The selected method's actual expressions, account bindings, cumulative memory write and preserved fields match this recurrence on both charts. Twenty complete runtime comparisons include nonuniform and same-day payments, unequal rational fractions, a tiny first payment, and a schedule leaving exactly $1/2^{64}$ in cash. The last residual is retained rather than rounded to zero. These finite checks support the selected implementation while the induction establishes the algebraic result.[28]

17.30.1 A date or an event identity can distinguish equal balances

Now compare a payment of 1 on 2 January with payments of $1/2$ on 2 and 3 January. Both leave cash $69/16$, unpaid coupon $35/8$ and cumulative payment 1. Their contract memory agrees. Their current dates, accepted event identities and dated gross flows do not.

A fresh instruction dated 2 January can follow the first history. It cannot follow the second, whose last accepted event is already dated 3 January. Similarly, an instruction using the second instalment's event identity can be fresh in the one-payment history and a replay in the split-payment history. The runtime rejects the replay without changing balances. Thus equal financial totals do not preserve the full set of permissible instructions.

[28] `data/accounting-master/coupon-schedule-invariant-20261027/` retains four method inspections, 517 assertions and eight controls. The runtime permits at most 64 commands per program, bounds its accumulated trace at 8 MiB, and imposes record-size and exact-rational limits. A 65-command schedule whose total equals available cash is refused by the command limit. Such a resource refusal is not economic infeasibility; satisfying one limit alone does not guarantee execution.

A common fresh payment of 1 on 4 January succeeds in both histories and leaves cash 53/16, claim 27/8 and cumulative payment 2. Numeric statements and activity totals agree, but complete reports retain different dated gross flows. This gives a useful restricted result. For the same original obligations and rules, equal cumulative payment, identical future amounts, dates no earlier than either current date, and identities unused in either history, the financial recurrence remains equal whenever bounded execution accepts. It does not make the complete histories equal or establish compatibility with every future method, observation or policy.[29]

17.30.2 Resume from the current book and report from the period opening

Execution and reporting need different dates here. The original period opening supplies the starting cash and empty current-period flow history for a cumulative report. The current book supplies the balances, last event date, used identities and contract memory for the next instruction. Replacing one with the other would either forget accepted events or misstate the reporting interval.

The present declared-program runner validates a fiscal opening at the period start with no flows. It therefore refuses a midperiod current book as its opening. The selected continuation uses the same typed event engine directly for one instruction, retains the actual prefix unchanged, and reports the resulting book against the original period opening. This is a checked single-step interface. A general program-resumption interface still needs explicit bindings for both books, the verified prefix and the remaining commands; it cannot be implemented by resetting a date or discarding prior flows.

17.31 Resuming selected payments without losing their history

A selected same-period runner now extends the checked coupon prefix through several supplied instructions. Its plan retains the original period opening for reports and the current book for execution. It also identifies the original source, selected method registry, verified prefix, complete future instruction list and resource limits. A checkpoint's cursor counts only accepted new instructions. The earlier coupon refusal remains in its archive; it is not replaced by the outcome of a later payment.

For a successful three-instruction suffix, processing one event per call, two per call or the whole suffix produces the same terminal checkpoint. The complete accepted journals, dated flows, current book and cumulative report agree with independent arithmetic and uninterrupted execution. This comparison also covers a later unaffordable payment, duplicate identity and backwards date. In each case, execution retains the accepted prefix and leaves the refused instruction first in the remaining list.[30]

An execution pause has a different meaning from a refusal. A call may consume its allowed number of instructions while another valid instruction remains; this returns a budget pause and can continue from the checked cursor. An economic or chronology refusal stops before the

[29] `data/accounting-master/coupon-continuation-scope-20261027/` retains 24 complete continuation comparisons, 514 assertions and ten controls on the four chart/action origins. Original refusal archives and complete reports are checked against the external identified prefix. Altering a prefix and assigning it a new identity does not bypass the independent comparison.

[30] `data/accounting-master/coupon-program-resume-20261027/` retains sixteen cases and 48 chunk comparisons, with 824 assertions and eighteen adverse controls. A separate audit adds 493 assertions, twelve controls and two exact byte-boundary comparisons. The first attempt is preserved: its comparator incorrectly required identical duplicate-refusal wrappers. The generic runner detects a duplicate command before calling the event engine; the selected runner retains the engine's duplicate-event diagnostic. Both reject the same instruction and preserve the same financial state.

offending event, which cannot be silently skipped on a subsequent call. Changing that event requires a newly identified plan. A complete, refused or resource-limited checkpoint remains unchanged when submitted again under its existing plan.

Resource limits apply to the entire selected suffix, not afresh to each chunk. The inherited command ceiling is 64 and the default trace allowance is 8 MiB. A deliberately smaller allowance exposes the boundary exactly: 82,883 bytes fit the first accepted trace row, but not the second. An allowance of 82,882 bytes accepts no row. The runner retains the next command and reports the unchanged accepted state in both cases. Splitting execution into additional calls does not enlarge the allowance. These limits govern this diagnostic implementation; they neither determine the financial shortfall nor prove that a larger economically feasible schedule is impossible.

The checkpoint's hash is necessary for identifying the supplied record but is insufficient for validating it. The selected interface recomputes the bounded accepted prefix from the identified source, registry and instruction list, then compares the complete checkpoint. This detects altered request metadata even when a narrower financial comparison would still find the same balances. It also detects a changed opening, skipped cursor, missing trace, erased refusal, modified report or enlarged plan budget. Recomputed validation does not append a second accepted event or duplicate a cash flow.

This evidence establishes a selected same-period partial-coupon interface. It does not migrate the general program runner, support every method or reporting-period transition, or establish durable exactly-once commits under concurrent writers. Repeated prefix computation is also a cost, not a scalability result. The distinction between original opening, accepted current book and remaining instructions nevertheless supplies a concrete starting point for broader resumption contracts without erasing the company's earlier history.

Chapter 18

Reference computations that distinguish a useful approximation from a wrong answer

A method can be tested most clearly when a small version of its target is known exactly. This chapter constructs such targets for cash bounds, breach probabilities, policy comparison and residual minimization. The calculations are analytic reference cases. They are not a benchmark ranking and do not supply empirical evidence about a real company. Their role is to expose errors that a large simulation can conceal behind apparently stable averages.

18.1 An exact distribution from a declared prior

Return to cash $C = 90 - 2\alpha$, and now deliberately assume α is uniform on $[0, 1]$. This assumption is not inferred from the financial statements. It creates a transparent probabilistic reference. Cash is uniform on $[88, 90]$, with mean 89 and variance $4/12 = 1/3$.

For a covenant requiring cash at least 89.4, breach occurs when $\alpha > 0.3$. Its probability is 0.7. The expected cash conditional on breach is the mean of the interval $[88, 89.4)$, namely 88.7. These quantities answer different questions: unconditional mean, breach frequency and severity conditional on breach.

A simulation with the right mean but the wrong breach probability is not accurate for covenant analysis. For example, a two-point approximation placing equal mass at 88 and 90 has mean 89 but breach probability one half. Matching the first moment does not preserve threshold events.

The exact distribution also gives quantiles. Under the usual left-continuous generalized-inverse convention, the p-quantile is $88 + 2p$ for $0 < p < 1$, with endpoint values defined by continuous extension. The convention should be explicit when discrete mass points are introduced, because quantile definitions can differ at jumps.

18.2 A discrete grid with known bias

Approximate α by equally weighted midpoints $(i - 1/2)/n$, for $i = 1, \ldots, n$. The mean of the grid remains one half, so mean cash is exactly 89 for every n. Breach frequency is the fraction of midpoints above 0.3. For $n = 4$, midpoints are 0.125, 0.375, 0.625 and 0.875, giving breach probability 0.75.

The mean check passes perfectly while the breach estimate differs from 0.7. More grid points need not reduce that error at every refinement. With $n = 5$, the strict breach event includes only 0.5, 0.7 and 0.9, so its estimate is 0.6. The absolute error rises from 0.05 at $n = 4$ to 0.1 at $n = 5$. At $n = 10$ the estimate is exactly 0.7, whereas $n = 11$ gives $8/11$. Alignment with the threshold matters even as the grid becomes finer.

A bound explains what refinement does guarantee. For a threshold $t \in [0, 1]$, let $p_n(t)$ be the fraction of the n midpoints strictly above t. A midpoint is at or below t precisely when $i \le nt + 1/2$. There are therefore $\lfloor nt + 1/2 \rfloor$ such points, and

$$p_n(t) = 1 - \frac{\lfloor nt + 1/2 \rfloor}{n}, \qquad \bigl|p_n(t) - (1 - t)\bigr| \le \frac{1}{2n}. \tag{18.1}$$

To obtain the bound, write $k = \lfloor nt + 1/2 \rfloor$. The defining inequalities $k \le nt + 1/2 < k + 1$ give $-1/2 \le nt - k < 1/2$. Dividing by n bounds $p_n(t) - (1 - t)$. Thus the maximum permitted error shrinks to zero, although the actual error can rise between successive grids. The result holds for this uniform midpoint rule and one threshold event; a convergence study must still examine the event actually being queried.

If the threshold lies exactly on a grid point, strict versus inclusive inequality changes the finite approximation. The continuous reference assigns zero probability to one point, but a discrete approximation assigns positive mass. This is a concrete reason to preserve boundary semantics during discretization.

A deterministic grid is not an independent random sample. Standard Monte Carlo standard errors do not describe its quadrature error. Its uncertainty should be assessed through analytic bounds or refinement, not by treating grid points as independent observations.

18.3 A nonlinear guard with a narrow region

Let α remain uniform on $[0, 1]$, but suppose a special covenant fails only when $|\alpha - 0.5| < 0.01$. The exact probability is 0.02. A coarse endpoint or midpoint grid can miss this region entirely.

An independent random sample of size 100 has probability 0.98^{100}, about 0.133, of observing no failures. A deterministic grid can have either zero or several hits depending on alignment. The two methods fail for different reasons: sampling variation versus systematic discretization geometry.

A query-aware method should detect the guard and refine around its roots. If the guard is known analytically, solving its boundary is more efficient than hoping random paths discover it. Sampling remains useful when the region is complex, but the model should exploit available structure.

The example also warns against dropping a region because its volume is small. A two per cent event can dominate expected loss if severity is large. Probability and consequence must be evaluated together.

18.4 A rare-loss mean

Suppose loss is zero with probability 0.999 and 1,000 with probability 0.001. Expected loss is 1, while the most common observed loss is zero. In 100 independent draws, the chance of no loss is 0.999^{100}, about 0.905. Most such short runs report a sample mean of zero.

The variance is $0.001(1000^2) - 1^2 = 999$. The standard error of the sample mean at $n = 100$ is $\sqrt{999/100}$, about 3.16. A single observed mean cannot support a precise estimate. A normal approximation is also questionable in a sample with very few tail events.

This target can test whether a simulator reports uncertainty honestly. A zero-event run should not declare zero risk. A rare-event method should be checked against the known probability and mean, including its weighting and interval procedure.

The example does not establish how rare real defaults are. Its parameters are chosen to make sampling failure visible. A real application needs an admitted event law and severity model, with model uncertainty separate from Monte Carlo error.

18.5 Importance sampling on the rare event

For the same two-state target, choose a proposal that samples the loss state with probability 0.1 instead of 0.001. The target-to-proposal weight is 0.01 in the loss state and $0.999/0.9 = 1.11$ in the zero state. The ordinary weighted loss contribution is therefore 10 in the loss state and zero otherwise.

Its expectation under the proposal is $0.1(10) = 1$, the correct target mean. Its variance is $0.1(10^2) - 1 = 9$, much smaller than 999 in this constructed example. The reduction follows because the proposal deliberately visits the only important rare state.

This is a mathematical comparison for a known finite distribution, not a general performance claim for importance sampling. In a high-dimensional model, a poor proposal can create extreme weights or miss support. A proposal tuned to one failure mode may neglect another. The finite case provides a correctness test for the weighting formula.

If the implementation averages the unweighted sampled loss, it obtains expectation 100 under the proposal, not target expectation 1. If it reports the proportion of sampled defaults as the target probability, it reports 0.1 instead of 0.001. These large errors can look stable as sample size grows because the wrong quantity is being estimated accurately.

18.6 Paired policy differences

Suppose two policies have costs $Y_A = Z + 2$ and $Y_B = Z + 1$, where Z is a common random operating cost. The true difference $Y_A - Y_B$ is exactly 1 in every world. Using the same draw of Z gives zero sampling variance for the difference. Independent draws add unnecessary noise.

This is an ideal illustration of common random numbers. In a real policy comparison, the policies can change transition probabilities or event timing. The coupling must preserve each policy's correct marginal distribution. Reusing a random seed alone does not guarantee that corresponding random draws represent the same economic shock after control flow diverges.

A stable shock identity can help: demand innovation for a named period and customer should remain associated with that economic event rather than the next call to a global random-number generator. Otherwise adding a harmless diagnostic draw can change all subsequent scenarios.

The paired comparison should also hold the initial history fixed. Subtracting policy A on one reconstructed company from policy B on another confounds policy and history. If the goal is a range of policy differences, optimize the paired difference over the same family.

18.7 A sampled minimum and a true minimum

Let $C(\alpha) = 90 - 2\alpha$ on $[0, 1]$. The true minimum is 88 at $\alpha = 1$. A sample whose largest value is 0.9 has minimum cash 88.2. This is an upper bound on the true minimum because the sample is a subset of the family.

If a payment requirement is 88.1, the sampled minimum suggests no sampled failure while the true family contains failure. An optimization or analytic monotonicity argument identifies it immediately. More random sampling may eventually approach the endpoint, but no finite sample alone proves the exact bound.

Conversely, a relaxed outer model allowing $\alpha \in [0, 1.1]$ gives minimum cash 87.8. That is a lower bound on the original minimum, but its witness is not a feasible original company. Inner and outer methods can bracket the answer. The bracket's width, not a single attractive number, describes unresolved computation.

18.8 A local optimizer misses another basin

Consider an illustrative cash function $C(x) = (x^2 - 1)^2 + 0.1x$ on $[-2, 2]$. It has regions near both negative and positive one where cash is relatively low. A local optimizer initialized near one basin may not find the global minimum.

The function is a numerical diagnostic, not a proposed cash-flow mechanism. Its purpose is to test whether a method labels a local result honestly. A global claim requires exhaustive interval bounds, verified comparison of all stationary and boundary candidates, or another suitable certificate.

In a financial model, analogous nonconvexity can arise from fixed issuance fees, default options, tax thresholds or discrete investment. Repeated random starts can find more candidates but do not automatically prove completeness. The result should distinguish found local solutions from a verified global bound.

A benchmark should include such a case only if the target profile admits nonconvexity. Requiring a linear solver to solve it would be an unfair capability test. Conversely, describing a general solver while testing only convex examples leaves a material gap.

18.9 Residual scaling can alter a compromise

Suppose an approximation cannot satisfy two residuals simultaneously: $r_1(\theta) = \theta$ and $r_2(\theta) = \theta - 1$. Minimizing $r_1^2 + r_2^2$ gives $\theta = 1/2$. If the first residual is multiplied by 100 because currency units changed, minimizing $10000\theta^2 + (\theta - 1)^2$ gives $\theta = 1/10001$.

The different result is caused by weighting, not new economic information. In a well-specified exact system with sufficient capacity, both residuals may be zero and scaling would not change the exact solution. Under approximation, inconsistency or finite optimization, scaling affects the compromise.

This diagnostic motivates reporting component residuals in original units and testing unit invariance. A normalized aggregate loss can be useful for training, but it should not hide a cash pricing error or constraint violation. A single loss threshold needs an economic interpretation.

18.10 Expected residual versus mean squared contribution

The earlier example $r = \theta - 1 + \theta\varepsilon$ with mean-zero unit-variance innovation gives desired squared expected residual $(\theta - 1)^2$ and single-draw squared objective $(\theta - 1)^2 + \theta^2$. Their minimizers are one and one half.

A reference implementation can evaluate both objectives analytically and by Monte Carlo. It should verify that independent paired draws estimate the former expectation and identical draws estimate the latter. The test should compare the target functions, not merely whether training loss decreases.

A gradient check can add another layer. The desired derivative is $2(\theta - 1)$; the other derivative is $4\theta - 2$. At $\theta = 1/2$, the second vanishes while the first is negative one. Automatic differentiation can return the correct derivative of the wrong implemented objective.

This case is particularly relevant to learned equilibrium conditions because it separates stochastic estimation from target definition. Increasing batch size makes the wrong objective estimate more precise; it does not repair the missing expectation structure.

18.11 A boundary optimum

Minimize $(a + 1)^2$ subject to $a \geq 0$. The answer is $a = 0$, objective 1. An interior first-order condition gives $a = -1$, objective zero, which is infeasible. A constrained solver should return the boundary with the appropriate multiplier.

A penalty objective $(a + 1)^2 + \rho \min(a, 0)^2$ has a negative optimum for every finite positive ρ, namely $a = -1/(1 + \rho)$. The violation shrinks as the penalty grows but is not exactly zero. Reporting the penalized optimum as a feasible original solution without checking the constraint is wrong.

Penalty methods can be useful algorithms. Their residual feasibility must be assessed against the original constraint and the required tolerance. A small violation may be immaterial for some economic quantities and decisive when it represents issuing more shares than authorized or paying cash that does not exist.

18.12 Designing a fair method comparison

A comparison should use the same evidence family, objective, policy class, timing, terminal condition and accuracy target. A faster method that omits a recovery term is not a faster solution of the original pricing model. A method trained on a smaller domain should not be compared as though it covers the full stress region.

The baseline ladder can include exact finite enumeration, a refined classical numerical method, the plain proposed approximation and any enhanced version. Tuning budgets and stopping criteria should be declared. A deliberately weak comparator can establish only a narrow improvement over that comparator.

For stochastic methods, use replications and uncertainty analysis appropriate to the claim. If several candidates pass hard validity screens but differences are not statistically resolved, describe them as viable under current evidence. Runtime, tail error and objective values from a few seeds remain descriptive unless the predeclared analysis supports ranking.

18.13 What to retain from a run

A run should preserve the model and data version, command, environment, seeds, hardware mode, wall time, outputs and the plan defining the question. It should record which diagnostics are validity vetoes, which trigger repair and which merely explain behavior.

The result note should identify whether a failure invalidated the harness, implementation, target or candidate. A candidate failing a planned stress can motivate a repair phase. It does not automatically reject the broader research direction. A broken harness, by contrast, prevents interpreting the candidate comparison.

For the present manuscript, these are proposed experiments and exact reference calculations. No stochastic ranking is claimed. The retained arithmetic script checks selected deterministic examples and records its scope. The package companion `companions/reference-computations` reproduces its complete checked values and adds the midpoint-boundary examples above. The all-threshold conclusion rests on the floor argument in equation (18.1), not on the number of rational examples tested. Future numerical runs should use these cases to establish semantics before making claims about scalable performance.

Chapter 19

What the literatures contribute, and where their assumptions stop

The simulation problem crosses several literatures because it combines inference from incomplete reports, accounting representation, dynamic corporate finance, sequential information and computational approximation. No inspected source supplies the entire system. The useful synthesis identifies the problem each source actually solves and the additional work needed for AccDSL. A source's notation or software can be valuable without becoming a universal model of a disclosed company.

19.1 Economic events and accounting views

McCarthy's REA model provides a useful architectural antecedent. Resources, events and agents, linked through stock-flow, duality and control relations, organise economic activity before particular accounting views are produced (McCarthy, 1982, pp. 560–570). Double entry can be one external view. Claims can be derived from imbalances or retained as base records when their uses warrant it. The paper also recognises the practical need for aggregation and the fact that measurement and periodicity are not settled by a database schema (McCarthy, 1982, pp. 568–576). These distinctions support an event-and-contract representation with journal views. They do not imply that disclosures identify a unique event history, or supply the likelihood of observing a disclosure.

19.2 Inverse problems and incomplete information

Stuart (2010) develops Bayesian inverse problems with explicit forward models, priors and analytical conditions. It is relevant because statements are observations generated by hidden economic histories under a reporting map. The source does not reconstruct company journals or justify a particular prior over contracts. Its contribution is a disciplined distinction between the observation relation and the assumptions needed for inference.

Libkin (2014, 2016) studies incomplete databases and answers across possible worlds. This perspective fits the requirement to preserve alternatives without assigning probabilities. A certain answer holds across admitted worlds; a possible answer has a witness. Translating that logic to finance requires a domain of economic histories and valid accounting relations, not merely null values in a table.

The two approaches are complementary. A feasible family expresses what has not been ruled

out. A probability measure expresses relative weight under a model. A Bayesian posterior can be computed on a family when priors and likelihoods are justified, but the family itself should not be discarded when those assumptions are uncertain.

19.3 Set identification and prior sensitivity

Giacomini and Kitagawa (2021, Sections 2.1–2.3) separates parameters identified by the observable distribution from remaining directions whose conditional priors are not revised by data. The retained technical reading uses the April 2020 working-paper version associated with the later publication. That version identity matters when citing equations and claims.

For company simulation, the immediate implication is conditional: first determine which hidden features the observation model identifies. A long history of interest expense may leave fixed-versus-floating terms unresolved when rates have not varied. Additional data of the same uninformative kind need not eliminate the ambiguity.

The local history family is not automatically the identified set of the source's statistical model. Finite compatibility with reports is weaker than equality of full observable laws. The manuscript therefore uses report-compatible histories unless the stronger identification argument has been established. This avoids borrowing a technical term to imply more evidence than exists.

19.4 Model averaging

Hoeting et al. (1999) surveys Bayesian model averaging, in which uncertainty across models is represented by posterior weights. This can be useful when alternative economic or observation models are explicitly specified. It also makes clear that averaging requires model priors and likelihoods, not branch counts.

The practical difficulty for AccDSL is defining model identity. Ten syntactic encodings of one contract should not receive ten times the weight of a single encoding of another. A prior over economic structures must be invariant to irrelevant representation choices or at least expose its dependence.

Model averaging can also conceal a decision-relevant minority case if the report shows only a mean. Conditional outcomes, tail measures and sensitivity to prior weights remain important. When weights are not defensible, the unweighted family is the appropriate retained representation.

19.5 Causal scenarios

Pearl (2009, Sections 3.2–3.4) distinguishes observational conditioning, interventions and counterfactual reasoning. This distinction matters whenever a financial scenario is described as the effect of a shock. A model fitted to periods with high rates need not identify the effect of changing rates while other mechanisms remain fixed.

AccDSL can support deterministic stress paths without claiming causal identification. Such a run answers a conditional mechanical question. A causal claim additionally needs structural equations, intervention semantics and assumptions about confounding and invariance.

Counterfactual comparisons require a coupling between factual and alternative worlds. Preserving the same hidden history and exogenous innovations is one modeled coupling. Independently drawing a new company for each policy changes the comparison. The causal literature supplies the conceptual distinction; the company-specific structure remains to be justified.

19.6 Corporate microsimulation

Shahnazarian (2004) links financial stocks through accounting bridges and separately specifies behavioral relationships for incorporated businesses. The inspected sections include stock movements, investment, disposal, depreciation and tax-related distinctions. The source's Swedish data and tax setting are not modern universal issuer defaults.

Its relevance is methodological: identities constrain movement while behavioral equations determine choices. A simulator can preserve that separation while using different data, accounting rules and policies. Importing estimated coefficients without target-specific evidence would be an unsupported transfer.

The source also illustrates why state includes more than current statement totals. Tax reserves, capital and other accumulated quantities affect future responses. AccDSL's state design should identify such memory by mechanism rather than simply copying a fixed list of variables from one model.

19.7 Accounting in system dynamics

Pierson (2020) develops accounting reporting within a system-dynamics model. Accrual, payment and reporting accumulation are separate, and the retained author model exposes equations for cash, receivables, payables and report timing. These are useful implementation concepts.

The source's payables payment rule uses a delay approximation, and its financing and inventory mechanisms have stated limits. A first-order payment delay is not an invoice-level due-date schedule. Reporting values expressed as annualized rates must be converted before comparison with period amounts.

The source therefore contributes a bridge between continuous simulation stocks and discrete reports. It does not establish a complete instrument-level financing engine or empirical validity for a particular company. AccDSL should reuse the separation of clocks while admitting more detailed schedules when the query requires them.

19.8 Agent-based stock-flow consistency

Caiani et al. (2016) models heterogeneous agents, loans and capital vintages with an explicit within-period sequence. The retained paper and selected official code show production, credit, payments, taxes and distributions connected through counterpart balance changes. This supports the importance of event ordering and financial counterparts.

The source begins from an artificial economy with its own initialization and calibration. That is not a reconstruction of an observed issuer from disclosures. Its reported simulation properties are author results, not experiments reproduced for AccDSL.

A particularly useful source-to-target warning concerns its operating-cash-flow label: the inspected definition subtracts scheduled principal repayments. That quantity can be appropriate for its credit-demand mechanism, but is not automatically the same as a financial-statement operating cash-flow category. Definitions must travel with values across adapters.

Selected code inspection is also bounded. The retained implementation revision exposes event switches, debt payment calculations and bankruptcy paths. It does not certify every formula, the entire runtime or correspondence with the paper-era executable. Before reuse, zero-rate schedules, timing, numerical domains and accounting adapters need targeted checks.

19.9 Financial planning and simultaneous equations

The retained Gentry working paper studies strategic financial planning from management inputs such as earnings-per-share growth, investment growth, leverage and financing costs (Gentry, 1974). It asks an inverse planning question about returns needed to achieve objectives. Its managerial distributions and independence assumptions are part of that model, not facts about every company.

The retained scan lacks a referenced derivation appendix in the inspected trailing pages, and correspondence with the later journal version has not been established. The manuscript therefore does not rely on an unchecked derivation as a validated algorithm. This is a source-specific limit, not a reason to dismiss the planning approach.

Gentry and Hoque (1990) organizes cash-flow components and estimates simultaneous behavioral equations. Its accounting reconciliation, definitions and behavioral relationships perform different jobs. The retained reading identifies a normalization inconsistency in one displayed cash term and records the local algebraic normalization explicitly. It does not silently claim a corrected edition.

The reported sample fits do not establish out-of-sample intervention validity. Firm-level data, a complete instrument matrix and code were not available for local replication. The useful comparison is architectural: a simultaneous system can propose related annual amounts, while a scenario engine still needs dated events, feasibility, information and accounting treatment.

19.10 Dynamic corporate finance

Strebulaev and Whited (2012) surveys dynamic corporate-finance models with investment, financing frictions and default. The risky-debt section makes debt pricing and equity value interdependent. This is essential for any model claiming endogenous financing terms.

The retained equations also illustrate why source inspection cannot stop at a citation. The face/proceeds convention mismatch in equations (3.26)–(3.27) must be reconciled before implementation. A positive-recovery calculation exposes it; a no-default example can conceal it. The present manuscript uses distinct cash and promise symbols and retains the source qualification.

Other simplifications require equally explicit transfer decisions. Signed net debt can collapse cash and borrowing. Present-valued tax benefits are not actual tax receipts. Aggregate issuance costs do not create legal share rights. A structural model can be internally useful under these assumptions while remaining insufficient for a full accounting simulator.

19.11 Bank funding and adjustment

Halaj (2016) models portfolio choices under uncertain funding and constraints. Behn et al. (2019) links bank behavior to capital and liquidity restrictions using its own supervisory-data setting and structural assumptions. These sources show that funding is constrained behavior, not an automatic balancing liability.

A corporate issuer does not inherit their bank-specific objective or regulatory measurements. Their estimated parameters are not identified from public company statements merely because the balance-sheet labels resemble those of banks. AccDSL must distinguish accounting, prudential and economic quantities.

The transferable design principle is a policy interface that receives state, information and

constraints jointly. Funding availability can change under stress and affect asset decisions. Applying that principle to another sector requires sector-specific rules and evidence, not a relabeled bank equation.

19.12 Contract algorithms

The retained ACTUS technical specification separates schedules, state transitions and payoffs (ACTUS Financial Research Foundation, 2026). This is a promising component boundary for contractual cash flows. It does not supply operating behavior, accounting recognition, consolidation or a justified future risk process.

The inspected specification and service code leave important adapter conventions to be resolved: timezone mapping, midnight interpretation, same-date event order, day-count implementation, currency scaling and risk observations. A service fallback to current wall-clock time plus a horizon is incompatible with reproducible historical simulation unless overridden explicitly.

The selected service requests a particular core dependency but does not establish a fully resolved, matched implementation and dictionary version. No runtime adoption follows from the inspection alone. A conditional algebraic correspondence for a simple borrower loan is useful evidence, but not complete executable conformance.

19.13 Sequential decisions under partial observation

Kaelbling et al. (1998) provides the belief-state formulation of planning under partial observation with specified transition and observation probabilities. It explains how information history can be summarized for control. The relevant sections were inspected in the retained author preprint, including rendered pages because the copy is scanned.

AccDSL adds two complications. Some uncertainty is unweighted and cannot be represented as a single probability belief without assumptions. Different actors can have different private information. A public analyst's belief is not necessarily the state on which management acts.

The POMDP formulation is therefore a method for an admitted probabilistic subproblem, not the universal semantics of all company uncertainty. It is especially useful for testing nonanticipativity, signal value and belief updates in finite reference cases.

19.14 Exact aggregation and knowledge compilation

Givan et al. (2003) gives equivalence and minimization results for MDPs, including policy-preserving stochastic bisimulation. Its fixed-action counterexample explains why equal outputs under preselected action sequences can be too weak. AccDSL's local extension preserves observations and requested outputs as part of a joint successor law.

Darwiche and Marquis (2002) maps representational tradeoffs among compiled logical languages. It supports sharing and query-aware structure while warning that succinctness and tractable operations trade off. It does not prove that arbitrary financial histories can be compiled cheaply.

Together these sources support a careful answer to state explosion: share common structure, preserve joint constraints and aggregate only under a scoped sufficiency condition. They do not support the claim that replacing a tree with a graph eliminates exponential worst cases.

19.15 Filtering and approximation

Doucet and Johansen (2008) explains particle filtering, smoothing, degeneracy and conditional analytic integration in the retained tutorial version. Its technical treatment motivates weighted particles and diagnostics when a state-space probability model is specified.

Unknown static contract parameters require additional care. Repeated resampling can remove types without any evidence ruling them out. A high effective sample size after resampling does not guarantee historical diversity. The retained feasible family should therefore remain available alongside the particle approximation.

Dupačová et al. (2000, Sections 2–4) studies probability-metric scenario reduction in the retained 2000 preprint. Theorem 3.1 solves redistribution for a fixed subset; the subsequent subset-selection algorithms are heuristics. The stability results depend on the stochastic program and its regularity, while the reported electrical-load experiment measures distribution distances in its own setting. It does not establish company default accuracy or preserve every adaptive policy. The local derivations in Chapter 9 explain the resulting expected-output and policy bounds, and the discontinuous-threshold counterexample shows their limits.

19.16 Dynamic ambiguity

Epstein and Schneider (2002) derives recursive multiple-priors preferences under stated axioms and rectangularity relative to an information filtration. The definition, theorem and relevant proof structure were inspected in the retained September 2002 preprint. The result concerns a particular preference representation.

Its relevance is the danger of using stagewise worst-case recursion with an uncertainty set that restricts cross-time dependence. Rectangular enlargement can change the admitted joint laws. A fixed unknown contract cannot silently become an adversary choosing a new type each period.

AccDSL should offer commitment, original-set and recursive robust formulations only with their semantics stated. The choice can alter policy and should remain visible. Dynamic consistency is not a label that makes every robust calculation equivalent.

19.17 Learned economic solvers

Maliar et al. (2021) formulates dynamic economic equations for deep learning, with careful distinctions among residual objectives and sampling constructions. Its definitions and algorithm are relevant to the supplied lecture's proposed solver. The double-sampling derivation in this part shows why the target objective matters before training begins.

Duarte and Fonseca (2026) is a newer structural-estimation working paper with jointly solved financing and value components, lagged-network training and local refinement. Its assumptions, state and pricing conventions differ from the lecture model. Its numerical organization is a candidate, not a verified AccDSL implementation or general global-optimality result.

Official code for these learned methods has not been comprehensively audited for this local integration. Consequently, the manuscript makes no code-equivalence, convergence or performance promotion claim. Any implementation decision must add that inspection and a target-specific numerical protocol.

19.18 Disclosure selection and the observation likelihood

Imagine equal prior probabilities of a good and bad type. If good types disclose with probability 0.9 and bad types with probability 0.1, then disclosure raises the probability of a good type to 0.9, while silence reduces it to 0.1. Treating a missing disclosure as no information would retain 0.5 in both cases. The discrepancy comes from omitting the selection mechanism, not from numerical approximation. These probabilities are an authored example, not estimates of actual issuer disclosure behaviour.

Let Y denote potentially observable information, M the pattern of observed components, $f_\theta(Y)$ its model and $g_\phi(M \mid Y)$ the publication or missingness mechanism. The likelihood of the realised data is

$$L(\theta, \phi) = \int f_\theta(y_{\text{obs}}, y_{\text{mis}}) g_\phi(m \mid y_{\text{obs}}, y_{\text{mis}})\, dy_{\text{mis}}. \tag{19.1}$$

Here the realised pattern m fixes which coordinates are observed; the integration is over the missing coordinates only. Discrete missing components use a sum instead. If, for the realised observation, $g_\phi(m \mid y)$ is constant over all compatible missing components, it factors out of this integral. Together with distinct parameter spaces this yields the relevant likelihood proportionality; a Bayesian factorisation also needs the appropriate prior independence. Seaman and coauthors distinguish these realised-MAR conditions from the everywhere-MAR conditions relevant to repeated-sampling comparisons (Seaman et al., 2013, Theorems 1–3, Section 5.4 and Appendix). These are sufficient conditions for specified inferential equivalences. They do not demonstrate that corporate missingness satisfies them.

19.19 When silence has an equilibrium interpretation

A separate local illustration makes non-disclosure endogenous. Let value V be uniform on $[0, 1]$. A manager knows V with probability $1/2$, independently of value; otherwise the manager cannot disclose it. Informed disclosure is truthful and costless, the manager maximizes the contemporaneous price, and risk-neutral investors price at conditional expected value. Suppose an informed manager discloses above cutoff a. Silence occurs with probability $(1 + a)/2$. Its unconditional value contribution is $1/4 + a^2/4$, giving silence price

$$P_{\text{silence}}(a) = \frac{1 + a^2}{2(1 + a)}. \tag{19.2}$$

An informed manager at the cutoff is indifferent, so $a = P_{\text{silence}}(a)$. Multiplying through and collecting terms gives $a^2 + 2a - 1 = 0$, hence $a = \sqrt{2} - 1 \simeq 0.4142$. Types above that value disclose and those below it prefer the silence price. The uninformed manager explains why silence does not identify the worst type. This derivation is a local example; the blocked Dye original is not claimed to have been reproduced.

Nor is favorable selection universal. Suijs models a firm that knows its return but is uncertain how investors compare it with an outside opportunity (Suijs, 2007, Sections 3–4, Proposition 2 and Appendices A–B). Under the stated conditions a profitable firm can disclose lower returns and withhold higher returns. Multiple partial-disclosure equilibria are possible. The implication for AccDSL is to represent alternative disclosure mechanisms and test their consequences, rather than hard-code one direction of selection.

19.20 What changes under an intervention

A mechanical rate stress replaces an input while holding specified other relations fixed. A conditional forecast computes outcomes among states where an observed rate has a chosen value. A causal intervention changes the mechanism generating the rate and then propagates the remaining structural relations. These calculations can disagree.

For a finite counterexample, let U be a fair binary variable and observe $X = U$, $Y = U$. An alternative structural model has $X = U$, $Y = X$. Both give exactly the same observed joint distribution. Under an intervention setting $X = 1$, the first model retains $\mathbb{E}[Y] = 1/2$ while the second gives $\mathbb{E}[Y] = 1$. Observational agreement does not identify the intervention response. A forecast labelled as causal therefore needs the structural assumptions that distinguish these possibilities, or must retain the set of responses. Source corrections and simulated observations also have different time meanings: a hypothetical future disclosure updates an actor inside that run, not the analyst's actual present evidence.

19.21 What remains to be researched

The survey is broad but not exhaustive. Important remaining questions include scenario-reduction guarantees for adaptive company policies, robust filtering with static unidentified parameters, strategic disclosure, multi-creditor restructuring, real-company policy estimation and empirical forecast evaluation. Some require additional primary sources; others require derivations and data specific to the target company.

The earlier Francis–Rowell simultaneous-equation lead remains technically uninspected because attempted routes did not provide the paper. It cannot be declared inferior or irrelevant on that basis. The missing Gentry derivation and source-version questions also remain explicit.

A source gap blocks the unsupported claim, not all independent work. The methods already derived here can be specified and checked while the literature queue continues. This keeps research honest without allowing an inaccessible paper to become a reason to stop every implementation task.

19.22 The resulting method proposal

The synthesis favors a joint history family, dated contract and ownership state, actor-specific information, explicit scenario semantics, feasible policy generation, event-to-accounting translation and query-driven computation. Each component has a distinct literature basis and a distinct local obligation.

No single source validates their composition. The worked cases and proposed tests supply initial evidence for the interfaces, while broader implementation and empirical studies remain necessary. The design should therefore preserve the origin and scope of each assumption instead of presenting the synthesis as an established off-the-shelf theory.

The next chapter turns this method proposal into concrete records and algorithms. Its engineering choices remain replaceable. What must survive replacement is the meaning of evidence, uncertainty, information, financing and result status developed throughout this part.

Chapter 20

Implementing the proposed methods without losing their meaning

The preceding chapters define what a result means. Implementation should preserve that meaning across extraction, reconstruction, simulation and reporting. The engineering choices in this chapter are proposals and can be replaced. The essential requirements are stable: evidence is not an assumption, a history family is not a sample, actor information is not global state, a financing request is not cash, and numerical failure is not default.

20.1 Evidence records

An evidence assertion needs a source identifier and exact location, value or text, concept, entity, dimensions, units, basis, period and precision. It also needs publication and ingestion information, extraction method and review status. A correction links to the assertion it revises instead of deleting it.

The record should preserve original text alongside normalized values. A normalized number can be checked against the source, and an unresolved phrase remains visible. A mapping from source label to model concept is a separate interpretation with its own support. This allows the mapping to change without pretending the source changed.

An evidence view selects assertions for a particular cutoff and purpose. It can represent as-published, currently restated or analyst-reviewed information. The view should be immutable once used for a frozen result. A new correction creates a new view and an explicit dependency change.

20.2 Assumptions as first-class inputs

An assumption record identifies the quantity or relation supplied, its scope, provenance and role. Examples include a fixed management payout policy, an offer price, a prior over debt types or a terminal liquidation rule. A convenient default should be marked as a hypothesis or baseline rather than disguised as observed evidence.

Assumptions can conflict. A policy allowing unlimited borrowing conflicts with an offer cap if both claim to govern the same action. The system should detect or expose such inconsistencies. Precedence rules must be explicit; silently favoring whichever input was loaded last is unsafe.

Sensitivity should vary assumptions by identity. If one assumption supplies both customer

delay and credit loss, changing only one derived field can produce an inconsistent variant. A structured assumption relation makes the intended intervention and its dependencies visible.

20.3 History families

A history family contains shared economic records, persistent unknowns, domains, constraints, interpretation alternatives and source support. It records the domain for which completeness is claimed and the method used to construct it. Optional witnesses are members, not substitutes for the family.

Continuous variables should retain units and bounds. Discrete choices should retain mutually exclusive premises where appropriate. Cross-time identity should connect a loan, customer or asset across periods. A new contract version is a dated event, not a reassignment of an old unknown.

The family can expose queries for feasibility, witnesses, possible answers, certain answers and bounds. A probability operation requires an attached measure record. The API should not offer an unconditional mean for an unweighted family without requesting or selecting an explicitly declared measure.

20.4 Contract and security state

A contract record identifies parties, currency, dates, obligations, rights and clauses. Its execution state records remaining principal, accrued amounts, past exercise and breach status. Accounting carrying values are separate measurements linked to the contract.

A security register identifies classes, units, holders or categories, rights and settlement status. It distinguishes authorized, subscribed, issued, outstanding, treasury and potentially issuable units where the profile needs them. Conversion links the extinguished claim to created shares.

Unknown clauses should remain typed unknowns or alternatives. An unsupported clause should block only claims depending on it, with an explicit reason. A simple loan profile can still calculate unaffected payments while refusing a general default or option valuation. This is more useful than either inventing a default clause or refusing every calculation.

20.5 Actor information

Each actor has an information specification: observable variables, private facts, report channels, lags and memory. Policies receive a view constructed from that specification. They should not receive the full simulated world and rely on programmers to remember which fields are forbidden.

A policy invocation records its decision time and input view. Tests can then check that changing unobserved future data leaves the action unchanged. A manager's private contract knowledge can be included without exposing it as actual public evidence.

Randomized policies need a declared randomization source independent of unavailable future information. Reusing a shock draw as a policy coin can create unintended dependence. Economic random variables should have stable identities and explicit coupling, not depend on incidental call order.

20.6 Scenario definitions

A scenario identifies external paths or distributions, interventions, horizon, event timing and policy treatment. It distinguishes a deterministic stress from observational conditioning and a causal intervention. A scenario can also define alternative future observation processes.

Shock units should be typed: a two-percentage-point rate increase differs from a two per cent proportional increase. Currency rates need orientation. Price levels differ from returns. The scenario layer should reject ambiguous conversions rather than rely on labels.

A scenario version should include the underlying risk-process and observation versions. Changing a future disclosure rule can change policy values even when the rate path is unchanged. The cache and provenance system must recognize that dependency.

20.7 Event proposals and admission

A policy proposes an action. The event service checks contractual rights, resource availability, counterparty acceptance and timing. An accepted action creates dated events; a rejected action returns a reason and leaves the state unchanged except for any explicitly modeled attempt cost.

Financing proposals should include requested cash or quantity, eligible offers and settlement conditions. The service calculates proceeds, fees, promises and ownership consequences. It does not infer unlimited supply from a balance identity.

Admission can be conditional over a history region. If an unknown covenant makes an offer feasible in some histories and infeasible in others, the service returns guarded outcomes. It should not choose one witness merely to avoid branching.

20.8 Ordering events

The scheduler orders economic events by effective time and declared precedence. Same-time events can still have dependencies: proceeds must arrive before they fund payment, and a reset can occur before or after accrual according to contract terms. A total ordering chosen arbitrarily by event identifier can change results.

A partial-order representation can express independent events and necessary precedences. If simultaneous equations are genuinely required, they should be passed to a solver rather than forced into a sequential order. The result should distinguish an unresolved ordering from numerical nonconvergence.

Time conversion needs explicit precision and timezone rules. A local midnight, 23:59:59 and next-day midnight are not interchangeable in a boundary-sensitive query. The existing exact timing work in the repository is a foundation; external contract adapters must preserve its distinctions or state their loss of precision.

20.9 Accounting rule application

After an admitted economic event, the accounting service selects a rule under the declared basis, jurisdiction, entity facts and effective period. It produces a complete posting bundle and supporting measurements. Missing authority or facts can yield conditional treatments or a typed refusal.

Posting bundles are validated atomically. They reconcile debits and credits, currencies, account contexts and event-specific quantities. An event identifier prevents duplicate application. Reversal and correction events refer to the original bundle rather than editing it invisibly.

The reporting service aggregates within a specified perimeter and period. It preserves source-to-event-to-journal-to-line provenance. A consolidated report can eliminate internal balances while retaining the underlying entity records for liquidity and legal-entity analysis.

20.10 Symbolic execution

A symbolic state contains expressions and constraints rather than one numeric company. An event whose rule is common across the region can update expressions directly. A guard partitions the region only when its truth changes the relevant event or output.

The executor should preserve coverage of the parent region and distinguish disjoint alternatives from overlapping descriptions. Overlap can duplicate probability mass if weights are attached. A symbolic union should therefore use semantic region identity rather than merely concatenating records.

Simplification can share subexpressions and eliminate variables under checked projection rules. Any approximation records an outer or inner relation, error target and scope. A witness extracted from an outer approximation must be validated before being presented as a possible history.

20.11 Optional probability measures

A measure record identifies support, weights or densities, parameterization, conditioning variables and source assumptions. It is attached to economic variables and histories, not to the number of stored nodes. Duplicate encodings should not change the measure.

Updates record likelihood and evidence identity. Reusing the same evidence twice should be detected where the model treats it as one observation. Correlated sources require a joint observation model or an explicit approximation.

Simulation can carry both the retained feasible family and an approximate measure. A particle filter's support loss does not delete the original family. A user can request probability estimates under one measure and robust bounds over the family without conflating them.

20.12 Query planning

A query names outputs, dates, actors, scenarios, policies and acceptable guarantees. The planner traces dependencies through contracts, guards, observations and reports. It selects exact symbolic calculation, optimization, enumeration or approximation according to the admitted profile.

The planner should prefer the smallest method that answers the question. An affine interest range does not require a million paths. A policy-sensitive signal problem cannot be answered by a fixed-path cash table. A worst-case query cannot be answered by an unlabeled Monte Carlo minimum.

Resource budgets are explicit. If the selected exact method cannot finish, the planner may return bounds or ask for an allowed approximation according to the request. It should not

silently downgrade a guarantee. Useful partial results include verified witnesses and certified safe regions with unresolved remainder.

20.13 Result records

A result identifies its target and scope, evidence and assumptions, conditional regions, values or formulas, measure if any, numerical status and provenance. It can contain exact values, intervals, distributions, witnesses and unresolved statuses together. Their semantics should be explicit.

Possible failure, certain failure and unresolved failure are different answers. A breach is different from default. An infeasible proposed action differs from a company with no feasible action in the modeled class. An unsupported mechanism differs from numerical failure. These distinctions should survive both machine output and reader-facing explanation.

Aggregates must retain unresolved regions or mass. If some paths fail numerically, reporting only successful paths can bias results. The system should state what portion of the admitted family or probability remains unevaluated and avoid renormalization unless that conditional target is explicitly requested.

20.14 A small request sequence

A user asks whether cash can fall below 14 after a rate and customer shock. The system selects an evidence view and reconstructs the four-history family of Chapter 13. It applies the stated events, obtains cash formula $33 - 18c - 2f$, and solves the threshold condition.

The result returns the concentrated floating witness with cash 13 and the other three outcomes. It states that no probabilities are attached. If the user adds a committed facility, the event service checks its timing and eligibility, creates a draw in the required region and recomputes debt and cash.

If a new source excludes the concentrated floating combination, the family version changes and the query's dependency triggers recomputation. The old result remains reproducible. This sequence demonstrates the intended workflow without requiring every possible company to be instantiated as a separate database.

20.15 DSL and typed-library authoring

A textual language can make scenario and contract definitions readable and reviewable. A typed library can offer direct programmatic construction. Both should lower to the same intermediate representation and be compared on the same tasks. The DSL is not justified merely by its existence.

Equivalent authored inputs should produce equivalent IR and results. Syntax convenience should not alter uncertainty or timing. A textual default that silently assumes a rate or probability would be especially dangerous because it can look like a declarative fact.

The existing repository includes language, expression, method and event work that should be reused where it matches these semantics. New records should be added only when an included operation cannot be expressed adequately. The design should avoid rebuilding a parallel incompatible representation for the new chapters.

20.16 Adapters to external components

An external contract engine can propose schedules and payoffs. An optimization package can solve a declared mathematical problem. A probabilistic library can approximate a posterior. Each adapter needs input and output semantics, version identity, numerical conventions and failure mapping.

The ACTUS inspection illustrates the boundary: contract events do not supply accounting journals, and local-time schedules do not automatically preserve exact timestamps. A double-valued payoff is not exact rational evidence merely because its printed decimal matches a fixture.

Adapters should be checked against small independent cases and preserve unsupported outcomes. A service exception must not become zero cash flow. A missing risk series must not silently carry forward an unrelated value. Dependency and license review remain necessary before selecting a runtime, but no external adoption is claimed by this manuscript.

20.17 Incremental implementation

The first vertical slice should support a bounded history family, exact event timing, simple debt and equity offers, actor information and coherent statements. It should include both successful and refused paths. The worked company and refinancing cases provide expected outcomes.

The next slice can add class rights, conversion, group transfers and persistent tax or asset state. Later work can add nonlinear pricing, partially observed policy optimization and controlled approximation. Each stage should preserve the full mission's open requirements rather than imply that a small slice closes whole-company representation.

A slice is complete only when its source-to-result behavior is checked and its unsupported scope is explicit. A collection of isolated module tests is useful but insufficient for composition. The critical interactions are where financing, information and accounting meet.

20.18 Review and release evidence

A release should preserve model and source versions, exact fixture results, numerical validation where applicable and rendered documentation. Engineering conformance, accounting authority, scientific validity and human comprehension remain separate evidence categories.

A clean PDF or passing arithmetic script does not establish product readiness. A reader-facing explanation must still make assumptions and derivations understandable. A human review can identify comprehension problems that mechanical checks miss; its absence does not prevent provisional drafting but does prevent claiming that acceptance occurred.

The stable documentation links should point to a built, identified candidate with an honest status. Historical manuscripts and hashes remain available. This allows the design to advance without rewriting evidence from earlier stages or confusing the newest document with a completed system.

Chapter 21

Groups, counterparties, and contagion across otherwise coherent statements

A company group is not a single bank account. Consolidation can eliminate internal balances for reporting while cash remains held by particular entities, currencies and custodians. Stress can also propagate through guarantees, collateral and common counterparties. These mechanisms make group simulation a natural test of whether the model preserves economic rights beyond accounting totals. The examples here stipulate transfer and priority rules; actual legal applicability requires source-specific authority.

21.1 Consolidated cash and entity cash

Consider a parent P and subsidiary S. P holds cash 5 and owes external debt 20 tomorrow. S holds cash 30 and owes external suppliers 8 the following day. The consolidated cash balance is 35. It exceeds the parent's debt maturity, but that fact does not establish that P can pay.

Suppose S may transfer at most 10 before tomorrow because the remainder is restricted or needed under a binding local rule. P can assemble 15 and has a shortfall of 5. A consolidated cash forecast that deducts 20 from 35 reports 15 remaining and misses the entity failure.

If a transfer of 15 is permitted instead, P can pay but S's remaining cash must still cover its own obligations. The transfer is not free group liquidity: it changes where resources are held and can create an intercompany claim or distribution. The event should be represented in both entities before consolidation.

21.2 Intercompany loan mechanics

Suppose S lends 15 to P. S records an intercompany receivable and cash outflow; P records cash and an intercompany payable. P then repays external debt 20 using its original 5 plus the loan. Consolidated cash falls from 35 to 15 and external debt falls by 20. Internal claims eliminate in the consolidated statement under the stipulated perimeter.

The internal loan nevertheless matters for future entity liquidity. P owes S according to its terms. If S later needs cash, it cannot spend a consolidated elimination entry. A model that deletes internal contracts after consolidation loses repayment dates and possible stress transmission.

Interest on the internal loan can affect entity tax and distributable earnings even when consolidated interest eliminates. Currency differences can create further effects. The general engine should retain legal-entity records and produce consolidation as a reporting view rather than using the consolidated view as its only state.

21.3 A guarantee creates contingent support

Suppose P guarantees S's external supplier obligation of 8. The guarantee produces no cash payment while S pays normally. If S fails, P may be required to pay under the stipulated terms. The group's reported balance sheet may not reveal the same exposure as a direct loan.

A scenario that stresses S can therefore affect P even when there is no planned intercompany transfer. The guarantee needs beneficiary, covered obligation, trigger, cap, expiry and settlement timing. A vague statement of support is not automatically an enforceable guarantee. Conversely, omitting a binding guarantee because it has no current principal balance understates contingent funding need.

If P's payment gives it a recovery claim against S, that successor claim should be recorded. Paying the guarantee and also leaving the supplier fully paid by S would double settle the obligation unless the model includes reimbursement or another event. Counterparty identities make this check possible.

21.4 A common collateral pool

Two loans may claim the same collateral. A model that assigns the full collateral value independently to each overstates recovery. Let collateral realize 50, with secured claims 40 and 30. Under an assumed first-ranking claim of 40 and second-ranking claim of 30, recoveries are 40 and 10.

If the claims rank equally and share proportionally, recoveries are 50(40/70) and 50(30/70). Both allocations conserve 50 but differ by creditor. Priority is an economic and legal fact, not an accounting balancing choice.

The collateral record should link each asset to encumbrances and ranking. Unknown ranking creates alternative recovery histories. An aggregate secured-debt amount does not identify whether collateral is shared, separate or overpledged. A stress result should preserve that ambiguity when it affects pricing or liquidity.

21.5 Collateral value and borrowing capacity

A borrowing-base facility can limit drawn debt to a fraction of eligible collateral. Suppose eligible receivables are 100 and the advance rate is 0.8, giving capacity 80. Drawn debt is 70. A customer default removes 20 from eligibility, reducing capacity to 64 and creating a borrowing-base deficiency of 6.

The credit loss has two effects: it reduces asset value and can trigger a cash repayment or collateral substitution requirement. A model that records only the impairment misses the liquidity consequence. A model that records a repayment without the contractual deficiency rule invents one.

Eligibility can depend on age, concentration, jurisdiction and disputes. Two receivable pools with equal net carrying value can have different borrowing capacity. The reconstruction may

need invoice cohorts or symbolic eligibility amounts even when the financial statement reports one total.

21.6 A margin call before cash income

A derivative can economically hedge a future exposure while requiring collateral today. Suppose a company has a fixed-rate asset whose economic value rises when rates fall and a derivative position that offsets that value movement but requires cash margin after an adverse mark. The ultimate hedge can reduce value risk while increasing short-term liquidity need.

The cash call depends on collateral agreement, thresholds, minimum transfer amounts, eligible assets and settlement timing. It cannot be inferred from the derivative's year-end fair value alone. A scenario that nets the derivative and underlying value change may correctly describe net economic exposure while missing gross cash settlement.

The simulator should track valuation, collateral posting and final payoff as distinct events. Posted collateral may remain an asset or claim under the applicable treatment; it is not necessarily an expense. Reusing the fair-value change as a cash flow can create double counting.

21.7 Currency trapped in the wrong account

Suppose P owes 10 foreign units and holds 100 domestic units. At an executable exchange rate of 8 domestic per foreign, conversion needs 80. Under a depreciation to 12, it needs 120. If conversion capacity is limited or markets close before payment, even a favorable reporting-currency total may not be usable.

Now suppose S holds 10 foreign units but cannot transfer them to P in time. Consolidated foreign cash equals the payment, yet P still faces a shortfall. A model should retain both currency and entity location. Netting by currency across the group can be as misleading as netting all currencies into one reporting amount.

A permitted currency swap or intercompany transfer can solve the problem, but it introduces counterparties, settlement risk and possible costs. The model should generate those transactions rather than assume frictionless conversion as an unnamed default.

21.8 Settlement risk

Two offsetting payments on the same date need not settle simultaneously. A company may pay one currency before receiving another. A counterparty failure between the legs can leave an exposure. A net daily cash amount does not describe the gross settlement sequence.

For a bounded example, P pays 80 domestic units at noon and expects 10 foreign units at 15:00. Its foreign debt is due at 14:00. Even if both legs settle as promised, the receipt is too late. A settlement service with payment-versus-payment or a bridge can change the sequence, but must be represented.

Exact intraday simulation is unnecessary for every query. It becomes necessary when ordering affects payment feasibility. The method should allow event resolution to increase around material boundaries without pretending that an annual model resolved them.

21.9 Cross-default and acceleration

A default on one obligation can trigger rights under another. The triggering event may be nonpayment, acceleration, insolvency or a threshold amount. A generic flag called default cannot substitute for the actual cross-default clause.

Suppose S misses 8 and P has a loan of 50 whose clause triggers only for subsidiary defaults exceeding 10. The event does not meet that threshold. If the clause instead aggregates several defaults totaling 12, it may trigger. The scope, aggregation period and cure rights matter.

Acceleration changes the payment calendar and can create a liquidity spiral. The model should record the trigger, exercise or automatic effect, new due date and notice requirements under the stipulated contract. It should not accelerate every group obligation whenever any branch is distressed.

21.10 A contagion sequence

Combine the borrowing-base example with a parent guarantee. A customer default reduces S's eligible receivables from 100 to 80. The facility demands repayment 6. S has cash 4 and cannot pay the full amount. Under the example's guarantee, P pays the remaining 2. P's cash falls from 5 to 3.

If P has an independent payment of 4 later that day, the support creates a parent shortfall of 1. Consolidated asset loss alone does not show the sequence. The guarantee transfers liquidity pressure across entities.

If the guarantee is capped at 1, S remains in arrears by 1 and P retains cash 4. The parent can meet its payment, but subsidiary distress persists. Different guarantee terms produce different locations and timing of default even when consolidated net assets are similar.

21.11 Behavioral contagion

Counterparties may react to distress before a contractual trigger. Suppliers can shorten terms, customers can delay orders, lenders can decline new credit and employees can leave. These responses are behavioral assumptions or estimated mechanisms. They should not be inferred automatically from a balance-sheet loss.

A scenario can include a supplier-term tightening rule conditional on a public rating event. The rule needs the event's timing and actor information. If suppliers have not observed the distress, immediate tightening may give them unavailable knowledge. If they have private signals, those should be specified.

Behavioral contagion can amplify losses, but an arbitrary amplification coefficient is not evidence. The model should expose sensitivity and distinguish contractual consequences from discretionary reactions. This separation makes the result more useful for identifying which mechanisms require empirical research.

21.12 Endogenous prices and fire sales

Selling assets to meet cash needs can depress their prices when the market is illiquid or several firms sell simultaneously. A fixed-price sale model may then overstate proceeds. An endogenous price rule creates feedback between liquidity need, quantity sold and realized value.

For illustration, let sale price per unit be $p(q) = 10 - 0.1q$ for $0 \leq q \leq 50$. Proceeds are $10q - 0.1q^2$, which increase on this interval but at a decreasing rate. Raising 200 requires solving the quadratic rather than dividing by the initial price 10.

The formula is a teaching assumption. It should not be treated as an estimated market impact curve. Its role is to expose how quantity and price can interact. Beyond the stated domain, the rule may be nonsensical. A solver should not extrapolate it to negative prices to force a solution.

21.13 Feedback across several sellers

If price depends on total sales by several entities, each policy affects the others' proceeds. Sequentially solving each entity at an unchanged price can be inconsistent. A simultaneous market-clearing or declared sequential mechanism is needed.

The agent-based stock-flow literature provides examples of interacting counterparties and explicit event order (Caiani et al., 2016). It does not imply that a specific market-impact function is valid for the company under study. The local mechanism must be stated and tested.

Multiple equilibria or no feasible clearing outcome can arise. A numerical failure to find one should remain distinct from a proved absence. If the model selects one equilibrium by a behavioral adjustment rule, that rule is part of the scenario and should be retained in provenance.

21.14 Consolidation changes after distress

A restructuring can change control, ownership or the reporting perimeter. A subsidiary may be sold, placed into a process that changes control, or receive new capital. The group statements then require a perimeter transition under the applicable basis.

The economic event should identify assets, liabilities, consideration and retained rights. The accounting layer determines derecognition, gains, losses and continuing interests. A generic default event that removes the entire subsidiary from consolidation without an authority-based control assessment is unsupported.

The scenario can remain useful with conditional perimeter outcomes if facts are unresolved. The result should show which cash and obligations remain in each legal entity and how each accounting interpretation reports them. A single consolidated number can conceal materially different legal exposures.

21.15 A group-state proposal

The state should include legal entities, ownership and control assessments, currency accounts, external and internal contracts, collateral pools, guarantees and transfer permissions. It should also retain actor-specific information and event calendars. Consolidated reporting is a derived view.

This does not require every query to load every field. Dependency-aware execution can select the relevant subgraph. A parent liquidity query follows transfers and guarantees; a consolidated revenue query follows perimeter and eliminations. The underlying representation preserves the links needed when the question changes.

Unknown links can be represented as alternatives. A guarantee's existence, cap or trigger may be unresolved. The query can report outcomes conditional on those terms and identify the source that would resolve them. Inventing no guarantee as a default would systematically favor benign outcomes.

21.16 Tests for group coherence

A transfer test should show equal and opposite cash movements before fees and consistent internal claims. A consolidation test should eliminate internal balances without deleting entity state. A collateral test should prevent recovery exceeding the shared pool. A timing test should prevent late cash from funding an earlier obligation.

A guarantee test should create the correct successor claim or loss under the stipulated rules and avoid double payment. A currency test should preserve native units through conversion and reporting. A cross-default test should distinguish threshold, cure and acceleration.

These tests are stronger than a final consolidated balance-sheet equality. They examine the mechanisms that can make a group fail despite apparently sufficient aggregate resources. Passing them establishes the specified small cases, while actual legal and regulatory applicability remains tied to source evidence.

21.17 What cannot be compressed away

For group liquidity, entity location, currency, transfer rights and dates are often essential. For recovery, collateral and priority are essential. For behavior, information and contractual discretion are essential. A compression that removes them may still reproduce current totals while changing stress outcomes.

The practical method is to retain a rich underlying state and derive query-specific summaries. A summary approved for one purpose can be refined for another. This avoids both extremes: duplicating the entire group for every possibility and collapsing the group into totals that cannot explain who can pay whom.

Chapter 22

When the model itself is incomplete: missingness, misspecification, and research priorities

Preserving all histories inside a model does not protect against a missing mechanism outside it. A feasible family can be exact and still omit a guarantee, a customer concentration or a financing restriction. This chapter examines how to represent that limitation without abandoning useful analysis. The aim is to make conditional conclusions precise and to direct further work toward omissions that can change decisions.

22.1 Unknown value versus absent mechanism

An unknown interest rate can be represented by a variable if the contract form is known. An unknown existence of a guarantee requires an alternative contract structure. A mechanism the language cannot express is a different limitation. These cases should not share one generic null.

A missing value within a supported form can often be bounded or inferred. A missing structural alternative can change the dimension and event logic of the family. An unsupported mechanism may require a new model component before any bound is meaningful. The result should identify which kind of gap it faces.

For example, allowing a loan rate between zero and twenty per cent does not cover a missed conversion right. A wider rate interval cannot reproduce the ownership change caused by conversion. Numerical sensitivity over existing parameters is not a substitute for structural coverage.

22.2 The disclosure package defines an evidence boundary

A company filing can incorporate other documents by reference or discuss risks without quantifying them. A reconstruction based only on primary statements may omit material terms in notes, exhibits or subsequent events. The evidence package should state which documents and passages were considered.

Coverage has stages: retrieval, parsing, interpretation and model admission. A page can be retrieved but not parsed; parsed but not understood; understood but outside the current executable profile. These distinctions matter for the completeness claim.

An unresolved passage should remain linked to relevant model areas. A guarantee discussion may block a claim that all contingent obligations are represented. It need not block a simple historical revenue reconciliation. Scope should be query-sensitive rather than all-or-nothing.

22.3 Missingness can be informative

A missing disclosure may result from immateriality, reporting exemptions, extraction failure or deliberate omission within a strategic disclosure model. These causes have different implications. Treating every missing value as zero assumes the most benign explanation. Treating every missing value as arbitrarily large can make analysis useless.

The first task is to identify the source of missingness where possible. If a table was not extracted from a scanned page, the remedy is data processing. If the filing does not disclose a contract term, the uncertainty is substantive. If a disclosure is legally required under certain facts, inference from absence requires actual authority and evidence about compliance; it cannot be assumed.

A probabilistic model of missingness can be added when justified. It should state whether missingness depends on hidden exposure or actor behavior. An assumption of missing at random is not a universal default for corporate disclosure. Without support, retain alternative explanations rather than fabricate a likelihood.

22.4 Conflicting sources

An annual report, interim filing and debt exhibit can disagree. The disagreement can reflect different dates, scopes, currencies, amendments or errors. A model that takes the latest number without reconciling context can discard valid historical information.

Source precedence should depend on the claim. A later amendment may supersede future contract terms while leaving past payments unchanged. A restated financial statement may revise a comparative amount. A management presentation may use a nonstandard metric rather than contradict a filing.

The evidence view should preserve both assertions and the interpretation linking them. If conflict remains unresolved, the family may contain alternative source-admission views or return an explicit contradiction. It should not average incompatible facts merely because both are numeric.

22.5 Measurement error and reporting discretion

A discrepancy between predicted and reported amounts can arise from rounding, estimation, extraction error or an incorrect economic history. A single Gaussian residual model collapses these mechanisms. It may be convenient statistically but can imply implausible support and independence.

Rounding often gives bounded intervals. Extraction errors can be discrete shifts in sign or scale. Accounting estimates can have structured dependence on assumptions and information. Reporting discretion may be strategic. The observation model should reflect the mechanism relevant to the claim.

A robust likelihood or contamination model can be considered, but its parameters need provenance and sensitivity analysis. It should not become a way to make every contradiction

disappear. A large residual may be evidence that the model omitted a transaction rather than merely noisy reporting.

22.6 Structural alternatives

Suppose revenue is modeled as price times quantity. A different model includes subscription renewals, usage charges and cancellation options. Both may fit historical totals. A demand shock can produce different future responses because the contract structures differ.

The family can include alternative structural models with shared observations and explicit scope. Comparing their conditional outcomes identifies model risk. Assigning posterior weights requires model priors and likelihoods; otherwise the alternatives remain unweighted.

A structural alternative should be concrete enough to simulate or bound. A vague statement that anything could happen is not useful. The modeler should identify plausible omitted mechanisms, their source support and the smallest discriminating evidence. This makes uncertainty actionable without claiming completeness.

22.7 Behavioral parameter uncertainty

A collection-delay parameter estimated from historical data can be uncertain and unstable. A recession can change customer behavior, credit terms or the company's enforcement policy. Treating the estimate as a fixed fact understates uncertainty.

Separate estimation uncertainty within a behavioral form from uncertainty about the form itself. A confidence interval around a constant collection rate does not cover a regime switch in which major customers stop paying. Conversely, a broad regime set may be unnecessarily pessimistic for a short, well-understood contract schedule.

The scenario should state whether behavior is held fixed, estimated conditionally, changed by a stress rule or optimized. Each treatment answers a different question. An empirical policy estimated under one financing regime may not transfer after covenants bind.

22.8 Extrapolation

A model calibrated in low-rate periods can be applied to high-rate stress. Contractual cash formulas may remain valid if terms are known, while behavioral relationships may not. The extrapolation status should be component-specific.

A floating coupon formula with a known cap can be evaluated outside historical rates because the contract supplies the rule. A regression of sales on rates may have no support in that region. A neural surrogate trained only below the cap may also fail even though the underlying contract formula is simple.

The model should distinguish extrapolation of an economic law, a behavioral estimate and a numerical approximation. Each requires different evidence. A global warning that the scenario is extreme is less useful than identifying which components are unsupported and how that affects the result.

22.9 Unknown unknowns and explicit coverage

A model cannot enumerate mechanisms it has not conceived. It can, however, maintain a coverage map organized by assets, obligations, rights, entities, timing and information. Review can ask whether material disclosures have a place in that map.

Stress testing the representation itself is useful. Introduce an unexpected guarantee, a delayed tax payment or a conversion clause and see whether the system can preserve it as unresolved rather than dropping it. A robust data model should fail visibly when a new mechanism is outside its profile.

This is different from claiming that a checklist proves completeness. The coverage map supports systematic discovery. It remains extensible as new cases arise. The manuscript's failure catalogue should be read in that spirit.

22.10 Sensitivity to the family boundary

Suppose a result is safe for every admitted history. The conclusion can still depend on the family's bounds. If floating share is limited to 0.4 by a weak assumption rather than a source, expanding it to one can change safety.

A useful analysis identifies active assumptions: those whose relaxation changes the conclusion or its bound. This can be done through alternative runs, dual information in suitable optimization problems or counterexample search. The method should not infer economic importance solely from a numerical coefficient.

An assumption may be inactive for one query and decisive for another. A maturity bound irrelevant to next month's interest can dominate a one-year refinancing stress. Sensitivity records should therefore be tied to query and horizon.

22.11 Partial identification of a decision

The evidence may not identify every parameter but still identify the preferred action. If one policy dominates another across all compatible histories, further historical detail is unnecessary for that comparison. Conversely, a narrow parameter interval can still straddle a decision threshold.

This distinction directs research effort. The goal is not always to recover the true journal uniquely. It may be enough to establish a robust decision or a bounded exposure. A history family is useful precisely because it can show which unresolved details matter.

The system should therefore report both parameter uncertainty and decision sensitivity. A large uncertainty set need not imply indecision, and a small set need not imply safety. The geometry relative to policies and thresholds is what matters.

22.12 Research priorities from counterexamples

A verified adverse witness can reveal the missing fact that drives risk. If the witness requires a near-term foreign-currency maturity, the next source target is a currency-maturity cross-classification. If it requires a guarantee, inspect the guarantee terms. If it requires refinancing refusal, investigate committed capacity and draw conditions.

The witness should be accompanied by competing interpretations and checked consequences. This is useful internal preparation before asking a professional a specific question. An unresolved label alone is not a review-ready brief.

The repository's internal-preparation requirement therefore fits the method: finish feasible source retrieval, calculations and competing interpretations first. No professional engagement or expenditure is authorized by the existence of an open accounting or legal question. Ordinary reader feedback is a separate need.

22.13 Model revision without losing history

When a missing mechanism is added, old results may change. Preserve the old model and result with their scope, then create a revised version. This allows users to distinguish a forecast error under the old model from a correction to the target definition.

A new guarantee can expand the family or add obligations. A new disclosure can narrow it. A corrected accounting mapping can change reported outcomes without changing economic cash. These revisions have different meanings and should not all be called data refreshes.

Dependency tracking should identify which results need recomputation. A source affecting one loan should not require rebuilding unrelated evidence, but its effects may propagate through liquidity, default and group guarantees. The dependency graph must follow economic consequences as well as direct formulas.

22.14 Communicating model risk

A useful explanation states the result, the assumptions that drive it and the specific unresolved mechanisms that could change it. It does not bury the conclusion under generic warnings. For example, "cash remains above 14 for all retained histories provided the facility remains drawable after customer default" identifies a meaningful condition.

If that condition is not established, say so directly and show the alternative result. Avoid presenting a conditional safe outcome as unconditional safety. Equally, do not describe every unresolved mechanism as inevitable disaster. The family should distinguish possible, necessary and unassessed consequences.

The reader should be able to see whether more evidence, a different policy or a better computation would resolve the issue. These are different next steps. A numerical approximation gap cannot be closed by another disclosure if the model already knows the relevant fact; a missing contract clause cannot be closed by more solver iterations.

22.15 A defensible stopping point

A query can be complete within a declared scope even when the entire company model is incomplete. Completion means that the requested target has been computed with the stated guarantees and that material excluded mechanisms are disclosed. It does not mean all uncertainty has disappeared.

A broader product-readiness claim requires broader coverage and evidence. The 150–300-page treatment is a documentation deliverable, not proof of that readiness. The manuscript remains a candidate whose assumptions, sources and proposed methods require continued implementation and human review.

This distinction permits useful progress. It avoids both false certainty and paralysis. The system can answer bounded questions accurately, retain uncertainty honestly and identify the concrete work needed to support stronger conclusions.

Chapter 23

Memory across reporting periods: taxes, vintages, arrears, and future rights

Two companies with the same current statements can differ because their past creates different future rights and obligations. A simulation that keeps only current totals forgets this memory. This chapter follows several small examples over more than one period. The examples use stipulated accounting and tax rules solely to expose state requirements. They are not claims about a particular jurisdiction's current law.

23.1 A loss carryforward

Assume a teaching tax system with rate 25 per cent, no refund for current losses, unlimited use of carried losses against future positive taxable income and no expiry. Let L_t be unused losses entering period t and T_t taxable income before using them. For positive T_t, used losses are $U_t = \min(L_t, T_t)$. Cash tax and next loss balance are

$$\text{Tax}_t = 0.25 \max(T_t - L_t, 0), \tag{23.1}$$

$$L_{t+1} = \max(L_t - \max(T_t, 0), 0) + \max(-T_t, 0). \tag{23.2}$$

The second equation covers both profit and loss periods under these simplified rules. A positive income reduces available losses; a negative income adds its magnitude.

Company A enters with losses 40 and company B with none. Both have the same current cash, assets and book equity because the example assumes no recognized deferred tax asset. Next period taxable income is 30. A pays no cash tax and retains losses 10. B pays 7.5. Current statements alone do not identify that difference.

In the following period taxable income is again 30. A uses its remaining 10 and pays 5; B pays 7.5. Over two periods, tax is 5 versus 15. Omitting loss memory overstates A's tax. Treating the original loss as an immediate refund would instead overstate its initial cash. Both errors can occur in simple effective-tax-rate forecasts.

23.2 Limits and expiry change the recurrence

Now impose a rule that losses can offset at most 50 per cent of positive taxable income in a period. With $L_0 = 40$ and $T_0 = 30$, usable losses are 15, tax is 3.75 and remaining losses are 25.

The earlier recurrence is no longer correct.

An expiry date requires vintage-specific loss balances. A single total 40 cannot determine which amount expires next year. Two companies with the same total can have different future tax capacity. The state may need a vector by expiry cohort, jurisdiction and restriction.

A change of ownership can affect utilization under some actual systems, but no such rule is assumed here without authority. The point is structural: tax rights depend on facts beyond a total balance. The model should admit the applicable rule and preserve the state it requires.

23.3 Deferred tax is a measurement, not the loss pool

A deferred tax asset may be recognized only under specified accounting criteria. Its carrying amount can differ from the mechanical tax value of losses. A valuation allowance or recognition restriction can change book equity without changing the legal loss pool.

Suppose the legal losses remain 40 while a recognized deferred tax asset is reduced from 10 to 4 under the chosen assessment. The reduction is not a cash tax payment. The future tax computation still needs the legal loss balance and applicable utilization rules.

Conversely, recognizing an asset of 10 does not create 10 of spendable cash. A liquidity scenario should use actual tax payment or refund dates. The accounting measurement and economic tax rights are linked but distinct states.

23.4 Asset vintages with the same carrying amount

Consider two machines under a simple straight-line model with zero residual value. Machine A originally cost 100 and has two of ten years remaining, so its carrying amount is 20 and annual depreciation 10. Machine B originally cost 40 and has one of two years remaining, also carrying 20 but annual depreciation 20. Equal current carrying amount does not imply equal next-year expense.

If the reported current depreciation is also constrained to match, combinations of several machines can still create ambiguity. For example, a portfolio can mix assets with different remaining lives whose total charge is the same. The next-period charge changes when a cohort reaches the end of life.

The state should retain acquisition or remaining-life cohorts where depreciation, maintenance or replacement depends on them. A single average life may be adequate for a smooth aggregate forecast but can miss a replacement wall. The approximation should be checked against the horizon and decision.

23.5 Replacement and operating capacity

An asset can be fully depreciated while still productive, or retain book value while becoming unusable. Depreciation is not a direct physical failure process. Replacing capital expenditure with depreciation assumes a relationship that must be justified separately.

Suppose A's machine can operate for two more years without replacement and B's fails after one year unless 30 is spent. Both enter with carrying amount 20. A financing shock in year two affects B's production but not A's in the same way. A model using only carrying amount cannot distinguish them.

A richer state includes capacity, maintenance, failure or retirement rule and replacement lead time. These can be uncertain. The history family may retain alternative vintage and condition profiles compatible with the same accounting data. Their scenario outcomes remain conditional rather than being collapsed into one average machine.

23.6 Arrears are not current-period expense again

Suppose a loan coupon of 5 accrues in period one but is not paid. Under the illustrative accrual treatment, interest expense and a payable increase by 5. In period two, paying the arrears reduces cash and the payable. It does not recognize the same interest expense again.

If penalty interest accrues on arrears, that is an additional contractual event with its own base, rate and dates. The state must retain unpaid amounts and payment allocation rules. A payment may be applied first to fees, interest or principal according to the admitted contract.

A model that stores only total debt can lose this allocation. Paying 5 might reduce principal when it should settle interest, altering future coupon calculations. The resulting statements may still balance, but the contract state becomes wrong. Multi-period replay exposes the error.

23.7 A payment allocation example

At a date, principal is 100, accrued interest 5 and fee arrears 2. A payment of 6 is made. Under the stipulated order fees first, then interest, then principal, fees become zero, interest becomes 1 and principal remains 100.

Under principal-first allocation, principal becomes 94 while interest and fees remain 7. Total obligation falls by 6 in both cases, but future interest and default status can differ. A total-debt bridge cannot distinguish the two. The allocation rule is part of the contract.

If the rule is unknown, both allocation histories may remain possible. A scenario should not choose the order that minimizes future cost unless management actually has that right. Creditor-controlled allocation cannot be optimized by the borrower without permission in the model.

23.8 Cure periods and clocks

A breach can start a cure period measured in business or calendar days. The state needs the trigger date, deadline, required remedy and whether cure has occurred. Merely retaining a Boolean breach flag loses the deadline.

Suppose a payment is missed on day 1 and can be cured within three calendar days, with the deadline defined at the end of day 4 under the stipulated convention. A receipt on day 4 can cure if settlement precedes the cutoff; one on day 5 cannot. An annual or monthly timestamp cannot resolve this distinction.

The actual contract may count days differently, include notices or exclude holidays. The simulator should not infer a convention from a generic date library. Exact boundary fixtures should accompany the admitted timing rule.

23.9 Options survive quiet periods

An option can remain out of the money for many reporting periods and then become decisive under a shock. A call right, put right, conversion feature or renewal option should not disappear because it had zero exercise activity historically.

Two loans can have identical coupon and principal histories while one permits early redemption at par and the other imposes a premium. A rate decline creates different refinancing choices. Current book debt and past interest do not identify the option.

A state summary can omit the option only for a query where its exercise cannot affect outputs over the horizon. That claim may rely on bounds showing the option never becomes relevant. Once the scenario or horizon changes, the summary must be reconsidered.

23.10 A refinancing option value example

Suppose a company owes a fixed payment of 110 next year. It can refinance today at a cost of 102 if it has a call right, and current cash is sufficient. Ignoring discounting for this one-period comparison, the right can save 8. Without the right, the old obligation remains.

If exercising requires a fee of 9, the saving disappears and the policy should not exercise under a cost-minimizing objective. If the fee is uncertain between 0 and 12, the exercise decision is conditional. A model that always redeems at par gives every borrower an option it may not possess.

The example does not price the option probabilistically. It shows a state-contingent decision consequence. Pricing would require a distribution or valuation operator over future rates and policies, along with the contractual exercise schedule.

23.11 Estimates carry information history

An impairment estimate can depend on forecasts and discount rates available at the reporting date. A later recovery in value does not mean the earlier estimate was necessarily an error. A restatement that corrects a mistaken fact is different from an estimate revised because new information arrived.

A historical simulation should preserve the information used for each estimate. If it recomputes every past estimate using today’s knowledge, it changes the as-reported history and can introduce look-ahead into validation. A current restated view can be useful, but should be named separately.

The economic asset state and accounting estimate can therefore have different version histories. A new estimate changes measurement without necessarily changing physical assets or cash. A model should track the relation rather than force every valuation change into an economic transaction.

23.12 Inventory age and obsolescence

Inventory with the same quantity and carrying value can have different remaining shelf life. A demand slowdown can make one cohort obsolete while another remains saleable. A pooled average age can conceal the threshold.

Suppose ten units expire next month and ten otherwise identical units expire next year. Both pools are carried at 50. If no sales occur for two months, the first may lose all saleable value under the stipulated physical rule while the second remains available. Historical cost equality does not preserve the response.

A cohort model can retain quantities by expiry and location without individual serial numbers when units within a cohort are interchangeable for the query. This is a justified aggregation opportunity. If a recall affects a particular batch, the cohort key must include batch identity or be refined.

23.13 Customer behavior has memory

A collection process can depend on invoice age, previous delinquency, dispute status and payment promises. A single receivable total or average collection ratio may omit these predictors. Two companies with equal receivables can have different next-month receipts.

A Markov model can be appropriate if the chosen state captures the relevant dependence under its assumptions. Calling the process Markov does not make a deficient state sufficient. If repayment depends on the duration of arrears, include age or a suitable duration state.

An empirical model should test whether the proposed summary predicts the relevant transitions and whether stress changes that relation. A purely contractual due-date model answers a different question from a behavioral collection model. Both can be retained with separate roles.

23.14 Horizon truncation

A policy can look safe over one year by issuing debt due just after the horizon. The closing cash is positive, but the future obligation remains. A terminal report should show that maturity and avoid treating its proceeds as permanent wealth.

Suppose a company borrows 20 at year-end and repays 22 one month later. A one-year cash objective rewards the 20 inflow unless terminal debt is valued or constrained. A policy optimizer can exploit this omission even if the event mechanics are correct.

Possible remedies include extending the horizon, adding a justified continuation value, requiring terminal funding coverage or reporting residual obligations separately. Each changes the decision problem in a stated way. No terminal rule is universally correct for all company questions.

23.15 A sufficient state is query-relative

For a one-day payment query, asset age may be irrelevant while settlement dates are essential. For a five-year capital plan, vintages and replacement needs matter. For tax cash, legal loss pools matter; for book equity, deferred tax measurement also matters.

The underlying representation should preserve these quantities where supported, while the active execution state can be reduced for the query. A sufficiency argument names the allowed scenarios, policies, outputs and horizon. It is not a permanent declaration that discarded details never matter.

This is how compact simulation can coexist with rich historical evidence. The system need not replay every old event at every step, but it must retain the memory that determines future events and the provenance needed to reconstruct it after revisions.

23.16 Implementation tests across periods

A loss-pool test should distinguish cash tax from deferred tax measurement and preserve expiry cohorts when required. A vintage test should produce different replacement paths from equal carrying amounts. An arrears test should prevent double expense and apply payments in the stipulated order.

A cure-clock test should distinguish receipts before and after the deadline. An option test should preserve unexercised rights through quiet periods. A horizon test should retain obligations beyond the final reporting date. Each case targets a specific missing state variable.

The tests should also verify legitimate aggregation. Identical loans with the same terms and dates can sometimes be combined additively. Identical inventory units can be pooled by cohort. The goal is sufficient memory, not maximal detail for its own sake.

Chapter 24

Evaluating a real-company simulator: what evidence would support its use

The exact cases in this part establish useful semantics, but a real-company simulator needs a broader evaluation. It must represent disclosures, preserve important uncertainty, generate coherent events and provide decision-relevant results. These objectives cannot be reduced to one extraction score or one forecast error. This chapter proposes an evaluation design and explains what each result would and would not establish. No empirical evaluation is claimed to have been performed here.

24.1 Define the unit of evaluation

A unit can be a source assertion, a contract, a reporting period, a company history family, a scenario query or a policy decision. Accuracy at one unit does not automatically transfer to another. Correctly extracting a debt total does not establish that maturity and option terms are represented.

A company-period case should identify its document package, entity perimeter, accounting basis, evidence cutoff and materiality scope. A scenario case adds shocks, policies, information and requested outputs. These definitions prevent evaluation from quietly changing the target after seeing results.

The unit should also preserve unresolved content. If a system refuses an unsupported instrument, that can be correct behavior within a bounded profile. The evaluation should distinguish honest refusal from silent omission, while still recording the coverage limitation for the full mission.

24.2 Source-to-model traceability

For each material modeled assertion, a reviewer should be able to reach the source passage and interpretation. Conversely, material source passages should have a disposition: represented, irrelevant to the declared query, unresolved or outside the executable profile.

This bidirectional trace catches different failures. A model assertion without a source may be an assumption or hallucination. A source passage without a disposition may be an omission. A correct numeric total can coexist with a missed covenant note that changes the stress result.

The evaluation should sample across difficult formats and concepts, including scanned notes, tables with overlapping dimensions, incorporated documents and revisions. A convenient set

of clean tables is insufficient evidence for whole-filing coverage. The selection procedure and omissions should be recorded.

24.3 Annotation and disagreement

Human annotations of financial meaning can disagree. A term may be ambiguous, a source may conflict, or the accounting treatment may depend on facts not disclosed. The reference set should permit unresolved or alternative annotations rather than force artificial unanimity.

Adjudication should record the competing interpretations and the evidence that resolves them. A majority vote is not necessarily an authority on a technical accounting or contractual question. For some cases the correct reference is a conditional family.

Ordinary comprehension review and expert technical review serve different purposes. A reader can identify unclear exposition without certifying accounting treatment. An accountant can assess a specific treatment without validating a probabilistic forecasting model. Evaluation should not collapse these roles into a generic human-approved label.

24.4 Historical reconstruction checks

A reconstruction should reproduce observed statements within admitted precision and context. It should also satisfy cross-statement bridges and contract constraints. These are necessary checks, but a model explicitly fitted to the reports can pass them while selecting the wrong hidden history.

Use withheld disclosures where possible. For example, reconstruct from primary statements and selected notes, then test whether the retained family includes a contract detail revealed in a withheld exhibit. The procedure must avoid using that detail indirectly through another source.

A family that excludes the revealed truth under correct source interpretation has failed coverage. A family that includes everything can pass inclusion trivially but be uninformative. Evaluation therefore needs both coverage and useful restriction, interpreted relative to the available evidence.

24.5 Synthetic truth and real uncertainty

Synthetic companies provide known complete histories and controlled disclosure maps. They allow tests of whether the system preserves all intended alternatives and rejects impossible ones. They are valuable for semantic coverage and adversarial construction.

Their limitation is that the generator defines the world. A system matching that generator may fail on mechanisms the generator omits. Synthetic results should not be presented as real-company empirical validity. The generator's assumptions and diversity need review.

Real disclosures supply realistic ambiguity but often lack a fully observed ground truth. Evaluation can then assess traceability, consistency, withheld facts, conditional predictions and decision consequences. The two evidence sources are complementary rather than interchangeable.

24.6 Forecast origins and leakage

A forecast should be reconstructed using information available at its origin. Later restatements, realized prices and revised macro data can leak into features or labels. The evidence store needs

dated versions to prevent this.

A rolling-origin evaluation can produce forecasts at several historical dates and compare them with subsequent observations. The model's parameter estimation and tuning must also respect the cutoff. Training on the full sample and then calling earlier predictions out of sample is incorrect.

Company events such as acquisitions or changes of reporting basis can alter comparability. The evaluation should bridge those changes or define a target that remains meaningful. A forecast error caused by a perimeter mismatch is not the same as an economic prediction error.

24.7 Conditional scenario evaluation

A deterministic stress scenario may never occur exactly. Its mechanical correctness can still be tested through contract formulas, accounting invariants and controlled cases. Empirical validation of conditional responses requires suitable observed variation or another identification strategy.

A model predicting cash under a stipulated 200-basis-point shock cannot be validated by comparing it with actual cash after a period containing many other shocks without accounting for those differences. Such a comparison may be descriptive, but it does not isolate the rate response.

The evaluation should distinguish unconditional forecasting, conditional stress mechanics and causal intervention claims. Each has a different comparator and evidence requirement. Calling all three simulation accuracy obscures the target.

24.8 Probabilistic forecasts

If the system supplies a predictive distribution, evaluate both calibration and informativeness with methods appropriate to the outcome. A forecast assigning broad mass everywhere may be calibrated in a weak sense while offering little value. A sharp forecast can be badly miscalibrated.

Proper scoring rules, discussed in Gneiting and Raftery (2007), provide a principled way to reward honest predictive distributions under their conditions. The chosen score should match the outcome type and decision context. A score on annual earnings does not automatically validate default probability or liquidity tails.

Probabilities conditional on one reconstructed history should be distinguished from probabilities averaged across histories. If history weights are subjective or ambiguous, report sensitivity. A good score under one prior does not establish that the prior is uniquely justified.

24.9 Set-valued forecasts

An unweighted family produces a set of possible outcomes. Evaluation can ask whether later observed outcomes fall within it and how wide or structured the set is. A rectangular interval display may obscure joint restrictions, so multivariate coverage should respect the actual feasible set.

If observed cash and interest each lie within their marginal intervals but violate the modeled relation $C + I = 95$, the joint model has failed for that observation. Marginal coverage alone would miss it. Conversely, a wide marginal interval may coexist with a precise relation that is

useful for a decision.

A set can be evaluated against a declared decision task: does it correctly identify policies that are robustly feasible or expose the history driving a failure? This should supplement, not replace, source and mathematical checks. A useful decision result can still rest on an incomplete family.

24.10 Policy evaluation

A policy should be evaluated under the information and action constraints it would actually face. A perfect-foresight benchmark can provide an upper bound in some settings, but it is not an implementable comparator. A no-action baseline may be useful but can be too weak to establish practical value.

A fair comparison includes a credible existing policy or classical method, with equivalent access to data and tuning. If the proposed method receives a richer contract representation, separate the benefit of representation from the benefit of its optimizer. Otherwise a solver improvement claim may actually be a data improvement.

Policy value estimates need uncertainty analysis and attention to rare failures. A higher mean payout can come with unacceptable liquidity risk. The primary criterion and vetoes should be declared before selecting a winner. Descriptive advantages from a few simulations should not be promoted to a default policy.

24.11 Ablations that answer real questions

An ablation can remove contract detail, actor information, persistent memory or joint constraints to test their contribution. The comparison should preserve other conditions. Removing several features at once can show that the bundle matters but not which mechanism explains the change.

For example, compare a full debt schedule with an aggregate debt balance while holding pricing and policy fixed. Then test the effect of adding endogenous refinancing. This sequence separates contract-state sufficiency from behavioral response.

Expected failures can be informative. If the aggregate model misses a maturity wall, that supports the need for timing in that query. It does not prove that every query requires instrument-level detail. The conclusion should match the demonstrated mechanism and scope.

24.12 Stress coverage

A validation set should include ordinary operations, near-boundary cases and severe but admitted stresses. Important dimensions include rates, currency, customer concentration, maturity, collateral, offer capacity, settlement delay and information timing.

A Cartesian product of every dimension may be infeasible or economically incoherent. Use the joint model to identify feasible combinations and targeted adversarial cases. Cover interactions known to matter, such as collateral loss reducing funding exactly when cash falls.

Coverage should record mechanisms tested and regions untested. A large path count is not a substitute. Ten thousand smooth scenarios can leave a single conversion or cure-period boundary unchecked.

24.13 Numerical reliability

For approximate methods, validation should compare against exact or highly controlled small references, assess convergence or refinement, and inspect downstream decisions. Solver tolerances should be translated into economic quantities where possible.

A residual below a threshold can still be unacceptable if it changes default classification. A slightly larger residual in a large nonbinding balance may be immaterial. The evidence contract should state which diagnostics veto use and which explain remaining error.

Repeat stochastic comparisons with appropriate seeds and uncertainty estimates. If candidates remain statistically indistinguishable after validity screens, report that. A method can be viable without being proven superior. Default promotion requires stronger evidence than an optional experimental path.

24.14 Failure accounting

Evaluation must count unsupported, refused and numerically failed cases. Excluding them from average error can make a fragile method look accurate on the easy subset. Report coverage and conditional performance together.

A refusal can be correct if the request exceeds the admitted profile, but excessive refusal limits usefulness. A numerical failure is different from a correctly identified economic shortfall. A result table should preserve these categories rather than assign all failures the same placeholder number.

If an approximation leaves probability mass unresolved, include that mass in risk interpretation. If the model is unweighted, report unresolved regions and the bounds they permit. Renormalizing completed cases is a different conditional target and should be labeled as such.

24.15 Reproducibility and independent checks

A reproducible run records source snapshots, model versions, assumptions, methods, seeds and environment. It also preserves exact commands and outputs. Reproduction verifies that the same computation can be obtained; it does not independently validate the model.

Independent checks should use different routes where feasible: algebra versus code, event-level versus aggregate reconciliation, or one solver versus a certified reference. Shared data and code can share errors. A second model reading the same incorrect assumption is not independent economic evidence.

Rendered document review is also distinct. It can reveal missing symbols, broken references and unreadable derivations. A clean build cannot establish that a reader understands the argument. The manuscript should remain provisional until actual human comprehension evidence is available.

24.16 A proposed staged evaluation

First establish exact semantics with the synthetic cases in this part. Then evaluate source-to-model coverage on a bounded real disclosure package, preserving unresolved content. Next test historical reconstruction against withheld facts and carefully defined reporting bridges.

Only after those stages should a forecast or policy study claim economic performance. Its plan must identify target companies, time cutoffs, baselines, tuning protocol, primary criterion and uncertainty analysis. A failure at an earlier stage can invalidate interpretation of the later results.

The stages are dependencies, not excuses to stop independent work. Source research, documentation and bounded implementation can continue while empirical data or human review remain pending. The full mission remains open until its specified evidence exists.

24.17 What a defensible conclusion sounds like

A strong narrow conclusion might state that a method reproduces all exact cash and ownership transitions in a declared profile and preserves the tested information restrictions. Another might state that a predictive distribution outperforms a named baseline under a predeclared score with uncertainty evidence on a stated sample.

Neither conclusion should be expanded into universal accounting compliance or company forecasting ability. A failed candidate can be rejected without rejecting the entire uncertainty-preserving approach. A successful small model can justify the next experiment without becoming a production default.

The evaluation program should make those distinctions easy to maintain. That is the practical counterpart of the manuscript's central idea: preserve what is known, preserve what remains possible, and make every stronger conclusion earn its evidence.

Chapter 25

Failures at the joins: how individually plausible methods produce a wrong simulation

Many serious errors occur where two reasonable methods meet. A contract engine can calculate coupons correctly while an accounting adapter treats them as expense under an incompatible basis. A particle filter can implement its weights correctly while a policy sees hidden future state. A consolidation routine can balance while a liquidity engine spends cash in the wrong entity. This chapter examines these joins and proposes checks that target the interaction rather than one isolated module.

25.1 Reconstruction followed by valuation

Suppose historical debt carrying amount is 100 and an analyst reconstructs face as 100. A valuation engine then prices the promise correctly. If the carrying amount included an unamortized discount or accrued interest, the promise was misidentified before pricing began. The valuation solver can return a precise answer to the wrong contract.

The interface should require a distinction among observed carrying amount, inferred face and supplied valuation terms. If face is unknown, pricing should operate conditionally over its feasible values or request a supported mapping. A test should include face 110, carrying amount 100 and proceeds different from both.

The repair is at the reconstruction boundary, not by adjusting the discount rate until price matches book value. Such calibration would hide the missing contractual amount and could distort every later scenario. A book-to-contract bridge should explain the difference with admitted movements and measurement rules.

25.2 Valuation followed by cash forecasting

A bond’s market value falls after rates rise. A cash forecast that treats the decline as a cash inflow because the liability became cheaper creates money unless the company actually repurchases or settles the debt. The valuation change is not itself a financing event.

If repurchase is allowed, it requires a market offer, available cash, quantity and settlement date. The company may realize an economic benefit by paying less than face, but it still spends cash

to do so. Accounting gain or loss depends on carrying amount and the applicable treatment.

A discriminating test changes market rates while holding contractual payments and policy fixed. Cash should not change merely from revaluation unless a contract or collateral clause creates a payment. A second test enables repurchase and checks the explicit transaction.

25.3 A hedge followed by accounting classification

An economic hedge can offset cash exposure while failing a particular hedge-accounting designation. Conversely, a qualifying designation does not guarantee perfect economic offset. A simulator that equates the two can misstate both cash and reported income.

The contract layer should calculate the underlying and derivative cash flows separately, including basis and timing. The accounting layer applies designation, effectiveness and measurement rules under the admitted basis. Unknown designation affects reporting, while unknown notional or reset terms affect economic exposure.

A test can use equal notionals but different reset dates. Annual net interest may appear small while interim collateral or coupon cash differs. Another can hold economic cash fixed and vary the accounting designation to show that reporting changes without inventing a new cash flow.

25.4 A credit loss followed by a funding rule

An impairment model may reduce receivable value after customer deterioration. A funding model may continue using the old gross receivable as collateral. Each module can pass its own local checks while the joined model overstates borrowing capacity.

The interface should distinguish accounting allowance, contractual eligibility and realized collection. A borrowing base may exclude overdue invoices regardless of allowance, or apply a separate haircut. These relations need explicit terms. The correct funding reduction is not automatically equal to the accounting impairment.

Use a case where an invoice becomes ineligible before an accounting loss is recognized, and another where an allowance changes but eligibility does not. The tests prevent accidental identification of two different quantities. The repair is a shared event and state relation, not a single reused field called net receivables.

25.5 A source update followed by a cached policy

A new note reveals a near-term maturity. The reconstruction updates debt timing, but a cached financing policy was optimized for a later maturity. Reusing it can produce an infeasible payment path while all displayed balances use the new source.

Cache invalidation must follow semantic dependencies. A policy depends on contract schedule, offers, information, horizon and objective. A hash of the displayed statement totals is insufficient because the totals may be unchanged.

A regression test changes only the maturity clause while preserving current balances. The policy and relevant scenario result should be recomputed or marked stale. Unaffected historical arithmetic may remain cached. This tests precise invalidation rather than indiscriminate recomputation.

25.6 A restatement followed by backtesting

A current data pipeline supplies restated historical earnings to a forecast evaluation that claims to use information available at the time. The forecasting code can be correct while the data view leaks later knowledge.

The evaluation needs source versions and publication cutoffs, including parameter-estimation inputs. A restatement can be used as a later target if that is the defined outcome, but it cannot silently become an earlier predictor.

A test should run an old forecast before and after adding a later restatement. The frozen as-published prediction remains identical, while a separately named restated analysis can change. This is a direct check of evidence semantics at the validation boundary.

25.7 A particle filter followed by robust optimization

A particle approximation may retain only a small subset of plausible contract types. A robust optimizer then treats those particles as the entire feasible family and reports a worst case. The filter approximated a distribution; it did not prove that discarded histories were impossible.

The optimizer should receive the retained family for robust queries, or explicitly report a sample-restricted worst case. Particle witnesses can supply an incumbent adverse case, while symbolic constraints or other methods bound the unexplored region.

A test deliberately gives a low-weight but feasible history a severe cash outcome. Resampling may remove it. The robust result should remain sensitive to it unless evidence excludes it. A probabilistic expected-value result can weight it according to the admitted measure, but that is a different target.

25.8 Scenario reduction followed by policy optimization

A reduced scenario set can preserve mean revenue while removing a signal pattern that changes the optimal financing decision. Optimizing on the reduced set can then produce a policy that fails on omitted paths.

Reduction should preserve or bound decision-relevant outcomes and information structure. A representative scenario is not merely a nearby vector of shocks. Its observation timing, feasible actions and threshold events matter.

A useful test has two close rate paths on opposite sides of a covenant and a refinancing action available only before the breach is observed. A reduction that merges them must either retain conditional behavior or report an approximation bound that covers the decision loss. Mean matching alone is insufficient.

25.9 A policy optimizer followed by event execution

An optimizer can choose an issue amount based on continuous shares and immediate settlement. The event engine enforces integer lots and a two-day settlement lag. The optimized plan is no longer feasible. Rounding and delaying it after optimization can alter the objective materially.

The optimization problem should include the same action domain and timing as execution, or explicitly account for a projection and its effect. A post hoc repair that borrows the residual

cash is another policy and needs its own feasibility.

A test can use a funding need just above a lot boundary and a maturity before settlement. It should reject the original plan or reoptimize under the correct constraints. Reporting the optimizer's objective before execution repair would overstate achieved value.

25.10 An accounting rule followed by an economic policy

A dividend policy based on book profit may distribute cash that is unavailable because profit includes noncash gains or uncollected sales. The accounting calculation can be correct; the policy is infeasible without a cash constraint.

Conversely, a cash-rich company can have accounting restrictions on distributions under the applicable rules. A cash-only policy can violate those restrictions. The feasible action set should combine cash, legal, contractual and class-right constraints where admitted.

A test should include positive profit with negative operating cash and another with positive cash but a binding distribution restriction. Neither a profit ratio nor a cash balance alone defines feasible dividends. The policy must receive the relevant state and rules jointly.

25.11 Consolidation followed by credit analysis

A consolidated group can eliminate an intercompany loan while a creditor's claim remains against one legal entity. A credit model using only consolidated net assets can assign support that the creditor cannot access.

The credit analysis needs borrower, guarantor, security and transfer relationships. Consolidated reporting remains useful but is not a legal recovery waterfall. The same assets can support different claims under different entity structures.

A fixture should compare two groups with identical consolidated statements but different guarantees or asset ownership. Recovery and payment capacity should differ where the stipulated rights differ. If the model returns the same result, it has probably compressed away a material legal-entity relation.

25.12 A default model followed by accounting close

A default indicator can trigger an accounting routine that zeros every liability and equity balance. This makes the balance sheet close neatly while erasing continuing obligations, recovery claims and operating assets.

The economic default event should create a process state. Accounting consequences follow actual or stipulated modification, impairment, settlement and extinguishment events under the basis. Default alone does not supply every measurement.

A test should leave unpaid debt outstanding during a cure period, then compare repayment, waiver and restructuring paths. Another should include delayed recovery. The system should preserve obligations until the admitted event changes them, even if a value function assigns zero to shareholders.

25.13 A pricing measure followed by a risk report

A debt price may be computed under a pricing measure, while a risk report asks for real-world default frequency. Reusing pricing probabilities as forecasts confuses valuation with statistical prediction.

The measure identity should travel with every expectation. A risk-neutral probability can be useful for price consistency, but does not automatically describe actual event frequency. A physical probability model can estimate frequency but requires a risk adjustment to price claims unless special assumptions apply.

A test with different physical and pricing probabilities should produce different expected losses while preserving the declared price. The report should refuse an unlabeled probability when the measure is ambiguous. This is a semantic type check with direct financial consequences.

25.14 A numerical tolerance followed by a legal threshold

A solver treats a constraint violation of 10^{-6} as acceptable. A downstream routine interprets the approximate value as exact compliance with a strict contractual threshold. The numerical convention has silently become a legal conclusion.

The result should carry an error interval or tolerance. If the interval crosses the threshold, classification is unresolved unless another calculation determines the sign. The contract's inclusive or strict comparison remains unchanged.

A test at equality and on both sides by less than the solver tolerance reveals this interaction. Exact rational cases can establish intended boundary semantics. Floating-point production methods then need an explicit policy for translating their error into a classification.

25.15 A language model followed by evidence admission

A language model can propose a plausible missing journal or summarize a contract. Those outputs are interpretations or hypotheses until supported. If they are inserted into the evidence store as facts, subsequent inference can become circular: the model later cites its own generated explanation.

The admission layer should retain source anchors and distinguish extracted text, normalized interpretation and generated suggestion. Unsupported suggestions can be useful search leads or candidate templates. They should not narrow the family merely because the text is fluent.

A test should submit a plausible but unanchored maturity date. The system should preserve it as a proposed assumption or reject factual admission. If a genuine source later supports it, the new assertion can narrow the family with traceable provenance.

25.16 A user interface followed by decision interpretation

An interface may display one attractive central path while hiding unresolved regions. Users can reasonably interpret it as the forecast even when the calculation produced a family. The presentation then overstates certainty without any numerical bug.

The default display should identify conditional paths, ranges, measures and unresolved cases in proportion to their importance. It should not label an arbitrary witness most likely. A central

estimate needs a stated selection rule or probability model.

A useful review asks whether the reader can identify the history driving a shortfall, the assumption that could remove it and the status of the computation. If those answers require searching internal logs, the explanation is insufficient even if the underlying data are preserved.

25.17 Composition contracts

Each boundary should specify input meaning, units, timing, information, allowed uncertainty and failure behavior. A contract engine returns obligations and payoffs; an accounting engine returns postings under a basis; a policy engine returns admissible actions; an inference engine returns families or measures.

These interfaces should be tested end to end with the worked cases. A local adapter can be mechanically correct while mapping the wrong concept. The test must compare the resulting economic and accounting consequences, not only schema validity.

Version compatibility also matters. A changed rule can alter event order or measurement. Results should identify the exact component versions and invalidate dependent caches. A stable interface name does not establish semantic compatibility across versions.

25.18 A complete rate, covenant and ownership path

An exact combined example shows why the financing steps must be dated. Two possible companies each start with cash 16 and loan face 100. One pays a fixed coupon of 5; the other pays 7 after the specified floating-rate reset. Each then makes an operating payment of 6. A covenant tests cash of at least 5 immediately after these payments and before financing. Breach forbids new borrowing but permits an equity cure. Management requires closing cash 8. These are assumed contract terms, not a claim about a standard covenant.

The fixed-rate company has cash $16 - 5 - 6 = 5$, meets the covenant and needs 3. A candidate debt issue with maximum face 5, conditional solvency probability 4/5, recovery 2 and gross risk-free return 21/20 has proceeds 88/21. Holding that quoted per-face price fixed for this conditional stress, issuing face 315/88 raises exactly 3. The floating-rate company has cash $16 - 7 - 6 = 3$, breaches the covenant and needs 5. Debt is unavailable; even an improper borrowing override at the maximum face would raise only $88/21 < 5$.

Suppose the permitted equity cure has fee 1, pre-issue economic equity value 20 and 100 incumbent shares. A gross subscription of 6 provides net cash 5. The post-issue equity value is 25, new investors own 6/25, and the common share price is 19/100. Their 600/19 new shares have value 6, while incumbents retain value 19. Both companies finish with cash 8, but their new debt and ownership differ. Merging them on closing cash would conceal those differences. Testing the covenant only after the cure would erase the earlier breach and wrongly admit the borrowing alternative.

This example holds quoted financing prices and the pre-issue equity valuation fixed to expose event ordering and allocation. It does not solve the feedback from financing to default and prices. A source correction introduces another interaction: withdrawing an unsupported fixed-rate classification can reopen the floating-rate history, invalidating an earlier single-history result. If management later receives a rate notice, its admissible policies can change even when the analyst's reported balances do not. Shared computations must therefore depend on evidence version, actor information, contracts and the requested outcomes, as well as current cash.

25.19 Executing promises, carrying amounts and ownership

The combined example can be executed through the existing typed company runtime once its state distinguishes contractual face from carrying amount. For the fixed-rate history, the authored initial-measurement convention posts cash of 3 and a liability carrying amount of 3, while contractual face increases by 315/88. The closing liability carrying amount is 103 and outstanding face is 9115/88. Subsequent effective-interest measurement and settlement require further methods; substituting carrying amount for the promise would lose the very exposure the example is intended to preserve.

For the floating-rate history, subscription proceeds of 6 increase cash and contributed capital; the direct issue cost of 1 reduces both. The typed state retains 100 incumbent shares and 600/19 new shares. Under this explicitly synthetic posting convention, both histories report assets of 112. The fixed history has liabilities of 103 and book equity of 9; the floating history has liabilities of 100 and book equity of 12. These book amounts differ from the economic equity values used to price the subscription. No reporting-standard admission follows from the balanced journals.

The reference adapter executes coupon payment, operations, covenant observation and financing in that order, using the unchanged common company executor. It rejects borrowing after the floating-rate breach, proceeds inconsistent with the supplied per-face quote, a zero issue price, duplicate issuance and payment without sufficient cash. A mutation that omits the share-count update still produces balanced accounts and completes the accounting path; the independent ownership comparison detects the error. Accounting conservation alone therefore cannot validate a financing method.

The same runtime also executes the earlier information example. Starting with cash 6, a fixed coupon 5 leaves 1, while a floating coupon 7 cannot be paid. A hedge costing 1 caps the coupon at 5. An informed manager hedges only the floating history and retains cash 1 or 0; a manager who must act identically in both histories can hedge both and retain 0. Withdrawing the earlier fixed-only evidence reopens the payment-refusal branch. The negative cash in the algebraic shortfall calculation denotes unmet funding; the runtime refuses the payment atomically and keeps the opening cash, rather than recording an impossible settlement. A missed-payment event and any ensuing default process would need separate declared methods.

25.20 Prioritizing interaction tests

The highest-value interactions are those that can create cash, remove obligations, leak information or drop adverse histories. They should be tested early because they can make every downstream chart look plausible. Positive-recovery pricing, settlement timing, ownership conversion and joint-history preservation are examples.

Other interactions can be staged according to the active company and query. A bank capital rule may be irrelevant to a simple industrial-company fixture but essential to a bank funding scenario. The full mission retains these domains without forcing every first implementation to solve all of them at once.

The catalogue remains extensible. Each newly discovered failure should identify the target, actual behavior, mechanism, discriminating case and repair. That practice turns failures into durable knowledge and prevents the same convenient shortcut from reappearing under another module name.

25.21 Credit schedules after financial write-off

Financial closure leaves a separate question about the history of scheduled receipts. The retained credit examples already record uncollected amounts and portions removed by write-off. A dated reconciliation now checks how those amounts relate to the original schedule, actual collections and any remaining modeled receipts. These quantities are nominal currency amounts. They are neither discounted carrying values nor an assessment of enforceable legal rights.

Let d denote a scheduled payment date and k the position in the recorded event sequence. Write O_d for the original scheduled amount, $\Delta_{k,d}$ for cumulative signed changes to the modeled receipt schedule, $C_{k,d}$ for collections, $U_{k,d}$ for uncollected amounts recorded when a due date is resolved, $W_{k,d}$ for portions removed by financial write-off, and $R_{k,d}$ for the remaining schedule. Each dated record satisfies

$$O_d + \Delta_{k,d} = C_{k,d} + U_{k,d} + W_{k,d} + R_{k,d}.$$

A collection transfers the scheduled amount at that date from R to C and U. A proportional write-off transfers the specified portion from R to W. A replacement schedule changes R and Δ by the same dated amount. Accrual changes the financial measurements but leaves this nominal relation unchanged. These observations derive the identity from the admitted events; they do not establish which events should be admitted for a real loan.

In the partial-write-off example, the original schedule is 66 on 1 January 2026 and 363/5 on 1 January 2027. The identified one-fifth portion removed on 1 July 2025 contains 66/5 and 363/25. Subsequent collections of 44 and 55 leave uncollected amounts 44/5 and 77/25. At the two dates separately,

$$66 = 44 + \frac{44}{5} + \frac{66}{5}, \qquad \frac{363}{5} = 55 + \frac{77}{25} + \frac{363}{25}.$$

Their sum is $693/5 = 99 + 297/25 + 693/25$. The full-write-off example instead retains the whole original schedule among the removed portions and keeps its zero-valued remaining entries. The record distinguishes those entries from missing dates.

The performing prepayment illustrates why the signed revision belongs in the identity. Under its supplied unchanged-terms finding, the modeled schedule changes from 55 and 121/2 at the two annual dates to 105 at the first date. The changes are therefore $+50$ and $-121/2$. The later receipt of 105 reconciles to that explicit replacement. Calling the replacement a legal amendment, or silently comparing the receipt with the old total, would change the meaning of the example.

The read-only companion checks all 57 stages of the nine retained histories against their original schedules and archives, with independently authored endpoints. It retains event order even when two events share a date, zero uncollected amounts, write-off portion identities, and the exact source and financial-result identities. Twenty-five controls reject missing or duplicated portions, changed dates, hidden replacements and unsupported assertions of forgiveness or accounting admission. The existing financial journals remain unchanged. Reproducible calculations and their limits are recorded in `data/accounting-master/credit-claim-continuity-20261027/result.md`.

All nine examples finish with a financially closed book. Adding a further event through the existing native interface produces `event_after_close`. That refusal describes the implemented interface; it does not determine whether a recovery can occur. A later recovery needs its own dated receipt evidence, allocation and supported accounting treatment. The present record preserves the earlier history while leaving forgiveness and later recovery unassessed.

25.22 What a later recovery would require

A receipt after financial write-off raises a new accounting question. The original EU source used by the retained examples treats write-off as a derecognition event (IFRS 9, paragraph 5.4.4), while its disclosure requirements explicitly contemplate written-off assets still subject to enforcement (IFRS 7, paragraphs 35F(e) and 35L). These provisions explain why the history of a removed portion should remain available. They do not determine the legal position of a particular borrower or supply evidence of a later receipt.

The distinction also matters when choosing a recognition rule. The IFRS Interpretations Committee's March 2019 decision on curing a credit-impaired financial asset addresses the adjustment of a recognised loss allowance, including discount unwinding. It requires the difference considered in that request to be presented as a reversal of impairment losses rather than interest revenue. Applying that answer to cash received after full write-off would require a further argument: the checked decision does not explicitly address that fact pattern. The retained purchased or originated credit-impaired cases also need their own applicability analysis. Exact inspected passages, source versions and unresolved questions are retained in `data/accounting-master/credit-recovery-scope-20261027/source-map.md`.

Before a later event can affect the book, the model needs the receipt's amount, currency, transaction and information dates, source availability and allocation. An allocation to a previously written-off portion must identify that portion; an allocation to a different supported claim must preserve that distinction. The applicable recognition and presentation treatment remains a separate proposal. Cash-flow classification also depends on the entity and the nature of the receipt, so a synthetic operating or investing label cannot answer it.

The first intake therefore records missing evidence explicitly. For each of the nine closed histories, it retains the complete dated nominal record and leaves the new receipt and its allocation unknown. Zero-valued historical components remain visible; an unknown new receipt is not replaced by zero. Actual prior collection identities submitted as new receipts are rejected because those collections are already represented. The checked intake has no financial effects and accepts no positive recovery in this version. Its 50 adverse controls test preservation and refusal boundaries, not recognition of an actual recovery. The reproducible result is in `data/accounting-master/credit-recovery-intake-20261027/result.md`.

This limited interface makes the remaining work concrete. A positive recovery method needs authentic receipt evidence, a supported allocation and a justified post-derecognition treatment. Those additions must preserve the original write-off finding and financial history, rather than retrospectively rewriting them to match a later outcome.

25.23 Issued and held results before dated reporting

An insurance calculation can preserve the right total while losing the relation between issued contracts and reinsurance held. The retained covered-loss example already calculates both sides under a supplied one-half quota and zero rates. Its estimate change reduces issued claims by 4 and expected held recoveries by 2. The issued loss reversal therefore contributes 4 to profit, while the held remeasurement contributes an expense of 2. The held loss-recovery component falls from 4 to 2 as a tracked component; that movement is not a second expense at the estimate-change stage.

A complete stage ledger now exposes the consequence through settlement. It starts with cash 247/2, an issued liability of 120, held fulfilment cash flows of 49/2 and held cost CSM of 6. Thus opening equity is

$$\frac{247}{2} + \frac{49}{2} + 6 - 120 = 34.$$

The estimate change raises equity to 36. Issued service subsequently releases revenue of 100 and the remaining loss component of 16 against claims expense of 114, adding service profit of 2. Held service records recovery income of 22, premium allocation of 53/2 and loss-recovery allocation of 2, for a loss of 13/2. Final equity is therefore

$$34 + (4 - 2) + 2 - \frac{13}{2} = \frac{63}{2}.$$

Cash reaches the same amount through the separate payment of 114 and receipt of 22. The held component's later service allocation remains present even though there is no additional component expense at estimate revision. Combining those two stages would conceal the distinction.

The second retained case reduces covered claims by 8, exhausts the held loss-recovery component and ends with cash and equity of 67/2. Its zero component and zero later release remain explicit. Both examples reconcile every account, journal, cash movement and component origin at opening, estimate change, issued service and held service. Their complete native results and independently authored intermediate balances agree; 22 adverse controls test double expenses, lost origins, net-only substitutions and unsupported admission.

These are ordered accounting stages, not dated fiscal reports. The source calculation supplies neither event dates nor separate identifiers for the issued and held groups. The existing annual reporting adapter accepts only the two retained issued-loss schedules; direct calls confirm that it does not admit these recovery-change examples. The stage ledger therefore leaves dates and separate group identities unknown and keeps that report-connection gap open. Its complete results and the consumer-call audit are retained in `data/accounting-master/insurance-recovery-stage-book-20261027/result.md` and `data/accounting-master/insurance-shared-group-audit-20261027/result.md`. A dated extension must supply chronology, group identity and a compatible fiscal opening before these stages can become a company-book execution.

Chapter 26

A fully solved risky-debt model before any neural approximation

A small structural model can make the financing equations completely transparent. This chapter solves one with two future asset states. It is deliberately restrictive: one period, one creditor, no tax, no fees initially, no bankruptcy costs, no interim distributions and risk-neutral pricing with zero risk-free rate. These assumptions isolate the relationship among promised repayment, proceeds, default, recovery and equity value. They are not a general corporate-finance specification.

26.1 Assets and claims

At the final date, assets available to creditors and shareholders are 60 in the low state and 140 in the high state, each with probability one half. A debt contract promises face $F \geq 0$. The creditor receives the smaller of face and available assets. Shareholders receive the positive residual.

$$D(A, F) = \min(F, A), \tag{26.1}$$

$$E(A, F) = \max(A - F, 0). \tag{26.2}$$

For every state, $D(A, F) + E(A, F) = A$. This conservation relation is exact under the stipulated priority and absence of costs. Default occurs when assets are insufficient for the promise, with equality treated as full payment.

Debt proceeds at issuance are the expected creditor payoff because the pricing rate is zero and risk neutrality is assumed. Economic equity value is the expected shareholder payoff. Therefore

$$P(F) = \tfrac{1}{2}\min(F, 60) + \tfrac{1}{2}\min(F, 140), \qquad V_E(F) = 100 - P(F). \tag{26.3}$$

The expected asset value 100 is allocated between the claims. Face does not equal proceeds once default becomes possible.

26.2 Solve the price schedule by regions

For $0 \leq F \leq 60$, debt is paid in both states, so $P(F) = F$. For $60 < F \leq 140$, the creditor receives 60 in the low state and F in the high state, so $P(F) = 30 + F/2$. For $F > 140$, the

creditor receives all assets in both states, so proceeds remain 100.

$$P(F) = \begin{cases} F, & 0 \le F \le 60, \\ 30 + F/2, & 60 < F \le 140, \\ 100, & F > 140. \end{cases} \tag{26.4}$$

The function is continuous but its slope changes. Promising another unit raises proceeds by one in the safe region, by one half in the intermediate region and by zero beyond the highest asset payoff. An unlimited face amount does not create unlimited funding.

Equity value is $100 - F$ in the first region, $70 - F/2$ in the second and zero in the third. These are economic values under the model. They are not a balance-sheet measurement rule. The model has not specified accounting carrying amounts.

26.3 A funding need of 80

Suppose an admitted transaction requires gross proceeds of 80. This need lies above the safe-debt capacity 60. Solve $30 + F/2 = 80$, giving face 100. In the low state the lender receives 60 and shareholders zero. In the high state the lender receives 100 and shareholders 40.

Expected lender payoff is 80, matching proceeds. Expected shareholder payoff is 20. The promised yield is $100/80 - 1 = 25$ per cent, although the risk-free rate is zero. That promised yield compensates for the modeled default loss; it is not the realized return in both states.

The low-state recovery is 60, or 60 per cent of face. If the model instead specifies recovery as a fixed fraction of face independent of available assets, it is a different recovery model and may violate conservation at some face values. Recovery assumptions cannot be swapped silently.

26.4 A funding need of 110

No face value raises 110 in this model because expected distributable assets are only 100 and the creditor cannot receive more than assets. A root finder searching for $P(F) = 110$ should report no solution under the model, or a validated infeasibility conclusion if its method establishes it.

A numerical optimizer that pushes face to a large upper bound has not solved the funding problem. Reporting that bound as issued debt would invent proceeds. The flat price region is a useful test of solver behavior when increasing the promise no longer raises cash.

The company could still have an alternative outside this model: new equity, collateral from another entity, a guarantee or a changed investment plan. The no-solution conclusion is conditional on the admitted financing channel and asset process. It is not proof that the real company has no option.

26.5 Fees and net cash

Add a fixed issuance fee of 5 paid at inception. To net 80, gross proceeds must be 85. Solving $30 + F/2 = 85$ gives face 110. Net cash is 80, expected creditor payoff 85, and expected equity payoff 15 under the unchanged asset distribution.

If the fee instead equals five per cent of gross proceeds, required gross proceeds are $80/0.95 = 1600/19$. Face is $2(1600/19 - 30) = 2060/19$, about 108.4211. The two cost structures give different promises.

If the fee is itself paid out of final assets rather than at inception, the asset payoff to claims changes and pricing must be recomputed. Merely subtracting the fee from current proceeds while also reducing final assets would charge it twice. The event dates determine which equation it enters.

26.6 Recovery costs

Suppose default destroys 10 of assets before creditor recovery. For $60 < F \leq 140$, low-state recovery becomes 50 and high-state payment remains F, so $P(F) = 25 + F/2$. Raising 80 now requires face 110.

At face 110, expected shareholder payoff is 15 and expected creditor payoff 80. Their sum is 95, below expected assets 100 by expected default cost 5. The conservation relation now includes the cost recipient or destroyed value: creditor payoff plus shareholder payoff plus default cost equals assets in each state.

A test that insists creditor plus shareholder value always equals 100 would be wrong after adding bankruptcy costs. Invariants must follow the model's resource accounting. Conversely, a model with no costs that loses value unexplained has a defect.

26.7 Endogenous investment changes the price

Suppose management can choose between project S with assets 100 in both states and project R with assets zero or 180 equally likely. Expected assets are 100 for S and 90 for R. Let face be 80. Under S, creditor payoff is 80 and equity payoff 20. Under R, creditor expected payoff is 40 and equity expected payoff 50.

Shareholders prefer R if they choose after debt proceeds are fixed and maximize their residual value. Total expected asset value is lower under R. A lender anticipating R prices face 80 at 40, not 80. Pricing as though S will be chosen while allowing shareholders to choose R is inconsistent.

A covenant restricting project choice, a different security structure or a commitment can change the feasible policy. The model must say whether project choice is committed before issuance, observed by lenders or chosen afterward. Timing determines the financing equilibrium.

26.8 A committed project comparison

If management commits credibly to S before issuance, face 80 raises 80. If it commits to R, face 80 raises 40. To raise 80 under R, face must satisfy $F/2 = 80$ while $F \leq 180$, giving face 160. Equity expected payoff is then 10.

The comparison is now between different promises financing the same proceeds under different committed projects. S leaves equity value 20; R leaves 10. Under the stated objective and no other benefits, S is preferred ex ante. This differs from the post-issuance asset-substitution choice at fixed face.

The example demonstrates why a structural model must distinguish issuance decisions from later policy. A single optimization over all variables can accidentally grant commitment that the actor does not possess. A sequential model must encode which choices are binding when the lender prices the debt.

26.9 A simple equity alternative

Suppose the company can instead sell a fraction u of the final asset claim for fair value under the original 60/140 distribution, with no fees. The price is $100u$. Raising 80 requires selling 80 per cent, leaving existing owners expected value 20.

This matches the expected residual value after issuing face 100 debt in the frictionless original example, but the state payoffs differ. With equity financing, old owners receive 12 in the low state and 28 in the high state. With debt financing, they receive zero and 40. Equal expected value does not imply equal risk or incentives.

Adding taxes, costs, control rights or different investor preferences can change the comparison. Those mechanisms must be specified rather than inferred from the equality in this simplified benchmark. The benchmark is valuable because it identifies exactly which additions drive a difference.

26.10 Unknown asset distribution

Suppose reports do not identify the low-state asset payoff and it may be 40 or 60, while the high payoff remains 140 and probabilities remain one half by assumption. For a promised face 100, proceeds are 70 or 80. A funding plan requiring 80 succeeds only in the second history.

If lenders know the asset type privately, they offer different prices across histories. If lenders share the analyst's uncertainty and use a prior, pricing requires that prior and the information structure. Averaging proceeds without stating who knows the type can misrepresent both the offer and management's choice.

An unweighted robust funding calculation asks whether a face can raise the required cash under every admitted pricing history. At face 120, proceeds are 80 in the low-asset history and 90 in the other. But the higher promise changes shareholders' payoffs. A robustly feasible financing plan need not maximize expected owner value under a selected prior.

26.11 Learning before issuance

A verified asset report arriving before issuance can identify the low-state payoff and narrow the price schedule. The same report arriving afterward cannot retroactively change settled proceeds, although it may affect secondary-market value or future refinancing.

The simulator should retain issuance-time information for the original price and current information for later valuation. Repricing the original cash receipt after new information would rewrite history. A market-value update is a separate measurement, not a new issuance event.

This distinction is particularly important in backtests of debt pricing. Using later realized recovery information to set the earlier issue price creates look-ahead. The model can use that information to evaluate the forecast, but not as an original input.

26.12 Two creditors, policy consistency and financing refusal

In a finite one-period model, policy consistency can be checked by enumeration. Specify a candidate policy π, the distribution of the terminal estate under that policy, the priority of each claim, and a pricing measure and discount factor. Allocate the estate once, calculate each

lender's expected discounted receipt, and use those receipts as candidate issue proceeds. At those fixed quoted proceeds, compare the issuer's payoffs from every feasible policy. Retain the candidate only when its conjectured policy is a best response and all funding and participation constraints hold. Ties must be retained. If no pure-policy pair passes, report that result for the declared class; neither a balancing loan nor an assumed mixed equilibrium repairs it automatically.

For a concrete local example, existing senior debt has face 40 and newly issued junior debt has face 80. A project costs 60. The safe policy produces estate 100; the risky policy produces estate 20 or 220 with equal probabilities. Both pricing and shareholder valuation use the stated risk-neutral probabilities and gross discount factor 21/20. Absolute priority in this illustration gives receipts

$$(s, j, e) = \big(\min(A, 40), \min((A - 40)^+, 80), (A - 120)^+\big). \tag{26.5}$$

Thus the safe estate pays $(40, 60, 0)$, and the risky estates pay $(20, 0, 0)$ and $(40, 80, 100)$. Each allocation sums to its estate. The candidate junior proceeds are 400/7 under the safe policy and 800/21 under the risky policy.

At a posted quote P, existing shareholders must subscribe $60 - P$ to cover the investment. Their net value from policy π is $\mathbb{E}[e_\pi]/(21/20) - (60 - P)$. They prefer risk at either quoted price. Consequently, the safe-policy quote is inconsistent with their best response: if risk is chosen, junior lenders receive present value 800/21 after advancing 400/7, a shortfall of 400/21. The risky quote is consistent within the pure-policy class, but requires shareholder cash 460/21. If shareholders have only 20, the project remains unfunded. The safe quote would need less cash, but its incentive problem remains. Even an enforceable safe-policy commitment would still need shareholder participation: here its zero terminal equity receipt does not cover their positive subscription. Liquidity, incentives, lender valuation and voluntary participation are distinct restrictions.

The same separation applies to restructuring. Given estate 70 and creditor outside values 50 and 20, an offer 45/25 conserves value but the senior creditor refuses. An offer 50/20 meets both constraints. A mandatory fee 1 reduces the available estate to69 and makes simultaneous participation impossible under these stated outside values. This is a feasibility result. Selecting a unique offer requires bargaining rights or another decision rule; it does not follow from total value conservation.

26.13 Why positive surplus can still end in liquidation

Hart and Moore's retained 1991 working paper makes the distinction between productive value and credible repayment particularly clear. The entrepreneur can withdraw human capital, the creditor can seize specified assets, and renegotiation follows an explicit offer game. Returns and liquidation values are deterministic, private savings have a particular seizure treatment, and one creditor controls the assets (Hart and Moore, 1991, Sections 2–3 and Appendix A1–A4). Under those assumptions, Lemma 1 derives the creditor's continuation reservation value and requires both future nonrepudiation and cash feasibility. The bargaining rule determines a result that cannot be obtained merely by balancing accounts.

In their Example 1, take two remaining operating receipts of 2 and 26 and liquidation rents of 8 in each period. The creditor's reservation value is 16, while the final credible payment is at most 13. The entrepreneur therefore needs to provide at least 3 before the final date. The first operating receipt provides only 2, so private wealth $w \geq 1$ is necessary. At $w = 1$, an immediate payment of 1 followed by payments 2 and 13 satisfies the cash and repayment conditions. At $w < 1$, it cannot. Liquidation occurs despite operating value 28 exceeding liquidation value 16

(Hart and Moore, 1991, Example 1, printed p. 16). The reference calculation reproduces this boundary with exact rational amounts. It is a useful counterexample to assuming that positive total surplus guarantees refinancing or restructuring.

Their repayment-path analysis also reinforces the distinction between an observed payment and its contractual explanation. A long-term contract and rolling short contracts can support the same path under the paper's conditions; the latter may additionally require access to outside refinancing (Hart and Moore, 1991, Proposition 2 and printed pp. 27–28). A future market closure can therefore distinguish histories that looked equivalent before the shock. For a multi-creditor company, the analyst must additionally specify priority, control and consent rights, collateral and intercreditor restrictions. The single-creditor result supplies neither an equity-issuance model nor a general legal restructuring procedure.

26.14 Reference-solver requirements

The finite model has a closed-form piecewise price schedule, so a numerical method can be checked at safe, risky, boundary and flat-price regions. Required points include face zero, 60, 100, 140 and above 140. Funding roots should be checked for needs below 60, between 60 and 100, exactly 100 and above 100.

A neural approximation can fit the schedule, but average error is not the only criterion. It should preserve monotonicity, upper funding capacity and boundary classification within stated error. A small positive price slope beyond 140 can falsely imply that an arbitrarily large face eventually raises 110. Downstream feasibility is the decisive check for that use.

A smooth approximation around the kinks can be useful for training, but final results should be evaluated against the original payoff. The source literature on learned economic solvers provides candidate algorithms; this exact model provides a local target-specific test. No training result is claimed here.

26.15 How the example answers the financing concern

Debt issuance is not a choice of a balance-sheet number. It is a contract whose cash price depends on future payoff and policy. Equity issuance is not merely negative dividends; it allocates future claims and potentially control. The finite model makes those relations explicit before numerical approximation.

For AccDSL, an external offer can be accepted as a conditional input, while an endogenous financing module can solve a stated pricing-policy problem. Both routes must keep face, proceeds, recovery, owner claims and information dates separate. This provides a defensible starting point for extending the supplied lecture rather than importing its conflicting convention into a larger simulator.

26.16 Conversion also changes the accounting question

The economic financing model specifies who receives cash and who chooses the project. Accounting for those events requires additional distinctions. In the supplied contingent-conversion extension, a creditor receives 120 shares alongside 40 existing shares. The resulting 75 per cent voting fraction explains the extension's assumed economic control rule. It does not by itself establish the date control begins for accounting, the holder's reporting perimeter, or the amount

to recognise in its accounts.

The following analysis uses Australian standards as a conditional design case. It does not select a reporting basis for an actual company. The retained editions and their application periods matter: the inspected AASB 10 compilation covers annual periods beginning in 2026 before 2027, and the inspected AASB 3 compilation covers starts from 1 July 2026 before 2027. The synthetic issue and conversion occur in 2027 and 2028, so a complete applicable edition and entity-scope chain remains necessary. The source paragraphs identify the questions that must be answered, while the missing facts remain unresolved.

26.16.1 Control can begin before the shares arrive

Under AASB 10, control requires power over the investee, exposure or rights to variable returns, and the ability to use that power to affect the returns. Power concerns existing rights giving the current ability to direct the activities that significantly affect those returns. Majority votes support that conclusion in the circumstances specified by the standard, but the votes must be substantive and contrary rights or arrangements can matter (Australian Accounting Standards Board, 2026c, paras. 6–18, B22–B28, B35–B37).

This is a question about rights and relevant decisions, not just a percentage. A lender's protective covenant can restrict a borrower without giving the lender power over its activities. Conversely, an apparently contingent decision right is not automatically protective. If the activities that significantly affect returns arise only after a particular event, the right to direct those activities can matter before the event occurs (Australian Accounting Standards Board, 2026c, paras. B26–B28, B53).

Potential voting rights introduce the same timing issue into a convertible instrument. AASB 10 requires consideration of substantive potential rights, including those arising from convertibles, options and forwards. Whether rights are substantive depends partly on whether they can be exercised when decisions about the relevant activities need to be made (Australian Accounting Standards Board, 2026c, paras. B24, B47–B50). It follows that the simulator needs a control assessment before conversion as well as after it. Setting the first control date equal to the share-issue timestamp would conceal this question.

The acquisition date for an in-scope business combination is the date control is obtained. It can precede or follow the legal closing date when the relevant facts support that timing. Consolidation similarly begins when control is obtained (Australian Accounting Standards Board, 2026c, para. 20) (Australian Accounting Standards Board, 2026h, paras. 8–9). Thus the economic conversion, the rights assessment and the accounting acquisition date require distinct records linked by evidence. A date cannot be supplied merely to make the annual schedule continuous.

26.16.2 Two control assessments do not determine the intervening history

Suppose the investor controls the company at both observation dates. Control may have continued throughout, or it may have been lost and later regained. Likewise, two observations of no control permit an intervening acquisition followed by a disposal. An unknown earlier assessment must preserve both acquisition and continued-control possibilities. The reporting period therefore needs a history of changes and evidence about the intervals between them; an empty event list does not establish that no change occurred.

The accounting consequences differ. Changes in ownership while control continues are equity transactions under the inspected AASB 10 paragraphs. Loss of control instead requires

the specified derecognition, retained-interest measurement, treatment of accumulated other comprehensive income and resulting gain or loss (Australian Accounting Standards Board, 2026c, paras. 20, 23, 25, B96, B98–B99). A later return to control raises a new acquisition assessment. The same closing voting fraction cannot select among these histories.

A boundary comparison exposed this distinction in the first conditional reference: unknown prior control incorrectly selected an acquisition route, and a known loss omitted the loss-of-control dependencies. The successor preserves all nine combinations of controlled, not controlled and unknown endpoints, together with possible intervening changes. Each interval remains unresolved until additional evidence narrows it. Its history classes preserve the alternatives relevant to these dependency questions; they do not enumerate every possible event sequence. A supplied measurement-reference string likewise identifies a dependency without establishing the measurement itself.

26.16.3 The holder's investment and the group's assets are different reports

In separate financial statements, AASB 127 permits investments in subsidiaries, joint ventures and associates to be accounted for at cost, under AASB 9, or using the equity method, subject to its requirements and consistent treatment within each investment category. Certain required fair-value treatments carry through to the separate statements (Australian Accounting Standards Board, 2022a, paras. 3–10, 11A). Choosing a cost policy does not provide the initial cost amount. The earlier price paid for a convertible claim cannot silently become the post-conversion measurement without the relevant transaction and measurement analysis.

For consolidated reporting, a parent normally consolidates controlled subsidiaries, with specified exemptions and exceptions. An investment entity generally measures qualifying subsidiary investments at fair value through profit or loss, while a non-investment-entity subsidiary providing services related to its parent's investment activities can fall within the consolidation exception to that exception (Australian Accounting Standards Board, 2026c, paras. 4–4B, 27–33). The investor's entity status, the subsidiary's activities and the complete exemption conditions therefore belong among the required facts. None is determined by the 75 per cent fraction.

Consolidation combines the relevant assets, liabilities, income, expenses and cash flows, eliminates the parent's investment against the relevant subsidiary equity, and removes intragroup transactions and balances. It also requires consistent policies and appropriate reporting dates (Australian Accounting Standards Board, 2026c, paras. B86–B93). Potential rights used in a control assessment do not automatically set the proportion allocated to non-controlling interests: the standard separately addresses existing ownership and instruments that already provide access to ownership returns (Australian Accounting Standards Board, 2026c, paras. B89–B91).

This creates a concrete implementation consequence. The existing selected group-period reference assumes full ownership and the same supplied control edges and membership at opening and closing; it supplies no non-controlling-interest method. A company acquired during that period violates the assumption. A correct continuation must distinguish the period before control, the acquisition measurements and the period consolidated afterward, rather than force the new subsidiary into the opening group or omit its acquisition effects.

26.16.4 A controlled project is not automatically a business combination

Obtaining control does not settle whether the acquired set is a business. AASB 3 distinguishes an in-scope business combination from acquiring an asset or group of assets, a common-control combination, and specified investment-entity acquisitions. An asset-group acquisition under

its scope paragraph does not create goodwill (Australian Accounting Standards Board, 2026h, paras. 2–3).

The optional concentration test can establish that an acquired set is not a business when substantially all of the gross-asset fair value is concentrated as specified. Failing that test does not establish that the set is a business; the further assessment remains necessary. The underlying analysis requires inputs and a substantive process, with different criteria depending on whether outputs are present (Australian Accounting Standards Board, 2026h, paras. B7A–B12D). A project described by a terminal cash payoff supplies neither the workforce and process evidence nor the fair-value composition needed to decide the question.

For an in-scope business combination, the acquisition method requires identifying the acquirer and control date, recognising and measuring the identifiable assets, liabilities and non-controlling interest, and determining goodwill or a bargain-purchase gain. The goodwill calculation uses the specified acquisition-date components, including any relevant previously held interest (Australian Accounting Standards Board, 2026h, paras. 4–10, 32–33). The synthetic funding price, a 5 per cent pricing discount and a balanced four-party journal do not supply these missing measurements. They remain separate economic evidence.

26.16.5 Keep dated authority and missing facts visible

The inspected AASB 18 consequential amendments change specified AASB 3 presentation and disclosure references, add title-related footnotes in AASB 10 and AASB 127, and alter accompanying AASB 10 guidance. They do not in that instrument rewrite the core control, business-definition or separate-policy paragraphs used here (Australian Accounting Standards Board, 2024a, Appendix D, pp. 57, 70, 101–102). That bounded observation cannot establish that no other amendment applies.

AASB 2024-4 defers specified sale-or-contribution amendments involving an investor and its associate or joint venture until annual periods beginning in 2028, with earlier application permitted. It is not a general acquisition-of-control rule (Australian Accounting Standards Board, 2024b, paras. 6–11). AASB 2026-3 changes a downstream associate/joint-venture fair-value election, while AASB 2026-4 contains entity-specific presentation and application changes and references AASB 2026-2 (Australian Accounting Standards Board, 2026d, p. 5) (Australian Accounting Standards Board, 2026f, paras. 1–3). Completing that entity and edition chain is still required before actual 2027 or 2028 admission.

The repaired conditional dependency reference consequently records supplied, dated assessments of rights, control, reporting scope and acquisition type. Unknown endpoints retain alternative control histories; missing measurements stay named requirements. It distinguishes a separate cost-policy choice from evidence of cost, and a positive supplied control assessment from actual source-backed admission. Its current results authorize no posting. This is the useful conclusion for scenario design: unresolved accounting alternatives must remain linked to the rights, dates, entities and measurements that could distinguish them, instead of being collapsed into whichever journal first balances.[1]

[1] Retained official sources, paragraph and physical-page anchors, the conditional analysis and refusal cases are in `data/accounting-master/conversion-control-scope-20261026/`. The original 21 scenarios are retained alongside its adverse boundary cases. The successor in `data/accounting-master/control-boundary-repair-20261026/` checks 30 cases and 23 invalid-input controls. These establish bounded dependency behavior, not actual accounting correctness or complete standards coverage.

26.17 A control history must fit the reporting method

A supplied history of control can answer the continuity question without yet supplying a method for consolidated reporting. The existing selected group-period calculation supports a stable perimeter and full ownership. A conditional history must therefore be checked against that scope before the simulator uses the calculation. Matching opening and closing control edges cannot replace the intervening history.

The checked connection carries the original history request, actor, exact interval and source identities into the group-report request. It identifies the same parent and subsidiary in both. It also requires an explicit relation between the history's reporting basis and the book-composition measurement basis. Similar labels do not establish that those two premises mean the same thing.

Time precision matters here too. The book period uses calendar dates, whereas the control history can use exact instants. The reference requires an explicit fixed-offset calendar interpretation and compares the resulting midnight boundaries exactly. A partial-day acquisition cannot be rounded into whole-period control merely because the annual report's date fields contain no time of day.

The outcome depends on the whole interval. Missing coverage, conflicting evidence or unresolved actor availability leave the control-history requirement unresolved. Continuous absence of control stays outside this parent–subsidiary reporting route, even if an ownership record says 100 per cent. A gain or loss requires changing-perimeter methods; this includes a loss followed by regain when the opening and closing assessments are identical. Continuous control with partial ownership requires non-controlling-interest methods beyond the present full-ownership reference.

Only continuous supplied control together with a separate full-ownership premise fits the existing method's control and ownership scope. Even that result is a candidate for further report checks. It supplies no missing legal-book report, acquisition measurement or accounting authority, and the scope check itself produces no consolidated statement. The original report can still refuse a request whose control scope fits but whose other inputs are absent or incompatible.

This separates two sources of failure. If a valid unchanged-period fixture is refused, the existing report or its inputs need diagnosis. If a history contains a mid-period acquisition and the stable-perimeter report refuses it, the refusal exposes a missing financial method. The next work must specify acquisition or loss accounting, dated member and flow allocation, ownership and non-controlling interests, and the required measurements. A source-history check cannot manufacture those amounts or authorize a journal merely because its identities and dates match.[2]

26.18 Acquiring control changes the measurement and the period

Suppose the financing arrangement eventually gives the creditor control of the debtor. The debtor's legal accounts do not thereby become the creditor's accounts, and the new shares do not supply an acquisition-date balance sheet. Consolidated reporting now needs to explain which assets and liabilities enter the group, at what amounts, and from which instant the acquired company's income belongs in the group result. Multiplying its annual statement by a

[2]The conditional history-to-group connection, complete expected routes and unchanged/changed-perimeter baseline checks are retained in `data/accounting-master/control-group-boundary-20261027/`. The selected comparison covers 23 cases and 15 refusal controls. Every scope result withholds source, accounting, posting and report authorization; it is preparation for financial-method composition.

75 per cent shareholding would answer none of those questions. It would both omit part of the controlled company's activity and bring in activity from before control.

The smallest continuation considered here is one gain of control in an ordinary business acquisition, with a single reporting currency, aligned reporting dates and no subsequent loss of control or change in ownership rights during the selected interval. These are limits of the first method. A purchase of assets that is not a business, a combination under common control, an investment-entity exception and separate financial statements require their own treatments. The business assessment and acquirer identification therefore precede the acquisition calculation. The acquisition date follows the assessed date of obtaining control, rather than the share-issue date by definition (Australian Accounting Standards Board, 2026h, paras. 2–9).

26.18.1 The acquisition balance needs its own measurements

Consider the consolidated acquisition worksheet at that date. Let A be the aggregate recognised acquired assets and L the aggregate assumed liabilities, both expressed as positive magnitudes in the same currency unit. Their amounts follow the applicable recognition and measurement rules, including item-specific exceptions; an unchanged legal carrying amount is not automatically the required acquisition amount. Let T denote measured consideration transferred, Q measured non-controlling interests, and P the acquisition-date fair value of a qualifying previously held equity interest. Under the ordinary business-acquisition branch, the excess described in AASB 3 paragraph 32 is

$$E = T + Q + P - (A - L). \tag{26.6}$$

The calculation compares the consideration and interests in the acquired business with its recognised net assets. A positive excess can determine goodwill only when the route and measurements support that recognition. A negative excess requires the reassessment specified in paragraphs 34–36 before a bargain-purchase gain is recognised. Computing $\max(E, 0)$ and putting the remainder into profit would skip that requirement (Australian Accounting Standards Board, 2026h, paras. 10–21, 24–36).

The distinction between T and P matters particularly in the conversion example. A creditor may hold debt immediately before obtaining shares. That debt is not, merely because it converts, the previously held equity interest addressed by paragraph 42. Nor does the issuer's transfer of a carrying amount from debt to equity determine the holder's acquisition consideration. The contract and its financial-instrument treatment must establish what is transferred and which effects belong to the business combination. Pre-existing relationships and separate transactions require their own assessment (Australian Accounting Standards Board, 2026h, paras. 42, 51–53 and B50–B53). The face amount, an economic price and a carrying amount cannot silently fill the same measurement slot.

Non-controlling interests introduce a further measurement choice. For qualifying present ownership interests, a proportional-share election gives $Q = q(A - L)$, where q is the non-controlling fraction. A fair-value election instead requires a distinct measurement of Q (Australian Accounting Standards Board, 2026h, para. 19). Substituting the proportional expression into equation (26.6) gives

$$E = T + P - (1 - q)(A - L). \tag{26.7}$$

For a conditional 75 per cent ordinary interest, $q = 1/4$ and the acquired net-asset contribution is $3(A - L)/4$. This algebra does not establish that a particular preference share or other right qualifies for the election. With full ownership, $q = 0$ and no non-controlling interest remains. An unspecified prior interest does not become $P = 0$; that zero requires an explicit no-prior-equity premise under the selected route.

26.18.2 Only the controlled interval contributes subsidiary results

After the acquisition worksheet is established, consolidation incorporates the controlled company's income and expenses from the date control is obtained. Those subsequent amounts follow the acquisition-date basis: for example, depreciation uses the acquired asset amount recognised in the consolidated statements (Australian Accounting Standards Board, 2026c, para. 20 and B86–B88). A legal-book expense may therefore need a group adjustment even when the legal journal itself is correct. The simulator must preserve the legal books and put the adjustment in the separate consolidation worksheet.

Write R for the subsidiary's legal profit over the exact post-control interval, and D for the signed additional expense required to express that interval on the group basis. Positive D reduces profit; negative D increases it. Let O be the legal other comprehensive income over the same interval and U its signed group adjustment. All four quantities have the same currency unit. The adjusted subsidiary contribution is

$$\pi = R - D, \qquad \omega = O + U. \tag{26.8}$$

This is a period and measurement reconciliation, not a forecast or a rule for dividing an annual total. A whole-year R cannot be prorated by elapsed months unless an independently supported method justifies that approximation. A journal recorded only by calendar date also needs an ordering premise when control changes within that date; rounding the control instant would conceal the ambiguity.

Under the deliberately limited assumption of constant ordinary present ownership rights, the owners' and non-controlling shares of π are $(1 - q)\pi$ and $q\pi$, and similarly for ω. Actual allocation must consider the relevant rights and the potential-voting-rights exception; the fraction of issued shares is not a universal allocation rule (Australian Accounting Standards Board, 2026c, B89–B95). Let V be distributions to non-controlling interests, expressed as a positive reduction, and M the other signed, supported changes in those interests within this limited branch. Their selected reconciliation is then

$$N_{\text{close}} = Q + q(R - D) + q(O + U) - V + M. \tag{26.9}$$

Starting with measured acquisition-date NCI, the equation adds its share of adjusted profit and OCI and then accounts for direct changes. A sufficiently large loss may make the closing amount negative; clipping that deficit at zero would contradict the attribution requirement (Australian Accounting Standards Board, 2026c, B94). The equation cannot establish whether the supplied M or adjustment population is complete. Ownership transactions with continued control, and a later loss of control, need their separate methods rather than an unexplained entry in M.

26.18.3 Cash and revisions retain their own dates

Acquisition also changes the cash included in the group. The conversion of debt to equity is a noncash transaction, but the acquired company may already hold cash. Its cash enters the consolidated perimeter even when the conversion itself creates no receipt or payment. The method must retain cash consideration, acquired cash and noncash consideration separately, with the applicable acquisition cash presentation net of acquired cash. Acquisitions and disposals are not netted against one another merely because both change control (Australian Accounting Standards Board, 2026b, paras. 39–44B). Cash transfers before control are not retrospectively intragroup flows; transfers afterwards require supported counterpart matching before elimination.

The acquisition worksheet must also distinguish new information about conditions at the acquisition date from later events. Provisional amounts and measurement-period adjustments follow their own rules, including the one-year maximum, rather than a general permission to rewrite the past (Australian Accounting Standards Board, 2026h, paras. 45–50). A revised measurement can therefore have an acquisition-date effect and a later information-availability date. The earlier actor view must remain reproducible, using the temporal distinctions developed in Chapter 17.

26.18.4 A transaction date does not select an edition

The conditional AASB analysis preserves several dates because they answer different questions. For example, the retained AASB 2026-2 compilation was prepared on 21 October 2026, has a compilation date of 31 December 2027, and generally applies to annual reporting periods beginning on or after 1 July 2029, with earlier application permitted under its terms. These dates cannot be substituted for one another. Its amendments to AASB 3 and AASB 10 concern application of the Conceptual Framework; entity classification and elections remain necessary inputs (Australian Accounting Standards Board, 2026e, paras. 3, 18–19, 25; compilation details). The inspected interaction with AASB 2026-4 also imposes a specific AASB 18 condition for early use of that amendment (Australian Accounting Standards Board, 2026f, paras. 3, 25).

Other branches remain visible. AASB 20's Appendix D adds a business-combination treatment for regulatory assets and liabilities, with its own application timing and an additional dependency when applied before AASB 18. The retained appendix alone does not establish the entity's eligibility or the complete transition treatment (Australian Accounting Standards Board, 2026g, Appendix D, amendments to AASB 3). AASB 1061's Appendix C changes tier wording relevant to AASB 10; it does not supply the full Tier 3 scope or measurement rules (Australian Accounting Standards Board, 2026a, Appendix C, amendment to AG1). Similarly, the inspected sale/contribution amendment chain affects a particular loss-of-control branch, rather than replacing every acquisition method (Australian Accounting Standards Board, 2024b, paras. 6–8). The dated authority matrix retains these alternatives and exact paragraph anchors. A future transaction's policy must be checked against its entity, annual period, elections and then-current applicable requirements.

26.18.5 What the selected reference establishes

The conditional expression reference binds the original control-history request and the exact gain-of-control instant before constructing these relations. Each amount remains an explicitly named unknown, with its role, currency, unit, timing, measurement basis and source identity. Future or unknown source availability remains unresolved for the selected actor. Missing assets, consideration, tax effects or adjustment methods cannot be supplied by the fact that an equation balances.

Independent coefficient maps check both NCI elections, qualifying prior equity, the full-ownership limit and the post-control reconciliations. Adverse requests test prior debt labelled as equity, pre-control profit leakage, sub-day rounding, inconsistent currency or ownership, source alteration and attempted insertion of an unmeasured face value. All source, accounting, posting and report authorization remains withheld.[3]

[3]The dated source review is retained in `data/accounting-master/acquisition-period-authority-20261026/`; its 13 primary PDFs have retained identities and inspected anchors. The symbolic design and independent expected expressions are in `data/accounting-master/acquisition-symbolic-20261027/`; the selected check covers 25 cases and 21 refusal controls. These deterministic checks establish the stated local behaviour, not source truth or complete acquisition accounting.

The next financial-method comparison must expand the aggregate unknowns into an identified item population, preserve every legal account, and reconcile a complete dated acquisition worksheet and period report against independently authored expectations. That comparison must include consideration and separate transactions, acquisition-basis expenses, NCI, cash, eliminations, taxes and revisions. The expressions developed here identify the measurements that comparison needs; they do not replace it.

26.19 A complete small acquisition makes the adjustments visible

The unknown amounts in the preceding section protect the real financing question from invented measurements. A separate synthetic example can now test whether the proposed accounting machinery combines those amounts correctly. Every amount in this example is stipulated in one currency unit. It is not a valuation of the creditor or debtor. The example assumes an ordinary business acquisition, a 75 per cent present ownership interest, aligned reporting periods, no prior interest or relationship, and a gain of control on 1 July 2025. Tax recognition and the measured tax amounts are supplied premises; the calculation tests their propagation.

At the start of the year the parent has cash 300 and capital 300. Immediately before acquisition, the subsidiary has cash 50, equipment 70, payables 20, contributed capital 80 and pre-control accumulated profit 20. Its net assets are 100. The parent pays 120 in cash and records an investment 120, leaving its own cash 180. Those legal accounts remain separate and balanced. The subsidiary's accumulated profit 20 belongs to the acquired equity; it does not become current post-control group profit.

The stipulated acquisition basis changes equipment from 70 to 90 and introduces a deferred-tax liability 5. Acquired recognised assets are therefore 140 and liabilities 25, giving net assets 115. With proportional non-controlling interests, $Q = 115/4$ and the positive excess in equation (26.6) is $120 + 115/4 - 115 = 135/4$. Conditional recognition of that excess is an explicit premise of this positive-excess example. The consolidated acquisition worksheet removes the parent's investment 120 and the subsidiary's acquired capital 80 and accumulated profit 20. It adds the equipment adjustment 20, the tax liability 5, goodwill 135/4 and NCI 115/4. Each entry retains its own origin; none is an unexplained balancing amount.

The resulting acquisition balance can be reconstructed from cash 230, equipment 90 and goodwill 135/4, against payables 20, deferred tax 5, parent capital 300 and NCI 115/4. The legal books have not changed. Only the separate consolidation worksheet contains the acquisition adjustments.

26.19.1 The closing report follows the controlled part of the year

After control the subsidiary receives external revenue 40 in cash and records legal depreciation 7. Its legal closing cash is 90 and equipment 63; its post-control legal profit is 33. The acquisition basis requires an additional depreciation 2 and a stipulated release of tax liability 1/2. Thus the group profit contribution is $33 - 2 + 1/2 = 63/2$. The original pre-control profit 20 is still excluded. This is a comparison of complete dated accounts, not a proportional allocation of a whole-year income number.

The closing group holds cash 270, equipment 81 and goodwill 135/4. Its payables are 20 and deferred-tax liability 9/2. Assets total 1539/4, liabilities 49/2 and equity 1441/4. Owners receive three quarters of profit, 189/8, while NCI receives 63/8. Closing owners' equity is 2589/8 and closing NCI 293/8; together they equal 1441/4. The expense and tax adjustments therefore

affect both the asset/liability measurements and the allocation of the controlled-period result.

Choosing fair-value NCI changes this example's acquisition measurement. If the stipulated Q is 40, goodwill is 45 and closing NCI 383/8; the same post-control profit still receives the stipulated 75/25 attribution. Full ownership instead removes NCI, gives goodwill 5 under the same total consideration, and assigns all profit to the owners. A further loss case adds a subsidiary operating expense 200 with a matching external payable. It leaves cash unchanged but produces closing NCI $-107/8$. The reference retains that deficit rather than truncating it, consistently with the attribution boundary already described (Australian Accounting Standards Board, 2026c, B94).

26.19.2 An equity-funded purchase need not consume cash

Hold total stipulated consideration at 120 and vary only its composition. A cash-only purchase uses cash 120 and no new parent equity. An equity-only purchase issues parent equity 120 and uses no cash. A mixed purchase pays cash 50 and issues equity 70. This controlled comparison isolates the accounting consequences of the supplied components; it does not establish that the securities have equivalent market values.

The subsidiary brings cash 50 into the consolidated perimeter in each case. Net acquisition cash is therefore -70, 50 or 0 for the cash-only, equity-only and mixed purchases. With the same later operating receipt 40, closing group cash is 270, 390 or 340. In the equity-only case, acquired cash 50 appears despite the absence of a purchase cash payment. Issuing shares does not create a fictitious cash receipt followed by a payment. The issued equity is retained as a separate noncash consideration component and increases parent capital, while the cash report retains cash paid and cash acquired separately (Australian Accounting Standards Board, 2026b, paras. 39–44B).

This newly issued parent equity must not be confused with the financing example's prior creditor claim. The cash/equity comparison assumes no prior relationship. Extinguishing or converting an existing debt requires the separate consideration, financial-instrument and pre-existing-relationship analysis discussed earlier. Renaming the debt as an equity payment would bypass that unresolved analysis.

26.19.3 A later measurement changes the worksheet and the information view

Suppose the synthetic acquisition equipment amount was provisional. A later proposed measurement changes it from 90 to 100, together with explicitly reviewed dependent amounts: the acquisition tax liability becomes 15/2, additional controlled-period depreciation becomes 3, and the tax release becomes 3/4. These values are supplied together; a revised equipment amount cannot silently inherit unrelated tax or expense assumptions. With proportional NCI, net assets become 245/2, NCI 245/8 and goodwill 225/8. Post-control profit becomes 123/4, owners' profit 369/16 and NCI profit 123/16. In the cash-only case, closing assets are 3105/8, liabilities 107/4 and equity 2891/8. Owners' equity 5169/16 plus NCI 613/16 gives the same total. All original legal accounts remain unchanged.

Those financial effects are conditional on the proposed information being eligible to revise acquisition-date measurements. A later event, a nonprovisional item, a different effective date or information outside the applicable measurement period cannot be treated as the same adjustment merely because the proposed worksheet balances (Australian Accounting Standards Board, 2026h, paras. 45–50). The selected reference receives those dispositions as explicit premises; it does not decide their truth.

Availability introduces a different question. In the example the new information is discovered on 15 January 2026 and reaches the selected actor on 1 February 2026. A 10 January query retains the original even if the exact delivery time is unknown, because delivery cannot precede discovery. A 20 January query likewise retains the original when the February delivery is known. At delivery, eligible supplied information can select the revised conditional report. If delivery is unknown after discovery, both information views may remain possible. If the supplied disposition is a later event, unknown delivery does not make retrospective acquisition adjustment an eligible financial alternative; only the original worksheet remains, with another-method work conditional on the actor receiving that evidence.

The original report must also survive an unsupported proposed method. An undiscovered revision might produce a negative excess that the current positive-excess reference cannot handle. It cannot make the earlier report invalid. Once the revision could be available and eligible, the system retains the original and the complete proposed request with its refusal and missing-method requirement. It does not select the original as if the new information had been resolved, and it does not fabricate a revised result. This distinction preserves history while making the next accounting-method obligation visible.

26.19.4 The comparison remains narrower than company accounting

The selected comparisons inspect every original account, acquisition item, adjustment origin, adjusted account and report field, including cash and parent/NCI equity movements. The repaired revision reference also rebinds the embedded source and control queries to the requested actor and cutoff. Earlier passing candidates were rejected when discriminating cases exposed a pre-discovery alternative, an ineligible adjustment under unknown access, and a future failed candidate that erased the original. Those failures remain in the evidence history.[4]

This remains a four-item synthetic acquisition with stipulated tax treatment, one currency, ordinary rights, one actor and one proposed revision. Prior interests, actual debt conversion, loss or later ownership changes, intragroup trade, general item mappings, foreign exchange, revision chains and complete statutory disclosures need further methods and evidence. The next comparison must add post-control intragroup transactions without leaking pre-control activity or losing the acquired measurement basis.

26.20 An internal sale becomes group profit only through an outside transaction

The acquisition worksheet now has a second task: keep two valid legal books from counting a transaction within the group as income from outside it. Continue the synthetic 75 per cent cash acquisition, but add inventory costing 20 and matching capital of 20 to the parent's opening balances. After obtaining control, the parent sells the whole lot to the subsidiary for 30 on credit. The parent records revenue 30, cost of sales 20 and a receivable 30. The subsidiary records inventory 30 and a payable 30. Each legal book balances, yet their sum includes a profit of 10 before any customer outside the group has bought the goods.

Consolidation removes the reciprocal receivable and payable in full. It also removes the internal revenue 30 against cost of sales 30, then charges the unrealised margin 10 to cost of sales and

[4]The complete cash-only and cash/new-parent-equity comparisons are retained in `data/accounting-master/acquisition-worksheet-20261027/` and `data/accounting-master/acquisition-consideration-20261027/`. The active revision evidence is `data/accounting-master/acquisition-revision-prefix-repair-20261027/`: 26 full cases and 15 refusal controls. Earlier revision profiles are preserved as superseded candidates after their counterexamples. All references withhold actual source, accounting, posting and report authorization.

reduces inventory by 10. After these two profit adjustments, the parent's original cost of sales 20 contributes zero, and the group carries inventory at its original cost of 20. Multiplying the adjustments by the parent's 75 per cent ownership would leave internal profit in group income. Full elimination and attribution of the subsidiary's result are separate operations (Australian Accounting Standards Board, 2026c, paras. B86–B89).

The original acquisition adjustments still apply. Equipment retains the acquired basis and the additional depreciation; the subsidiary's pre-control profit remains outside current group profit. The internal sale takes place after control, so it needs no retrospective reassignment of a pre-control transaction. If the subsidiary subsequently pays the 30, its cash payment and the parent's cash receipt cancel as separate intragroup flows. Settlement removes the reciprocal balances from the legal books, but does not realise the inventory profit for the group.

26.20.1 Removing profit does not remove an external tax obligation

Tax depends on more than the consolidated carrying amount. AASB 112 compares that amount with the appropriate tax base, which may depend on consolidated or separate tax returns. Its illustrative examples include intragroup profit eliminated from inventory without an equivalent adjustment for tax purposes (Australian Accounting Standards Board, 2023, para. 11 and illustrative examples, B11). The accounting rule does not tell us which tax law applies to this company.

To expose this dependency, stipulate that the parent's margin of 10 is taxed now at one quarter, that the subsidiary keeps a tax base of 30, and that the same rate applies when it uses the deduction. The parent therefore has a current tax expense and external payable of 5/2. The group inventory has carrying amount 20 and tax base 30, leaving a deductible temporary difference of 10. If the supplied recoverability conditions permit recognition, the deferred tax asset is also 5/2. Its income offsets the current expense in this example; the external payable remains. If recognition is explicitly withheld, the asset is zero and group profit is lower by 5/2. The reference retains these alternatives separately. If the tax treatment is unknown, it retains the pretax worksheet and requires the missing tax method before selecting a final report.

These are supplied tax and attribution premises. Recognition requires the appropriate taxable-profit assessment, and measurement depends on enacted or substantively enacted rates and expected recovery (Australian Accounting Standards Board, 2023, paras. 24–29, 46–51 and 58). In this downstream example the parent's current tax and the consolidation tax adjustment are attributed to parent owners. That premise does not resolve upstream transactions or every group tax arrangement. The retained source editions support the conditional design; they do not select an actual entity's reporting package or tax law.

With the deferred asset recognised, the unsold-inventory case closes with assets 1629/4, liabilities 27 and equity 1521/4. Parent equity is 2749/8 and NCI is 293/8. Group cash remains 270, whether the internal payable is settled or outstanding. Inventory and the extra opening parent capital account for 20 of the change from the acquisition-only case; the separate tax asset and payable account for 5/2 on each side. No balancing amount is needed.

26.20.2 An outside sale releases only the corresponding part of the margin

Now let the subsidiary sell half the lot to an outside customer for 20 cash before the year closes. The example stipulates a qualifying customer contract, transfer of control and a fixed price, with no returns or additional obligations. A cash receipt alone would not establish those facts (Australian Accounting Standards Board, 2022c, paras. 9, 31–38 and 47). The subsidiary recognises revenue 20 and inventory cost 15. Inventory cost is expensed when the related revenue

is recognised (Australian Accounting Standards Board, 2021, para. 34).

The group still eliminates the full internal sale of 30. Only half of its margin remains in unsold inventory, so the closing unrealised-profit adjustment is 5. The remaining group inventory is 10, while the sold half costs the group 10. The outside sale thus adds group pretax profit of 10: revenue 20 less the original group cost 10. Of this increment, the subsidiary's own sale contributes 5 and the release of the parent's margin contributes 5.

Continue the supplied tax premises. The parent's current tax remains 5/2; the subsidiary's profit of 5 adds current tax 5/4. Only the unsold margin supports the remaining deferred asset, now 5/4. Relative to the acquisition case without inventory trading, the group earns an additional after-tax profit of 15/2. Half arises in the subsidiary after its current tax, and half belongs to parent owners after the downstream consolidation adjustments. Ordinary NCI therefore participates in the subsidiary's 15/4 increment, not in the parent's released margin.

The full closing comparison gives group profit 39: owners receive 483/16 and NCI receives 141/16. Assets are 416, liabilities are 113/4, and equity is 1551/4. Parent equity of 5603/16 plus NCI of 601/16 gives the same total. Cash is 290, including the outside receipt of 20. That operating receipt remains in the group cash report even if the separate internal payment of 30 is also made.

26.20.3 The dated journals make the closing report explainable

The comparison retains every legal entry, including the supplied opening trial balances, the explicitly stipulated pre-control receipt and interim close, acquisition consideration, controlled-period receipts and depreciation, internal sale and settlement, outside sale and current taxes. Replaying these dated journals recovers the opening, acquisition and closing accounts. Their origins distinguish an outside receipt from a reciprocal internal payment; the consolidation entries remain in a separate worksheet. A set of closing balances could not establish that history on its own.

No resale, half resale and full resale are compared under each selected NCI election and consideration mix. Unknown tax preserves the missing dependency instead of completing the report with zero. These results remain a one-lot, same-period downstream example with supplied revenue, tax and ordinary-rights premises.[5] Carrying unsold inventory into another year creates a new question: the next report must release profit from opening group equity and carry its tax basis, rather than repeat the original acquisition or eliminate a sale that occurred in the prior year. That requires a separate two-year comparison.

26.21 The next year starts from closed group equity

Suppose the first year ends with the entire lot still inside the group. The next report cannot eliminate that year's internal sale as if it had happened again. The parent's legal margin has been closed into retained profit, and the group's opening inventory already excludes it. The first task is therefore to carry the two legal books and the consolidation adjustments into the new year without changing the preceding report.

The legal close transfers each entity's income and expense balances into retained profit. The group close also transfers the acquired-basis depreciation and related tax effect into the subsidiary's

[5] The complete downstream comparison is retained in `data/accounting-master/acquisition-intragroup-20261027/`; the dated journal and resale comparison is in `data/accounting-master/acquisition-intragroup-realisation-20261027/`. Both preserve full independent account, journal and report expectations and keep source, posting and report authorisation false.

retained result. NCI's first-year profit is then moved out of owners' retained profit and added to the NCI already recognised at acquisition. In the unsold-inventory example, group opening equity remains 1521/4: parent owners hold 2749/8 and NCI holds 293/8. Carrying the original acquisition NCI alone would omit its first-year profit; adding the whole closing NCI again would count part of it twice.

A separate opening worksheet records the account-by-account difference between the closed legal books and those closed group accounts. It retains the investment/equity elimination, acquired equipment basis, goodwill, inventory margin and tax adjustments, with their original journal sources. This is a carried balance, not another acquisition. Its complete accounts must reproduce the first year's closing assets, liabilities, cash and both equity interests before any second-year transaction is evaluated.

26.21.1 Releasing a carried margin changes current cost of sales

In the second year, let the subsidiary sell the full lot outside the group for 40 cash, under the same supplied revenue-recognition premises. Its legal cost is 30, but the group opening inventory is 20. The legal sale removes 30 from the subsidiary's inventory; the opening group adjustment still contains a reduction of 10. Releasing that carried adjustment debits inventory 10 and credits current cost of sales 10. The group therefore reports cost 20 and zero closing inventory. Debiting opening retained profit a second time would duplicate an adjustment already present in the opening worksheet.

Under the supplied tax assumptions, the subsidiary's current tax on its legal profit is 5/2, and the carried inventory deferred tax asset of 5/2 reverses. The inventory transaction consequently contributes after-tax group profit 15. The subsidiary contributes 15/2 after its current tax; the remaining 15/2 is the parent-attributed release of the downstream margin, net of the deferred tax reversal. These attribution premises remain explicit. An upstream seller would require a different comparison.

26.21.2 Other acquired assets continue to affect the year's result

The equipment did not disappear at the inventory sale. A second-year comparison that freezes it would omit a known accounting dependency. Continue the synthetic example with separately supplied full-year legal depreciation 14, additional acquired-basis depreciation 4 and a tax-liability release of 1. The amounts extend the first year's half-year pattern, but are declared schedules, not inferred useful lives or tax deductions. AASB 10 requires the subsidiary's consolidated expenses to retain the acquisition basis; it does not determine these particular amounts (Australian Accounting Standards Board, 2026c, para. B88).

Equipment contributes a net expense of 17: legal depreciation 14 plus the additional 4, less tax income 1. Combined with inventory profit 15, second-year group profit is therefore -2. The subsidiary-attributed result is $15/2 - 17 = -19/2$, so ordinary NCI receives $-19/8$. Parent owners receive three quarters of that result plus the parent-attributed margin release, giving 3/8. Their sum is -2. The negative NCI result is retained rather than clipped (Australian Accounting Standards Board, 2026c, para. B94).

The closing group holds cash 310, equipment 63 and goodwill 135/4, with no remaining inventory or inventory deferred tax asset. Assets total 1627/4. Liabilities are 57/2: the original external payable 20, acquired deferred tax liability 7/2, and unpaid current taxes of 5. Equity is consequently 1513/4, equal to opening equity 1521/4 less the current loss of 2. The earlier tax payable has not vanished, and no payment has been invented to make the roll-forward balance.

The reference compares complete closed openings, the carried adjustment worksheet, second-year legal and group journals, closing accounts and cash origins. No-sale and partial-sale cases retain the appropriate inventory and tax amounts while still recognising the supplied equipment expenses. Unknown tax continuation retains the prior report and proposed request but selects no second-year final report.[6] This remains a one-lot downstream comparison with supplied continuing control, ordinary rights and financial schedules. The next discriminating case must place the seller inside the partially owned subsidiary and check which profit adjustments then belong to NCI.

26.22 When the subsidiary earns the internal margin

Now let the subsidiary sell the inventory to the parent. The group must still eliminate the full internal sale, but the profit being removed was earned in the company partly owned by NCI. Applying the minority percentage to that company's unadjusted legal profit would allocate profit that the group has not yet earned from an outside customer.

Keep the acquisition on 1 July 2025 unchanged. On 1 September, after control has begun, the subsidiary buys inventory for 20 cash from an outside vendor. On 1 October it sells the lot to the parent for 30 on credit. Buying the lot after control matters: inventory is not inserted into the acquired net assets, and the original goodwill and acquisition NCI remain unchanged. The subsidiary's cash falls by 20. This external payment survives consolidation even if the parent later settles the internal payable of 30. Settlement creates a payment by the parent and a receipt by the subsidiary; both internal cash flows are eliminated in full.

Before any outside resale, the parent's legal inventory is 30 and the subsidiary has legal trading profit 10. Consolidation removes the reciprocal receivable and payable, eliminates internal revenue 30, and reduces inventory by the unrealised margin 10. Group inventory is therefore 20. Neither the inventory adjustment nor the reciprocal-balance elimination is multiplied by the parent's 75% ownership. Full elimination precedes the allocation of the subsidiary's adjusted earnings between parent owners and NCI (Australian Accounting Standards Board, 2026c, paras. B86, B89–B94).

26.22.1 Tax recognition and attribution answer different questions

Continue with explicitly supplied separate tax bases and 25% rates. The subsidiary owes current tax 5/2 on its legal margin. If the parent's inventory tax base remains 30, the difference from the group carrying amount 20 can support a deferred tax asset of 5/2 when the recognition conditions are met. The comparison assumes those conditions in one branch and explicitly withholds recognition in another. The tax base and recognition conditions remain distinct from the inventory elimination (Australian Accounting Standards Board, 2023, paras. 11, 24 and illustrative example B11).

For this example, attribute the inventory elimination and its deferred tax effect to the subsidiary seller's earnings. This is a supplied attribution premise; the cited paragraphs do not themselves determine the example's tax sharing. With recognised deferred tax, removing profit 10 and adding tax income 5/2 offsets the subsidiary's after-tax legal margin 15/2. Its consolidated profit returns to 63/2, the acquired-basis operating result already derived. NCI receives one quarter, or 63/8, and parent owners receive 189/8.

[6] The full comparison and protected inputs are retained in `data/accounting-master/acquisition-intragroup-carry-20261027/`. The pre-execution audit repaired the omitted equipment schedule and avoided a duplicate debit to opening retained profit before running the candidate.

The group now holds cash 250, inventory 20, equipment 81, goodwill 135/4 and the inventory deferred tax asset 5/2. Total assets are 1549/4; liabilities are 27, including the unpaid current tax. Equity is 1441/4. If no inventory deferred tax asset is recognised, the current tax remains an expense. Group profit becomes 29, of which NCI receives 29/4 and parent owners 87/4. Assigning that entire tax expense to parent owners would be wrong under the stated subsidiary-attribution premise. If tax is unresolved, the reference retains the pretax worksheet and the outstanding tax questions but produces no final financial report.

26.22.2 Outside resale releases the subsidiary's margin

Let the parent sell a fraction r of the lot outside the group on 1 December for $40r$ cash, where this finite comparison permits $r = 0$, $1/2$ or 1. Its legal cost is $30r$ and its current tax is $5r/2$, so its own after-tax trading contribution is $15r/2$. The subsidiary's original margin 10 is now realised only in proportion r. Write d for the remaining recognised inventory deferred tax asset: $d = 5(1 - r)/2$ in the recognised branch and zero in the unrecognised branch. Under the supplied attribution, the two trading contributions and resulting NCI profit are

$$\pi_S^{\text{trade}} = 10r - \frac{5}{2} + d, \qquad \pi_P^{\text{trade}} = \frac{15r}{2}, \qquad \pi_{\text{NCI}} = \frac{1}{4}\left(\frac{63}{2} + \pi_S^{\text{trade}}\right).$$

The subsidiary contribution combines realised margin, its unchanged current tax, and the deferred benefit still recognised for unsold inventory. With recognised deferred tax it simplifies to $15r/2$. Half resale therefore adds 15/4 of profit to each company: group profit is 39 and NCI profit is 141/16. The group holds cash 270 and inventory 10; total assets are 396, liabilities 113/4 and equity 1471/4. The external vendor payment and outside customer receipt remain visible separately in the cash record.

Equal recognised-tax trading contributions happen to make several profit totals match the downstream example. That numerical coincidence is not a reason to forget which company earned the margin. The unrecognised-tax branch, the separate entity accounts and the carried opening equity distinguish the two directions. Complete legal journals, account worksheets, cash origins and attribution results are compared across the finite cases.[7] The remaining question is how the subsidiary-attributed margin and NCI move through retained profit when the outside sale occurs in a later year.

26.23 Carrying the subsidiary's margin into the next year

At the first upstream year end, suppose the parent still holds the whole lot. The subsidiary's legal profit is 81/2: operating profit 33 plus internal margin 10 less current tax 5/2. Closing its legal income accounts therefore puts 81/2 into post-control retained profit. The parent has earned no trading profit. These legal retained amounts cannot simply be copied into group equity.

With the inventory deferred tax asset recognised, the subsidiary's consolidated result is 63/2. The group close leaves 189/8 of that result in owners' retained profit and adds 63/8 to acquisition NCI of 115/4. Opening NCI is consequently 293/8. The account-by-account opening bridge reconciles these closed group accounts with the closed legal books, preserving the investment elimination, acquired basis, inventory adjustment and tax. It does not acquire the subsidiary again. Without the inventory deferred tax asset, the corresponding subsidiary result is 29:

[7] The exact upstream comparisons, malformed-request controls and protected sources are in `data/accounting-master/acquisition-intragroup-upstream-20261027/`. The dates, lot, customer-control facts, rights and tax premises are synthetic; no actual source or accounting admission is inferred.

owners retain 87/4 and closing NCI is 36. The tax recognition decision has already affected the equity that must be carried forward.

26.23.1 The parent's sale releases profit attributed to the subsidiary

In year two the parent sells the full lot outside the group for 40 cash. Its legal books recognise cost 30 and current tax 5/2, leaving its own after-tax trading profit 15/2. The group opening inventory adjustment still reduces the parent's inventory by 10. Releasing it requires a debit to the parent's inventory and a credit to the subsidiary's current cost of sales. The debit removes the carried adjustment after the legal sale; the credit locates the realised margin in the subsidiary seller's earnings. A second charge to opening retained profit would duplicate the opening bridge.

Where the inventory deferred tax asset was recognised, its 5/2 reversal is also attributed to the subsidiary under the supplied premise. The subsidiary's trading contribution is therefore 15/2. Continue the explicit equipment schedules: legal depreciation 14, additional acquired-basis depreciation 4, and acquired tax-liability release 1. They contribute a net subsidiary expense of 17, so its total second-year result is −19/2. NCI receives −19/8; parent owners receive 3/8 after adding the parent's own trading result. Group profit is −2. The negative NCI result remains part of the allocation (Australian Accounting Standards Board, 2026c, paras. B88, B94).

Without a recognised inventory deferred tax asset, there is no asset to reverse. The subsidiary instead contributes margin 10 less equipment expense 17, a loss of 7. NCI receives −7/4 and parent owners receive 9/4; group profit is 1/2. This branch earned 5/2 less in the first year and 5/2 more in the second. After full realisation, the two branches therefore reach the same closing equity under the stipulated tax assumptions. Their yearly earnings are different.

Both branches end with cash 290, equipment 63, goodwill 135/4, and no inventory or inventory deferred tax asset. Assets are 1547/4. Liabilities of 57/2 include the original payable 20, remaining acquired deferred tax liability 7/2, and unpaid current taxes 5. Equity is 1433/4, split between owners 324 and NCI 137/4. The prior tax payable is carried; no settlement is invented to obtain these balances.

Complete opening and closing accounts distinguish the attribution even where aggregate profits coincide. Partial resale retains the appropriate margin and tax; an unresolved opening or continuing tax method retains the prior result and proposed continuation without a second-year final report.[8] These acquisition examples all started with no prior interest in the subsidiary. When an investor already holds an interest before obtaining control, the opening investment, acquisition consideration and any remeasurement must be distinguished before the same consolidation logic can be applied.

26.24 An investment already held before control

An investor may already own shares when it obtains control of a business. Suppose the old holding is carried at 35, its acquisition-date fair value is 40, and the investor pays 80 for additional shares. These are three different amounts. The old carrying value describes the opening books; 40 measures the previously held equity at the acquisition date; 80 is the new consideration. Treating 35 as new consideration would misstate the acquisition calculation. Treating the increase from 35 to 40 as cash would invent a receipt.

[8]The upstream carry comparisons, complete expected accounts and rejection controls are retained in `data/accounting-master/acquisition-intragroup-upstream-carry-20261027/`. Continuing control, ordinary rights, depreciation and tax schedules are supplied conditions. Actual source and accounting admission remain unresolved.

For an acquisition achieved in stages, the previously held equity interest is remeasured at acquisition-date fair value. The resulting gain or loss enters profit or loss or other comprehensive income as appropriate; prior OCI follows the basis that would apply to a direct disposal of that interest (Australian Accounting Standards Board, 2026h, paras. 41–42). That rule makes the prior accounting classification consequential. It cannot be selected merely by observing that the investor did not previously control the investee.

26.24.1 No control does not establish the prior accounting method

In the example, the old holding represents 20% of voting power. AASB 128 presumes significant influence at 20% or more unless its absence can be clearly demonstrated. Board representation, participation in policy decisions and other relationships can matter (Australian Accounting Standards Board, 2025, paras. 5–8). Joint control is a separate question. The simulator therefore needs explicit relationship assertions before it can use the restricted AASB 9 route below. A percentage and a no-control assertion are insufficient.

For this synthetic case, stipulate both no significant influence and no joint control immediately before acquisition, together with an AASB 9 fair-value- through-profit-or-loss classification. These are supplied premises, including the rebuttal of the influence presumption, rather than findings about an actual company. An unknown or positive influence or joint-control assertion leaves the method unresolved in this finite example. The particular FVTPL election available to specified venture-capital and similar investors is not presumed (Australian Accounting Standards Board, 2025, para. 18).

Separate financial statements require their own accounting-method choice: AASB 127 identifies cost, AASB 9 and the equity method as alternatives within its conditions (Australian Accounting Standards Board, 2022a, para. 10). Here, explicitly supply an AASB 9/FVTPL separate-book schedule and a fair value of 120 for the combined investment both at acquisition and at closing. The equality $120 = 40 + 80$ in this example is not a valuation rule for a controlling holding. The combined measurement is independently supplied. Also stipulate zero current and deferred tax on the prior-interest gain. This premise leaves the acquired equipment's existing tax schedule intact.

26.24.2 The prior gain belongs to the parent and uses no cash

The parent opens with cash 265, the old investment 35 and capital 300. Immediately before the additional purchase it debits investment 5 and credits its own remeasurement profit 5. It then pays 80 for a further 55% interest: investment is debited 80 and cash credited 80. The resulting 75% holding and control are separately stipulated. The parent's acquisition-date books now contain cash 185 and investment 120. The first entry is noncash; only the second entry pays for new shares.

Retain the earlier subsidiary's acquisition-date recognised assets 140 and liabilities 25, together with the proportional-NCI election and ordinary ownership rights. Identifiable net assets are $140 - 25 = 115$. NCI is therefore $115/4$, while the acquisition calculation combines new consideration 80, previously held equity at fair value 40, and that NCI. Goodwill is

$$80 + 40 + \frac{115}{4} - 115 = \frac{135}{4}.$$

This is the acquisition-method calculation under the supplied measurements (Australian Accounting Standards Board, 2026h, para. 32). The old carrying value 35 does not enter it. The parent investment of 120 is eliminated against the acquired equity and the acquisition adjustments; the parent's gain of 5 remains its own profit.

Continue the earlier subsidiary operating and equipment schedules, without an intragroup inventory sale. Its acquired-basis post-control profit is 63/2. The parent's prior gain is added separately when group profit is calculated:

$$\Pi_{\text{group}} = 5 + \frac{63}{2} = \frac{73}{2}, \qquad \Pi_{\text{owners}} = 5 + \frac{3}{4}\frac{63}{2} = \frac{229}{8}, \qquad \Pi_{\text{NCI}} = \frac{1}{4}\frac{63}{2} = \frac{63}{8}.$$

All three quantities are profits for the illustrated period, in the same monetary units as the accounts. Owners receive the whole parent gain and their share of subsidiary earnings. NCI participates only in the subsidiary result under the supplied ordinary rights. Allocating a quarter of the parent's remeasurement gain to NCI would change the stated economic interest.

The complete closing worksheet contains cash 275, equipment 81 and goodwill 135/4, giving assets 1559/4. Liabilities are 49/2 and equity 1461/4. Owners' closing equity is 2629/8; acquisition NCI 115/4 plus its earnings 63/8 gives closing NCI 293/8. These sum to total equity, which also equals assets less liabilities. The complete legal journals and consolidation worksheet retain the origins of the cash movements; the remeasurement never appears as a cash flow.

Changing the old carrying value to 40 or 45 produces zero gain or a loss of 5 at the same fair value 40. In these comparisons opening cash changes to 260 or 255 so that opening capital remains 300; this is an explicit change in the synthetic starting position. The new cash payment remains 80, and acquisition NCI and goodwill remain unchanged. Group profit becomes 63/2 or 53/2, respectively. This comparison isolates the signed prior-interest result without attributing it to the subsidiary's outside shareholders.

26.24.3 An equity-FVOCI election changes where the gain appears

The same difference between prior fair value and carrying value need not enter profit. For an eligible equity investment with the irrevocable OCI election, AASB 9 sends the applicable fair-value change to OCI. Amounts presented there are not subsequently recycled to profit or loss; a transfer within equity is permitted (Australian Accounting Standards Board, 2026i, para. 5.7.5 and B5.7.1). Applying the direct-disposal basis in the acquisition rule must preserve that distinction.

Write C for the old carrying value, P for its acquisition-date fair value, and O for accumulated prior OCI. The conditional current update is $P - C$. Under the eligible equity-FVOCI branch it enters OCI, leaving the reserve $O + P - C$. An explicit transfer choice either retains that reserve or transfers it within equity, with no recycling gain in profit. The FVTPL branch instead puts $P - C$ in parent pretax profit and requires zero prior OCI in this finite specification. These symbolic presentation branches do not supply a complete FVOCI tax or legal-book worksheet. Missing classification, relationship, measurement, tax or timely source premises remain unresolved.[9]

A debt claim held before a conversion is not previously held equity simply because the conversion produces shares. Its extinguishment or other settlement and any pre-existing relationship require their own transaction analysis; AASB 3 separates qualifying acquisition consideration from separate transactions (Australian Accounting Standards Board, 2026h, paras. 51–52 and B50–B53). That distinction determines the next extension of the financing example.

[9]The preserved expressions, repaired relationship scope and complete FVTPL worksheet are recorded in `data/accounting-master/acquisition-prior-interest-20261027/`, `data/accounting-master/acquisition-prior-interest-scope-20261027/`, and `data/accounting-master/acquisition-prior-worksheet-20261027/`. The earlier expression algebra alone did not establish classification scope. The repaired scope records supplied assertions, not actual source admission.

26.25 One debt exchange, two sets of accounts

Suppose a company extinguishes debt of 100 by issuing its own shares, valued at 70, to the lender. The borrower removes a liability and issues equity; the lender removes a financial asset and receives a different asset. Neither side receives cash. A gain for one party and a loss for the other can arise, but the two parties need their own classifications, measurements and journals. The shares received also raise a separate question about the lender's subsequent relationship with the borrower.

26.25.1 The debtor must first establish the scope of the exchange

AASB Interpretation 19 addresses a debtor's renegotiated financial liability settled by issuing equity. It does not address creditor accounting. It also excludes an existing shareholder acting as an owner, common control before and after the exchange together with equity-distribution or contribution substance, and share settlement in accordance with the liability's original terms (Australian Accounting Standards Board, 2020, paras. 2–3). An original contractual conversion therefore cannot enter this debtor example merely because it produces shares.

The compound conditions matter. An existing shareholding alone does not say that the creditor acts as an owner in this transaction. Common control alone does not establish the specified equity substance. The conditional scope calculation keeps yes, no and unknown assertions separate. A known exclusion remains effective even when another condition is unknown; otherwise missing scope facts leave the method unresolved. The same discipline applies to source availability at the stated information cutoff.

For the finite example, supply a renegotiated, whole-liability extinguishment within the Interpretation's scope, a financial liability and an issue of own equity. The borrower opens with cash 30, equipment 100, debt 100 and contributed capital 30. There is no other activity, interest, transaction cost, current or deferred tax, OCI or cash leg during the illustrated interval. Each absence is an explicit synthetic premise.

When the issued equity's fair value can be reliably measured, the debtor uses that fair value. Only when it cannot be reliably measured does the Interpretation direct measurement by reference to the extinguished liability's fair value (Australian Accounting Standards Board, 2020, paras. 6–7). Carrying amount is not a substitute for missing fair value. At the extinguishment date, the difference between the liability carrying amount and consideration paid enters profit or loss (Australian Accounting Standards Board, 2020, paras. 5, 9).

At the supplied equity value 70, the debtor debits debt 100, credits newly issued equity 70 and credits gain 30. Closing cash remains 30 and equipment 100; liabilities are zero. Equity of 130 comprises opening capital 30, new equity 70 and current profit 30. Operating, investing and financing cash flows are all zero, while the noncash debt-for-equity exchange is identified separately. The Interpretation's conceptual analysis using an issue for cash followed by repayment does not establish actual cash movements in this example (Australian Accounting Standards Board, 2020, BC13, BC25).

An equity value of 100 gives no gain; 120 gives a loss of 20. If equity fair value explicitly cannot be measured reliably and liability fair value is independently supplied as 60, the measured equity is 60 and the gain is 40. All four cases leave assets and total equity at 130 but change the division between contributed equity and earnings. A partial exchange needs an allocation between the extinguished portion and any modification of the remaining debt; it is outside this complete whole-debt worksheet (Australian Accounting Standards Board, 2020, paras. 8, 10–11).

26.25.2 The creditor must separate impairment from disposal

Now consider the lender's books. Supply a loan measured at amortised cost, with the relevant business-model and contractual-cash-flow conditions, followed by expiry of the entire contractual claim. These are distinct accounting premises (Australian Accounting Standards Board, 2026i, paras. 3.2.3, 4.1.2). For the shares received, stipulate no control, significant influence or joint control, an AASB 9 FVTPL classification and no OCI election. No-control alone would again be insufficient to establish that investment method.

The creditor opens with cash 30, gross loan 100, no loss allowance and capital 130. Before the exchange, an independently supplied expected-loss assessment requires an allowance of 30. The creditor debits impairment expense 30 and credits the allowance 30, leaving a net loan balance of 70. An allowance update belongs in impairment profit or loss; it must not disappear into a later net disposal calculation (Australian Accounting Standards Board, 2026i, paras. 5.2.2, 5.5.8).

At exchange, shares with supplied fair value 70 replace that net claim. The creditor debits shares 70 and the allowance 30, and credits gross loan 100. There is no disposal gain or loss. Total profit for the interval is a loss of 30, entirely from impairment. Closing cash 30 plus shares 70 gives assets 100; opening capital 130 less the loss gives equity 100. Cash flows remain zero. The complete record shows both journals and the intermediate assessed loan, so a correct closing total cannot conceal an omitted impairment step.

To see why this matters, supply a different assessment: allowance zero before receiving the same shares worth 70. The disposal then gives a loss of 30 and the total loss is again 30. This is a different conditional accounting history. The illustration does not infer expected losses from the share value or certify either assessment for an actual borrower. With allowance zero, shares worth 100 or 120 instead produce zero disposal result or a gain of 20. With allowance 30 and shares worth 60, impairment loss 30 is followed by disposal loss 10. Whole-asset derecognition and amortised-cost gains and losses have their own requirements (Australian Accounting Standards Board, 2026i, paras. 3.2.12, 5.7.2).

Initial share recognition also needs a measurement distinct from the debtor's fallback. The example supplies share fair value and initial transaction price as equal, with unchanged fair value at closing. A difference between initial fair value and transaction price invokes the additional conditions in AASB 9; the finite worksheet leaves it unresolved rather than automatically recognising a day-one gain (Australian Accounting Standards Board, 2026i, paras. 5.1.1–5.1.1A, B5.1.2A). Missing share value cannot be replaced by the debtor's liability value. Zero tax, fees, interest, other activity and cash legs remain supplied premises.

The matched 70, 100 and 120 examples give opposite total debtor and creditor results under their specified carrying amounts, share values and absence of other effects. That agreement does not make the accounting methods identical. It is the complete component accounts, measurement premises and dated source bindings that permit the comparison.[10]

If the new shares confer control, the creditor's noncontrolling worksheet no longer supplies the acquisition accounting. The debt settlement, consideration for the business and any pre-existing relationship must be distinguished before goodwill or consolidated statements can be measured. The next question is how to preserve that separation through the control-changing exchange.

[10]Scope truth tables and complete conditional statements are retained in `data/accounting-master/acquisition-debt-relationship-20261027/`, `data/accounting-master/debtor-conversion-worksheet-20261027/`, and `data/accounting-master/creditor-conversion-worksheet-20261027/`. The creditor plan was repaired before execution to make impairment and initial measurement explicit. Unknown premises retain the request without a report; actual source and accounting admission remain open.

26.26 Separating consideration from the other parts of an exchange

Obtaining control through a debt exchange raises a question that the separate debtor and creditor worksheets cannot answer alone: which amounts belong to the acquisition of the business? A transaction may also settle an existing relationship or pay for services. Before measuring goodwill, the company must distinguish consideration for the acquired business from effects accounted for separately. The acquisition guidance requires that separation and discusses the reasons for a transaction, who initiated it and its timing; no one indicator by itself settles the classification (Australian Accounting Standards Board, 2026h, paras. 51–53 and B50–B53).

Consider an explicitly supplied example in one currency at one measurement date. Four identified components are assigned values of 80 for acquisition consideration, 20 for separate debt settlement, 5 for acquisition costs and 3 for another separately accounted transaction. Their inventory sums to 108, while the declared consideration bucket contains only 80. The difference is not a residual to be forced into goodwill. Nor does this example infer the 20 settlement amount from a loan's face or carrying value: both the amount and its assigned purpose are supplied premises.

The finite calculation gives each component a distinct identity and retains its source, availability date, amount and assigned accounting purpose. It requires an explicit assertion that the supplied inventory is complete for the question. Reordering the four components leaves the same records and sums. An unknown settlement amount is not zero; missing allocation, coverage or source-timing premises leave the requested totals unresolved. Relabelling debt as a previously held equity interest or as acquisition consideration cannot establish the missing accounting premise.

These checks establish arithmetic conditional on the supplied partition. They do not determine the appropriate measured allocation for the intended conversion-to-control transaction. The inspected acquisition paragraphs explain separation and settlement of pre-existing relationships, but do not by themselves establish that loan-specific allocation. The remaining work is to support the relationship treatment, transaction assignments and measurement basis before constructing goodwill and acquired-basis statements.[11]

[11] `data/accounting-master/conversion-control-transaction-20261027/` retains the source inquiry, 33 finite cases including all 24 orderings of the four components, 21 refusal controls and 351 assertions. The calculation produces no goodwill, journals or consolidated report. The saved source inquiry distinguishes an irrelevant supplemental document from unavailable further support; neither is evidence that applicable guidance does not exist.

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